

ONLINE SUPPLEMENT TO “INFERENCE FOR HETEROGENEOUS EFFECTS USING LOW-RANK ESTIMATION OF FACTOR SLOPES”

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ABSTRACT. This document contains the estimation algorithm for the multivariate case, additional simulations, and all proofs.

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APPENDIX A. ESTIMATION ALGORITHM FOR THE MULTIVARIATE CASE

Consider

$$\begin{aligned} y_{it} &= \sum_{r=1}^R x_{it,r} \theta_{it,r} + \alpha'_i g_t + u_{it}, \quad i = 1, \dots, N, \quad t = 1, \dots, T. \\ \theta_{it,r} &= \lambda'_{i,r} f_{t,r}. \end{aligned}$$

where $(x_{it,1}, \dots, x_{it,R})'$ is an R -dimensional vector of covariates. Each coefficient $\theta_{it,r}$ admits a factor structure with $\lambda_{i,r}$ and $f_{t,r}$ as the “loadings” and “factors.” The factors and loadings are $\theta_{it,r}$ specific, but are allowed to have overlap. Here $(R, \dim(\lambda_{i,1}), \dots, \dim(\lambda_{i,R}))$ are all assumed fixed.

Suppose $x_{it,r} = \mu_{it,r} + e_{it,r}$. We further rely on a known structure under which μ_{it} can be estimated well. For instance, we present results under the assumption that $\mu_{it,r} = l'_{i,r} w_{t,r}$ also admits a factor structure. After partialing out $\mu_{it,r}$, the model can be written as

$$\dot{y}_{it} = \sum_{r=1}^R e_{it,r} \lambda'_{i,r} f_{t,r} + \alpha'_i g_t + u_{it}, \quad \dot{y}_{it} = y_{it} - \sum_{r=1}^R \mu_{it,r} \theta_{it,r}$$

As such, it is straightforward to extend the estimation algorithm from the univariate case presented in the main paper to the multivariate case.

Let X_r be the $N \times T$ matrix of $x_{it,r}$, and let Θ_r be the $N \times T$ matrix of $\theta_{it,r}$. We first estimate the low rank matrices Θ_r using penalized nuclear-norm regression. We then apply sample splitting, and employ steps 2-4 to iteratively estimate $(f_{t,r}, \lambda_{i,r})$. The formal algorithm is stated as follows.

Algorithm A.1. Estimate $\theta_{it,r}$ as follows.

Step 1. Estimate the number of factors. Run nuclear-norm penalized regression:

$$(\widetilde{M}, \{\widetilde{\Theta}_r\}) := \arg \min_{M, \{\Theta_r\}} \|Y - M - \sum_{r=1}^R X_r \odot \Theta_r\|_F^2 + \nu_0 \|M\|_n + \sum_{r=1}^R \nu_r \|\Theta_r\|_n.$$

Estimate $K_r = \dim(\lambda_{i,r})$ for $r = 1, \dots, R$, $K_0 = \dim(\alpha_i)$ as

$$\widehat{K}_r = \sum_i 1\{\psi_i(\widetilde{\Theta}_r) \geq (\nu_r \|\widetilde{\Theta}_r\|)^{1/2}\}, \quad \widehat{K}_0 = \sum_i 1\{\psi_i(\widetilde{M}) \geq (\nu_0 \|\widetilde{M}\|)^{1/2}\}.$$

Step 2. Estimate the structure $x_{it,r} = \mu_{it,r} + e_{it,r}$.

In the factor model, use the PC estimator to obtain $(\widehat{l'_{i,r} w_{t,r}}, \widehat{e}_{it,r})$ for all $i = 1, \dots, N$, $t = 1, \dots, T$ and $r = 1, \dots, R$.

Step 3: Sample splitting. Randomly split the sample into $\{1, \dots, T\}/\{t\} = I \cup I^c$, so that $|I|_0 = [(T-1)/2]$. Denote the $N \times |I|_0$ matrices of $(y_{is}, x_{is,r})$ for observations $s \in I$ as $Y_I, X_{I,r}$. Estimate the low-rank matrices $\{\Theta_r\}$ and M as in step 1, with (Y, X_r) replaced with $(Y_I, X_{I,r})$, and obtain $(\widetilde{M}_I, \widetilde{\Theta}_{I,r})$.

Let $\widetilde{\Lambda}_{I,r} = (\widetilde{\lambda}_{1,r}, \dots, \widetilde{\lambda}_{N,r})'$ be the $N \times \widehat{K}_r$ matrix whose columns are defined as $\sqrt{N}$ times the first $\widehat{K}_r$ eigenvectors of $\widetilde{\Theta}_{I,r} \widetilde{\Theta}_{I,r}'$. Let $\widetilde{A}_I = (\widetilde{\alpha}_1, \dots, \widetilde{\alpha}_N)'$ be the $N \times \widehat{K}_0$ matrix whose columns are defined as $\sqrt{N}$ times the first $\widehat{K}_0$ eigenvectors of $\widetilde{M}_I \widetilde{M}_I'$.

Step 4. Estimate the “partialled-out” components.

Substitute in $(\widetilde{\alpha}_i, \widetilde{\lambda}_{i,r})$, and define

$$(\widetilde{f}_{s,r}, \widetilde{g}_s) := \arg \min_{f_{s,r}, g_s} \sum_{i=1}^N (y_{is} - \widetilde{\alpha}'_i g_s - \sum_{r=1}^R x_{is,r} \widetilde{\lambda}'_{i,r} f_{s,r})^2, \quad s \in I^c \cup \{t\}.$$

and

$$(\dot{\lambda}_{i,r}, \dot{\alpha}_i) = \arg \min_{\lambda_{i,r}, \alpha_i} \sum_{s \in I^c \cup \{t\}} (y_{is} - \alpha'_i \widetilde{g}_s - \sum_{r=1}^R x_{is,r} \lambda'_{i,r} \widetilde{f}_{s,r})^2, \quad i = 1, \dots, N.$$

Step 5. Estimate $(f_{t,r}, \lambda_{i,r})$ for inference.

For all $s \in I^c \cup \{t\}$, let

$$(\widehat{f}_{I,s,r}, \widehat{g}_{I,s}) := \arg \min_{f_{s,r}, g_s} \sum_{i=1}^N (\widehat{y}_{is} - \widetilde{\alpha}'_i g_s - \sum_{r=1}^R \widehat{e}_{is,r} \widetilde{\lambda}'_{i,r} f_{s,r})^2.$$

Fix $i \leq N$,

$$(\hat{\lambda}_{I,i,r}, \hat{\alpha}_{I,i}) = \arg \min_{\lambda_{i,r}, \alpha_i} \sum_{s \in I^c \cup \{t\}} (\hat{y}_{is} - \alpha'_i \hat{g}_{I,s} - \sum_{r=1}^R \hat{e}_{is,r} \lambda'_{i,r} \hat{f}_{I,s,r})^2.$$

where $\hat{y}_{is} = y_{is} - \widehat{l'_{i,r} w_{s,r} \dot{\lambda}'_{i,r} \tilde{f}_{s,r}}$, $\hat{e}_{is,r} = x_{is,r} - \widehat{l'_{i,r} w_{s,r}}$ in the factor model.

Step 6. Estimate $\theta_{it,r}$. Repeat steps 3-5 with I and I^c exchanged, and obtain $(\hat{\lambda}_{I^c,i,r}, \hat{f}_{I^c,s,r} : s \in I \cup \{t\}, i \leq N, r \leq R)$. Define

$$\hat{\theta}_{it,r} := \frac{1}{2} [\hat{\lambda}'_{I,i,r} \hat{f}_{I,t,r} + \hat{\lambda}'_{I^c,i,r} \hat{f}_{I^c,t,r}].$$

The asymptotic variance can be estimated by $\hat{v}_{\lambda,r} + \hat{v}_{f,r}$, where

$$\begin{aligned} \hat{v}_{\lambda,r} &= \frac{1}{2N} (\hat{\lambda}'_{I,r,\mathcal{G}} \hat{V}_{\lambda,1}^{I-1} \hat{V}_{\lambda,2}^I \hat{V}_{\lambda,1}^{I-1} \hat{\lambda}_{I,r,\mathcal{G}} + \hat{\lambda}'_{I^c,r,\mathcal{G}} \hat{V}_{\lambda,1}^{I^c-1} \hat{V}_{\lambda,2}^{I^c} \hat{V}_{\lambda,1}^{I^c-1} \hat{\lambda}_{I^c,r,\mathcal{G}}) \\ \hat{v}_f &= \frac{1}{2T|\mathcal{G}|_0} (\hat{f}_{I,t,r} \hat{V}_{I,f} \hat{f}_{I,t,r} + \hat{f}_{I^c,t,r} \hat{V}_{I^c,f} \hat{f}_{I^c,t,r}) \\ \hat{V}_{S,f} &= \frac{1}{|\mathcal{G}|_0 |S|_0} \sum_{s \notin S} \sum_{i \in \mathcal{G}} \hat{\Omega}_{S,i,r} \hat{f}_{S,s,r} \hat{f}'_{S,s,r} \hat{\Omega}_{S,i,r} \hat{e}_{is,r}^2 \hat{u}_{is}^2 \\ \hat{V}_{\lambda,1}^S &= \frac{1}{N} \sum_j \hat{\lambda}_{S,r,j} \hat{\lambda}'_{S,r,j} \hat{e}_{jt,r}^2 \\ \hat{V}_{\lambda,2}^S &= \frac{1}{N} \sum_j \hat{\lambda}_{S,r,j} \hat{\lambda}'_{S,r,j} \hat{e}_{jt,r}^2 \hat{u}_{jt}^2 \\ \hat{\lambda}_{S,r,\mathcal{G}} &= \frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \hat{\lambda}_{S,r,i}, \quad \hat{\Omega}_{S,i,r} = \left(\frac{1}{|S|_0} \sum_{s \in S} \hat{f}_{S,s,r} \hat{f}'_{S,s,r} \right)^{-1} \left(\frac{1}{T} \sum_{s=1}^T \hat{e}_{is,r}^2 \right)^{-1} \end{aligned}$$

It is also straightforward to extend the univariate asymptotic analysis to the multivariate case, and establish the asymptotic normality for $\hat{\theta}_{it,r}$. The proof techniques are the same, subject to more complicated notation. Therefore, our proofs below focus on the univariate case.

APPENDIX B. FURTHER EMPIRICAL RESULTS

We present numeric summaries of results obtained by breaking the sample period into three time spans in Table B.1. Specifically, we consider the three periods defined as the first quarter of 1990 through the last quarter of 1993, the first quarter of 1994 through the last quarter of 1999, and the first quarter of 2000 through the second quarter of 2006. Within each window for each of the high- and low-minimum group, we report several quantities. As a simple summary of the

estimated effect sizes, we report the time series average of the time specific within group means in column $\hat{\theta}_{\mathcal{G}}$ - i.e. we report $\frac{1}{|S|_0} \sum_{t \in S} \hat{\theta}_{\mathcal{G},t}$ where $\hat{\theta}_{\mathcal{G},t} = \frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \hat{\theta}_{it}$ and S denotes one of the three time windows.

To summarize statistical significance of the estimated average effects, we compute the fraction of time periods within each time window for which the estimate of the group-level average effect is negative and the corresponding 95% pointwise confidence interval excludes zero and the fraction of observations for which the estimate of the group level average effect is positive and the corresponding 95% pointwise confidence interval excludes zero broken out by group. Specifically, for each $\mathcal{G} \in \{\text{Low}, \text{High}\}$ and for S denoting each of the time windows, we compute

$$\begin{aligned} P_{\mathcal{G}}^{+} &= \frac{1}{|S|_0} \sum_{t \in S} 1\{\hat{\theta}_{\mathcal{G},t} > 0\} 1\{|\hat{\theta}_{\mathcal{G},t}/s.e.(\hat{\theta}_{\mathcal{G},t})| > 1.96\} \\ P_{\mathcal{G}}^{-} &= \frac{1}{|S|_0} \sum_{t \in S} 1\{\hat{\theta}_{\mathcal{G},t} < 0\} 1\{|\hat{\theta}_{\mathcal{G},t}/s.e.(\hat{\theta}_{\mathcal{G},t})| > 1.96\} \end{aligned}$$

where $s.e.(\hat{\theta}_{\mathcal{G},t})$ denotes the estimated standard error of $\hat{\theta}_{\mathcal{G},t}$.

Similarly, we summarize significance of the effects at the individual level by reporting the fraction of (i, t) combinations within each window where the estimated θ_{it} is positive and significant and the fraction of (i, t) combinations where the estimated θ_{it} is negative and significant:

$$\begin{aligned} Q_{\mathcal{G}}^{+} &= \frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \frac{1}{|S|_0} \sum_{t \in S} 1\{\hat{\theta}_{it} > 0\} 1\{|\hat{\theta}_{it}/s.e.(\hat{\theta}_{it})| > 1.96\} \\ Q_{\mathcal{G}}^{-} &= \frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \frac{1}{|S|_0} \sum_{t \in S} 1\{\hat{\theta}_{it} < 0\} 1\{|\hat{\theta}_{it}/s.e.(\hat{\theta}_{it})| > 1.96\}. \end{aligned}$$

Looking at Table B.1, we see that the estimated average effect in both groups is negative during the 1990 to 1993 period. The average in the high-minimum wage group suggests a relatively small negative effect (-0.017), while the estimated average effect in the low-minimum-wage groups suggests a more substantial negative impact of the minimum wage on youth employment (-0.083). However, estimated average effects and individual state effects are also largely insignificant over this period. Over the period from 1994 to 1999, we see small estimated overall average effects, but these obscure substantial volatility in estimated effects over this period. Both the high and low minimum wage groups have substantial fractions of positive and pointwise significant (25% and 33% respectively) and negative and

TABLE B.1. Summary of Estimated Minimum Wage Effects within Groups and Time Windows

	1990 through 1993			1994 through 1999			2000 through 2006		
	P_G^+	P_G^-	$\hat{\theta}_G$	P_G^+	P_G^-	$\hat{\theta}_G$	P_G^+	P_G^-	$\hat{\theta}_G$
High min-wage	0	0	-0.017	0.250	0.166	-0.007	0.192	0.076	0.125
Low min-wage	0	0.062	-0.083	0.333	0.208	0.012	0.269	0.076	0.650
	Q_G^+	Q_G^-		Q_G^+	Q_G^-		Q_G^+	Q_G^-	
High min-wage	0.021	0.014		0.073	0.084		0.063	0.029	
Low min-wage	0.024	0.036		0.208	0.158		0.187	0.051	

Note: This table reports summaries of group level average effects within time windows. P_G^+ and P_G^- respectively denote the fraction of time periods within the specified window where the estimated group-level average effect is positive and statistically significant at the 5% level and the fraction of time periods within the specified window where the estimated group-level average effect is negative and statistically significant at the 5% level. $\hat{\theta}_G$ denotes the within time window average of the group-level average effects. Q_G^+ and Q_G^- respectively denote the fraction of (state, time) combinations within the specified window where the estimated θ_{it} is positive and statistically significant at the 5% level and where the estimated θ_{it} is negative and statistically significant at the 5% level.

pointwise significant (17% and 21% respectively) estimated effects. Individual effects are, of course, noisier than the group-level averages, but we also see similar patterns in significance of the state-level estimates. Estimated effects in the last time block are still relatively volatile with modest amounts of positive and negative pointwise significant estimates. However, at least on average, the estimates point to relatively large positive average effects of the minimum wage within both groups over this period. Overall, these patterns are consistent with there being substantial heterogeneity in the coefficient on the minimum wage, with the minimum wage having both statistically significant large positive and large negative effects on youth employment depending on the time period.

APPENDIX C. POLICY RELEVANT TESTS

We consider four potentially policy relevant tests in this section. The last two tests are also presented in the main text.

$$\text{Section C.2: } H_0 : \bar{\theta}_{\mathcal{G},t_1} = \bar{\theta}_{\mathcal{G},t_2}$$

$$\text{Section C.3: } H_0 : \theta_{1,t} = \dots = \theta_{N_0,t}, \quad \forall i \in \mathcal{G}, \forall t \leq T.$$

Section C.4: $H_0 : \bar{\theta}_{\mathcal{G}_1,t} = \dots = \bar{\theta}_{\mathcal{G}_J,t}.$

Section C.5: $H_0 : \theta_{it} = 0, \quad \forall i \leq N.$

C.1. Simulations. First, we illustrate testing for homogeneity of average effects across groups. That is, we consider testing

$$H_0^1 : \bar{\theta}_{\mathcal{G}_1,t_0} = \dots = \bar{\theta}_{\mathcal{G}_J,t_0}$$

where $\mathcal{G}_1, \dots, \mathcal{G}_J$ are pre-specified groups of units, t_0 is a given time point of interest, and $\bar{\theta}_{\mathcal{G}_j,t_0}$ denotes the average of θ_{it_0} in group $\mathcal{G}_j$. The second example we consider tests the null hypothesis

$$H_0^2 : \theta_{i,t_0} = 0, \forall i$$

at a given time point of interest t_0 . As noted in Section 3.4, this latter hypothesis is equivalent to testing $f_{t_0,1} = 0$. In our simulation examples, we set $t_0 = 1$.

As in the preceding simulations, we generate outcomes as

$$y_{it} = \alpha'_i g_t + x_{it,1} \theta_{it} + x_{it,2} \beta_{it} + u_{it}$$

where $\theta_{it} = \lambda'_{i,1} f_{t,1}$ and $\beta_{it} = \lambda'_{i,2} f_{t,2}$. β_{it} , α_i , and g_t are generated as in the previous simulation sections. The design for θ_{it} deviates somewhat from that used in previous simulations in that we impose the relevant null hypothesis for testing H_0^1 or H_0^2 when generating data so that we can look at size of the corresponding tests. We also generate data under structured deviations from the relevant null to provide some evidence about power of the tests.

For testing the null of homogeneity of group level average effects, we specify $\bar{\lambda}_{\mathcal{G}_j}$ as

$$\bar{\lambda}_{\mathcal{G}_j} = \begin{cases} 1 & \text{under the null} \\ 1 + Z_j N^{-a} & \text{under the alternative} \end{cases}, \quad j = 1, \dots, J$$

where $Z_j \in \mathcal{N}(0,1)$ and the parameter a measures the deviation from the null when generating data under the alternative. We set $J = 10$ and pre-specify each group to contain N/J elements. We then generate loadings as

$$\lambda_{i,1} = \bar{\lambda}_{\mathcal{G}_j}, \text{ for all } i \in \mathcal{G}_j,$$

and generate factors as in the designs in the previous sections.

TABLE C.1. Rejection Frequencies

		Test of Group Homogeneity			
N	T	null	$a = 3/4$	$a = 1/2$	$a = 1/4$
50	50	0.047	0.132	1.000	1.000
100	100	0.047	0.230	0.990	1.000
200	200	0.039	0.790	1.000	1.000
		Test of Joint Significance			
N	T	null	$a = 1/2$	$a = 1/4$	$a = 0$
50	50	0.038	0.090	0.384	0.972
100	100	0.054	0.144	0.874	1.000
200	200	0.057	0.214	0.940	1.000

Note: This table reports rejection frequencies for 5% level tests. The column “null” reports size of the test. The remaining columns give power against alternatives as defined in the main text. Results are based on 1000 simulation replications.

To generate data for examining tests of H_0^2 , we impose null and local alternatives on $f_{1,1}$:

$$f_{1,1} = \begin{cases} 0 & \text{under the null} \\ N^{-a} & \text{under the alternative} \end{cases}.$$

Again, a measures the deviation from the null when generating data under the alternative. We generate the factors in the remaining time periods and the loadings as in the previous simulation sections.

To test the hypotheses, we use the chi-square tests presented in Theorem 3.2. Table C.1 reports rejection frequencies of 5% level tests across 1000 simulation replications. The top panel, labeled “Test of Group Homogeneity,” provides results for testing H_0^1 ; and the bottom panel, labeled “Test of Joint Significance,” provides results for testing H_0^2 . Overall, the results show good size control and reasonable power for both tests.

C.2. Group effect dynamics of two periods. Consider testing

$$H_0 : \bar{\theta}_{\mathcal{G},t_1} = \bar{\theta}_{\mathcal{G},t_2}.$$

for two given periods t_1 and t_2 .

While this test involves two time periods t_1, t_2 , we do not have to run the algorithm twice to respectively estimate them. Instead, a simple modification of Algorithm 2.2 yields unbiased estimated θ_{it} at both t_1, t_2 . The main modification is to replace t by the two periods when doing sample splitting. The modified algorithm is as follows,

Algorithm C.1. Estimate $\theta_{i,t_1}, \theta_{i,t_2}$ as follows.

Step 1: Same as Algorithm 2.2

Step 2: Sample splitting. Randomly split the sample into $\{1, \dots, T\} / \{t_1, t_2\} = I \cup I^c$, so that $|I|_0 = [(T - 2)/2]$. Denote respectively by Y_I, X_I as the $N \times |I|_0$ matrices of (y_{is}, x_{is}) for observations $s \in I$. Obtain $(\tilde{\Lambda}, \tilde{A})$ on the two subsamples as in Algorithm 2.2.

Step 3-6. Same as Algorithm 2.2, with $\{t_1, t_2\}$ in place of $\{t\}$. Obtain $(\hat{\lambda}_{I,i}, \hat{f}_{I,t}), (\hat{\lambda}_{I^c,i}, \hat{f}_{I^c,t})$ for $t \in \{t_1, t_2\}$, and hence

$$\begin{aligned}\hat{\theta}_{it} &:= \frac{1}{2}[\hat{\lambda}'_{I,i}\hat{f}_{I,t} + \hat{\lambda}'_{I^c,i}\hat{f}_{I^c,t}], \quad t \in \{t_1, t_2\}, \\ \hat{\theta}_{\mathcal{G},t} &:= \frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \hat{\theta}_{it}.\end{aligned}$$

We shall focus on the case $f_{t_1} \neq f_{t_2}$ for the two periods. If $f_{t_1} = f_{t_2}$, then the problem is not just testing the equality for a particular group, but would be for all individuals. In that case, the test statistic would be simply based on $\hat{f}_{t_1} - \hat{f}_{t_2}$, whose asymptotic null distribution is also easy to derive.

Theorem C.1. Suppose $N = o(T^2)$, $N = o(T^{3/2}|\mathcal{G}|_0)$ and $\|f_{t_1} - f_{t_2}\| > c > 0$. Then under the null $H_0 : \bar{\theta}_{\mathcal{G},t_1} = \bar{\theta}_{\mathcal{G},t_2}$

$$\hat{\sigma}_{NT}^{-1}(\hat{\theta}_{\mathcal{G},t_1} - \hat{\theta}_{\mathcal{G},t_2}) \rightarrow^d \mathcal{N}(0, 1)$$

where

$$\begin{aligned}\hat{v}_\lambda &= \frac{1}{2}\hat{\lambda}'_{I,\mathcal{G}}\hat{V}_{\lambda 1,I}^{-1}\hat{V}_{\lambda 2,I}^{-1}\hat{\lambda}_{I,\mathcal{G}} + \frac{1}{2}\hat{\lambda}'_{I^c,\mathcal{G}}\hat{V}_{\lambda 1,I^c}^{-1}\hat{V}_{\lambda 2,I^c}^{-1}\hat{\lambda}_{I^c,\mathcal{G}} \\ \hat{v}_{t_1,t_2} &= \frac{1}{2}(\hat{f}_{I,t_1} - \hat{f}_{I,t_2})'\hat{V}_{I,f}(\hat{f}_{I,t_1} - \hat{f}_{I,t_2}) + \frac{1}{2}(\hat{f}_{I^c,t_1} - \hat{f}_{I^c,t_2})'\hat{V}_{I^c,f}(\hat{f}_{I^c,t_1} - \hat{f}_{I^c,t_2}) \\ \hat{\sigma}_{NT,S}^2 &= \frac{1}{T|\mathcal{G}|_0}\hat{v}_{t_1,t_2} + \frac{2}{N}\hat{v}_\lambda.\end{aligned}$$

Proof. The asymptotic analysis of $\hat{\theta}_{i,t_1}$ and $\hat{\theta}_{i,t_2}$ is the same as that of $\hat{\theta}_{i,t}$. Thus the same expansion (F.7) holds for $\{t_1, t_2\}$ in place of $\{t\}$. Therefore, for $\omega_{it} = e_{it}u_{it}$,

and $X := \frac{1}{\sqrt{T|\mathcal{G}|_0}} \sum_{i \in \mathcal{G}} \sum_{s=1}^T \Omega_i f_s \omega_{is}$,

$$\begin{aligned} \widehat{\theta}_{\mathcal{G},t_1} - \widehat{\theta}_{\mathcal{G},t_2} &= \frac{1}{\sqrt{T|\mathcal{G}|_0}} (f_{t_1} - f_{t_2})' X + \frac{1}{\sqrt{N}} \bar{\lambda}'_{\mathcal{G}} V_{\lambda 1}^{-1} \frac{1}{\sqrt{N}} \sum_{i=1}^N \lambda_i (\omega_{i,t_1} - \omega_{i,t_2}) + O_P(C_{NT}^{-2}) \\ &:= \zeta_1 + \zeta_2 + O_P(C_{NT}^{-2}). \end{aligned}$$

Let

$$\begin{aligned} \tilde{\sigma}_{NT,1}^2 &= \frac{1}{T|\mathcal{G}|_0} (f_{t_1} - f_{t_2})' V_{f,\mathcal{G}} (f_{t_1} - f_{t_2}), \quad \tilde{\sigma}_{NT,2}^2 = \frac{2}{N} \bar{\lambda}'_{\mathcal{G}} V_{\lambda} \bar{\lambda}_{\mathcal{G}}. \\ \tilde{\sigma}_{NT}^2 &= \tilde{\sigma}_{NT,1}^2 + \tilde{\sigma}_{NT,2}^2. \end{aligned}$$

We now apply the same argument as the proof of Theorem 3 in Bai (2003) to claim the asymptotic normality of $(\zeta_1 + \zeta_2)$. We have $(\tilde{\sigma}_{NT,1}^{-1} \zeta_1, \tilde{\sigma}_{NT,2}^{-1} \zeta_2) \rightarrow^d \mathcal{N}(0, I_2)$. By the almost sure representation there exist random vectors $(\xi_1, \xi_2) =^d (\tilde{\sigma}_{NT,1}^{-1} \zeta_1, \tilde{\sigma}_{NT,2}^{-1} \zeta_2)$, and $(Z_1, Z_2) =^d \mathcal{N}(0, I_2)$ such that $(\xi_1, \xi_2) \rightarrow (Z_1, Z_2)$ almost surely. Now

$$\begin{aligned} \frac{\zeta_1 + \zeta_2}{\tilde{\sigma}_{NT}} &=^d \frac{\tilde{\sigma}_{NT,1} \xi_1 + \tilde{\sigma}_{NT,2} \xi_2}{\tilde{\sigma}_{NT}} \\ &= \frac{\tilde{\sigma}_{NT,1} Z_1 + \tilde{\sigma}_{NT,2} Z_2}{\tilde{\sigma}_{NT}} + \frac{\tilde{\sigma}_{NT,1}(\xi_1 - Z_1) + \tilde{\sigma}_{NT,2}(\xi_2 - Z_2)}{\tilde{\sigma}_{NT}} \\ &=^d \mathcal{N}(0, 1) + o_P(1) \rightarrow^d \mathcal{N}(0, 1). \end{aligned}$$

Next, $|\tilde{\sigma}_{NT}^{-1}| \leq |(\frac{2}{N} \bar{\lambda}'_{\mathcal{G}} V_{\lambda} \bar{\lambda}_{\mathcal{G}})^{-1/2}| = O(\sqrt{N})$. So $\tilde{\sigma}_{NT}^{-1} O_P(C_{NT}^{-2}) = o_P(1)$ as long as $N = o(T^2)$. This implies

$$\tilde{\sigma}_{NT}^{-1} (\widehat{\theta}_{\mathcal{G},t_1} - \widehat{\theta}_{\mathcal{G},t_2}) \rightarrow^d \mathcal{N}(0, 1).$$

To prove $(\widehat{\sigma}_{NT,S}^2 - \tilde{\sigma}_{NT,S}^2) / \tilde{\sigma}_{NT,S}^2 = o_P(1)$, we first show $\frac{1}{T|\mathcal{G}|_0} \Delta / \tilde{\sigma}_{NT,S}^2 V_{f,\mathcal{G}} = o_P(1)$ where

$$\Delta := \widehat{v}_{t_1,t_2} - (f_{t_1} - f_{t_2})' V_{f,\mathcal{G}} (f_{t_1} - f_{t_2}),$$

by following the proof for the consistency of $V_{f,\mathcal{G}}$ in Section F.4.2. Note that here the consistency of $\widehat{v}_{t_1,t_2}$ is no longer sufficient, because it is likely $f_{t_1} = f_{t_2}$ under the null, in which case $\tilde{\sigma}_{NT}^2 = O(N^{-1})$. Then we need to show

$$\frac{N}{T|\mathcal{G}|_0} \Delta = o_P(1).$$

To obtain a sharper bound for Δ , we refine the proof of Section F.4.2. We assume serial homoskedasticity and redefine, for $\hat{\sigma}_i^2 = \frac{1}{T} \sum_{s=1}^T \hat{e}_{is}^2$, $\hat{\sigma}_{ui}^2 = \frac{1}{T} \sum_{s=1}^T \hat{u}_{is}^2$,

$$\hat{V}_f = \hat{S}_f^{-1} \frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \hat{\sigma}_{ui}^2 \hat{\sigma}_i^{-2}, \quad \tilde{V}_f = H_f'^{-1} S_f^{-1} \frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \mathbb{E} u_{is}^2 \sigma_i^{-2} H_f^{-1}.$$

Then $\|\hat{V}_f - \tilde{V}_f\| = O_P(C_{NT}^{-1})$ and $\|\tilde{V}_f - H_f'^{-1} V_{f,\mathcal{G}} H_f^{-1}\| = O_P(\frac{1}{\sqrt{T|\mathcal{G}|_0}})$. Hence $\hat{f}_{I,t_1} - \hat{f}_{I,t_2}' \hat{V}_{I,f} (\hat{f}_{I,t_1} - \hat{f}_{I,t_2}) - (f_{t_1} - f_{t_2})' V_{f,\mathcal{G}} (f_{t_1} - f_{t_2}) = O_P(C_{NT}^{-1})$, implying $\Delta = O_P(C_{NT}^{-1})$. Hence $\frac{N}{T|\mathcal{G}|_0} \Delta = o_P(1)$ holds so long as $N = o(T^{3/2}|\mathcal{G}|_0)$.

Next, $\frac{1}{N}(\hat{v}_\lambda - \bar{\lambda}'_\mathcal{G} V_\lambda \bar{\lambda}_\mathcal{G}) / \tilde{\sigma}_{NT,S}^2 = o_P(1)$ is straightforward, where the consistency of $\hat{v}_\lambda$ is sufficient. Hence $(\hat{\sigma}_{NT,S}^2 - \tilde{\sigma}_{NT,S}^2) / \tilde{\sigma}_{NT,S}^2 = o_P(1)$. Therefore under the null

$$\hat{\sigma}_{NT}^{-1}(\hat{\theta}_{\mathcal{G},t_1} - \hat{\theta}_{\mathcal{G},t_2}) \rightarrow^d \mathcal{N}(0, 1).$$

□

C.3. Group homogeneity test. Fix a group $\mathcal{G}$ of interest. For notational simplicity, write $N_0 = |\mathcal{G}|_0$. Consider testing homogeneity of coefficients with the same group:

$$H_0 : \theta_{1,t} = \dots = \theta_{N_0,t}, \quad \forall i \in \mathcal{G}, \forall t \leq T.$$

For instance, one may be interested in testing whether the minimum wage effect is the same within the group of high (or low) minimum wage states.

Using the structure $\theta_{i,t} = \lambda_i' f_t$ and maintaining invertibility of $\frac{1}{T} \sum_t f_t f_t'$, we have

$$\lambda_i = (\frac{1}{T} \sum_t f_t f_t')^{-1} \frac{1}{T} \sum_t f_t \theta_{it}.$$

Hence, the null is equivalent to testing

$$H_0 : \lambda_1 = \dots = \lambda_{N_0}, \quad \forall i \in \mathcal{G}.$$

Let $\hat{\lambda}_i = 0.5(\hat{\lambda}_{I,i} + \hat{\lambda}_{I^c,i})$ be the average of the estimated loadings obtained from the two subsamples, and let $\hat{\lambda} = \frac{1}{N_0} \sum_{i \in \mathcal{G}} \hat{\lambda}_i$. Naturally, we can test the null by comparing $\hat{\lambda}_i$ and $\hat{\lambda}$. As the number of members in the group may be large, testing using an aggregated statistic, such as $\sum_{i=1}^N \|\hat{\lambda}_i - \hat{\lambda}\|^2$ may lose power if the number of violations is small. As such, we consider the max-type test:

$$\max_{k \leq K} \max_{i \leq N} \frac{|(\hat{\lambda}_i - \hat{\lambda})_k|}{\hat{\sigma}_{ik}}$$

The test statistic is standardized by the estimated asymptotic variance of $(\widehat{\lambda}_i - \widehat{\lambda})_k$, given by

$$\widehat{\sigma}_{ik}^2 = (\widehat{\Gamma}_i)_{kk}, \quad \widehat{\Gamma}_i = \frac{1}{2T} \sum_{s \notin I} \widehat{\Omega}_{I,i} \widehat{f}_{I,s} \widehat{f}_{I,s}' \widehat{\Omega}_{I,i} \widehat{e}_{is}^2 \widehat{u}_{is}^2 + \frac{1}{2T} \sum_{s \notin I^c} \widehat{\Omega}_{I^c,i} \widehat{f}_{I^c,s} \widehat{f}_{I^c,s}' \widehat{\Omega}_{I^c,i} \widehat{e}_{is}^2 \widehat{u}_{is}^2.$$

Here $(a)_k$ denotes the k th element of the K -vector a , and $(A)_{kk}$ denotes the k th diagonal element of the $K \times K$ matrix A .

The asymptotic null hypothesis is derived in the following theorem. Recall that in this theorem

$$\widehat{\lambda}_i = \frac{1}{2}(\widehat{\lambda}_{I,i} + \widehat{\lambda}_{I^c,i}).$$

Theorem C.2. *Suppose $N_0 \rightarrow \infty$, $T = o(N^2)$ and*

$$\mathbb{E} \max_{it} \|\Omega_i f_s e_{is} u_{is}\|^2 \log^2 N = O(T).$$

Then under $H_0 : \theta_{1,t} = \dots = \theta_{N_0,t}$, $\forall i \in \mathcal{G}, \forall t \leq T$, we have

$$\max_{ik} \left| \sqrt{T} \frac{(\widehat{\lambda}_i - \widehat{\lambda})_k}{\widehat{\sigma}_{ik}} \right| \rightarrow^d \max_{ik} |Z_{ik}|$$

where Z_{ik} is the k^{th} element of $Z_i \in \mathbb{R}^K$, and $(Z_1, \dots, Z_N)$ are jointly normal, satisfying: $\mathbb{E} Z_i = 0$,

$$\mathbb{E} Z_i Z_j' = \begin{cases} \Sigma_i, & i = j \\ 0, & i \neq j \end{cases}.$$

Here Σ_i is the correlation matrix of $H_1' \frac{1}{\sqrt{T}} \sum_s \Omega_i f_s e_{is} u_{is}$.

The asymptotic null distribution can be simulated from $Z_1, \dots, Z_N$, with Σ_i replaced by the correlation matrix of $\widehat{\Gamma}_i$.

Proof. Let

$$\zeta_{is} := \Omega_i f_s e_{is} u_{is}.$$

Fix $S \in \{I, I^c\}$. By Proposition F.2,

$$\widehat{\lambda}_{S,i} - H_{S,f}'^{-1} \lambda_i = H_{S,f}'^{-1} \frac{1}{T_0} \sum_{s \notin S} \zeta_{is} + O_P(C_{NT}^{-2}).$$

From (F.3) in the proof of Proposition F.1,

$$H_{S,f} = (\bar{H}_3 + \mathbf{I}) H_1^{-1}$$

where H_1 is non-random, only depending on Λ (defined in step 2 of the proof of Proposition E.1). Also, $\bar{H}_3 = o_P(\log^{-1/2} N)$. This implies $\|H'_{S,f}{}^{-1} - H'_1\| = o_P(\log^{-1/2} N)$, and H_1 is non-random, not depending on the specific $S \in \{I, I^c\}$.

Let

$$\Delta_{S,i} = \frac{1}{T_0} \sum_{s \notin S} \zeta_{is}, \quad \Delta_{NT} = o_P(1) \frac{1}{N_0} \sum_{i \in \mathcal{G}} \Delta_{I,i} + o_P(1) \frac{1}{N_0} \sum_{i \in \mathcal{G}} \Delta_{I^c,i}.$$

Then uniformly in i ,

$$\hat{\lambda}_{S,i} - H'_{S,f}{}^{-1} \lambda_i = H'_1 \frac{1}{T_0} \sum_{s \notin S} \zeta_{is} + (H'_{S,f}{}^{-1} - H'_1) \Delta_{S,i} + O_P(C_{NT}^{-2}).$$

Hence $\hat{\lambda}_i = 0.5(\hat{\lambda}_{I,i} + \hat{\lambda}_{I^c,i})$, $\hat{\lambda} = \frac{1}{N_0} \sum_{i \in \mathcal{G}} \hat{\lambda}_i$ and $\bar{\lambda} = \frac{1}{N_0} \sum_{i \in \mathcal{G}} \lambda_i$ satisfy

$$\begin{aligned} \hat{\lambda}_i - 0.5(H'_{I,f}{}^{-1} + H'_{I^c,f}{}^{-1}) \lambda_i &= H'_1 \frac{1}{T} \sum_{s=1}^T \zeta_{is} + (H'_{S,f}{}^{-1} - H'_1)(\Delta_{I^c,i} + \Delta_{I,i}) + O_P(C_{NT}^{-2}). \\ \hat{\lambda} - 0.5(H'_{I,f}{}^{-1} + H'_{I^c,f}{}^{-1}) \bar{\lambda} &= H'_1 \frac{1}{N_0 T} \sum_s \sum_{i \in \mathcal{G}} \zeta_{is} + \Delta_{NT} + O_P(C_{NT}^{-2}). \end{aligned}$$

Under the null, $\lambda_i = \bar{\lambda}$ for all $i \in \mathcal{G}$. Thus, uniformly in i ,

$$\hat{\lambda}_i - \hat{\lambda} = H'_1 \frac{1}{T} \sum_{s=1}^T \zeta_{is} + (H'_{S,f}{}^{-1} - H'_1)(\Delta_{I^c,i} + \Delta_{I,i}) + O_P(C_{NT}^{-2} + \frac{1}{\sqrt{N_0 T}}).$$

We apply the concentration inequality from Lemma A.8 of Chernozhukov et al. (2015): The serial independence assumption implies

$$\max_i \|\Delta_{S,i}\| \leq O_P\left(\sqrt{\frac{\log N}{T}} + \frac{\sqrt{\mathbb{E} \max_{it} \|\zeta_{it}\|^2 \log N_0}}{T}\right).$$

Because $\|H'_{S,f}{}^{-1} - H'_1\| = o_P(\log^{-1/2} N)$ and $\mathbb{E} \max_{it} \|\zeta_{it}\|^2 \log^2 N = O(T)$,

$$\sqrt{T} \max_i \|(H'_{S,f}{}^{-1} - H'_1)(\Delta_{I^c,i} + \Delta_{I,i})\| \leq o_P(1).$$

Also, $N_0 \rightarrow \infty$ and $T = o(N^2)$.

Let ι'_k be the k^{th} canonical basis in $\mathbb{R}^K$: $\iota'_k a$ takes the k^{th} element of a vector a .

Let $\sigma_{ik}^2 = \iota'_k H'_1 \frac{1}{T} \sum_s \text{Var}(\zeta_{is}) H_1 \iota_k$, which is estimated by

$$\hat{\sigma}_{ik}^2 = \iota'_k \left[\frac{1}{2T} \sum_{s \notin I} \hat{\Omega}_{I,i} \hat{f}_{I,s} \hat{f}'_{I,s} \hat{\Omega}_{I,i} \hat{e}_{is}^2 \hat{u}_{is}^2 + \frac{1}{2T} \sum_{s \notin I^c} \hat{\Omega}_{I^c,i} \hat{f}_{I^c,s} \hat{f}'_{I^c,s} \hat{\Omega}_{I^c,i} \hat{e}_{is}^2 \hat{u}_{is}^2 \right] \iota_k$$

satisfying $\max_{ik} |\hat{\sigma}_{ik} - \sigma_{ik}| = o_P(1/\log^{1/2} N)$. Hence, uniformly in $i \leq N_0, k \leq K$,

$$\sqrt{T} \frac{\iota'_k(\hat{\lambda}_i - \hat{\lambda})}{\hat{\sigma}_{ik}} = A_{ik} + o_P(1), \quad A_{ik} := \frac{\iota'_k H'_1 \frac{1}{\sqrt{T}} \sum_{s=1}^T \zeta_{is}}{\sigma_{ik}}. \quad (\text{C.1})$$

By Corollary 2.1 of Chernozhukov et al. (2013),

$$\max_{ki} \left| \sqrt{T} \frac{\iota'_k(\hat{\lambda}_i - \hat{\lambda})}{\hat{\sigma}_{ik}} \right| = \max_{ik} |A_{ik}| + o_P(1) \rightarrow^d \max_{ik} |Z_{ik}|$$

where for $Z_i \in \mathbb{R}^K = (Z_{i1} \dots Z_{iK})'$,

$$\begin{pmatrix} Z_1 \\ \vdots \\ Z_N \end{pmatrix}_{NK \times 1} \sim \mathcal{N}(0, \Sigma_Z), \quad \Sigma_Z = \begin{pmatrix} \Sigma_1 & \dots & 0 \\ & \ddots & \\ 0 & & \Sigma_N \end{pmatrix},$$

and Σ_i is the correlation matrix of $H'_1 \frac{1}{\sqrt{T}} \sum_s \zeta_{is}$.

□

C.4. Test of group homogeneity: Theorem 3.2(i). We now study the test of

$$H_0 : \bar{\theta}_{\mathcal{G}_1, t} = \dots = \bar{\theta}_{\mathcal{G}_J, t}.$$

Under the null, it is assumed that $\bar{\lambda}_{\mathcal{G}_1} = \dots = \bar{\lambda}_{\mathcal{G}_J}$. Let $\hat{\theta}_{\mathcal{G}_j, t}$ be the estimated $\bar{\theta}_{\mathcal{G}_j, t}$. Because J is assumed to be fixed, the expansion (F.7) also holds jointly for $\{\hat{\theta}_{\mathcal{G}_j, t} : j \leq J\}$. Therefore,

$$\hat{\theta}_{\mathcal{G}_j, t} - \bar{\theta}_{\mathcal{G}_j, t} = \frac{1}{|\mathcal{G}_j|_0 \sqrt{T}} f'_t \sum_{i \in \mathcal{G}_j} \Omega_i m_i + \frac{1}{\sqrt{N}} \bar{\lambda}'_{\mathcal{G}_j} A + O_P(C_{NT}^{-2}).$$

Putting together, jointly we have

$$\Delta := \begin{pmatrix} \hat{\theta}_{\mathcal{G}_1, t} - \bar{\theta}_{\mathcal{G}_1, t} \\ \vdots \\ \hat{\theta}_{\mathcal{G}_J, t} - \bar{\theta}_{\mathcal{G}_J, t} \end{pmatrix} = \frac{1}{\sqrt{T}} [I_J \otimes f'_t] M + \frac{1}{\sqrt{N}} \bar{\Lambda} A_t + O_P(C_{NT}^{-2}) \quad (\text{C.2})$$

where

$$M = \begin{pmatrix} \frac{1}{|\mathcal{G}_1|_0} \sum_{i \in \mathcal{G}_1} \Omega_i m_i \\ \vdots \\ \frac{1}{|\mathcal{G}_J|_0} \sum_{i \in \mathcal{G}_J} \Omega_i m_i \end{pmatrix}, \quad \bar{\Lambda} = \begin{pmatrix} \bar{\lambda}'_{\mathcal{G}_1} \\ \vdots \\ \bar{\lambda}'_{\mathcal{G}_J} \end{pmatrix}$$

$$A_t = V_{\lambda 1}^{-1} \frac{1}{\sqrt{N}} \sum_{i=1}^N \lambda_i e_{it} u_{it}, \quad m_i = \frac{1}{\sqrt{T}} \sum_{s=1}^T f_s e_{is} u_{is}. \quad (\text{C.3})$$

We further assume that if i, j belong two groups, then $e_{it} u_{it}$ are independent given F and groups are mutually disjoint. Then $\text{Var}(\frac{1}{\sqrt{T}}[I_J \otimes f'_t]M|F)$ is a block diagonal matrix:

$$\text{Var}(\frac{1}{\sqrt{T}}[I_J \otimes f'_t]M|F) = \frac{1}{T} \begin{pmatrix} \frac{1}{|\mathcal{G}_1|_0} f'_t V_{f, \mathcal{G}_1} f_t & 0 & \cdots \\ & \ddots & \\ \cdots & 0 & \frac{1}{|\mathcal{G}_J|_0} f'_t V_{f, \mathcal{G}_J} f_t \end{pmatrix} := \frac{1}{T} D$$

where $V_{f, \mathcal{G}_j} = \frac{1}{T} \sum_{s=1}^T \text{Var} \left(\frac{1}{\sqrt{|\mathcal{G}_j|_0}} \sum_{i \in \mathcal{G}_j} \Omega_i f_s e_{is} u_{is} \middle| F \right)$. Additionally, let the $(J-1) \times J$ matrix selection matrix be:

$$\Xi = \begin{pmatrix} 1 & -1 & 0 & \cdots \\ 0 & 1 & -1 & 0 & \cdots \\ \vdots & & & & \\ 0 & \cdots & & 1 & -1 \end{pmatrix}, \quad \widehat{D}_S = \begin{pmatrix} \frac{1}{|\mathcal{G}_1|_0} \widehat{f}'_{S,t} \widehat{V}_{fS, \mathcal{G}_1} \widehat{f}_{S,t} & 0 & \cdots \\ & \ddots & \\ \cdots & 0 & \frac{1}{|\mathcal{G}_J|_0} \widehat{f}'_{S,t} \widehat{V}_{fS, \mathcal{G}_J} \widehat{f}_{S,t} \end{pmatrix}$$

for $S = I$ and I^c . Let $\widehat{D} = \frac{1}{2}(\widehat{D}_I + \widehat{D}_{I^c})$. We note that $\Xi \bar{\Lambda} = 0$ and $\Xi(\bar{\theta}_{\mathcal{G}_1, t}, \dots, \bar{\theta}_{\mathcal{G}_J, t})' = 0$ under the null. Thus,

$$\begin{aligned} \sqrt{T}(\Xi D \Xi')^{-1/2} \Xi \Delta &= \sqrt{T}(\Xi D \Xi')^{-1/2} \Xi \frac{1}{\sqrt{T}} [I_J \otimes f'_t] M \\ &\quad + \sqrt{T}(\Xi D \Xi')^{-1/2} \Xi O_P(C_{NT}^{-2}). \end{aligned} \quad (\text{C.4})$$

It remains to show three statements:

- (i) $\sqrt{T}(\Xi D \Xi')^{-1/2} \Xi O_P(C_{NT}^{-2}) = o_P(1)$.
- (ii) $\sqrt{T}(\Xi D \Xi')^{-1/2} \Xi \frac{1}{\sqrt{T}} [I_J \otimes f'_t] M | F \rightarrow^d \mathcal{N}(0, I)$.
- (iii) $[(\Xi \widehat{D} \Xi')^{-1/2} - (\Xi D \Xi')^{-1/2}](\Xi D \Xi')^{1/2} = o_P(1)$.

Recall $C_{NT}^{-2} = O(1/N + 1/T)$, $\lambda_{\min}(\Xi \Xi') \geq c$, and $\min_{j \leq J} \lambda_{\min}(V_{f, \mathcal{G}_j}) > c$. Hence,

$$\|(\Xi D \Xi')^{-1/2}\| \leq \sqrt{\frac{1}{\lambda_{\min}(\Xi \Xi') \min_{j \leq J} (\frac{1}{|\mathcal{G}_j|_0} f'_t V_{f, \mathcal{G}_j} f_t)}} \leq O_P(\|f_t\|^{-1}) \sqrt{\max_{j \leq J} |\mathcal{G}_j|_0}.$$

This implies (i) when both $\max_{j \leq J} |\mathcal{G}_j|_0 = o(T)$ and $T \max_{j \leq J} |\mathcal{G}_j|_0 = o(N^2)$. In addition, it is straightforward to check the multivariate Lindeberg-Feller condition, so that (ii) holds. As for (iii), let $\widehat{B} = \Xi \widehat{D} \Xi'$ and $B = \Xi D \Xi'$. Both matrices are

positive definite and

$$\|\widehat{B} - B\| = O_P\left(\frac{1}{\min_{j \leq J} |\mathcal{G}_j|_0}\right) \max_j \|\widehat{f}_t' \widehat{V}_{f, \mathcal{G}_J} \widehat{f}_t - f_t' V_{f, \mathcal{G}_J} f_t\| = o_P\left(\frac{1}{\min_{j \leq J} |\mathcal{G}_j|_0}\right).$$

Hence, with the assumption that $\max_j |\mathcal{G}_j|_0 / \min_j |\mathcal{G}_j|_0 = O(1)$,

$$\begin{aligned} \|\widehat{B}^{-1/2} - B^{-1/2}\| &\leq \frac{1}{\sqrt{\lambda_{\max}^{-1}(\widehat{B})} + \sqrt{\lambda_{\max}^{-1}(B)}} \|\widehat{B}^{-1} - B^{-1}\| \\ &\leq \sqrt{\lambda_{\min}(B)} \|\widehat{B} - B\| \|\widehat{B}^{-1}\| \|B^{-1}\|, \end{aligned}$$

so $[\widehat{B}^{-1/2} - B^{-1/2}]B^{1/2} = o_P(\max_j |\mathcal{G}_j|_0 / \min_j |\mathcal{G}_j|_0)^{3/2} = o_P(1)$. This gives (iii). (C.4) and (i), (ii) together imply $\sqrt{T}(\Xi \widehat{D} \Xi')^{-1/2} \Xi \Delta \rightarrow^d \mathcal{N}(0, I)$. This, combined with (iii), implies under the null,

$$\sqrt{T}(\Xi \widehat{D} \Xi')^{-1/2} \Xi \begin{pmatrix} \widehat{\theta}_{\mathcal{G}_1, t} \\ \vdots \\ \widehat{\theta}_{\mathcal{G}_J, t} \end{pmatrix} \rightarrow^d \mathcal{N}(0, I_{J-1}).$$

This leads to the desired asymptotic null distribution:

$$T(\widehat{\theta}_{\mathcal{G}_1, t}, \dots, \widehat{\theta}_{\mathcal{G}_J, t})' \Xi' (\Xi \widehat{D} \Xi')^{-1} \Xi \begin{pmatrix} \widehat{\theta}_{\mathcal{G}_1, t} \\ \vdots \\ \widehat{\theta}_{\mathcal{G}_J, t} \end{pmatrix} \rightarrow^d \chi^2(J-1).$$

C.5. Test of joint significance in a given time period:: Theorem 3.2(ii).

Recall that the factor estimator in the proof below is defined as

$$\widehat{f}_t = \frac{1}{2}(\widehat{f}_{I, t} + \widehat{f}_{I^c, t}).$$

Proof. Consider testing:

$$H_0 : \theta_{it} = 0 \text{ for all } i.$$

With the factor slope structure, we assume that H_0 is equivalent to

$$H'_0 : f_t = 0.$$

This becomes a more tractable low dimensional testing problem.

By Lemma C.1, under H'_0 , conditionally on F , when $N = o(T^2)$,

$$\sqrt{N}(H_1^{-1} V_\lambda H_1'^{-1})^{-1/2} \frac{1}{2}(\widehat{f}_{I, t} + \widehat{f}_{I^c, t}) \rightarrow^d \mathcal{N}(0, I), \quad (\text{C.5})$$

which then also holds unconditionally using the dominated convergence theorem.

Next, the eigenvalues of $H_1^{-1}V_\lambda H_1'^{-1}$ are bounded away from zero, thus (C.5) still holds with $\widehat{V}_\lambda$ in place of $H_1^{-1}V_\lambda H_1'^{-1}$. This yields

$$N \frac{1}{2} (\widehat{f}_{I,t} + \widehat{f}_{I^c,t})' \widehat{V}_\lambda^{-1} \frac{1}{2} (\widehat{f}_{I,t} + \widehat{f}_{I^c,t}) \rightarrow^d \chi^2(\dim(f_t)).$$

□

For $\widehat{f}_t = \frac{1}{2}(\widehat{f}_{I,t} + \widehat{f}_{I^c,t})$, we have the following lemma:

Lemma C.1. *Recall H_1 given in step 2 of the proof of Proposition E.1, which is nonrandom conditionally on F . Suppose $N = o(T^2)$, then whether or not $f_t = 0$,*

$$\sqrt{N}(H_1^{-1}V_\lambda H_1'^{-1})^{-1/2}(\widehat{f}_t - H_1^{-1}f_t) \rightarrow^d \mathcal{N}(0, I)$$

and we have $\|\widehat{V}_\lambda - H_1^{-1}V_\lambda H_1'^{-1}\| = o_P(1)$.

Proof. By Proposition F.1, for both $S = I$ and $S = I^c$,

$$\widehat{f}_{S,t} - H_{S,f}f_t = H_{S,f}V_{\lambda 1}^{-1} \frac{1}{N} \sum_i \lambda_i e_{it} u_{it} + O_P(C_{NT}^{-2}).$$

The above expansion still holds with $\widehat{f}_{S,t}$ and $H_{S,f}$ replaced by $\frac{1}{2}(\widehat{f}_{I,t} + \widehat{f}_{I^c,t})$ and $\bar{H}_f = \frac{1}{2}H_{I,f} + \frac{1}{2}H_{I^c,f}$. Thus $\widehat{f}_t := \frac{1}{2}(\widehat{f}_{I,t} + \widehat{f}_{I^c,t})$ yields:

$$\sqrt{N}(\widehat{f}_t - \bar{H}_f f_t) = \bar{H}_f V_{\lambda 1}^{-1} \frac{1}{\sqrt{N}} \sum_i \lambda_i e_{it} u_{it} + o_P(1).$$

We derive the probability limit of $\bar{H}_f$. In the proof of Proposition F.1, $H_{S,f} := (\bar{H}_3 + I)H_1^{-1} = H_1^{-1} + o_P(1)$ because $\bar{H}_3 = o_P(1)$. In addition, H_1 only depends on (Λ, F) . Thus $\|\bar{H}_f - H_1^{-1}\| = o_P(1)$, and H_1^{-1} is nonrandom conditional on F . Thus,

$$\sqrt{N}(H_1^{-1}V_\lambda H_1'^{-1})^{-1/2}(\widehat{f}_t - H_1^{-1}f_t) \rightarrow^d \mathcal{N}(0, I).$$

Additionally, the consistency proof in Section F.4.1 shows $\|\widehat{V}_{\lambda 2,S} - H_f'^{-1}V_{\lambda 2}H_f^{-1}\| = o_P(1)$ and $\|\widehat{V}_{\lambda 1,S} - H_f'^{-1}V_{\lambda 1}H_f^{-1}\| = o_P(1)$ for both $S = I$ and $S = I^c$. Therefore, for

$$\widehat{V}_\lambda := \frac{1}{2}\widehat{V}_{\lambda 1,I}^{-1}\widehat{V}_{\lambda 2,I}\widehat{V}_{\lambda 1,I}^{-1} + \frac{1}{2}\widehat{V}_{\lambda 1,I^c}^{-1}\widehat{V}_{\lambda 2,I^c}\widehat{V}_{\lambda 1,I^c}^{-1},$$

we have $\|\widehat{V}_\lambda - H_1^{-1}V_\lambda H_1'^{-1}\| = o_P(1)$.

□

APPENDIX D. PROOF OF PROPOSITION 2.1: COMPUTATIONAL
CONVERGENCE

Recall that $\Theta_{k+1} = S_{\tau\nu_1/2}(\Theta_k - \tau A_k)$, where

$$A_k = X \odot (X \odot \Theta_k - Y + M_k).$$

By Lemma D.2, set $\Theta = \tilde{\Theta}$ and $M = \tilde{M}$, and replace k with subscript m ,

$$F(\hat{\Theta}, \hat{M}) - F(\Theta_{m+1}, M_{m+1}) \geq \frac{1}{\tau} \left(\|\Theta_{m+1} - \hat{\Theta}\|_F^2 - \|\Theta_m - \hat{\Theta}\|_F^2 \right).$$

Let $m = 1, \dots, k$, and sum these inequalities up. Since $F(\Theta_{m+1}, M_{m+1}) \geq F(\Theta_{k+1}, M_{k+1})$ by Lemma D.1,

$$\begin{aligned} kF(\hat{\Theta}, \hat{M}) - kF(\Theta_{k+1}, M_{k+1}) &\geq kF(\hat{\Theta}, \hat{M}) - \sum_{m=1}^k F(\Theta_{m+1}, M_{m+1}) \\ &\geq \frac{1}{\tau} \left(\|\Theta_{k+1} - \hat{\Theta}\|_F^2 - \|\Theta_1 - \hat{\Theta}\|_F^2 \right) \geq -\frac{1}{\tau} \|\Theta_1 - \hat{\Theta}\|_F^2. \quad \square \end{aligned}$$

The above proof depends on the following lemmas.

Lemma D.1. *We have (i)*

$$\begin{aligned} \Theta_{k+1} &= \arg \min_{\Theta} p(\Theta, \Theta_k, M_k) + \nu_1 \|\Theta\|_n, \\ p(\Theta, \Theta_k, M_k) &:= \tau^{-1} \|\Theta_k - \Theta\|_F^2 - 2\text{tr}((\Theta_k - \Theta)' A_k). \end{aligned}$$

(ii) For any $\tau \in (0, 1/\max x_{it}^2)$,

$$F(\Theta, M_k) \leq p(\Theta, \Theta_k, M_k) + \nu_1 \|\Theta\|_n + \nu_2 \|M_k\|_n + \|Y - M_k - X \odot \Theta_k\|_F^2.$$

(iii) $F(\Theta_{k+1}, M_{k+1}) \leq F(\Theta_{k+1}, M_k) \leq F(\Theta_k, M_k)$.

Proof. (i) We have $\|\Theta_k - \tau A_k - \Theta\|_F^2 = \|\Theta_k - \Theta\|_F^2 + \tau^2 \|A_k\|_F^2 - 2\text{tr}[(\Theta_k - \Theta)' A_k] \tau$.

Thus,

$$\begin{aligned} &\arg \min_{\Theta} \|\Theta_k - \tau A_k - \Theta\|_F^2 + \tau \nu_1 \|\Theta\|_n \\ &= \arg \min_{\Theta} \|\Theta_k - \Theta\|_F^2 - 2\text{tr}[(\Theta_k - \Theta)' A_k] \tau + \tau \nu_1 \|\Theta\|_n \\ &= \arg \min_{\Theta} \tau \cdot p(\Theta, \Theta_k, M_k) + \tau \nu_1 \|\Theta\|_n. \end{aligned}$$

It is well known that $\Theta_{k+1} = S_{\tau\nu_1/2}(\Theta_k - \tau A_k)$ is the solution to the first problem in the above equalities (Ma et al., 2011). This proves (i).

(ii) Note that for $\Theta_k = (\theta_{k,it})$ and $\Theta = (\theta_{it})$, and any $\tau^{-1} > \max_{it} x_{it}^2$, we have

$$\|X \odot (\Theta_k - \Theta)\|_F^2 = \sum_{it} x_{it}^2 (\theta_{k,it} - \theta_{it})^2 < \tau^{-1} \|\Theta_k - \Theta\|_F^2.$$

Then,

$$\begin{aligned} F(\Theta, M_k) &= \|Y - M_k - X \odot \Theta\|_F^2 + \nu_2 \|M_k\|_n + \nu_1 \|\Theta\|_n \\ &= \|Y - M_k - X \odot \Theta_k\|_F^2 + \|X \odot (\Theta_k - \Theta)\|_F^2 - 2\text{tr}[A'_k(\Theta_k - \Theta)] \\ &\quad + \nu_2 \|M_k\|_n + \nu_1 \|\Theta\|_n \\ &\leq \|Y - M_k - X \odot \Theta_k\|_F^2 + \tau^{-1} \|\Theta_k - \Theta\|_F^2 - 2\text{tr}[A'_k(\Theta_k - \Theta)] \\ &\quad + \nu_2 \|M_k\|_n + \nu_1 \|\Theta\|_n \\ &= p(\Theta, \Theta_k, M_k) + \|Y - M_k - X \odot \Theta_k\|_F^2 + \nu_2 \|M_k\|_n + \nu_1 \|\Theta\|_n. \end{aligned}$$

(iii) By definition, $p(\Theta_k, \Theta_k, M_k) = 0$. Thus,

$$\begin{aligned} F(\Theta_{k+1}, M_{k+1}) &= \|Y - X \odot \Theta_{k+1} - M_{k+1}\|_F^2 + \nu_1 \|\Theta_{k+1}\|_n + \nu_2 \|M_{k+1}\|_n \\ &\stackrel{(a)}{\leq} \|Y - X \odot \Theta_{k+1} - M_k\|_F^2 + \nu_1 \|\Theta_{k+1}\|_n + \nu_2 \|M_k\|_n \\ &= F(\Theta_{k+1}, M_k) \\ &\stackrel{(b)}{\leq} p(\Theta_{k+1}, \Theta_k, M_k) + \nu_1 \|\Theta_{k+1}\|_n + \nu_2 \|M_k\|_n + \|Y - M_k - X \odot \Theta_k\|_F^2 \\ &\stackrel{(c)}{\leq} p(\Theta_k, \Theta_k, M_k) + \nu_1 \|\Theta_k\|_n + \nu_2 \|M_k\|_n + \|Y - M_k - X \odot \Theta_k\|_F^2 \\ &= F(\Theta_k, M_k). \end{aligned}$$

(a) is due to the definition of M_{k+1} ; (b) is due to (ii); (c) is due to (i). □

Lemma D.2. For any $\tau \in (0, 1/\max x_{it}^2)$, any (Θ, M) and any $k \geq 1$,

$$F(\Theta, M) - F(\Theta_{k+1}, M_{k+1}) \geq \frac{1}{\tau} (\|\Theta_{k+1} - \Theta\|_F^2 - \|\Theta_k - \Theta\|_F^2).$$

Proof. The proof is similar to that of Lemma 2.3 of Beck and Teboulle (2009), with the extension that M_{k+1} is updated after Θ_{k+1} . The key difference here is that, while an update to M_{k+1} is added to the iteration, we show that the lower bound does not depend on M_{k+1} or M_k . Therefore, the convergence property of the algorithm depends mainly on the step of updating Θ .

Let $\partial\|A\|_n$ be an element that belongs to the subgradient of $\|A\|_n$. Note that $\partial\|A\|_n$ is convex in A . Also, $\|Y - X \odot \Theta - M\|_F^2$ is convex in (Θ, M) , so for any Θ, M , we have the following three inequalities:

$$\|Y - X \odot \Theta - M\|_F^2 \geq \|Y - X \odot \Theta_k - M_k\|_F^2 - 2\text{tr}[(\Theta - \Theta_k)'(X \odot (Y - X \odot \Theta_k - M_k))]$$

$$\begin{aligned}
& -2\text{tr}[(M - M_k)'(Y - X \odot \Theta_k - M_k)] \\
\nu_1 \|\Theta\|_n & \geq \nu_1 \|\Theta_{k+1}\|_n + \nu_1 \text{tr}[(\Theta - \Theta_{k+1})' \partial \|\Theta_{k+1}\|_n] \\
\nu_2 \|M\|_n & \geq \nu_2 \|M_k\|_n + \nu_2 \text{tr}[(M - M_k)' \partial \|M_k\|_n].
\end{aligned}$$

In addition,

$$\begin{aligned}
-F(\Theta_{k+1}, M_{k+1}) & \geq -F(\Theta_{k+1}, M_k) \\
& \geq -p(\Theta_{k+1}, \Theta_k, M_k) - \nu_1 \|\Theta_{k+1}\|_n - \nu_2 \|M_k\|_n - \|Y - M_k - X \odot \Theta_k\|_F^2,
\end{aligned}$$

where the two inequalities are due to Lemma D.1. Summing the above inequalities,

$$\begin{aligned}
& F(\Theta, M) - F(\Theta_{k+1}, M_{k+1}) \geq (A) \\
(A) & := -2\text{tr}[(\Theta - \Theta_k)'(X \odot (Y - X \odot \Theta_k - M_k))] - 2\text{tr}[(M - M_k)'(Y - X \odot \Theta_k - M_k)] \\
& + \nu_1 \text{tr}[(\Theta - \Theta_{k+1})' \partial \|\Theta_{k+1}\|_n] + \nu_2 \text{tr}[(M - M_k)' \partial \|M_k\|_n] - p(\Theta_{k+1}, \Theta_k, M_k).
\end{aligned}$$

We now simplify (A). Since $k \geq 1$, both M_k and Θ_{k+1} should satisfy the KKT condition. By Lemma D.1, they are

$$\begin{aligned}
0 & = \nu_1 \partial \|\Theta_{k+1}\|_n - \tau^{-1} 2(\Theta_k - \Theta_{k+1}) + 2A_k \\
0 & = \nu_2 \partial \|M_k\|_n - 2(Y - X \odot \Theta_k - M_k).
\end{aligned}$$

Plugging in, we have

$$\begin{aligned}
(A) & = \tau^{-1} 2\text{tr}[(\Theta - \Theta_{k+1})'(\Theta_k - \Theta_{k+1})] - \tau^{-1} \|\Theta_k - \Theta_{k+1}\|_F^2 \\
& = \frac{1}{\tau} (\|\Theta_{k+1} - \Theta\|_F^2 - \|\Theta_k - \Theta\|_F^2).
\end{aligned}$$

□

APPENDIX E. NUCLEAR-NORM MATRIX ESTIMATORS

Recall that $\widetilde{M}_S$ and $\widetilde{\Theta}_S$ respectively are the estimated low-rank matrices obtained by nuclear-norm penalized estimation on sample $S \in \{I, I^c, \{1, \dots, T\}\}$.

Proposition E.1. *Suppose $2\|X \odot U\| < (1 - c)\nu_1$, $2\|U\| < (1 - c)\nu_0$ and $\nu_0 \asymp \nu_1$ and $\nu_1 \asymp \nu_2 = \sqrt{N + T}$. Then for $S \in \{I, I^c, \{1, \dots, T\}\}$ (i)*

$$\frac{1}{NT} \|\widetilde{M}_S - M_S\|_F^2 = O_P\left(\frac{1}{N} + \frac{1}{T}\right) = \frac{1}{NT} \|\widetilde{\Theta}_S - \Theta_S\|_F^2.$$

(ii) *Additionally with Assumption 3.2, there are square matrices H_{S1}, H_{S2} , so that*

$$\frac{1}{N} \|\widetilde{A}_S - AH_{S1}\|_F^2 = O_P\left(\frac{1}{N} + \frac{1}{T}\right), \quad \frac{1}{N} \|\widetilde{\Lambda}_S - \Lambda H_{S2}\|_F^2 = O_P\left(\frac{1}{N} + \frac{1}{T}\right).$$

(iii) Furthermore,

$$P(\widehat{K}_1 = K_1, \quad \widehat{K}_2 = K_2) \rightarrow 1.$$

Proposition E.1 extends the usual low-rank results to the multi-dimensional case. The assumption that $\nu_1 \asymp \nu_2 = \sqrt{N+T}$ can be verified by Lemma E.1 which shows the score is of the order $\sqrt{N+T}$. The convergence result (ii) using the subsamples serve for post-SVT inference after sample-splitting. The proof is divided in the following subsections.

E.1. Level of the Score.

Lemma E.1. $\|X \odot U\| = O_P(\sqrt{N+T}) = \|U\|$. In addition, the chosen ν_0, ν_1 in Section 2.4 are $O_P(\sqrt{N+T})$.

Proof. The assumption that Ω_{NT} contains independent sub-Gaussian columns ensures that, by the eigenvalue-concentration inequality for sub-Gaussian random vectors (Theorem 5.39 of Vershynin (2010)):

$$\|\Omega_{NT}\Omega'_{NT} - \mathbb{E}\Omega_{NT}\Omega'_{NT}\| = O_P(\sqrt{NT} + N).$$

In addition, let w_i be the $T \times 1$ vector of $\{x_{it}u_{it} : t \leq T\}$. We have, for each (i, j, t, s) ,

$$\mathbb{E}(w_i w'_j)_{s,t} = \mathbb{E}x_{it}x_{js}u_{it}u_{js} = \begin{cases} \mathbb{E}x_{it}^2 u_{it}^2, & i = j, t = s \\ 0, & \text{otherwise} \end{cases}$$

due to the conditional cross-sectional and serial independence in u_{it} . Then for the (i, j) -th entry of $\mathbb{E}\Omega_{NT}\Omega'_{NT}$,

$$\begin{aligned} (\mathbb{E}\Omega_{NT}\Omega'_{NT})_{i,j} &= (\mathbb{E}(X \odot U)(X \odot U)')_{i,j} \\ &= \mathbb{E}w'_i w_j = \text{tr}(\mathbb{E}w_j w'_i) = \begin{cases} \sum_{t=1}^T \mathbb{E}x_{it}^2 u_{it}^2, & i = j \\ 0, & i \neq j. \end{cases} \end{aligned}$$

Hence $\|\mathbb{E}\Omega_{NT}\Omega'_{NT}\| \leq O(T)$. This implies $\|\Omega_{NT}\Omega'_{NT}\| \leq O(T + N)$. Hence $\|X \odot U\| = \|\Omega_{NT}\| \leq O_P(\max\{\sqrt{N}, \sqrt{T}\})$. The rate for $\|U\|$ follows from the same argument. The second claim that ν_0, ν_1 satisfy the same rate constraint follows from the same argument, by replacing U with Z , and Assumption 3.3 is still satisfied by Z and $X \odot Z$.

□

E.2. Useful Claims. The proof of Proposition E.1 uses some claims that are proved in the following lemma. Let us first recall the notations. Define $U_2 D_2 V_2' = \Theta$ and $U_1 D_1 V_1' = M$ as the singular value decompositions of the true values Θ and M . Further decompose, for $j = 1, 2$,

$$U_j = (U_{j,r}, U_{j,c}), \quad V_j = (V_{j,r}, V_{j,c})$$

Here $(U_{j,r}, V_{j,r})$ corresponds to the nonzero singular values, while $(U_{j,c}, V_{j,c})$ corresponds to the zero singular values. In addition, for any $N \times T/2$ matrix Δ , let

$$\mathcal{P}_j(\Delta) = U_{j,c} U_{j,c}' \Delta V_{j,c} V_{j,c}', \quad \mathcal{M}_j(\Delta) = \Delta - \mathcal{P}_j(\Delta).$$

Here $U_{j,c} U_{j,c}'$ and $V_{j,c} V_{j,c}'$ respectively are the projection matrices onto the columns of $U_{j,c}$ and $V_{j,c}$. Therefore, $\mathcal{M}_1(\cdot)$ and $\mathcal{M}_2(\cdot)$ can be considered as the projection matrices onto the “low-rank” spaces of Θ and M respectively, and $\mathcal{P}_1(\cdot)$ and $\mathcal{P}_2(\cdot)$ are projections onto their orthogonal spaces.

Lemma E.2 (claims). *Results below also apply to $\mathcal{P}_2$, Θ , and samples on I^c . For any matrix Δ ,*

- (i) $\|\mathcal{P}_1(\Delta) + M\|_n = \|\mathcal{P}_1(\Delta)\|_n + \|M\|_n$.
- (ii) $\|\Delta\|_F^2 = \|\mathcal{M}_1(\Delta)\|_F^2 + \|\mathcal{P}_1(\Delta)\|_F^2$
- (iii) $\text{rank}(\mathcal{M}_1(\Delta)) \leq 2K_1$, where $K_1 = \text{rank}(M)$.
- (iv) $\|\Delta\|_F^2 = \sum_j \sigma_j^2$ and $\|\Delta\|_n^2 \leq \|\Delta\|_F^2 \text{rank}(\Delta)$, with σ_j as the singular values of Δ .
- (v) For any Δ_1, Δ_2 , $|\text{tr}(\Delta_1 \Delta_2)| \leq \|\Delta_1\|_n \|\Delta_2\|_n$, Here $\|\cdot\|$ denotes the operator norm.

Proof. (i) Note that $M = U_{1,r} D_{1,r} V_{1,r}'$ where $D_{1,r}$ are the subdiagonal matrix of nonzero singular values. The claim follows from Lemma 2.3 of Recht et al. (2010).

(ii) Write

$$U_1' \Delta V_1 = \begin{pmatrix} A & B \\ C & U_{1,c}' \Delta V_{1,c} \end{pmatrix} = \begin{pmatrix} A & B \\ C & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 0 & U_{1,c}' \Delta V_{1,c} \end{pmatrix} := H_2 + H_1.$$

Then $\mathcal{P}_1(\Delta) = U_1 H_1 V_1'$ and $\mathcal{M}_1(\Delta) = U_1 H_2 V_1'$. So

$$\|\mathcal{P}_1(\Delta)\|_F^2 = \text{tr}(U_1 H_1 V_1' V_1 H_1' U_1') = \text{tr}(H_1 H_1') = \|H_1\|_F^2.$$

Similarly, $\|\mathcal{M}_1(\Delta)\|_F^2 = \|H_2\|_F^2$. So

$$\|H_2\|_F^2 + \|H_1\|_F^2 = \|U'_1 \Delta V_1\|_F^2 = \|\Delta\|_F^2.$$

(iii) This is Lemma 1 of Negahban and Wainwright (2011).

(iv) The first is a basic equality, and the second follows from the Cauchy-Schwarz inequality.

(v) Let $UDV' = \Delta_1$ be the singular value decomposition of Δ_1 , then

$$|\text{tr}(\Delta_1 \Delta_2)| = \left| \sum_i D_{ii} (V' \Delta_2 U)_{ii} \right| \leq \max_i |(V' \Delta_2 U)_{ii}| \sum_i D_{ii} \leq \|\Delta_1\|_n \|\Delta_2\|.$$

E.3. The RSC condition. We now prove primitive conditions for the Restricted strong convexity (RSC) condition: Define

$$\mathcal{C}(c_1, c_2) = \left\{ (\Delta_1, \Delta_2) : \|\mathcal{P}_1(\Delta_1)\|_n + \|\mathcal{P}_2(\Delta_2)\|_n \leq c_1 \|\mathcal{M}_1(\Delta_1)\|_n + c_1 \|\mathcal{M}_2(\Delta_2)\|_n, \right. \\ \left. \frac{1}{NT} \|\Delta_1\|_F^2 + \frac{1}{NT} \|\Delta_2\|_F^2 \geq \frac{c_2}{\sqrt{NT}} \right\}$$

The RSC condition says

$$\|\Delta_1 + \Delta_2 \odot X\|_F^2 \geq \kappa_c \|\Delta_1\|_F^2 + \kappa_c \|\Delta_2\|_F^2 - (N + T)B.$$

This is to be proved in Lemma E.3.

We shall consider the general structure of x_{it} :

$$x_{it} = \mu_{it} + e_{it}$$

where μ_{it} is the mean part of x_{it} , which can be either deterministic or random. We shall denote the conditional expectation $\mathbb{E}_\mu(\cdot) = \mathbb{E}(\cdot | \mu_{it} : t \leq T, i \leq N)$ when μ_{it} is random. Correspondingly, denote by $P_\mu(\cdot)$ as the conditional probability.

Lemma E.3. *For any $c_1 > 0$, there are $\kappa_c, c_2, B > 0$, uniformly for $(\Delta_1, \Delta_2) \in \mathcal{C}(c_1, c_2)$,*

$$\|\Delta_1 + \Delta_2 \odot X\|_F^2 \geq \kappa_c \|\Delta_1\|_F^2 + \kappa_c \|\Delta_2\|_F^2 - (N + T)B$$

provided that the following two conditions hold:

(i) *Variation in X : for all $i \leq N, t \leq T$, all eigenvalues of Σ_{it} are bounded from below by a nonrandom constant $c_0 > 0$ almost surely, where*

$$\Sigma_{it} = \begin{pmatrix} 1 & \mathbb{E}_\mu x_{it} \\ \mathbb{E}_\mu x_{it} & \mathbb{E}_\mu x_{it}^2 \end{pmatrix} = \begin{pmatrix} 1 & \mu_{it} \\ \mu_{it} & \mu_{it}^2 + \mathbb{E}_\mu e_{it}^2 \end{pmatrix}$$

(ii) *Concentration inequality: Define*

$$\mathcal{B}(x) := \{(\Delta_1, \Delta_2) \in \mathcal{C}(c_1, c_2), \frac{1}{NT} \|\Delta_1\|_F^2 + \frac{1}{NT} \|\Delta_2\|_F^2 \leq x\}.$$

For any $C_0 > 0$, $x > 0$ and sufficiently large constant $B > 0$, there are constants $C', c' > 0$ that only depend on C_0 , almost surely for $\{\mu_{it}\}$,

$$P_\mu \left(\sup_{(\Delta_1, \Delta_2) \in \mathcal{B}(x)} \left| \frac{1}{NT} \sum_{i=1}^N \sum_{t=1}^T (Z_\mu(\Delta)_{it} - \mathbb{E}_\mu Z(\Delta)_{it}) \right| > \frac{N+T}{NT} B + C_0 x \right) \leq C' \exp(-c' NT x^2) \quad (\text{E.1})$$

where $Z_\mu(\Delta)_{it} := (x_{it} \Delta_{2,it} + \Delta_{1,it})^2$.

Condition (ii) of Lemma E.3 requires a concentration inequality over $\mathcal{B}(x)$, which is an intersection of the low-rank set with

$$\left\{ (\Delta_1, \Delta_2) : \frac{c_2}{\sqrt{NT}} \leq \frac{1}{NT} \|\Delta_1\|_F^2 + \frac{1}{NT} \|\Delta_2\|_F^2 \leq x \right\}. \quad (\text{E.2})$$

This condition is often verifiable using more primitive conditions on restricted strong convexity sets, e.g., Lemma 14 of Klopp (2014) and result (32) of Negahban and Wainwright (2012). Below, we provide one set of primitive conditions to verify it. When $x < \frac{c_2}{\sqrt{NT}}$, $\mathcal{B}(x)$ is an empty set, so (E.1) is trivially satisfied. Thus, we focus on $x > \frac{c_2}{\sqrt{NT}}$.

Lemma E.4. *Suppose*

(i') *for all (i, t) $|\Delta_{1,it}| + |\Delta_{2,it}| < C$ when $(\Delta_1, \Delta_2) \in \mathcal{B}(x)$ and $|x_{it}| < C$;*

(ii') *x_{it} is independent across (i, t) , conditioning on $\{\mu_{it} : i \leq N, t \leq T\}$.*

Then the concentration inequality (E.1) and thus condition (ii) of Lemma E.3 holds.

Remark E.1. The primitive condition (i') requires the estimation errors Δ_1 and Δ_2 have bounded entrywise $\|\cdot\|_\infty$ -norm. We can always restrict the parameter space so that the regularization problem is optimized on sufficiently large compact sets to satisfy this condition. Additionally, for the iid Bernoulli regressors x_{it} as in the matrix completion problem, it is shown that the estimation error of the nuclear-norm penalized regression satisfies this condition (Theorem 1 of Chen et al. (2019)). Note that we require $|x_{it}| < C$ for technical reasons, which may be restrictive given that x_{it} has a factor structure (factors and loadings should be bounded). This is potentially extendable to variables with unbounded support but sub-Gaussian tails.

Condition (ii') in the above Lemma is imposed *conditioning on* μ_{it} , which is important, as it allows factor models on x_{it} . For instance, when $x_{it} = l'_i w_t + e_{it}$, $\mu_{it} = l'_i w_t$. This assumption then says that conditioning on $l'_i w_t$, x_{it} and thus the residual e_{it} is independent across (i, t) .

One may also add an additional $\|\cdot\|_\infty$ -norm constraint to the nuclear-norm penalized regression, such as $\|M\|_\infty < C$ and $\|\Theta\|_\infty < C$. Note that C may be arbitrarily large and independent of (N, T) , so *neither* C is an additional tuning parameter *nor* the $\|\cdot\|_\infty$ -constraint is an additional regularization in the usual sense. The $\|\cdot\|_\infty$ -constraint is simply a formalization to require that elements in the estimator do not diverge.

Proof. Proof of Lemma E.3

We use the standard peeling argument. For notational simplicity, write

$$\Delta = (\Delta_1, \Delta_2), \quad X(\Delta) := \|\Delta_1 + \Delta_2 \odot X\|_F^2.$$

Let $c_0 = \min_{it} \psi_{\min}(\Sigma_{it})$. Then

$$\mathbb{E}_\mu X(\Delta) = \sum_{it} (\Delta_{1,it}, \Delta_{2,it}) \Sigma_{it} (\Delta_{1,it}, \Delta_{2,it})' \geq c_0 \|\Delta_1\|_F^2 + c_0 \|\Delta_2\|_F^2.$$

Define, for sufficiently large $B > 0$,

$$\mathcal{E}_{NT}(\Delta) := \{|X(\Delta) - \mathbb{E}_\mu X(\Delta)| > \frac{1}{2} \mathbb{E}_\mu X(\Delta) + (N + T)B\}.$$

We aim to show $P(\exists \Delta \in \mathcal{C}(c_1, c_2), \mathcal{E}_{NT}(\Delta)) \rightarrow 0$. Once this is achieved, then $P(\forall \Delta \in \mathcal{C}(c_1, c_2), \mathcal{E}_{NT}(\Delta)^c) \rightarrow 1$. On the event $\mathcal{E}_{NT}(\Delta)^c$, the RSC condition holds for $\kappa_c = c_0/2$, because

$$X(\Delta) \geq \frac{1}{2} \mathbb{E}_\mu X(\Delta) - (N + T)B \geq \kappa_c \|\Delta_1\|_F^2 + \kappa_c \|\Delta_2\|_F^2 - (N + T)B.$$

To prove $P(\exists \Delta \in \mathcal{C}(c_1, c_2), \mathcal{E}_{NT}(\Delta)) \rightarrow 0$, let

$$\Gamma_l = \{\Delta \in \mathcal{C}(c_1, c_2) : 2^l v_{NT} \leq \frac{1}{NT} \mathbb{E}_\mu X(\Delta) \leq 2^{l+1} v_{NT}\}$$

where $v_{NT} = \frac{B}{\sqrt{NT}}$, and $l \in \mathbb{N}$. We let $c_2 = 2c_0^{-1}B$ in the definition of $\mathcal{C}(c_1, c_2)$. Then, for $\Delta \in \mathcal{C}(c_1, c_2)$, we have

$$\frac{1}{NT} \|\Delta_1\|_F^2 + \frac{1}{NT} \|\Delta_2\|_F^2 \geq 2c_0^{-1} v_{NT}.$$

Then $\frac{1}{NT}\mathbf{E}_\mu X(\Delta) \geq \frac{1}{NT}c_0\|\Delta_1\|_F^2 + \frac{1}{NT}c_0\|\Delta_2\|_F^2 \geq 2v_{NT}$. Hence there is $l \in \mathbb{N}$ so that $\Delta \in \Gamma_l$ as long as $\Delta \in \mathcal{C}(c_1, c_2)$. Thus $\mathcal{C}(c_1, c_2) \subset \cup_{l=1}^\infty \Gamma_l$.

In addition, if $\Delta \in \Gamma_l$, then $\mathcal{E}_{NT}(\Delta)$ implies

$$|X(\Delta) - \mathbf{E}_\mu X(\Delta)| - (N+T)B > \frac{1}{2}\mathbf{E}_\mu X(\Delta) \geq \frac{1}{2}NT2^l v_{NT} = \frac{1}{4}NT2^{l+1}v_{NT}$$

and

$$\frac{1}{NT}\|\Delta_1\|_F^2 + \frac{1}{NT}\|\Delta_2\|_F^2 \leq \frac{c_0^{-1}}{NT}\mathbf{E}_\mu X(\Delta) \leq c_0^{-1}2^{l+1}v_{NT}.$$

This also implies $\Gamma_l \subset \mathcal{B}(x_l)$ with $x_l := c_0^{-1}2^{l+1}v_{NT}$. Hence for c' that only depends on c_0 (but not on B), (E.1) yields

$$\begin{aligned} & P(\exists \Delta \in \mathcal{C}(c_1, c_2), \mathcal{E}_{NT}(\Delta)) = \mathbf{E}P_\mu(\exists \Delta \in \mathcal{C}(c_1, c_2), \mathcal{E}_{NT}(\Delta)) \\ & \leq \mathbf{E} \sum_{l=1}^\infty P_\mu\left(\sup_{\Delta \in \mathcal{B}(x_l)} |X(\Delta) - \mathbf{E}_\mu X(\Delta)| > \frac{1}{4}NT2^{l+1}v_{NT} + (N+T)B\right) \\ & = \mathbf{E} \sum_{l=1}^\infty P_\mu\left(\sup_{\Delta \in \mathcal{B}(x_l)} \left|\frac{1}{NT} \sum_{it} (Z_\mu(\Delta)_{it} - \mathbf{E}_\mu Z(\Delta)_{it})\right| > \frac{c_0}{4}x_l + \frac{(N+T)B}{NT}\right) \\ & \leq \mathbf{E} \sum_{l=1}^\infty C' \exp(-c'4^{l+1}B^2) \leq \frac{\exp(-16c'B^2)}{1 - \exp(-16c'B^2)} < \epsilon \end{aligned}$$

for any arbitrarily small $\epsilon > 0$, where the last inequality holds for sufficiently large B .

Proof of Lemma E.4

Let

$$Z := \sup_{(\Delta_1, \Delta_2) \in \mathcal{B}(x)} \left| \frac{1}{NT} \sum_{i=1}^N \sum_{t=1}^T (Z_\mu(\Delta)_{it} - \mathbf{E}_\mu Z(\Delta)_{it}) \right|$$

where we recall $Z_\mu(\Delta)_{it} := (x_{it}\Delta_{2,it} + \Delta_{1,it})^2$.

We first bound $\mathbf{E}_\mu Z$, then bound the tail probability of $Z > \mathbf{E}_\mu Z$.

As for $\mathbf{E}_\mu Z$, for any $\Delta \in \mathcal{B}(x) \subset \mathcal{C}(c_1, c_2)$,

$$\begin{aligned} \|\Delta_1\|_n + \|\Delta_2\|_n &= \|\mathcal{P}_1(\Delta_1) + \mathcal{M}_1(\Delta_1)\|_n + \|\mathcal{P}_2(\Delta_2) + \mathcal{M}_2(\Delta_2)\|_n \\ &\leq (1+c_1)\|\mathcal{M}_1(\Delta_1)\|_n + (1+c_1)\|\mathcal{M}_2(\Delta_2)\|_n \\ &\leq (1+c_1)\sqrt{\text{rank}(M)}\|\mathcal{M}_1(\Delta_1)\|_F + (1+c_1)\sqrt{\text{rank}(\Theta)}\|\mathcal{M}_2(\Delta_2)\|_F \\ &\leq C(\|\Delta_1\|_F + \|\Delta_2\|_F) \leq C\sqrt{2}\sqrt{NTx}, \end{aligned}$$

where the first and third inequalities follow from the definition of $\mathcal{B}(x)$ and $\mathcal{C}(c_1, c_2)$ while the second inequality follows from Lemma E.2. Thus,

$$\|\Delta_1\|_n + \|\Delta_2\|_n \leq C^* \sqrt{NTx}, \quad (\text{E.3})$$

where C^* only depends c_1 and the ranks of the low rank matrices M and Θ taking their true values. Next, let ϵ_{it} be an iid Rademacher sequence. Let $G_1 = (\epsilon_{it})_{N \times T}$ and $G_2 = (\epsilon_{it}x_{it})_{N \times T}$, and note both are $N \times T$ matrices. Then, by the expected operator norm for sub-Gaussian matrices, (e.g., Exercise 4.4.6 of Vershynin (2018)), $\mathbb{E}_\mu \|G_1\| + \mathbb{E}_\mu \|G_2\| \leq C(\sqrt{N+T})$ where C is nonrandom. Then,

$$\begin{aligned} \mathbb{E}_\mu Z &\stackrel{(i)}{\leq} 2\mathbb{E}_\mu \sup_{(\Delta_1, \Delta_2) \in \mathcal{B}(x)} \left| \frac{1}{NT} \sum_{i=1}^N \sum_{t=1}^T Z_\mu(\Delta)_{it} \epsilon_{it} \right| \\ &\stackrel{(ii)}{\leq} C\mathbb{E}_\mu \sup_{(\Delta_1, \Delta_2) \in \mathcal{B}(x)} \left| \frac{1}{NT} \sum_{i=1}^N \sum_{t=1}^T \epsilon_{it} x_{it} \Delta_{2,it} \right| + C\mathbb{E}_\mu \sup_{(\Delta_1, \Delta_2) \in \mathcal{B}(x)} \left| \frac{1}{NT} \sum_{i=1}^N \sum_{t=1}^T \epsilon_{it} \Delta_{1,it} \right| \\ &= C\mathbb{E}_\mu \sup_{(\Delta_1, \Delta_2) \in \mathcal{B}(x)} \left| \frac{1}{NT} \text{tr}(G_1 \Delta'_1) \right| + C\mathbb{E}_\mu \sup_{(\Delta_1, \Delta_2) \in \mathcal{B}(x)} \left| \frac{1}{NT} \text{tr}(G_2 \Delta'_2) \right| \\ &\leq C\mathbb{E}_\mu \sup_{(\Delta_1, \Delta_2) \in \mathcal{B}(x)} \frac{1}{NT} \|G_1\| \|\Delta_1\|_n + C\mathbb{E}_\mu \sup_{(\Delta_1, \Delta_2) \in \mathcal{B}(x)} \frac{1}{NT} \|G_2\| \|\Delta_2\|_n \\ &\leq C \frac{\sqrt{N+T}}{NT} \sup_{(\Delta_1, \Delta_2) \in \mathcal{B}(x)} \|\Delta_1\|_n + C \frac{\sqrt{N+T}}{NT} \sup_{(\Delta_1, \Delta_2) \in \mathcal{B}(x)} \|\Delta_2\|_n \\ &\stackrel{(iii)}{\leq} C \frac{\sqrt{N+T}}{NT} \sqrt{NTx} = 2C \sqrt{\frac{N+T}{NTC_0}} \sqrt{\frac{C_0}{4}} x \leq \frac{C_0 x}{8} + B \frac{N+T}{NT}. \end{aligned} \quad (\text{E.4})$$

Inequality (i) follows from the standard symmetrization argument. Inequality (ii) uses the contraction inequality (e.g., (2.3) of Koltchinskii (2011)), which requires $|\Delta_{it}| < C$ and $|x_{it}| < C$ for all (i, t) . Inequality (iii) follows from (E.3) for $\Delta \in \mathcal{B}(x)$. The last inequality holds for $B > \frac{2C^2}{C_0}$.

We now bound the tail probability of $Z > \mathbb{E}_\mu Z$. The condition $|\Delta_{it}| < C$ and $|x_{it}| < C$ implies there is $C_x > 0$ with

$$(x_{it} \Delta_{2,it} + \Delta_{1,it})^2 < C_x.$$

Also, conditionally on $\{\mu_{it}\}$, x_{it} are independent across (i, t) . We can thus apply the Massart inequality (e.g., Theorem 14.2 of Bühlmann and van der Geer (2011)) to reach $P_\mu(Z \geq \mathbb{E}_\mu Z + t) \leq \exp(-\frac{NTt^2}{8C_x^2})$ for any $t > 0$. Now set $t = \frac{7C_0 x}{8}$.

$$P_\mu \left(Z > \frac{N+T}{NT} B + C_0 x \right) \leq P_\mu \left(Z > \mathbb{E}_\mu Z - \frac{C_0 x}{8} + C_0 x \right) = P_\mu \left(Z > \mathbb{E}_\mu Z + \frac{7C_0 x}{8} \right)$$

$$\leq \exp\left(-\frac{NTt^2}{8C_x^2}\right) = \exp\left(-\frac{49C_0^2}{512C_x^2}NTx^2\right).$$

The first inequality follows from (E.4). Note that C_x does not depend on B . $\square$

E.4. Proof of Proposition E.1: Convergence of $\tilde{\Theta}_S, \tilde{M}_S$. In the proof below, we set $S = I$. That is, we consider estimation using only the data with $t \in I$. We set $T_0 = |I|_0$. The proof carries over to $S = I^c$ or $S = I \cup I^c$. We suppress the subscript S for notational simplicity. Let $\Delta_1 = \tilde{M} - M$ and $\Delta_2 = \tilde{\Theta} - \Theta$. Then

$$\|Y - \tilde{M} - X \odot \tilde{\Theta}\|_F^2 = \|\Delta_1 + X \odot \Delta_2\|_F^2 + \|U\|_F^2 - 2\text{tr}[U'(\Delta_1 + X \odot \Delta_2)].$$

Note that $\text{tr}(U'(X \odot \Delta_2)) = \text{tr}(\Delta_2'(X \odot U))$. Thus, by claim (v),

$$\begin{aligned} |2\text{tr}[U'(\Delta_1 + X \odot \Delta_2)]| &\leq 2\|U\|\|\Delta_1\|_n + 2\|X \odot U\|\|\Delta_2\|_n \\ &\leq (1-c)\nu_0\|\Delta_1\|_n + (1-c)\nu_1\|\Delta_2\|_n. \end{aligned}$$

Thus, $\|Y - \tilde{M} - X \odot \tilde{\Theta}\|_F^2 \leq \|Y - M - X \odot \Theta\|_F^2$ (evaluated at the true parameters) implies

$$\begin{aligned} \|\Delta_1 + X \odot \Delta_2\|_F^2 + \nu_0\|\tilde{M}\|_n + \nu_1\|\tilde{\Theta}\|_n &\leq (1-c)\nu_0\|\Delta_1\|_n + (1-c)\nu_1\|\Delta_2\|_n \\ &\quad + \nu_0\|M\|_n + \nu_1\|\Theta\|_n. \end{aligned}$$

Now

$$\begin{aligned} \|\tilde{M}\|_n &= \|\Delta_1 + M\|_n = \|M + \mathcal{P}_1(\Delta_1) + \mathcal{M}_1(\Delta_1)\|_n \\ &\geq \|M + \mathcal{P}_1(\Delta_1)\|_n - \|\mathcal{M}_1(\Delta_1)\|_n \\ &= \|M\|_n + \|\mathcal{P}_1(\Delta_1)\|_n - \|\mathcal{M}_1(\Delta_1)\|_n, \end{aligned}$$

where the last equality follows from claim (i). A similar lower bound applies to $\|\tilde{\Theta}\|_n$. Therefore,

$$\begin{aligned} &\|\Delta_1 + X \odot \Delta_2\|_F^2 + c\nu_0\|\mathcal{P}_1(\Delta_1)\|_n + c\nu_1\|\mathcal{P}_2(\Delta_2)\|_n \\ &\leq (2-c)\nu_0\|\mathcal{M}_1(\Delta_1)\|_n + (2-c)\nu_1\|\mathcal{M}_2(\Delta_2)\|_n. \end{aligned} \tag{E.5}$$

In the case where U is Gaussian, $\|U\|$ and $\|X \odot U\| \asymp \max\{\sqrt{N}, \sqrt{T}\}$; while in the more general case, we set $\nu_0 \asymp \nu_1 \asymp \max\{\sqrt{N}, \sqrt{T}\}$. Inequality (E.5) implies there is $c_1 > 0$,

$$\|\mathcal{P}_1(\Delta_1)\|_n + \|\mathcal{P}_2(\Delta_2)\|_n \leq c_1\|\mathcal{M}_1(\Delta_1)\|_n + c_1\|\mathcal{M}_2(\Delta_2)\|_n.$$

We now apply Lemma E.3: Recall the set

$$\mathcal{C}(c_1, c_2) = \left\{ (\Delta_1, \Delta_2) : \|\mathcal{P}_1(\Delta_1)\|_n + \|\mathcal{P}_2(\Delta_2)\|_n \leq c_1 \|\mathcal{M}_1(\Delta_1)\|_n + c_1 \|\mathcal{M}_2(\Delta_2)\|_n, \right. \\ \left. \|\Delta_1\|_F^2 + \|\Delta_2\|_F^2 \geq c_2 \sqrt{NT} \right\}.$$

Consider two cases:

Case 1: $\|\Delta_1\|_F^2 + \|\Delta_2\|_F^2 \leq c_2 \sqrt{NT}$. Then $\|\Delta_1\|_F^2 + \|\Delta_2\|_F^2 \leq O_P(\nu_0^2 + \nu_1^2)$.

Case 2: $\|\Delta_1\|_F^2 + \|\Delta_2\|_F^2 \geq c_2 \sqrt{NT}$. Then $(\Delta_1, \Delta_2) \in \mathcal{C}(c_1, c_2)$, so by Lemma E.3, for some constants $\kappa, \eta > 0$, with large probability

$$\|\Delta_1 + \Delta_2 \odot X\|_F^2 \geq \kappa \|\Delta_1\|_F^2 + \kappa \|\Delta_2\|_F^2 - (N + T)\eta.$$

Then (E.5) and the proven claims show that for a generic $C > 0$,

$$\begin{aligned} \|\Delta_1\|_F^2 + \|\Delta_2\|_F^2 &\leq C\nu_0 \|\mathcal{M}_1(\Delta_1)\|_n + C\nu_1 \|\mathcal{M}_2(\Delta_2)\|_n + O_P(N + T) \\ &\stackrel{\text{claim (iv)}}{\leq} C\nu_0 \|\mathcal{M}_1(\Delta_1)\|_F \sqrt{\text{rank}(\mathcal{M}_2(\Delta_1))} \\ &\quad + C\nu_1 \|\mathcal{M}_2(\Delta_2)\|_F \sqrt{\text{rank}(\mathcal{M}_2(\Delta_2))} + O_P(N + T) \\ &\stackrel{\text{claim (iii)}}{\leq} C\nu_0 \|\mathcal{M}_1(\Delta_1)\|_F \sqrt{2K_1} + C\nu_1 \|\mathcal{M}_2(\Delta_2)\|_F \sqrt{2K_2} + O_P(N + T) \\ &\stackrel{\text{claim (ii)}}{\leq} C\nu_0 \|\Delta_1\|_F + C\nu_1 \|\Delta_2\|_F + O_P(N + T) \\ &\leq C \max\{\nu_0, \nu_1\} \sqrt{\|\Delta_1\|_F^2 + \|\Delta_2\|_F^2} + O_P(N + T). \end{aligned}$$

Thus, $\|\Delta_1\|_F^2 + \|\Delta_2\|_F^2 \leq O_P(\nu_0^2 + \nu_1^2)$.

E.5. Proof of Proposition E.1: Convergence of $\tilde{\Lambda}_S, \tilde{A}_S$.

Proof. The proof proceeds in the following steps.

Step 1: Bound the eigenvalues

Replace $\nu_0^2 + \nu_1^2$ with $O_P(N + T)$, then

$$\|\tilde{\Theta}_S - \Theta_S\|_F^2 = O_P(N + T).$$

Let $S_f = \frac{1}{T_0} \sum_{t \in I} f_t f_t'$, $\Sigma_f = \frac{1}{T} \sum_{t=1}^T f_t f_t'$ and $S_\Lambda = \frac{1}{N} \Lambda' \Lambda$. Let $\psi_{I,1}^2 \geq \dots \geq \psi_{I,K_1}^2$ be the K_1 nonzero eigenvalues of $\frac{1}{NT_0} \Theta_I \Theta_I' = \frac{1}{N} \Lambda S_f \Lambda'$. Let $\tilde{\psi}_1^2 \geq \dots \geq \tilde{\psi}_{K_2}^2$ be the first K_2 nonzero singular values of $\frac{1}{NT_0} \tilde{\Theta}_I \tilde{\Theta}_I'$. Also, let ψ_j^2 be the j th largest eigenvalue of $\frac{1}{N} \Lambda \Sigma_f \Lambda'$. Note that $\psi_1^2, \dots, \psi_{K_1}^2$ are the same as the eigenvalues of $\Sigma_f^{1/2} S_\Lambda \Sigma_f^{1/2}$. Hence by Assumption 3.2, there are constants $c_1, \dots, c_{K_1} > 0$, such that

$$\psi_j^2 = c_j, \quad j = 1, \dots, K_1.$$

Then by Weyl's theorem, for $j = 1, \dots, \min\{T_0, N\}$, with the assumption that $\|S_f - \Sigma_f\| = O_P(\frac{1}{\sqrt{T}})$, $|\psi_{I,j}^2 - \psi_j^2| \leq \frac{1}{N} \|\Lambda(S_f - \Sigma_f)\Lambda'\| \leq O(1)\|S_f - \Sigma_f\| = O_P(\frac{1}{\sqrt{T}})$. This also implies $\|\Theta_I\| = \psi_{I,1}\sqrt{NT_0} = \sqrt{(c_1 + o_P(1))T_0N}$.

Still by Weyl's theorem, for $j = 1, \dots, \min\{T_0, N\}$,

$$\begin{aligned} |\tilde{\psi}_j^2 - \psi_{I,j}^2| &\leq \frac{1}{NT_0} \|\tilde{\Theta}_I \tilde{\Theta}_I' - \Theta_I \Theta_I'\| \\ &\leq \frac{2}{NT_0} \|\Theta_I\| \|\tilde{\Theta}_I - \Theta_I\| + \frac{1}{NT_0} \|\tilde{\Theta}_I - \Theta_I\|^2 = O_P\left(\frac{1}{\sqrt{N}} + \frac{1}{\sqrt{T}}\right). \end{aligned}$$

implying

$$|\tilde{\psi}_j^2 - \psi_j^2| = O_P\left(\frac{1}{\sqrt{N}} + \frac{1}{\sqrt{T}}\right).$$

Then for all $j \leq K_1$, with probability approaching one,

$$\begin{aligned} |\psi_{j-1}^2 - \tilde{\psi}_j^2| &\geq |\psi_{j-1}^2 - \psi_j^2| - |\psi_j^2 - \tilde{\psi}_j^2| \geq (c_{j-1} - c_j)/2 \\ |\tilde{\psi}_j^2 - \psi_{j+1}^2| &\geq |\psi_j^2 - \psi_{j+1}^2| - |\tilde{\psi}_j^2 - \psi_j^2| \geq (c_j - c_{j+1})/2 \end{aligned} \quad (\text{E.6})$$

with $\psi_{K_1+1}^2 = c_{K_1+1} = 0$ because $\Theta_I \Theta_I'$ has at most K_1 nonzero eigenvalues.

Step 2: Characterize the eigenvectors

Next, we show that there is a $K_1 \times K_1$ matrix H_1 , so that the columns of $\frac{1}{\sqrt{N}}\Lambda H_1$ are the first K_1 eigenvectors of $\Lambda \Sigma_f \Lambda'$. Let $L = S_\Lambda^{1/2} \Sigma_f S_\Lambda^{1/2}$. Let R be a $K_1 \times K_1$ matrix whose columns are the eigenvectors of L . Then $D = R'LR$ is a diagonal matrix of the eigenvalues of L that are distinct nonzeros according to Assumption 3.2. Let $H_1 = S_\Lambda^{-1/2}R$. Then

$$\begin{aligned} \frac{1}{N} \Lambda \Sigma_f \Lambda' \Lambda H_1 &= \Lambda S_\Lambda^{-1/2} S_\Lambda^{1/2} \Sigma_f S_\Lambda^{1/2} S_\Lambda^{1/2} H_1 = \Lambda S_\Lambda^{-1/2} R R' L R \\ &= \Lambda H_1 D. \end{aligned}$$

Now $\frac{1}{N}(\Lambda H_1)' \Lambda H_1 = H_1' S_\Lambda H_1 = R' R = I$. So the columns of $\Lambda H_1 / \sqrt{N}$ are the eigenvectors of $\Lambda \Sigma_f \Lambda'$, corresponding to the eigenvalues in D .

Importantly, the rotation matrix H_1 , by definition, depends only on S_Λ, Σ_f , which is time-invariant, and does not depend on the sample split.

Step 3: Prove convergence

We first assume $\hat{K}_1 = K_1$. The proof of the consistency is given in step 4 below. Once this is true, then the following argument can be carried out conditional on

the event $\widehat{K}_1 = K_1$. Apply the Davis-Kahan sin-theta inequality, and by (E.6),

$$\begin{aligned} & \left\| \frac{1}{\sqrt{N}} \widetilde{\Lambda}_I - \frac{1}{\sqrt{N}} \Lambda H_1 \right\|_F \leq \frac{\frac{1}{N} \|\Lambda \Sigma_f \Lambda' - \frac{1}{T_0} \widetilde{\Theta}_I \widetilde{\Theta}_I'\|}{\min_{j \leq K_2} \min\{|\psi_{j-1}^2 - \widetilde{\psi}_j^2|, |\widetilde{\psi}_j^2 - \psi_{j+1}^2|\}} \\ & \leq O_P(1) \frac{1}{N} \|\Lambda \Sigma_f \Lambda' - \frac{1}{T_0} \widetilde{\Theta}_I \widetilde{\Theta}_I'\| \\ & \leq O_P(1) \frac{1}{N} \|\Lambda(\Sigma_f - S_f) \Lambda'\| + \frac{1}{NT_0} \|\Theta_I \Theta_I' - \widetilde{\Theta}_I \widetilde{\Theta}_I'\| = O_P\left(\frac{1}{\sqrt{T}} + \frac{1}{\sqrt{N}}\right). \end{aligned}$$

Step 4: Prove $P(\widehat{K}_1 = K_1) = 1$.

Note that $\psi_j(\widetilde{\Theta}) = \widetilde{\psi}_j \sqrt{NT}$. By Step 1, for all $j \leq K_1$, $\widetilde{\psi}_j^2 \geq c_j - o_P(1) \geq c_j/2$ with probability approaching one. Also, $\widetilde{\psi}_{K_1+1}^2 \leq O_P(T^{-1/2} + N^{-1/2})$, implying that

$$\min_{j \leq K_1} \psi_j(\widetilde{\Theta}) \geq c_{K_1} \sqrt{NT}/2, \quad \max_{j > K_1} \psi_j(\widetilde{\Theta}) \leq O_P(T^{-1/4} + N^{-1/4}) \sqrt{NT}.$$

In addition, $\nu_0^{1/2-\epsilon} \|\widetilde{\Theta}\|^{1/2+\epsilon} \asymp (\sqrt{N+T})^{1/2-\epsilon} (\sqrt{NT})^{1/2+\epsilon}$ for some small $\epsilon \in (0, 1)$,

$$\min_{j \leq K_1} \psi_j(\widetilde{\Theta}) \geq \nu_0^{1/2-\epsilon} \|\widetilde{\Theta}\|^{1/2+\epsilon}, \quad \max_{j > K_1} \psi_j(\widetilde{\Theta}) \leq o_P(1) \nu_0^{1/2-\epsilon} \|\widetilde{\Theta}\|^{1/2+\epsilon}.$$

This proves the consistency of $\widehat{K}_1$.

The proof of convergence for $\widetilde{A}_I$ and consistency of $\widehat{K}_2$ follow from the same argument. □

APPENDIX F. PROOF OF THEOREM 3.1

In the factor model

$$x_{it} = l'_i w_t + e_{it},$$

we write $\widehat{e}_{it} = x_{it} - \widehat{l'_i w_t}$, $\widehat{\mu}_{it} = \widehat{l'_i w_t}$ and $\mu_{it} = l'_i w_t$.

Let

$$C_{NT} = \min\{\sqrt{N}, \sqrt{T}\}.$$

First recall that $(\widetilde{f}_s, \dot{\lambda}_i)$ are computed as the preliminary estimators in step 3. The main technical requirement of these estimators is that their estimation effects are negligible, specifically, there is a rotation matrix H_1 that is independent of the sample splitting, for each fixed $t \notin I$,

$$\left\| \frac{1}{N} \sum_j (\dot{\lambda}_j - H_1' \lambda_j) e_{jt} \right\|^2 = O_P(C_{NT}^{-4}),$$

$$\frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \left\| \frac{1}{T} \sum_{s \notin I} f_s (\tilde{f}_s - H_1^{-1} f_s)' \mu_{is} e_{is} \right\|^2 = O_P(C_{NT}^{-4}).$$

These are given in Lemmas G.5 and G.6 for the factor model.

F.1. Behavior of $\hat{f}_t$. Recall that for each $t \notin I$,

$$(\hat{f}_{I,t}, \hat{g}_{I,t}) := \arg \min_{f_t, g_t} \sum_{i=1}^N (\hat{y}_{it} - \tilde{\alpha}'_i g_t - \hat{e}_{it} \tilde{\lambda}'_i f_t)^2.$$

For notational simplicity, we simply write $\hat{f}_t = \hat{f}_{I,t}$ and $\hat{g}_t = \hat{g}_{I,t}$, but keep in mind that $\tilde{\alpha}$ and $\tilde{\lambda}$ are estimated through low rank estimation on data I . Note that $\tilde{\lambda}_i$ consistently estimates λ_i up to a rotation matrix H_1' , so $\hat{f}_t$ is consistent for $H_1^{-1} f_t$. However, as we shall explain below, it is difficult to establish the asymptotic normality for $\hat{f}_t$ centered at $H_1^{-1} f_t$. Instead, we obtain a new centering quantity, and obtain an expansion for

$$\sqrt{N}(\hat{f}_t - H_f f_t)$$

with a new rotation matrix H_f that is also independent of t . For the purpose of inference for θ_{it} , this is sufficient.

Let $\hat{w}_{it} = (\tilde{\lambda}'_i \hat{e}_{it}, \tilde{\alpha}'_i)'$, and $\hat{B}_t = \frac{1}{N} \sum_i \hat{w}_{it} \hat{w}'_{it}$. Define $w_{it} = (\lambda'_i e_{it}, \alpha'_i)'$, and

$$\hat{Q}_t = \frac{1}{N} \sum_i \hat{w}_{it} (\mu_{it} \lambda'_i f_t - \hat{\mu}_{it} \tilde{\lambda}'_i \tilde{f}_t + u_{it}).$$

We have

$$\begin{aligned} \begin{pmatrix} \hat{f}_t \\ \hat{g}_t \end{pmatrix} &= \hat{B}_t^{-1} \frac{1}{N} \sum_i \hat{w}_{it} (y_{it} - \hat{\mu}_{it} \tilde{\lambda}'_i \tilde{f}_t) \\ &= \begin{pmatrix} H_1^{-1} f_t \\ H_2^{-1} g_t \end{pmatrix} + \hat{B}_t^{-1} \hat{S}_t \begin{pmatrix} H_1^{-1} f_t \\ H_2^{-1} g_t \end{pmatrix} + \hat{B}_t^{-1} \hat{Q}_t \end{aligned} \quad (\text{F.1})$$

where

$$\hat{S}_t = \frac{1}{N} \sum_i \begin{pmatrix} \tilde{\lambda}_i \hat{e}_{it} (\lambda'_i H_1 e_{it} - \tilde{\lambda}'_i \hat{e}_{it}) & \tilde{\lambda}_i \hat{e}_{it} (\alpha'_i H_2 - \tilde{\alpha}'_i) \\ \tilde{\alpha}_i (\lambda'_i H_1 e_{it} - \tilde{\lambda}'_i \hat{e}_{it}) & \tilde{\alpha}_i (\alpha'_i H_2 - \tilde{\alpha}'_i) \end{pmatrix}.$$

Note that the “upper block” of $\hat{B}_t^{-1} \hat{S}_t$ is not first-order negligible. Essentially, this is due to the fact that the moment condition

$$\frac{\partial}{\partial \lambda_i} \mathbb{E} \lambda_i e_{it} (\dot{y}_{it} - \alpha'_i g_t - e_{it} \lambda'_i f_t) \neq 0,$$

so is not “Neyman orthogonal” with respect to λ_i . In Lemma G.7 below, we show that $\widehat{B}_t$ and $\widehat{S}_t$ both converge in probability to block diagonal matrices that are independent of t . Thus, g_t and f_t are “orthogonal”, and

$$\widehat{B}_t^{-1}\widehat{S}_t = \begin{pmatrix} \bar{H}_3 & 0 \\ 0 & \bar{H}_4 \end{pmatrix} + o_P(N^{-1/2}).$$

Then (F.1) implies that

$$\widehat{f}_t = H_f f_t + \text{upper block of } \widehat{B}_t^{-1}\widehat{Q}_t.$$

Therefore $\widehat{f}_t$ converges to f_t up to a new rotation matrix H_f , which equals H_1^{-1} up to an $o_P(1)$ term $\bar{H}_3 H_1^{-1}$. While the effect of $\bar{H}_3$ is not negligible, it is “absorbed” into the new rotation matrix. As such, we are able to establish the asymptotic normality for $\sqrt{N}(\widehat{f}_t - H_f f_t)$.

Proposition F.1. *For each fixed $t \notin I$, we have*

$$\widehat{f}_t - H_f f_t = H_f \left(\frac{1}{N} \sum_i \lambda_i \lambda_i' \mathbb{E} e_{it}^2 \right)^{-1} \frac{1}{N} \sum_i \lambda_i e_{it} u_{it} + O_P(C_{NT}^{-2}).$$

Proof. Define

$$B = \frac{1}{N} \sum_i \begin{pmatrix} H_1' \lambda_i \lambda_i' H_1 \mathbb{E} e_{it}^2 & 0 \\ 0 & H_2' \alpha_i \alpha_i' H_2 \end{pmatrix}, \quad S = \frac{1}{N} \sum_i \begin{pmatrix} H_1' \lambda_i (\lambda_i' H_1 - \widetilde{\lambda}_i') \mathbb{E} e_{it}^2 & 0 \\ 0 & \widetilde{\alpha}_i' (\alpha_i' H_2 - \widetilde{\alpha}_i') \end{pmatrix}.$$

Both B, S are independent of t due to the stationarity of e_{it}^2 . S depends on the sample splitting through $(\widetilde{\lambda}_i, \widetilde{\alpha}_i)$.

From (F.1),

$$\begin{aligned} \begin{pmatrix} \widehat{f}_t \\ \widehat{g}_t \end{pmatrix} &= (B^{-1}S + \mathbb{I}) \begin{pmatrix} H_1^{-1} f_t \\ H_2^{-1} g_t \end{pmatrix} + B^{-1} \frac{1}{N} \sum_i \begin{pmatrix} H_1' \lambda_i e_{it} \\ H_2' \alpha_i \end{pmatrix} u_{it} \\ &\quad + \sum_{d=1}^6 A_{dt}, \text{ where} \\ A_{1t} &= (\widehat{B}_t^{-1} \widehat{S}_t - B^{-1}S) \begin{pmatrix} H_1^{-1} f_t \\ H_2^{-1} g_t \end{pmatrix} \\ A_{2t} &= (\widehat{B}_t^{-1} - B^{-1}) \frac{1}{N} \sum_i \begin{pmatrix} H_1' \lambda_i e_{it} \\ H_2' \alpha_i \end{pmatrix} u_{it} \\ A_{3t} &= \widehat{B}_t^{-1} \frac{1}{N} \sum_i \begin{pmatrix} \widetilde{\lambda}_i' \widehat{e}_{it} - H_1' \lambda_i e_{it} \\ \widetilde{\alpha}_i - H_2' \alpha_i \end{pmatrix} u_{it} \end{aligned}$$

$$\begin{aligned}
A_{4t} &= \widehat{B}_t^{-1} \frac{1}{N} \sum_i \begin{pmatrix} \widetilde{\lambda}_i \widehat{e}_{it} - H'_1 \lambda_i e_{it} \\ \widetilde{\alpha}_i - H'_2 \alpha_i \end{pmatrix} (\mu_{it} \lambda'_i f_t - \widehat{\mu}_{it} \dot{\lambda}'_i \widetilde{f}_t) \\
A_{5t} &= (\widehat{B}_t^{-1} - B^{-1}) \frac{1}{N} \sum_i \begin{pmatrix} H'_1 \lambda_i e_{it} \\ H'_2 \alpha_i \end{pmatrix} (\mu_{it} \lambda'_i f_t - \widehat{\mu}_{it} \dot{\lambda}'_i \widetilde{f}_t) \\
A_{6t} &= B^{-1} \frac{1}{N} \sum_i \begin{pmatrix} H'_1 \lambda_i e_{it} \\ H'_2 \alpha_i \end{pmatrix} (\mu_{it} \lambda'_i f_t - \widehat{\mu}_{it} \dot{\lambda}'_i \widetilde{f}_t).
\end{aligned} \tag{F.2}$$

Note that $B^{-1}S$ is a block-diagonal matrix, with the upper block being $\bar{H}_3$:

$$\begin{aligned}
\bar{H}_3 &:= H_1^{-1} \left(\frac{1}{N} \sum_i \lambda_i \lambda'_i \mathbb{E} e_{it}^2 \right)^{-1} \frac{1}{N} \sum_i \lambda_i (\lambda'_i H_1 - \widetilde{\lambda}'_i) \mathbb{E} e_{it}^2 \\
H_{I,f} &:= (\bar{H}_3 + \mathbf{I}) H_1^{-1}.
\end{aligned} \tag{F.3}$$

For notational simplicity, we write $H_f = H_{I,f}$.

For fixed $t \notin I$, in the factor model, in Lemma G.9 we show that $\sum_{d=2}^5 A_{dt} = O_P(C_{NT}^{-2})$, the “upper block” of A_{6t} , $\frac{1}{N} \sum_i \lambda_i e_{it} (\widehat{l}'_i w_t \lambda'_i f_t - \widehat{l}'_i \widehat{w}_t \dot{\lambda}'_i \widetilde{f}_t) = O_P(C_{NT}^{-2})$ and the upper block of A_{1t} is $O_P(C_{NT}^{-2})$. Therefore,

$$\widehat{f}_t = H_f f_t + H_1^{-1} \left(\frac{1}{N} \sum_i \lambda_i \lambda'_i \mathbb{E} e_{it}^2 \right)^{-1} \frac{1}{N} \sum_i \lambda_i e_{it} u_{it} + O_P(C_{NT}^{-2}).$$

Given that $\bar{H}_3 = O_P(C_{NT}^{-1})$ we have $H_1^{-1} = H_f + O_P(C_{NT}^{-1})$. By $\frac{1}{N} \sum_i \lambda_i e_{it} u_{it} = O_P(N^{-1/2})$,

$$\widehat{f}_t = H_f f_t + H_f \left(\frac{1}{N} \sum_i \lambda_i \lambda'_i \mathbb{E} e_{it}^2 \right)^{-1} \frac{1}{N} \sum_i \lambda_i e_{it} u_{it} + O_P(C_{NT}^{-2}). \tag{F.4}$$

□

F.2. Behavior of $\widehat{\lambda}_i$. Recall that for fixed $i \leq N$,

$$(\widehat{\lambda}_{I,i}, \widehat{\alpha}_{I,i}) = \arg \min_{\lambda_i, \alpha_i} \sum_{s \notin I} (\widehat{y}_{is} - \alpha'_i \widehat{g}_{I,s} - \widehat{e}_{is} \lambda'_i \widehat{f}_{I,s})^2.$$

For notational simplicity, we simply write $\widehat{\lambda}_i = \widehat{\lambda}_{I,i}$ and $\widehat{\alpha}_i = \widehat{\alpha}_{I,i}$, but keep in mind that $\widetilde{\alpha}$ and $\widetilde{\lambda}$ are estimated through low rank estimation on data I . Write

$$T_0 = |I^c|_0.$$

$$(B^{-1}S + \mathbf{I}) \begin{pmatrix} H_1^{-1} & 0 \\ 0 & H_2^{-1} \end{pmatrix} := \begin{pmatrix} H_f & 0 \\ 0 & H_g \end{pmatrix}$$

$$(F.5) \quad \widehat{D}_i = \frac{1}{T_0} \sum_{s \notin I} \begin{pmatrix} \widehat{f}_s \widehat{f}'_s \widehat{e}_{is}^2 & \widehat{f}_s \widehat{g}'_s \widehat{e}_{is} \\ \widehat{g}_s \widehat{f}'_s \widehat{e}_{is} & \widehat{g}_s \widehat{g}'_s \end{pmatrix}, \quad D_i = \frac{1}{T_0} \sum_{s \notin I} \begin{pmatrix} H_f f_s f'_s H'_f \mathbf{E} e_{is}^2 & 0 \\ 0 & H_g g_s g'_s H'_g \end{pmatrix}$$

Importantly, $B^{-1}S$ is a block-diagonal matrix, with the upper block being $\bar{H}_3$, so $H_f = (\bar{H}_3 + \mathbf{I})H_1^{-1}$.

Proposition F.2. Fix a group $\mathcal{G} \subset \{1, \dots, N\}$. Write $\Omega_i := (\frac{1}{T} \sum_{s=1}^T f_s f'_s \mathbf{E} e_{is}^2)^{-1}$.

$$\frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} (\widehat{\lambda}_{I,i} - H_f'^{-1} \lambda_i) = H_f'^{-1} \frac{1}{|\mathcal{G}|_0 T_0} \sum_{i \in \mathcal{G}} \sum_{s \notin I} \Omega_i f_s e_{is} u_{is} + O_P(C_{NT}^{-2}).$$

Proof. By definition and (F.2),

$$\begin{aligned} \begin{pmatrix} \widehat{\lambda}_i \\ \widehat{\alpha}_i \end{pmatrix} &= \widehat{D}_i^{-1} \frac{1}{T_0} \sum_{s \notin I} \begin{pmatrix} \widehat{f}_s \widehat{e}_{is} \\ \widehat{g}_s \end{pmatrix} (y_{is} - \widehat{\mu}_{is} \dot{\lambda}'_i \widetilde{f}_s) \\ &= \begin{pmatrix} H_f'^{-1} \lambda_i \\ H_g'^{-1} \alpha_i \end{pmatrix} + D_i^{-1} \frac{1}{T_0} \sum_{s \notin I} \begin{pmatrix} H_f f_s e_{is} \\ H_g g_s \end{pmatrix} u_{is} + \sum_{d=1}^6 R_{di}, \quad \text{where,} \\ R_{1i} &= (\widehat{D}_i^{-1} - D_i^{-1}) \frac{1}{T_0} \sum_{s \notin I} \begin{pmatrix} H_f f_s e_{is} \\ H_g g_s \end{pmatrix} u_{is} \\ R_{2i} &= \widehat{D}_i^{-1} \frac{1}{T_0} \sum_{s \notin I} \begin{pmatrix} H_f f_s (\widehat{e}_{is} - e_{is}) u_{is} \\ 0 \end{pmatrix} \\ R_{3i} &= (\widehat{D}_i^{-1} - D_i^{-1}) \frac{1}{T_0} \sum_{s \notin I} \begin{pmatrix} \widehat{f}_s \widehat{e}_{is} \\ \widehat{g}_s \end{pmatrix} (\mu_{is} \lambda'_i f_s - \widehat{\mu}_{is} \dot{\lambda}'_i \widetilde{f}_s) \\ R_{4i} &= D_i^{-1} \frac{1}{T_0} \sum_{s \notin I} \begin{pmatrix} \widehat{f}_s \widehat{e}_{is} \\ \widehat{g}_s \end{pmatrix} (\mu_{is} \lambda'_i f_s - \widehat{\mu}_{is} \dot{\lambda}'_i \widetilde{f}_s) \\ R_{5i} &= \widehat{D}_i^{-1} \frac{1}{T_0} \sum_{s \notin I} \begin{pmatrix} \widehat{e}_{is} \mathbf{I} & 0 \\ 0 & \mathbf{I} \end{pmatrix} \begin{pmatrix} \widehat{f}_s - H_f f_s \\ \widehat{g}_s - H_g g_s \end{pmatrix} u_{is} \\ R_{6i} &= \widehat{D}_i^{-1} \frac{1}{T_0} \sum_{s \notin I} \begin{pmatrix} \widehat{f}_s \widehat{e}_{is} \\ \widehat{g}_s \end{pmatrix} (\lambda'_i H_f^{-1}, \alpha'_i H_g^{-1}) \begin{pmatrix} \widehat{e}_{is} \mathbf{I} & 0 \\ 0 & \mathbf{I} \end{pmatrix} \begin{pmatrix} \widehat{f}_s - H_f f_s \\ \widehat{g}_s - H_g g_s \end{pmatrix} \quad (F.6) \end{aligned}$$

Let r_{di} be the upper blocks of R_{di} . Lemma G.10 shows $\frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \|\widehat{D}_i^{-1} - D_i^{-1}\|^2 = O_P(C_{NT}^{-2})$. So $\frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \|R_{1i}\| = O_P(C_{NT}^{-2})$. Lemma G.2 shows $\frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \|\frac{1}{T} \sum_{s \notin I} f_s (\widehat{e}_{is} - e_{is}) u_{is}\|^2 = O_P(C_{NT}^{-4})$, so $\frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \|R_{2i}\| = O_P(C_{NT}^{-2})$.

By Lemma G.12,

$$\begin{aligned} \frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \left\| \frac{1}{T} \sum_{s \notin I} \widehat{f}_s \widehat{e}_{is} (\mu_{it} \lambda'_i f_s - \widehat{\mu}_{it} \dot{\lambda}'_i \widetilde{f}_s) \right\|^2 &= O_P(C_{NT}^{-4}) \\ \frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \left\| \frac{1}{T} \sum_{s \notin I} \widehat{g}_s (\mu_{it} \lambda'_i f_s - \widehat{\mu}_{it} \dot{\lambda}'_i \widetilde{f}_s) \right\|^2 &= O_P(C_{NT}^{-2}). \end{aligned}$$

So $\frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \|R_{3i}\| = O_P(C_{NT}^{-2}) = \frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \|r_{4i}\|$. Next, by Lemma G.13, $\frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \|r_{5i}\| = O_P(C_{NT}^{-2})$. By Lemma G.14, $\frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \|r_{6i}\| = O_P(C_{NT}^{-2})$. Therefore,

$$\sum_{d=1}^6 \frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \|r_{di}\| = O_P(C_{NT}^{-2}).$$

Now write $S_{f,I} = \frac{1}{T_0} \sum_{s \notin I} f_s f'_s$ and $S_f = \frac{1}{T} \sum_{s=1}^T f_s f'_s$,

$$\begin{aligned} \Omega_{I,i} &:= \left(\frac{1}{T_0} \sum_{s \notin I} f_s f'_s \mathbb{E} e_{is}^2 \right)^{-1} = S_{f,I}^{-1} (\mathbb{E} e_{is}^2)^{-1} \\ \Omega_i &:= \left(\frac{1}{T_0} \sum_{s=1}^T f_s f'_s \mathbb{E} e_{is}^2 \right)^{-1} = S_f^{-1} (\mathbb{E} e_{is}^2)^{-1}. \end{aligned}$$

By assumption, $S_{f,I} = S_f + O_P(T^{-1/2})$. It remains to bound, for $\omega_{is} = e_{is} u_{is}$,

$$\begin{aligned} &\left\| \frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} [\Omega_{I,i} - \Omega_i] \frac{1}{T_0} \sum_{s \notin I} f_s \omega_{is} \right\|^2 = \left\| \frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \Omega_{I,i} (S_{f,I} - S_f) \Omega_i \frac{1}{T_0} \sum_{s \notin I} f_s \omega_{is} \mathbb{E} e_{is}^2 \right\|^2 \\ &\leq \|S_{f,I} - S_f\|^2 \|S_f^{-1}\| \|S_{f,I}^{-1}\| \left\| \frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \frac{1}{T_0} \sum_{s \notin I} f_s \omega_{is} (\mathbb{E} e_{is}^2)^{-1} \right\|_F^2 \leq O_P(T^{-2} |\mathcal{G}|_0^{-1}). \end{aligned}$$

This finishes the proof. □

F.3. Proof of normality of $\widehat{\theta}_{it}$. Fix $t \notin I$. Suppose T is odd and $|I|_0 = |I^c|_0 = (T-1)/2$. Let

$$\begin{aligned} Q_I &:= \frac{1}{|\mathcal{G}|_0 |I|_0} \sum_{i \in \mathcal{G}} \sum_{s \notin I} \Omega_i f_s e_{is} u_{is} = O_P(C_{NT}^{-1} |\mathcal{G}|_0^{-1/2}), \\ Q_{I^c} &:= \frac{1}{|\mathcal{G}|_0 |I^c|_0} \sum_{i \in \mathcal{G}} \sum_{s \notin I^c} \Omega_i f_s e_{is} u_{is} = O_P(C_{NT}^{-1} |\mathcal{G}|_0^{-1/2}), \\ Q &:= \frac{1}{|\mathcal{G}|_0 T} \sum_{i \in \mathcal{G}} \sum_{s=1}^T \Omega_i f_s e_{is} u_{is}, \end{aligned}$$

$$J_t := \left(\frac{1}{N} \sum_i \lambda_i \lambda'_i \mathbf{E} e_{it}^2 \right)^{-1} \frac{1}{N} \sum_i \lambda_i e_{it} u_{it} = O_P(C_{NT}^{-1})$$

$$\bar{\lambda}_{\mathcal{G}} := \frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \lambda_i.$$

$Q_I = O_P(\frac{1}{\sqrt{T|\mathcal{G}|_0}}) = Q_{I^c}$ follows from $\max_i \sum_j |\mathbf{Cov}(u_{is}e_{is}, u_{js}e_{js}|F)| < C$, and $\max_i \|\Omega_i\| \leq \lambda_{\min}^{-1}(\frac{1}{T} \sum_s f_s f'_s)(\min_i \mathbf{E} e_{is}^2)^{-1} < C$. By Propositions F.1 and F.2, $\hat{f}_t - H_f f_t = H_f J_t + O_P(C_{NT}^{-2})$ and $\frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} (\hat{\lambda}_{I,i} - H_f'^{-1} \lambda_i) = H_f'^{-1} Q_I + O_P(C_{NT}^{-2})$. Therefore, for $t \notin I$,

$$\frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} (\hat{\lambda}_{I,i} \hat{f}_{I,t} - \lambda'_i f_t) = f'_t Q_I + \bar{\lambda}'_{\mathcal{G}} J_t + O_P(C_{NT}^{-2}).$$

Exchanging I with I^c , we have, for $t \notin I^c$,

$$\frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} (\hat{\lambda}_{I^c,i} \hat{f}_{I^c,t} - \lambda'_i f_t) = f'_t Q_{I^c} + \bar{\lambda}'_{\mathcal{G}} J_t + O_P(C_{NT}^{-2}).$$

Note that the fixed $t \notin I \cup I^c$, so we have

$$\begin{aligned} \frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} (\hat{\theta}_{it} - \theta_{it}) &= f'_t \frac{Q_I + Q_{I^c}}{2} + \bar{\lambda}'_{\mathcal{G}} J_t + O_P(C_{NT}^{-2}) \\ &= f'_t \frac{1}{|\mathcal{G}|_0 T} \sum_{i \in \mathcal{G}} \left[\sum_{s \in I^c \cup \{t\}} \Omega_i f_s e_{is} u_{is} + \sum_{s \in I \cup \{t\}} \Omega_i f_s e_{is} u_{is} \right] + \bar{\lambda}'_{\mathcal{G}} J_t \\ &\quad + O_P(C_{NT}^{-2}) \\ &= f'_t Q + \bar{\lambda}'_{\mathcal{G}} J_t + O_P(C_{NT}^{-2}) \\ &= \frac{1}{\sqrt{T|\mathcal{G}|_0}} \zeta_{NT} + \frac{1}{\sqrt{N}} \xi_{NT} + O_P(C_{NT}^{-2}), \end{aligned} \tag{F.7}$$

where

$$\begin{aligned} \zeta_{NT} &= f'_t \frac{1}{\sqrt{|\mathcal{G}|_0 T}} \sum_{i \in \mathcal{G}} \sum_{s=1}^T \Omega_i f_s e_{is} u_{is}, \\ \xi_{NT} &= \bar{\lambda}'_{\mathcal{G}} V_{\lambda 1}^{-1} \frac{1}{\sqrt{N}} \sum_i \lambda_i e_{it} u_{it} \\ V_{\lambda 1} &= \frac{1}{N} \sum_i \lambda_i \lambda'_i \mathbf{E} e_{it}^2. \end{aligned}$$

Next, we compute $\mathbf{Cov}(\xi_{NT}, \zeta_{NT}|F)$. Let $\mathcal{A} = \{F, G, W, U\}$. By Assumption 3.3, $\{e_{it}\}$ is conditionally independent across t given $\{F, G, W, U\}$. Hence for $s \neq t$,

and any $i, j \leq N$, $\mathbb{E}(e_{is}e_{jt}|\mathcal{A}) = 0$. Thus for $\omega_{it} = u_{it}e_{it}$ and any $s \neq t$ we have

$$\mathbb{E}(\omega_{is}\omega_{jt}|F) = \mathbb{E}\{\mathbb{E}(e_{is}e_{jt}|\mathcal{A})u_{is}u_{jt}|F\} = 0$$

Therefore

$$\begin{aligned} \text{Cov}(\xi_{NT}, \zeta_{NT}|F) &= \frac{1}{\sqrt{N}} \frac{1}{\sqrt{|\mathcal{G}|_0 T}} \sum_j \sum_{i \in \mathcal{G}} \sum_{s=1}^T \bar{\lambda}'_{\mathcal{G}} V_{\lambda 1}^{-1} \lambda_j f'_t \Omega_i f_s \mathbb{E}(\omega_{is}\omega_{jt}|F) \\ &= \frac{1}{\sqrt{|\mathcal{G}|_0 T N}} \sum_j \sum_{i \in \mathcal{G}} \bar{\lambda}'_{\mathcal{G}} V_{\lambda 1}^{-1} \lambda_j f'_t \Omega_i f_t \mathbb{E}(\omega_{it}\omega_{jt}|F) = o_P(1). \end{aligned}$$

Next, we use a similar argument as the proof of Theorem 3 in Bai (2003) to claim that

$$\frac{\frac{1}{\sqrt{T|\mathcal{G}|_0}}\zeta_{NT} + \frac{1}{\sqrt{N}}\xi_{NT}}{\Sigma_{NT}^{1/2}} \rightarrow^d \mathcal{N}(0, 1), \quad (\text{F.8})$$

where

$$\Sigma_{NT} = \frac{1}{T|\mathcal{G}|_0} f'_t V_{f, \mathcal{G}} f_t + \frac{1}{N} \bar{\lambda}'_{\mathcal{G}} V_{\lambda} \bar{\lambda}_{\mathcal{G}}.$$

Let $A_1 = (f'_t V_{f, \mathcal{G}} f_t)^{-1/2} \zeta_{NT}$ and $A_2 = (\bar{\lambda}'_{\mathcal{G}} V_{\lambda} \bar{\lambda}_{\mathcal{G}})^{-1/2} \xi_{NT}$. By Assumption 3.4,

$$(A_1, A_2) \rightarrow^d \mathcal{N}(0, I).$$

By the almost sure representation (Pollard, 2012), there exist random vectors $(\xi_1, \xi_2) =^d (A_1, A_2)$, and $(Z_1, Z_2) =^d \mathcal{N}(0, I)$ such that $(\xi_1, \xi_2) \rightarrow (Z_1, Z_2)$ almost surely. Now

$$\begin{aligned} \frac{\frac{1}{\sqrt{T|\mathcal{G}|_0}}\zeta_{NT} + \frac{1}{\sqrt{N}}\xi_{NT}}{\Sigma_{NT}^{1/2}} &= \frac{\frac{1}{\sqrt{T|\mathcal{G}|_0}}(f'_t V_{f, \mathcal{G}} f_t)^{1/2} A_1 + \frac{1}{\sqrt{N}}(\bar{\lambda}'_{\mathcal{G}} V_{\lambda} \bar{\lambda}_{\mathcal{G}})^{1/2} A_2}{\Sigma_{NT}^{1/2}} \\ &=^d \frac{\frac{1}{\sqrt{T|\mathcal{G}|_0}}(f'_t V_{f, \mathcal{G}} f_t)^{1/2} \xi_1 + \frac{1}{\sqrt{N}}(\bar{\lambda}'_{\mathcal{G}} V_{\lambda} \bar{\lambda}_{\mathcal{G}})^{1/2} \xi_2}{\Sigma_{NT}^{1/2}} \\ &= \frac{\frac{1}{\sqrt{T|\mathcal{G}|_0}}(f'_t V_{f, \mathcal{G}} f_t)^{1/2} Z_1 + \frac{1}{\sqrt{N}}(\bar{\lambda}'_{\mathcal{G}} V_{\lambda} \bar{\lambda}_{\mathcal{G}})^{1/2} Z_2}{\Sigma_{NT}^{1/2}} \\ &\quad + \frac{\frac{1}{\sqrt{T|\mathcal{G}|_0}}(f'_t V_{f, \mathcal{G}} f_t)^{1/2} (\xi_1 - Z_1) + \frac{1}{\sqrt{N}}(\bar{\lambda}'_{\mathcal{G}} V_{\lambda} \bar{\lambda}_{\mathcal{G}})^{1/2} (\xi_2 - Z_2)}{\Sigma_{NT}^{1/2}} \\ &=^d \mathcal{N}(0, 1) + o_P(1) \rightarrow^d \mathcal{N}(0, 1). \end{aligned}$$

This proves (F.8).

It remains to show the remainder term $\Sigma_{NT}^{-1/2} C_{NT}^{-2} = o_P(1)$. We recall $C_{NT} = \min\{\sqrt{N}, \sqrt{T}\}$. In fact,

$$\Sigma_{NT}^{-1/2} C_{NT}^{-2} \leq O_P\left(\frac{1}{T|\mathcal{G}|_0} + \frac{1}{N}\right)^{-1/2} \left(\frac{1}{T} + \frac{1}{N}\right).$$

Case 1: $|\mathcal{G}|_0 = o(\min\{N/T, T\})$. Then $T|\mathcal{G}|_0 = o(N)$. Hence

$$\Sigma_{NT}^{-1/2} C_{NT}^{-2} \leq O_P\left(\frac{1}{T} + \frac{1}{N}\right) \sqrt{T|\mathcal{G}|_0} \leq o_P(1) + O_P\left(\sqrt{\frac{|\mathcal{G}|_0}{T}}\right) = o_P(1).$$

Case 2: $N = O(T|\mathcal{G}|_0)$ and $N = o(T^2)$. Then

$$\Sigma_{NT}^{-1/2} C_{NT}^{-2} \leq O_P\left(\frac{1}{T} + \frac{1}{N}\right) \sqrt{N} = o_P(1) + O_P\left(\frac{\sqrt{N}}{T}\right) = o_P(1).$$

In either case, $\Sigma_{NT}^{-1/2} O_P(C_{NT}^{-2}) = o_P(1)$. Therefore, $\Sigma_{NT}^{-1/2} \frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} (\hat{\theta}_{it} - \theta_{it}) \rightarrow^d \mathcal{N}(0, 1)$.

F.4. Covariance estimation. Suppose $C_{NT}^{-1} \max_{is} e_{is}^2 u_{is}^2 = o_P(1)$. We now consider estimating Σ_{NT} in the general case, under the assumption that $e_{it} u_{it}$ are cross sectionally independent given F . Then

$$V_{\lambda 2} = \frac{1}{N} \sum_i \lambda_i \lambda_i' \mathbb{E}(e_{it}^2 u_{it}^2 | F), \quad V_{f, \mathcal{G}} = \frac{1}{T|\mathcal{G}|_0} \sum_{s=1}^T \sum_{i \in \mathcal{G}} \Omega_i f_s f_s' \Omega_i \mathbb{E}(e_{is}^2 u_{is}^2 | F)$$

The main goal is to consistently estimate the above two quantities. For notational simplicity, we assume $\dim(f_t) = \dim(\alpha_i) = 1$.

F.4.1. Consistency for V_{λ} . We have $\hat{V}_{\lambda 2, I} := \frac{1}{N} \sum_i \hat{\lambda}_{I, i} \hat{\lambda}_{I, i}' \hat{e}_{it}^2 \hat{u}_{it, I}^2$, where $\hat{u}_{it, I} = y_{it} - x_{it} \hat{\lambda}_{I, i}' \hat{f}_{I, t} - \hat{\alpha}_{I, i}' \hat{g}_{I, t}$.

We remove the subscript “ I ” for notational simplicity.

Step 1. Show $\|\hat{V}_{\lambda 2, I} - \tilde{V}_{\lambda 2}\| = o_P(1)$ where $\tilde{V}_{\lambda 2} = H_f'^{-1} \frac{1}{N} \sum_i \lambda_i \lambda_i' e_{it}^2 u_{it}^2 H_f^{-1}$.

By Lemma G.14 $\max_i [\|\hat{\lambda}_i - H_f'^{-1} \lambda_i\| + \|\hat{\alpha}_i - H_g'^{-1} \alpha_i\|] = o_P(1)$. By Lemma G.17,

$$\frac{1}{N} \sum_i |\hat{u}_{it} - u_{it}|^2 (1 + e_{it}^2) \leq \max_{it} |\hat{u}_{it} - u_{it}|^2 O_P(1) = o_P(1).$$

We therefore have

$$\begin{aligned} \|\hat{V}_{\lambda 2} - \tilde{V}_{\lambda 2}\| &\leq \frac{1}{N} \sum_i (\hat{\lambda}_i - \lambda_i)^2 \hat{e}_{it}^2 \hat{u}_{it}^2 + \frac{4}{N} \sum_i |(\hat{\lambda}_i - \lambda_i) \lambda_i| (\hat{u}_{it} - u_{it})^2 \max_i \hat{e}_{it}^2 \\ &\quad + \frac{4}{N} \sum_i (\hat{\lambda}_i - \lambda_i) \lambda_i \hat{e}_{it}^2 u_{it}^2 + \frac{2}{N} \sum_i \lambda_i^2 (\hat{e}_{it}^2 - e_{it}^2) (\hat{u}_{it} - u_{it})^2 \end{aligned}$$

$$\begin{aligned}
& + \frac{2}{N} \sum_i \lambda_i^2 (\hat{e}_{it}^2 - e_{it}^2) u_{it}^2 + \frac{2}{N} \sum_i \lambda_i^2 e_{it}^2 (\hat{u}_{it} - u_{it}) u_{it} + \frac{1}{N} \sum_i \lambda_i^2 e_{it}^2 (\hat{u}_{it} - u_{it})^2 \\
& \leq o_P(1).
\end{aligned}$$

Step 2. Show $\|\tilde{V}_{\lambda 2} - H_f'^{-1} V_{\lambda 2} H_f^{-1}\| = o_P(1)$.

It suffices to show $\text{Var}(\frac{1}{N} \sum_i \lambda_i^2 e_{it}^2 u_{it}^2 | F) \rightarrow 0$ almost surely. Almost surely in F ,

$$\text{Var}\left(\frac{1}{N} \sum_i \lambda_i^2 e_{it}^2 u_{it}^2 | F\right) \leq \frac{\max_i \|\lambda_i\|^2}{N} \max_{i \leq N} \mathbb{E}(e_{it}^4 u_{it}^4 | F) \rightarrow 0$$

given that $\max_{i \leq N} \mathbb{E}(e_{it}^4 u_{it}^4 | F) < C$ almost surely.

The consistency for $V_{\lambda 1}$ follows from the same argument.

Therefore

$$\hat{\lambda}_{I, \mathcal{G}}' \hat{V}_{\lambda 1, I}^{-1} \hat{V}_{\lambda 2, I} \hat{V}_{\lambda 1, I}^{-1} \hat{\lambda}_{I, \mathcal{G}} - \bar{\lambda}_{\mathcal{G}}' V_{\lambda} \bar{\lambda}_{\mathcal{G}} = o_P(1)$$

F.4.2. *Consistency of V_f .* Write $\hat{\Omega}_{I, i} = (\frac{1}{T_0} \sum_{s \notin I} \hat{f}_{I, s} \hat{f}_{I, s}')^{-1} \hat{\sigma}_i^{-2}$ for $\hat{\sigma}_i^2 = \frac{1}{T} \sum_{s=1}^T \hat{e}_{is}^2$. $\hat{V}_{f, I} = \frac{1}{T_0 |\mathcal{G}|_0} \sum_{s \notin I} \sum_{i \in \mathcal{G}} \hat{\Omega}_{I, i} \hat{f}_{I, s} \hat{f}_{I, s}' \hat{\Omega}_{I, i} \hat{e}_{is}^2 \hat{u}_{is}^2$. We aim to show the consistency of $\hat{V}_{f, I}$. We remove the subscript “ I ” for notational simplicity. Then simply write

$$\begin{aligned}
\hat{V}_f &= \hat{S}_f^{-1} \frac{1}{T |\mathcal{G}|_0} \sum_s \sum_{i \in \mathcal{G}} \hat{f}_s \hat{f}_s' \hat{e}_{is}^2 \hat{u}_{is}^2 \hat{\sigma}_i^{-4} \hat{S}_f^{-1} \\
\tilde{V}_f &= H_f'^{-1} S_f^{-1} \frac{1}{T |\mathcal{G}|_0} \sum_s \sum_{i \in \mathcal{G}} f_s f_s' e_{is}^2 u_{is}^2 \sigma_i^{-4} S_f^{-1} H_f^{-1}
\end{aligned}$$

where $\sigma_i^2 = \mathbb{E} e_{it}^2$.

Step 1. Show $\|\hat{V}_f - \tilde{V}_f\| = o_P(1)$. First note that

$$\hat{S}_f - H_f S_f H_f' = \frac{1}{T} \sum_t \hat{f}_t \hat{f}_t' - H_f f_t f_t' H_f' = O_P(T^{-1} + N^{-1/2}).$$

We show $\|\frac{1}{T |\mathcal{G}|_0} \sum_s \sum_{i \in \mathcal{G}} \hat{f}_s \hat{f}_s' \hat{e}_{is}^2 \hat{u}_{is}^2 \hat{\sigma}_i^{-4} - H_f \frac{1}{T |\mathcal{G}|_0} \sum_s \sum_{i \in \mathcal{G}} f_s f_s' e_{is}^2 u_{is}^2 \sigma_i^{-4} H_f'\|^2 = o_P(1)$.

Ignoring H_f , it is bounded by

$$\begin{aligned}
& O_P(C_{NT}^{-2}) \max_s \left\| \frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \hat{e}_{is}^2 \hat{u}_{is}^2 \hat{\sigma}_i^{-4} \right\|^2 + O_P(C_{NT}^{-2}) \frac{1}{T |\mathcal{G}|_0} \sum_s \sum_{i \in \mathcal{G}} \|f_s f_s' \hat{u}_{is}^2 \hat{\sigma}_i^{-4}\|^2 \\
& + \frac{1}{T |\mathcal{G}|_0} \sum_s \sum_{i \in \mathcal{G}} (\hat{u}_{is} - u_{is})^2 \frac{1}{T |\mathcal{G}|_0} \sum_s \sum_{i \in \mathcal{G}} \|f_s f_s' e_{is}^2 (\hat{u}_{is} + u_{is}) \hat{\sigma}_i^{-4}\|^2 + O_P(1) \max_i |\hat{\sigma}_i^4 - \sigma_i^4|^2
\end{aligned}$$

By Lemma G.16, $\max_{is} |\hat{u}_{is} - u_{is}| = o_P(1)$, $\max_i |\hat{\sigma}_i^2 - \sigma_i^2| = o_P(1)$, $C_{NT}^{-1} \max_{is} e_{is}^2 u_{is}^2 = o_P(1)$. Thus, the above is $o_P(1)$.

Step 2. Show $\|\tilde{V}_{f, \mathcal{G}} - H_f'^{-1} V_{f, \mathcal{G}} H_f^{-1}\| = O_P(\frac{1}{\sqrt{T |\mathcal{G}|_0}})$.

It suffices to prove $\text{Var}(\frac{1}{T|\mathcal{G}|_0} \sum_s \sum_{i \in \mathcal{G}} f_s^2 e_{is}^2 u_{is}^2 \sigma_i^{-4} | F) \rightarrow 0$ almost surely. It is in fact bounded by

$$\frac{1}{T^2 |\mathcal{G}|_0^2} \sum_s \sum_{i \in \mathcal{G}} f_s^4 \mathbb{E}(e_{is}^4 u_{is}^4 | F) \max_i \sigma_i^{-8} = O_P(\frac{1}{T|\mathcal{G}|_0})$$

given that $\frac{1}{T} \sum_t \|f_t\|^4 < C$ almost surely. Thus,

$$\hat{f}_{I,t} \hat{V}_{I,f} \hat{f}_{I,t} - f_t' V_{f,g} f_t = O_P(\frac{1}{T} + \frac{1}{\sqrt{N}} + \frac{1}{\sqrt{T|\mathcal{G}|_0}}) + o_P(1) = o_P(1).$$

APPENDIX G. TECHNICAL LEMMAS

G.1. The effect of $\hat{e}_{it} - e_{it}$ in the factor model. Here we present intermediate results when x_{it} admits a factor structure:

$$x_{it} = l_i' w_t + e_{it}.$$

Let $\hat{w}_t$ be the PC estimator of w_t . Then $\hat{l}_i' = \frac{1}{T} \sum_s x_{is} \hat{w}_s'$ and $\hat{e}_{it} - e_{it} = l_i' H_x (\hat{w}_t - H_x^{-1} w_t) + (\hat{l}_i' - l_i' H_x) \hat{w}_t$.

Lemma G.1. (i) $\max_{it} |\hat{e}_{it} - e_{it}| = O_P(\phi_{NT})$, where

$$\phi_{NT} := (C_{NT}^{-2} (\max_i \frac{1}{T} \sum_s e_{is}^2)^{1/2} + b_{NT,4} + b_{NT,5}) (1 + \max_{t \leq T} \|w_t\|) + (b_{NT,1} + C_{NT}^{-1} b_{NT,2} + b_{NT,3}).$$

Thus, $\max_{it} |\hat{e}_{it} - e_{it}| \max_{it} |e_{it}| = O_P(1)$.

(ii) All terms below are $O_P(C_{NT}^{-2})$ for a fixed t : $\frac{1}{N} \sum_i (\hat{e}_{it} - e_{it})^4$, $\frac{1}{N} \sum_i (\hat{e}_{it} - e_{it})^2$, and $\frac{1}{N} \sum_i (\hat{e}_{it} - e_{it})^2 e_{it}^2$, $\frac{1}{N} \sum_i \lambda_i \lambda_i' (\hat{e}_{it} - e_{it}) e_{it}$, $\frac{1}{N} \sum_i \lambda_i (\hat{e}_{it} - e_{it}) u_{it}$.

(iii) $\frac{1}{N} \sum_i \lambda_i \alpha_i' (\hat{e}_{it} - e_{it}) = O_P(C_{NT}^{-1})$ for a fixed t .

Proof. Below we first simplify the expansion of $\hat{e}_{it} - e_{it}$. Let $K_3 = \dim(l_i)$. Let $\mathcal{Q}$ be a diagonal matrix consisting of the reciprocal of the first K_3 eigenvalues of $XX'/(NT)$. Let

$$\begin{aligned} \zeta_{st} &= \frac{1}{N} \sum_i (e_{is} e_{it} - \mathbb{E} e_{is} e_{it}), & \eta_t &= \frac{1}{N} \sum_i l_i e_{it}, \\ \sigma^2 &= \frac{1}{N} \sum_i \mathbb{E} e_{it}^2. \end{aligned}$$

For the PC estimator, there is a rotation matrix $\bar{H}_x$, by (A.1) of Bai (2003), (which can be simplified due to the serial independence in e_{it})

$$\begin{aligned}\hat{w}_t - \bar{H}_x w_t &= \mathcal{Q} \frac{\sigma^2}{T} (\hat{w}_t - \bar{H}_x w_t) + \mathcal{Q} [\frac{\sigma^2}{T} \bar{H}_x + \frac{1}{TN} \sum_{is} \hat{w}_s l'_i e_{is}] w_t \\ &\quad + \mathcal{Q} \frac{1}{T} \sum_s \hat{w}_s \zeta_{st} + \mathcal{Q} \frac{1}{T} \sum_s \hat{w}_s w'_s \eta_t.\end{aligned}$$

Move $\mathcal{Q} \frac{\sigma^2}{T} (\hat{w}_t - \bar{H}_x w_t)$ to the left hand side (LHS); then the LHS becomes $(I - \mathcal{Q} \frac{\sigma^2}{T}) (\hat{w}_t - \bar{H}_x w_t)$. Note that $\|\mathcal{Q}\| = O_P(1)$, so $\mathcal{Q}_1 := (I - \mathcal{Q} \frac{\sigma^2}{T})^{-1}$ exists with eigenvalues all converging to one. Then multiplying $\mathcal{Q}_1$ on both sides, we reach

$$\begin{aligned}\hat{w}_t - \bar{H}_x w_t &= \mathcal{Q}_1 \mathcal{Q} [\frac{\sigma^2}{T} \bar{H}_x + \frac{1}{TN} \sum_{is} \hat{w}_s l'_i e_{is}] w_t \\ &\quad + \mathcal{Q}_1 \mathcal{Q} \frac{1}{T} \sum_s \hat{w}_s \zeta_{st} + \mathcal{Q}_1 \mathcal{Q} \frac{1}{T} \sum_s \hat{w}_s w'_s \eta_t.\end{aligned}$$

Next, after moving $\mathcal{Q}_1 \mathcal{Q} [\frac{\sigma^2}{T} \bar{H}_x + \frac{1}{TN} \sum_{is} \hat{w}_s l'_i e_{is}] w_t$ to the LHS combined with $-\bar{H}_x w_t$, the LHS becomes $\hat{w}_t - H_x^{-1} w_t$, where $H_x^{-1} = (I + \mathcal{Q}_1 \mathcal{Q} \frac{\sigma^2}{T}) \bar{H}_x + \mathcal{Q}_1 \mathcal{Q} \frac{1}{TN} \sum_{is} \hat{w}_s l'_i e_{is}$, and $\mathcal{Q}_1 \mathcal{Q} \frac{1}{TN} \sum_{is} \hat{w}_s l'_i e_{is} = o_P(1)$. Thus, the eigenvalues of H_x^{-1} converge to those of $\bar{H}_x$, which are well known to be bounded away from both zero and infinity (Bai, 2003). Finally, letting $R_1 = \mathcal{Q}_1 \mathcal{Q}$ and $R_2 = \mathcal{Q}_1 \mathcal{Q} \frac{1}{T} \sum_s \hat{w}_s w'_s$, we reach

$$\hat{w}_t - H_x^{-1} w_t = R_1 \frac{1}{T} \sum_s \hat{w}_s \zeta_{st} + R_2 \eta_t \quad (\text{G.1})$$

with $\|R_1\| + \|R_2\| = O_P(1)$.

Also, $\hat{l}'_i = \frac{1}{T} \sum_s x_{is} \hat{w}'_s$, for $\mathcal{Q}_3 = -H_x (R_1 \frac{1}{T^2} \sum_{m,s \leq T} \zeta_{ms} \hat{w}_m \hat{w}'_s + R_2 \frac{1}{T} \sum_s \eta_s \hat{w}'_s) = O_P(C_{NT}^{-2})$,

$$\begin{aligned}\hat{e}_{it} - e_{it} &= l'_i H_x (\hat{w}_t - H_x^{-1} w_t) + (\hat{l}'_i - l'_i H_x) \hat{w}_t \\ &= l'_i H_x (\hat{w}_t - H_x^{-1} w_t) + \frac{1}{T} \sum_s e_{is} (\hat{w}'_s - w'_s H_x^{-1'}) \hat{w}_t \\ &\quad + \frac{1}{T} \sum_s e_{is} w'_s H_x^{-1'} \hat{w}_t + l'_i H_x \frac{1}{T} \sum_s (H_x^{-1} w_s - \hat{w}_s) \hat{w}'_s \hat{w}_t \\ &= \frac{1}{T} \sum_s e_{is} w'_s H_x^{-1'} \hat{w}_t + \frac{1}{T^2} \sum_{s,m \leq T} e_{is} \zeta_{ms} \hat{w}'_m R'_1 \hat{w}_t + \frac{1}{T} \sum_s e_{is} \eta'_s R'_2 \hat{w}_t \\ &\quad + l'_i \mathcal{Q}_3 \hat{w}_t + l'_i H_x R_1 \frac{1}{T} \sum_s \hat{w}_s \zeta_{st} + l'_i H_x R_2 \eta_t.\end{aligned} \quad (\text{G.2})$$

(i) We first show that $\max_{t \leq T} \|\hat{w}_t - H_x^{-1} w_t\| = O_P(1)$. Define

$$\begin{aligned} b_{NT,1} &= \max_{t \leq T} \left\| \frac{1}{NT} \sum_{is} w_s (e_{is} e_{it} - \mathbb{E} e_{is} e_{it}) \right\| \\ b_{NT,2} &= \left(\max_{t \leq T} \frac{1}{T} \sum_s \left(\frac{1}{N} \sum_i e_{is} e_{it} - \mathbb{E} e_{is} e_{it} \right)^2 \right)^{1/2} \\ b_{NT,3} &= \max_{t \leq T} \left\| \frac{1}{N} \sum_i l_i e_{it} \right\| \\ b_{NT,4} &= \max_i \left\| \frac{1}{T} \sum_s e_{is} w_s \right\| \\ b_{NT,5} &= \max_i \left\| \frac{1}{NT} \sum_{js} l_j (e_{js} e_{is} - \mathbb{E} e_{js} e_{is}) \right\| \end{aligned}$$

Then

$$\begin{aligned} \max_{t \leq T} \left\| \frac{1}{T} \sum_s \hat{w}_s \zeta_{st} \right\| &\leq O_P(1) \max_{t \leq T} \left\| \frac{1}{T} \sum_s w_s \zeta_{st} \right\| + O_P(C_{NT}^{-1}) \left(\max_{t \leq T} \frac{1}{T} \sum_s \zeta_{st}^2 \right)^{1/2} \\ &= O_P(b_{NT,1} + C_{NT}^{-1} b_{NT,2}). \end{aligned}$$

Then, by assumption, $\max_{t \leq T} \|\hat{w}_t - H_x^{-1} w_t\| = O_P(b_{NT,1} + C_{NT}^{-1} b_{NT,2} + b_{NT,3}) = O_P(1)$. As such, $\max_{t \leq T} \|\hat{w}_t\| = O_P(1) + \max_{t \leq T} \|w_t\|$.

In addition,

$$\begin{aligned} \max_i \left\| \frac{1}{T^2} \sum_{m \leq T} \sum_{s \leq T} e_{is} \zeta_{ms} \hat{w}_m' \right\| &\leq O_P(C_{NT}^{-1}) \left(\max_i \frac{1}{T} \sum_{s \leq T} e_{is}^2 \right)^{1/2} \left(\frac{1}{T^2} \sum_{s, t \leq T} \zeta_{st}^2 \right)^{1/2} \\ &+ O_P(1) \left(\max_i \frac{1}{T} \sum_s e_{is}^2 \right)^{1/2} \left(\frac{1}{T} \sum_s \left\| \frac{1}{TN} \sum_{jt} (e_{jt} e_{js} - \mathbb{E} e_{jt} e_{js}) w_t \right\|^2 \right)^{1/2} \\ &= O_P(C_{NT}^{-2}) \left(\max_i \frac{1}{T} \sum_s e_{is}^2 \right)^{1/2}. \end{aligned}$$

So $\max_{it} |\hat{e}_{it} - e_{it}| = O_P(\phi_{NT})$, where

$$\phi_{NT} := (C_{NT}^{-2} \left(\max_i \frac{1}{T} \sum_s e_{is}^2 \right)^{1/2} + b_{NT,4} + b_{NT,5}) (1 + \max_{t \leq T} \|w_t\|) + (b_{NT,1} + C_{NT}^{-1} b_{NT,2} + b_{NT,3}).$$

(ii) Let $a \in \{1, 2, 4\}$, $b \in \{0, 1, 2\}$, and c_i be a bounded constant sequence c_i .

Consider, up to a $O_P(1)$ multiplier that is independent of (t, i) ,

$$\begin{aligned} \frac{1}{N} \sum_i c_i e_{it}^b (\hat{e}_{it} - e_{it})^a &= \frac{1}{N} \sum_i c_i e_{it}^b \left(\frac{1}{T} \sum_s e_{is} w_s \right)^a \hat{w}_t^a \\ &+ \frac{1}{N} \sum_i c_i e_{it}^b \left(\frac{1}{T^2} \sum_{s, m \leq T} e_{is} \zeta_{ms} \hat{w}_m \right)^a \hat{w}_t^a \end{aligned}$$

$$\begin{aligned}
& + \frac{1}{N} \sum_i c_i e_{it}^b \left(\frac{1}{T} \sum_s e_{is} \eta'_s \right)^a \widehat{w}_t^a + \frac{1}{N} \sum_i c_i e_{it}^b l_i^a \mathcal{Q}_3^a \widehat{w}_t^a \\
& + \frac{1}{N} \sum_i c_i e_{it}^b l_i^a \left(\frac{1}{T} \sum_s \widehat{w}_s \zeta_{st} \right)^a + \frac{1}{N} \sum_i c_i e_{it}^b l_i^a \eta_t^a. \quad (\text{G.3})
\end{aligned}$$

Note that

$$\begin{aligned}
\frac{1}{T} \sum_s \widehat{w}_s \zeta_{st} & \leq O_P(1) \frac{1}{TN} \sum_{is} w_s (e_{is} e_{it} - \mathbb{E} e_{is} e_{it}) \\
& + \left(\frac{1}{T} \sum_s \left(\frac{1}{N} \sum_i (e_{is} e_{it} - \mathbb{E} e_{is} e_{it}) \right)^2 \right)^{1/2} O_P(C_{NT}^{-1}) = O_P(C_{NT}^{-2}) \\
\frac{1}{N} \sum_i e_{it}^b \left(\frac{1}{T} \sum_s e_{is} w_s \right)^a & = O(T^{-a/2}), \quad a = 2, 4, \quad b = 0, 2 \\
\frac{1}{N} \sum_i \left(\frac{1}{T^2} \sum_{s, m \leq T} e_{is} \zeta_{ms} \widehat{w}_m \right)^4 & \leq \left(\frac{1}{T^2} \sum_{m, s \leq T} \zeta_{ms}^2 \right)^2 \left(\frac{1}{T} \sum_m w_m^2 \right)^2 \frac{1}{N} \sum_i \left(\frac{1}{T} \sum_s e_{is}^2 \right)^2 \\
& + \frac{1}{N} \sum_i \left(\frac{1}{T} \sum_{m \leq T} \left(\frac{1}{T} \sum_{s \leq T} e_{is} \zeta_{ms} \right)^2 \right)^2 O_P(C_{NT}^{-4}) = O_P(C_{NT}^{-4}) \\
\frac{1}{N} \sum_i \left(\frac{1}{T^2} \sum_{s, m \leq T} e_{is} \zeta_{ms} \widehat{w}_m \right)^2 & \leq \frac{1}{N} \sum_i \frac{1}{T} \sum_m \left(\frac{1}{T} \sum_s e_{is} \zeta_{ms} \right)^2 \frac{1}{T} \sum_m w_m^2 \\
& + \frac{1}{N} \sum_i \frac{1}{T} \sum_{m \leq T} \left(\frac{1}{T} \sum_{s \leq T} e_{is} \zeta_{ms} \right)^2 O_P(C_{NT}^{-2}) = O_P(C_{NT}^{-2}) \\
\frac{1}{N} \sum_i \left(\frac{1}{T} \sum_s e_{is} \eta'_s \right)^4 & \leq \frac{1}{N} \sum_i \left(\frac{1}{T} \sum_s e_{is}^2 \right)^2 \left(\frac{1}{T} \sum_t \eta_t^2 \right)^2 = O_P(C_{NT}^{-4}) \\
\frac{1}{N} \sum_i e_{it}^2 \left(\frac{1}{T} \sum_s e_{is} \eta'_s \right)^2 & = O_P(C_{NT}^{-2}) \\
\frac{1}{N} \sum_i e_{it}^2 l_i^2 \left(\frac{1}{T} \sum_s \widehat{w}_s \zeta_{st} \right)^2 & \leq \frac{1}{N} \sum_i e_{it}^2 l_i^2 \left(\frac{1}{T} \sum_s w_s \zeta_{st} \right)^2 + O_P(C_{NT}^{-2}) = O_P(C_{NT}^{-2}) \\
\frac{1}{T} \sum_t \left(\frac{1}{T} \sum_s \widehat{w}_s \zeta_{st} \right)^2 & \leq O_P(C_{NT}^{-4}) \\
\frac{1}{N} \sum_i c_i e_{it} \left(\frac{1}{T} \sum_s e_{is} w_s \right) & = O_P(1) \left(\frac{1}{N^2 T^2} \sum_{ij \leq N} \sum_{sl \leq T} c_i c_j \mathbb{E} w_s w_l \mathbb{E}(e_{it} e_{jt} e_{is} e_{jl} | W) \right)^{1/2} \\
& = O_P(C_{NT}^{-2}) \\
\frac{1}{N} \sum_i c_i e_{it} \left(\frac{1}{T^2} \sum_{s, m \leq T} e_{is} \zeta_{ms} \widehat{w}_m \right) & \leq O_P(C_{NT}^{-2}) + \frac{1}{N} \sum_i c_i e_{it} \left(\frac{1}{T} \sum_{s \leq T} \frac{1}{T} \sum_{m \leq T} e_{is} \zeta_{ms} w_m \right) \\
& \leq O_P(C_{NT}^{-2}) + O_P(1) \left(\frac{1}{T} \sum_{s \leq T} \mathbb{E} \left(\frac{1}{T} \sum_{m \leq T} \zeta_{ms} w_m \right)^2 \right)^{1/2} \\
& = O_P(C_{NT}^{-2}).
\end{aligned}$$

where the last equality follows from the following:

$$\begin{aligned}
& \frac{1}{T} \sum_{s \leq T} \mathbb{E} \left(\frac{1}{T} \sum_{m \leq T} \zeta_{ms} w_m \right)^2 = O(T^{-2}) + \frac{1}{T} \sum_{s \leq T} \frac{1}{T^2} \sum_{t \neq s} \mathbb{E} w_s w_t \text{Cov}(\zeta_{ss}, \zeta_{ts} | W) \\
& + \frac{1}{T} \sum_{s \leq T} \frac{1}{T} \sum_{m \neq s} \frac{1}{T} \sum_{t \leq T} \mathbb{E} w_m w_t \text{Cov}(\zeta_{ms}, \zeta_{ts} | W) \\
& = O(T^{-2}) + \frac{1}{T} \sum_{s \leq T} \frac{1}{T} \sum_{t \neq s} \frac{1}{T} \mathbb{E} w_t^2 \frac{1}{N^2} \sum_{ij} \mathbb{E}(e_{is} e_{js} | W) \mathbb{E}(e_{it} e_{jt} | W) = O(C_{NT}^{-4}).
\end{aligned}$$

With the above results ready, we can proceed proving (ii), (iii) as follows.

Now for $a = 4, b = 0, c_i = 1$, up to a $O_P(1)$ multiplier,

$$\begin{aligned}
\frac{1}{N} \sum_i (\widehat{e}_{it} - e_{it})^4 &= \frac{1}{N} \sum_i \left(\frac{1}{T} \sum_s e_{is} w_s \right)^4 + \frac{1}{N} \sum_i \left(\frac{1}{T^2} \sum_{s, m \leq T} e_{is} \zeta_{ms} \widehat{w}_m \right)^4 \\
&+ \frac{1}{N} \sum_i \left(\frac{1}{T} \sum_s e_{is} \eta'_s \right)^4 + \mathcal{Q}_3^4 + \left(\frac{1}{T} \sum_s \widehat{w}_s \zeta_{st} \right)^4 + \eta_t^4 \\
&\leq O_P(C_{NT}^{-4}).
\end{aligned}$$

For $a = 2, b = 0, c_i = 1$,

$$\begin{aligned}
\frac{1}{N} \sum_i (\widehat{e}_{it} - e_{it})^2 &\leq \frac{1}{N} \sum_i \left(\frac{1}{T} \sum_s e_{is} w_s \right)^2 + \frac{1}{N} \sum_i \left(\frac{1}{T^2} \sum_{s, m \leq T} e_{is} \zeta_{ms} \widehat{w}_m \right)^2 \\
&+ \frac{1}{N} \sum_i \left(\frac{1}{T} \sum_s e_{is} \eta'_s \right)^2 + \mathcal{Q}_3^2 + \left(\frac{1}{T} \sum_s \widehat{w}_s \zeta_{st} \right)^2 + \eta_t^2 \\
&\leq O_P(C_{NT}^{-2}).
\end{aligned}$$

For $a = 2, b = 2, c_i = 1$,

$$\begin{aligned}
\frac{1}{N} \sum_i e_{it}^2 (\widehat{e}_{it} - e_{it})^2 &= \frac{1}{N} \sum_i e_{it}^2 \left(\frac{1}{T} \sum_s e_{is} w_s \right)^2 + \frac{1}{N} \sum_i e_{it}^2 \left(\frac{1}{T^2} \sum_{s, m \leq T} e_{is} \zeta_{ms} \widehat{w}_m \right)^2 \\
&+ \frac{1}{N} \sum_i e_{it}^2 \left(\frac{1}{T} \sum_s e_{is} \eta'_s \right)^2 + \mathcal{Q}_3^2 + \left(\frac{1}{T} \sum_s \widehat{w}_s \zeta_{st} \right)^2 + \eta_t^2 \\
&\leq O_P(C_{NT}^{-2}).
\end{aligned}$$

Next, let $a = b = 1$ and c_i be any element of $\lambda_i \lambda'_i$,

$$\begin{aligned}
\frac{1}{N} \sum_i c_i e_{it} (\widehat{e}_{it} - e_{it}) &= \frac{1}{N} \sum_i c_i e_{it} \left(\frac{1}{T} \sum_s e_{is} w_s \right) + \frac{1}{N} \sum_i c_i e_{it} \left(\frac{1}{T^2} \sum_{s, m \leq T} e_{is} \zeta_{ms} \widehat{w}_m \right) \\
&+ \frac{1}{N} \sum_i c_i e_{it} \left(\frac{1}{T} \sum_s e_{is} \eta'_s \right) + \frac{1}{N} \sum_i c_i e_{it} l_i \mathcal{Q}_3
\end{aligned}$$

$$\begin{aligned}
& + \frac{1}{N} \sum_i c_i e_{it} l_i \left(\frac{1}{T} \sum_s \widehat{w}_s \zeta_{st} \right) + \frac{1}{N} \sum_i c_i e_{it} l_i \eta_t \\
& \leq O_P(C_{NT}^{-2}).
\end{aligned}$$

Next, ignoring an $O_P(1)$ multiplier,

$$\begin{aligned}
\frac{1}{N} \sum_i \lambda_i (\widehat{e}_{it} - e_{it}) u_{it} & \leq \frac{1}{N} \sum_i \lambda_i \frac{1}{T} \sum_s e_{is} w_s u_{it} \\
& + \frac{1}{N} \sum_i \lambda_i \frac{1}{T^2} \sum_{s,m \leq T} e_{is} \zeta_{ms} \widehat{w}_m u_{it} + \frac{1}{N} \sum_i \lambda_i \frac{1}{T} \sum_s e_{is} \eta_s u_{it} \\
& + \frac{1}{N} \sum_i \lambda_i l'_i \mathcal{Q}_3 u_{it} + \frac{1}{N} \sum_i \lambda_i l_i u_{it} \frac{1}{T} \sum_s \widehat{w}_s \zeta_{st} + \frac{1}{N} \sum_i \lambda_i l_i \eta_t u_{it} \\
& \leq O_P(C_{NT}^{-2}).
\end{aligned}$$

(iii) Let $a = 1, b = 0$ and c_i be any element of $\lambda_i \alpha'_i$,

$$\begin{aligned}
\frac{1}{N} \sum_i c_i (\widehat{e}_{it} - e_{it}) & = \frac{1}{N} \sum_i c_i \left(\frac{1}{T} \sum_s e_{is} w_s \right) + \frac{1}{N} \sum_i c_i \left(\frac{1}{T^2} \sum_{s,m \leq T} e_{is} \zeta_{ms} \widehat{w}_m \right) \\
& + \frac{1}{N} \sum_i c_i \left(\frac{1}{T} \sum_s e_{is} \eta'_s \right) + \mathcal{Q}_3 + \left(\frac{1}{T} \sum_s \widehat{w}_s \zeta_{st} \right) + \eta_t \\
& \leq O_P(C_{NT}^{-1}),
\end{aligned}$$

where the dominating term is $\eta_t = O_P(C_{NT}^{-1})$.

□

Lemma G.2. Assume $\max_{it} |e_{it}| C_{NT}^{-1} = O_P(1)$ and $\mathbb{E} e_{it}^8 < C$.

Let c_i be a non-random bounded sequence.

(i) $\max_{t \leq T} \frac{1}{N} \sum_i (\widehat{e}_{it} - e_{it})^4 = O_P(1 + \max_{t \leq T} \|w_t\|^4 + b_{NT,2}^4) C_{NT}^{-4} + O_P(b_{NT,1}^4 + b_{NT,3}^4)$.

$$\begin{aligned}
& \max_{t \leq T} \frac{1}{N} \sum_i (\widehat{e}_{it} - e_{it})^2 e_{it}^2 \leq O_P(1 + \max_{t \leq T} \|w_t\|^2 + b_{NT,2}^2) \max_{it} |e_{it}|^2 C_{NT}^{-2} \\
& + O_P(b_{NT,1}^2 + b_{NT,3}^2) \max_{t \leq T} \frac{1}{N} \sum_i e_{it}^2. \\
& \max_t \left| \frac{1}{N} \sum_i c_i (\widehat{e}_{it} - e_{it}) e_{it} \right| \leq O_P(1 + \max_{t \leq T} \|w_t\| + b_{NT,2}) \max_{it} |e_{it}| C_{NT}^{-1} \\
& + \max_{t \leq T} \left\| \frac{1}{N} \sum_i c_i e_{it} \right\|_F O_P(b_{NT,1} + b_{NT,3}). \\
& \max_t \left| \frac{1}{N} \sum_i c_i (\widehat{e}_{it} - e_{it})^2 \right| \leq O_P(1 + \max_{t \leq T} \|w_t\|^2 + b_{NT,2}^2) C_{NT}^{-2} + O_P(b_{NT,1}^2 + b_{NT,3}^2). \\
& \max_{t \leq T} \frac{1}{N} \sum_i (\widehat{e}_{it} - e_{it})^2 \leq O_P(1 + \max_{t \leq T} \|w_t\|^2 + b_{NT,2}^2) C_{NT}^{-2} + O_P(b_{NT,1}^2 + b_{NT,3}^2) \\
& \max_t \left| \frac{1}{N} \sum_i c_i (\widehat{e}_{it} - e_{it}) \right| \leq O_P(1 + \max_{t \leq T} \|w_t\|) C_{NT}^{-1} + O_P(b_{NT,1} + b_{NT,3} + C_{NT}^{-1} b_{NT,2}). \\
& \max_i \frac{1}{T} \sum_t (\widehat{e}_{it} - e_{it})^2 \leq O_P(b_{NT,4}^2 + b_{NT,5}^2 + C_{NT}^{-2}).
\end{aligned}$$

(ii) All terms below are $O_P(C_{NT}^{-2})$: $\frac{1}{NT} \sum_{it} (\widehat{e}_{it} - e_{it})^4$

$\frac{1}{NT} \sum_{it} (\hat{e}_{it} - e_{it})^2$, $\frac{1}{NT} \sum_{it} e_{it}^2 (\hat{e}_{it} - e_{it})^2$, $\frac{1}{T} \sum_t |\frac{1}{N} \sum_i c_i (e_{it} - \hat{e}_{it})|^2 m_t^2$, where $m_t^2 = 1 + \frac{1}{|\mathcal{G}|_0} \sum_{j \in \mathcal{G}} \|e_{jt} f_t\|^2 (\|f_t\| + \|g_t\|)^2$.

(iii) All terms below are $O_P(C_{NT}^{-4})$: $\frac{1}{T} \sum_t |\frac{1}{N} \sum_i c_i (\hat{e}_{it} - e_{it}) e_{it}|^2$, $\frac{1}{T} \sum_t |\frac{1}{N} \sum_i c_i (\hat{e}_{it} - e_{it})^2|^2$, $\frac{1}{T} \sum_t |\frac{1}{N} \sum_i c_i (\hat{e}_{it} - e_{it}) u_{it}|^2$, $\frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \|\frac{1}{T} \sum_{s \notin I} f_s (\hat{e}_{is} - e_{is}) u_{is}\|^2$, $\frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \|\frac{1}{T} \sum_{s \notin I} f_s e_{is} (\hat{e}_{is} - e_{is}) \lambda'_i f_s\|^2$, $\frac{1}{|\mathcal{G}|_0} \sum_{j \in \mathcal{G}} \|\frac{1}{T_0} \sum_{t \in I^c} \frac{1}{N} \sum_i c_i f_t^2 e_{jt} (e_{it} - \hat{e}_{it})\|^2$

Proof. (i) First, in the proof of Lemma G.1(i), we showed $\max_{t \leq T} \|\hat{w}_t\| \leq O_P(1 + \max_{t \leq T} \|w_t\|)$. By the proof of Lemma G.1(ii),

$$\begin{aligned} \max_{t \leq T} \frac{1}{N} \sum_i (\hat{e}_{it} - e_{it})^4 &\leq \frac{1}{N} \sum_i \left(\frac{1}{T} \sum_s e_{is} w_s \right)^4 \max_{t \leq T} \hat{w}_t^4 + \frac{1}{N} \sum_i \left(\frac{1}{T^2} \sum_{s, m \leq T} e_{is} \zeta_{ms} \hat{w}_m \right)^4 \max_{t \leq T} \hat{w}_t^4 \\ &+ \frac{1}{N} \sum_i \left(\frac{1}{T} \sum_s e_{is} \eta'_s \right)^4 \max_{t \leq T} \hat{w}_t^4 + \mathcal{Q}_3^4 \max_{t \leq T} \hat{w}_t^4 + \max_{t \leq T} \left(\frac{1}{T} \sum_s \hat{w}_s \zeta_{st} \right)^4 + \max_{t \leq T} \eta_t^4 \\ &\leq O_P(1 + \max_{t \leq T} \|w_t\|^4 + b_{NT,2}^4)(C_{NT}^{-4}) + O_P(b_{NT,1}^4 + b_{NT,3}^4). \end{aligned}$$

Next,

$$\begin{aligned} \max_{t \leq T} \frac{1}{N} \sum_i e_{it}^2 (\hat{e}_{it} - e_{it})^2 &= \max_{t \leq T} \frac{1}{N} \sum_i e_{it}^2 \left(\frac{1}{T} \sum_s e_{is} w_s \right)^2 \max_{t \leq T} \hat{w}_t^2 + \max_{t \leq T} \frac{1}{N} \sum_i e_{it}^2 l_i^4 \mathcal{Q}_3^2 \max_{t \leq T} \hat{w}_t^2 \\ &+ \max_{t \leq T} \frac{1}{N} \sum_i e_{it}^2 \left(\frac{1}{T^2} \sum_{s, m \leq T} e_{is} \zeta_{ms} \hat{w}_m \right)^2 \max_{t \leq T} \hat{w}_t^2 \\ &+ \max_{t \leq T} \frac{1}{N} \sum_i e_{it}^2 \left(\frac{1}{T} \sum_s e_{is} \eta'_s \right)^2 \max_{t \leq T} \hat{w}_t^2 \\ &+ \max_{t \leq T} \frac{1}{N} \sum_i e_{it}^2 l_i^2 \left(\frac{1}{T} \sum_s \hat{w}_s \zeta_{st} \right)^2 + \max_{t \leq T} \frac{1}{N} \sum_i e_{it}^2 l_i^2 \eta_t^2 \\ &\leq O_P(1 + \max_{t \leq T} \|w_t\|^2 + b_{NT,2}^2) \max_{it} |e_{it}|^2 C_{NT}^{-2} \\ &+ O_P(b_{NT,1}^2 + b_{NT,3}^2) \max_{t \leq T} \frac{1}{N} \sum_i e_{it}^2. \end{aligned}$$

$$\begin{aligned} \frac{1}{N} \sum_i c_i e_{it} (\hat{e}_{it} - e_{it}) &\leq \max_{t \leq T} \frac{1}{N} \sum_i c_i e_{it} \left(\frac{1}{T} \sum_s e_{is} w_s \right) \hat{w}_t \\ &+ \max_{t \leq T} \frac{1}{N} \sum_i c_i e_{it} \left(\frac{1}{T^2} \sum_{s, m \leq T} e_{is} \zeta_{ms} \hat{w}_m \right) \hat{w}_t \\ &+ \max_{t \leq T} \frac{1}{N} \sum_i c_i e_{it} \left(\frac{1}{T} \sum_s e_{is} \eta'_s \right) \hat{w}_t + \max_{t \leq T} \frac{1}{N} \sum_i c_i e_{it} l_i \mathcal{Q}_3 \max_{t \leq T} \hat{w}_t \\ &+ \max_{t \leq T} \frac{1}{N} \sum_i c_i e_{it} l_i \left(\frac{1}{T} \sum_s \hat{w}_s \zeta_{st} \right) + \max_{t \leq T} \frac{1}{N} \sum_i c_i e_{it} l_i \max_{t \leq T} \eta_t \\ &\leq O_P(1 + \max_{t \leq T} \|w_t\| + b_{NT,2}) \max_{it} |e_{it}| C_{NT}^{-1} \end{aligned}$$

$$\begin{aligned}
& + \max_{t \leq T} \left\| \frac{1}{N} \sum_i c_i e_{it} l_i \right\|_{FOP} (b_{NT,1} + b_{NT,3}). \\
\max_{t \leq T} \frac{1}{N} \sum_i c_i (\hat{e}_{it} - e_{it})^2 & \leq \frac{1}{N} \sum_i c_i \left(\frac{1}{T} \sum_s e_{is} w_s \right)^2 \max_{t \leq T} \hat{w}_t^2 + \max_{t \leq T} \left(\frac{1}{T} \sum_s \hat{w}_s \zeta_{st} \right)^2 + \max_{t \leq T} \eta_t^2 \\
& + \frac{1}{N} \sum_i c_i \left(\frac{1}{T^2} \sum_{s,m \leq T} e_{is} \zeta_{ms} \hat{w}_m \right)^2 \max_{t \leq T} \hat{w}_t^2 \\
& + \frac{1}{N} \sum_i c_i \left(\frac{1}{T} \sum_s e_{is} \eta'_s \right)^2 \max_{t \leq T} \hat{w}_t^2 + \max_{t \leq T} \mathcal{Q}_3^2 \hat{w}_t^2 \\
& \leq O_P(1 + \max_{t \leq T} \|w_t\|^2 + b_{NT,2}^2) C_{NT}^{-2} + O_P(b_{NT,1}^2 + b_{NT,3}^2), \\
\max_{t \leq T} \frac{1}{N} \sum_i c_i (\hat{e}_{it} - e_{it}) & \leq \frac{1}{N} \sum_i c_i \left(\frac{1}{T} \sum_s e_{is} w_s \right) \max_{t \leq T} \hat{w}_t \\
& + \frac{1}{N} \sum_i c_i \left(\frac{1}{T^2} \sum_{s,m \leq T} e_{is} \zeta_{ms} \hat{w}_m \right) \max_{t \leq T} \hat{w}_t \\
& + \frac{1}{N} \sum_i c_i \left(\frac{1}{T} \sum_s e_{is} \eta'_s \right) \max_{t \leq T} \hat{w}_t + \mathcal{Q}_3 \max_{t \leq T} \hat{w}_t \\
& + \max_{t \leq T} \left(\frac{1}{T} \sum_s \hat{w}_s \zeta_{st} \right) + \max_{t \leq T} \eta_t \\
& \leq O_P(1 + \max_{t \leq T} \|w_t\|) C_{NT}^{-2} + O_P(b_{NT,1} + b_{NT,3} + C_{NT}^{-1} b_{NT,2}).
\end{aligned}$$

Finally,

$$\begin{aligned}
\max_i \frac{1}{T} \sum_t (\hat{e}_{it} - e_{it})^2 & \leq \max_i \frac{1}{T} \sum_t \hat{w}_t^2 \left(\frac{1}{T} \sum_s e_{is} w'_s H_x^{-1'} \right)^2 + \max_i \frac{1}{T} \sum_t \hat{w}_t^2 \left(\frac{1}{T^2} \sum_{s,m \leq T} e_{is} \zeta_{ms} \hat{w}'_m R'_1 \right)^2 \\
& + \max_i \frac{1}{T} \sum_t \hat{w}_t^2 \left(\frac{1}{T} \sum_s e_{is} \eta'_s R'_2 \right)^2 + \max_i \frac{1}{T} \sum_t \hat{w}_t^2 (l'_i \mathcal{Q}_3)^2 \\
& + \max_i \frac{1}{T} \sum_t (l'_i H_x R_1 \frac{1}{T} \sum_s \hat{w}_s \zeta_{st})^2 + \max_i \frac{1}{T} \sum_t \eta_t^2 (l'_i H_x R_2)^2 \\
& \leq O_P(b_{NT,4}^2 + b_{NT,5}^2 + C_{NT}^{-2}).
\end{aligned}$$

(ii) Note that $\max_{t \leq T} \|\hat{w}_t\|^2 = O_P(1) + O_P(1) \max_{t \leq T} \|w_t\|^2 \leq O_P(1) + o_P(C_{NT})$, where the last inequality follows from the assumption that $\max_{t \leq T} \|w_t\|^2 = o_P(C_{NT})$.

$$\begin{aligned}
\frac{1}{NT} \sum_{it} (\hat{e}_{it} - e_{it})^4 & \leq \frac{1}{T} \sum_t \|\hat{w}_t\|^2 \max_{t \leq T} \|\hat{w}_t\|^2 \left[\frac{1}{N} \sum_i \left(\frac{1}{T} \sum_s e_{is} w_s \right)^4 + O_P(1) \left(\frac{1}{T^2} \sum_{s,m \leq T} \zeta_{ms}^2 \right)^2 \right] \\
& + \frac{1}{T} \sum_t \|\hat{w}_t\|^2 \max_{t \leq T} \|\hat{w}_t\|^2 \left[\frac{1}{N} \sum_i \left(\frac{1}{T} \sum_s e_{is} \eta_s \right)^4 + \mathcal{Q}_3^4 \right] \\
& + \frac{1}{T} \sum_t \left(\frac{1}{T} \sum_s \zeta_{st}^2 \right)^2 O_P(C_{NT}^{-4}) + \frac{1}{T} \sum_t \eta_t^4 + \frac{1}{T} \sum_t \left(\frac{1}{T} \sum_s w_s \zeta_{st} \right)^4
\end{aligned}$$

$$\begin{aligned}
&\leq (O_P(1) + o_P(C_{NT}))C_{NT}^{-4} + O_P(C_{NT}^{-2}) \\
&\quad + O_P(1)\frac{1}{T}\sum_t \frac{1}{T^4}\sum_{s,k,l,m \leq T} \mathbb{E}w_l w_m w_k w_s \mathbb{E}(\zeta_{st}\zeta_{kt}\zeta_{lt}\zeta_{mt}|W) \\
&= O_P(C_{NT}^{-2}).
\end{aligned}$$

Similarly, $\frac{1}{NT}\sum_{it}(\hat{e}_{it} - e_{it})^2 = O_P(C_{NT}^{-2})$.

$$\begin{aligned}
\frac{1}{NT}\sum_{it} e_{it}^2(\hat{e}_{it} - e_{it})^2 &\leq \frac{1}{NT}\sum_{it} e_{it}^2 \hat{w}_t^2 \left(\frac{1}{T}\sum_s e_{is}\eta_s\right)^2 + \frac{1}{NT}\sum_{it} e_{it}^2 \hat{w}_t^2 \left(\frac{1}{T}\sum_s e_{is}w_s\right)^2 \\
&\quad + \frac{1}{NT}\sum_{it} e_{it}^2 \hat{w}_t^2 \left(\frac{1}{T^2}\sum_{s,m \leq T} e_{is}\zeta_{ms}\hat{w}_m\right)^2 + \frac{1}{NT}\sum_{it} e_{it}^2 \hat{w}_t^2 (l'_i \mathcal{Q}_3)^2 \\
&\quad + \frac{1}{NT}\sum_{it} e_{it}^2 (l'_i \frac{1}{T}\sum_s \hat{w}_s \zeta_{st})^2 + \frac{1}{NT}\sum_{it} e_{it}^2 (l'_i \eta_t)^2 = O_P(C_{NT}^{-2}).
\end{aligned}$$

Next,

$$\begin{aligned}
\frac{1}{T}\sum_t \left(\frac{1}{N}\sum_i c_i(e_{it} - \hat{e}_{it})\right)^2 m_t^2 &\leq \frac{1}{T}\sum_t m_t^2 \hat{w}_t^2 \left(\frac{1}{N}\sum_i c_i \frac{1}{T}\sum_s e_{is}w_s\right)^2 \\
&\quad + \frac{1}{T}\sum_t m_t^2 \hat{w}_t^2 \left(\frac{1}{N}\sum_i c_i \frac{1}{T^2}\sum_{s,m \leq T} e_{is}\zeta_{ms}\hat{w}_m\right)^2 \\
&\quad + \frac{1}{T}\sum_t m_t^2 \hat{w}_t^2 \left(\frac{1}{N}\sum_i c_i \frac{1}{T}\sum_s e_{is}\eta_s\right)^2 \\
&\quad + \frac{1}{T}\sum_t m_t^2 \hat{w}_t^2 \mathcal{Q}_3^2 + \frac{1}{T}\sum_t m_t^2 \left(\frac{1}{T}\sum_s \hat{w}_s \zeta_{st}\right)^2 + \frac{1}{T}\sum_t m_t^2 \eta_t^2 \\
&= O_P(C_{NT}^{-2}).
\end{aligned}$$

(iii)

$$\begin{aligned}
&\frac{1}{T}\sum_t \left(\frac{1}{N}\sum_i c_i(\hat{e}_{it} - e_{it})e_{it}\right)^2 \\
&\leq \max_{t \leq T} \|\hat{w}_t - H_x w_t\|^2 \frac{1}{T}\sum_t \left[\left(\frac{1}{N}\sum_i c_i \frac{1}{T}\sum_s e_{is}w_s e_{it}\right)^2 + \frac{1}{T}\sum_m \left(\frac{1}{N}\sum_i c_i e_{it} \frac{1}{T}\sum_{s \leq T} e_{is}\zeta_{ms}\right)^2\right] \\
&\quad + \frac{1}{T}\sum_t w_t^2 \left[\left(\frac{1}{N}\sum_i c_i \frac{1}{T}\sum_s e_{is}w_s e_{it}\right)^2 + \frac{1}{T}\sum_m \left(\frac{1}{N}\sum_i c_i e_{it} \frac{1}{T}\sum_{s \leq T} e_{is}\zeta_{ms}\right)^2\right] \\
&\quad + \frac{1}{T}\sum_t \hat{w}_t^2 \left(\frac{1}{N}\sum_i c_i \frac{1}{T}\sum_s e_{is}\eta_s e_{it}\right)^2 + \frac{1}{T}\sum_t \hat{w}_t^2 \left(\frac{1}{N}\sum_i c_i l'_i e_{it}\right)^2 \mathcal{Q}_3^2 \\
&\quad + \frac{1}{T}\sum_t \left(\frac{1}{N}\sum_i c_i l_i e_{it}\right)^2 \frac{1}{T}\sum_s \zeta_{st}^2 + \frac{1}{T}\sum_t \eta_t^2 \left(\frac{1}{N}\sum_i c_i l_i e_{it}\right)^2 = O_P(C_{NT}^{-4}).
\end{aligned}$$

Next, to bound $\frac{1}{T} \sum_t (\frac{1}{N} \sum_i c_i (\hat{e}_{it} - e_{it})^2)^2$, we first bound $\frac{1}{T} \sum_t \hat{w}_t^4$. By (G.1),

$$\begin{aligned} \frac{1}{T} \sum_t \hat{w}_t^4 &\leq O_P(1) + \frac{1}{T} \sum_t \|\hat{w}_t - H_x^{-1} w_t\|^4 \\ &\leq O_P(1) + \frac{1}{T} \sum_t \|\eta_t\|^4 + \frac{1}{T} \sum_t \left\| \frac{1}{T} \sum_s w_s \zeta_{st} \right\|^4 + \frac{1}{T} \sum_t \left(\frac{1}{T} \sum_s \zeta_{st}^2 \right)^2 O_P(C_{NT}^{-4}) \\ &= O_P(1). \end{aligned}$$

Thus

$$\begin{aligned} &\frac{1}{T} \sum_{t \notin I} \left(\frac{1}{N} \sum_i c_i (\hat{e}_{it} - e_{it})^2 \right)^2 \\ &\leq \frac{1}{T} \sum_{t \notin I} \hat{w}_t^4 \left(\frac{1}{N} \sum_i c_i \left(\frac{1}{T} \sum_s e_{is} w_s \right)^2 \right)^2 + \frac{1}{T} \sum_{t \notin I} \hat{w}_t^4 \left(\frac{1}{N} \sum_i c_i \left(\frac{1}{T} \sum_s e_{is} \eta_s \right)^2 \right)^2 \\ &\quad + \frac{1}{T} \sum_{t \notin I} \hat{w}_t^4 \mathcal{Q}_3^4 + \frac{1}{T} \sum_{t \notin I} \eta_t^4 + \frac{1}{T} \sum_{t \notin I} \left(\frac{1}{T} \sum_s w_s \zeta_{st} \right)^4 + \frac{1}{T} \sum_{t \notin I} \left(\frac{1}{T} \sum_s \zeta_{st}^2 \right)^2 O_P(C_{NT}^{-4}) \\ &\quad + \frac{1}{T} \sum_{t \notin I} \hat{w}_t^4 \left(\frac{1}{N} \sum_i c_i \frac{1}{T} \sum_{m \leq T} \left(\frac{1}{T} \sum_{s \leq T} e_{is} \zeta_{ms} \right)^2 \right)^2 = O_P(C_{NT}^{-4}). \end{aligned}$$

Also, up to an $O_P(1)$ as a product term,

$$\begin{aligned} &\frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \left\| \frac{1}{T} \sum_{t \notin I} f_t (\hat{e}_{it} - e_{it}) u_{it} \right\|^2 \\ &\leq \frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \left\| \frac{1}{T} \sum_{t \notin I} f_t u_{it} \hat{w}_t \right\|^2 \left(\frac{1}{T^2} \sum_{s, m \leq T} e_{is} \zeta_{ms} \hat{w}_m \right)^2 + \left\| \frac{1}{T} \sum_s e_{is} w_s \right\|^2 + \left\| \frac{1}{T} \sum_s e_{is} \eta_s \right\|^2 + \|\mathcal{Q}_3\|^2 \\ &\quad + \frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \left\| \frac{1}{T} \sum_{t \notin I} f_t u_{it} l_i \eta_t \right\|^2 + \frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \left\| \frac{1}{T} \sum_{t \notin I} f_t u_{it} \frac{1}{T} \sum_s \hat{w}_s \zeta_{st} \right\|^2. \end{aligned}$$

It is easy to see each term except for the last one is $O_P(C_{NT}^{-4})$. To see the last one, note that it is bounded by

$$\begin{aligned} &O_P(1) \frac{1}{T} \sum_s (\hat{w}_s - H_x w_s)^2 \frac{1}{T} \sum_s \frac{1}{T} \sum_{t \notin I} \zeta_{st}^2 + O_P(1) \frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \left\| \frac{1}{T} \sum_{t \notin I} f_t u_{it} \frac{1}{T} \sum_s w_s \zeta_{st} \right\|^2 \\ &\leq O_P(C_{NT}^{-4}) + O_P(1) \frac{1}{|\mathcal{G}|_0} \sum_{j \in \mathcal{G}} \left\| \frac{1}{T} \sum_{t \notin I} f_t u_{jt} \frac{1}{T} \sum_s w_s \frac{1}{N} \sum_i (e_{is} e_{it} - \mathbb{E} e_{is} e_{it}) \right\|^2 \\ &= O_P(C_{NT}^{-4}) \end{aligned}$$

since e_{it} is conditionally serially independent given (U, W, F) and u_{it} is conditionally serially independent given (E, W, F) .

The conclusion that $\frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \left\| \frac{1}{T} \sum_{s \notin I} f_s e_{is} (\hat{e}_{is} - e_{is}) \lambda_i' f_s \right\|^2 = O_P(C_{NT}^{-4})$ follows similarly, due to $\max_j \sum_{i \leq N} |(\mathbb{E} e_{it} e_{jt} | F, W)| < C$.

Next, for $a := \frac{1}{|\mathcal{G}|_0} \sum_{j \in \mathcal{G}} \frac{1}{N} \sum_i \left\| \frac{1}{T_0} \sum_{t \in I^c} c_i f_t^2 e_{jt} \widehat{w}_t \right\|^2 = O_P(C_{NT}^{-2})$,

$$\begin{aligned}
& \frac{1}{|\mathcal{G}|_0} \sum_{j \in \mathcal{G}} \left\| \frac{1}{T_0} \sum_{t \in I^c} \frac{1}{N} \sum_i c_i f_t^2 e_{jt} (e_{it} - \widehat{e}_{it}) \right\|^2 \\
\leq & a \frac{1}{N} \sum_i \left[\left\| \frac{1}{T} \sum_s e_{is} w_s \right\|^2 + \left\| \frac{1}{T^2} \sum_{s, m \leq T} e_{is} \zeta_{ms} \widehat{w}'_m \right\|^2 + \left\| \frac{1}{T} \sum_s e_{is} \eta_s \right\|^2 + \|\mathcal{Q}_3\|^2 \right] \\
& + \frac{1}{|\mathcal{G}|_0} \sum_{j \in \mathcal{G}} \left\| \frac{1}{T_0} \sum_{t \in I^c} f_t^2 e_{jt} \frac{1}{T} \sum_s \widehat{w}_s \frac{1}{N} \sum_i (e_{is} e_{it} - \mathbf{E} e_{is} e_{it}) \right\|^2 \\
& + \frac{1}{|\mathcal{G}|_0} \sum_{j \in \mathcal{G}} \left\| \frac{1}{N} \sum_i l_i \frac{1}{T_0} \sum_{t \in I^c} f_t^2 (e_{jt} e_{it} - \mathbf{E} e_{jt} e_{it}) \right\|^2 + \frac{1}{|\mathcal{G}|_0} \sum_{j \in \mathcal{G}} \left\| \frac{1}{N} \sum_i l_i \frac{1}{T_0} \sum_{t \in I^c} f_t^2 \mathbf{E} e_{jt} e_{it} \right\|^2 \\
= & O_P(C_{NT}^{-4}).
\end{aligned}$$

□

G.2. Behavior of preliminary estimation. Recall that

$$(\widetilde{f}_s, \widetilde{g}_s) := \arg \min_{f_s, g_s} \sum_{i=1}^N (y_{is} - \widetilde{\alpha}'_i g_s - x_{is} \widetilde{\lambda}'_i f_s)^2, \quad s \notin I.$$

and

$$(\dot{\lambda}_i, \dot{\alpha}_i) = \arg \min_{\lambda_i, \alpha_i} \sum_{s \notin I} (y_{is} - \alpha'_i \widetilde{g}_s - x_{is} \lambda'_i \widetilde{f}_s)^2, \quad i = 1, \dots, N.$$

The goal of this section is to show that the effect of the preliminary estimation is negligible. Specifically, we aim to show, for each fixed $t \notin I$,

$$\begin{aligned}
\left\| \frac{1}{N} \sum_j (\dot{\lambda}_j - H_1' \lambda_j) e_{jt} \right\|^2 &= O_P(C_{NT}^{-4}), \\
\frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \left\| \frac{1}{T} \sum_{s \notin I} f_s (\widetilde{f}_s - H_1^{-1} f_s)' \mu_{is} e_{is} \right\|^2 &= O_P(C_{NT}^{-4}).
\end{aligned}$$

Throughout the proof below, we treat $|I^c| = T$ instead of $T/2$ to avoid keeping the constant “2”. In addition, for notational simplicity, we write $\widetilde{\Lambda} = \widetilde{\Lambda}_I$ and $\widetilde{A} = \widetilde{A}_I$ by suppressing the subscripts, but we should keep in mind that $\widetilde{\Lambda}$ and $\widetilde{A}$ are estimated on data D_I as defined in step 2. In addition, let $\mathbf{E}_I$ and $\mathbf{Var}_I$ be the conditional expectation and variance, given D_I . Recall that X_s is the vector of x_{is} fixing $s \leq T$, and $M_{\widetilde{\alpha}} = I_N - \widetilde{A}(\widetilde{A}'\widetilde{A})^{-1}\widetilde{A}'$; X_i is the vector of x_{is} fixing $i \leq N$, and $M_{\widetilde{g}} = I - \widetilde{G}(\widetilde{G}'\widetilde{G})^{-1}\widetilde{G}'$, for $\widetilde{G}$ as the $|I^c|_0 \times K_1$ matrix of $\widetilde{g}_s$. Define $\widetilde{F}$ similarly. Let L denote $N \times K_3$ matrix of l_i , so $X_s = Lw_s + e_s$. Also let W be $T \times K_3$ matrix of w_t .

Define

$$\begin{aligned}
\tilde{D}_{fs} &= \frac{1}{N} \tilde{\Lambda}' \text{diag}(X_s) M_{\tilde{\alpha}} \text{diag}(X_s) \tilde{\Lambda} \\
D_{fs} &= \frac{1}{N} \Lambda' (\text{diag}(X_s) M_{\alpha} \text{diag}(X_s)) \Lambda \\
\bar{D}_{fs} &= \frac{1}{N} \Lambda' \mathbb{E}((\text{diag}(e_s) M_{\alpha} \text{diag}(e_s)) \Lambda) + \frac{1}{N} \Lambda' (\text{diag}(Lw_s) M_{\alpha} \text{diag}(Lw_s)) \Lambda \\
\tilde{D}_{\lambda i} &= \frac{1}{T} \tilde{F}' \text{diag}(X_i) M_{\tilde{g}} \text{diag}(X_i) \tilde{F} \\
D_{\lambda i} &= \frac{1}{T} F' (\text{diag}(X_i) M_g \text{diag}(X_i)) F \\
\bar{D}_{\lambda i} &= \frac{1}{T} F' \mathbb{E}(\text{diag}(E_i) M_g \text{diag}(E_i)) F + \frac{1}{T} F' (\text{diag}(Wl_i) M_g \text{diag}(Wl_i)) F
\end{aligned}$$

By stationarity, D_f does not depend on s .

Lemma G.3. Suppose $\max_{it} e_{it}^2 + \max_{t \leq T} \|w_t\|^2 = o_P(C_{NT})$, $\max_{t \leq T} \mathbb{E} \|w_t\|^4 = O(1)$. $\frac{1}{N} \sum_{ij} |\text{Cov}(e_{is}, e_{js} | w_s)| < \infty$ and $\|\mathbb{E} e_s e_s'\| < \infty$. $\frac{1}{N^3} \sum_{ijkl} \text{Cov}(e_{is} e_{js}, e_{ks} e_{ls}) < C$. Also, there is $c > 0$, so that $\min_s \min_j \psi_j(D_{fs}) > c$. Then

- (i) $\max_s \|\tilde{D}_{fs}^{-1}\| = O_P(1)$.
- (ii) $\frac{1}{T} \sum_{s \notin I} \|\tilde{D}_{fs}^{-1} - (H_1' \bar{D}_{fs} H_1)^{-1}\|^2 = O_P(C_{NT}^{-2})$.

Proof. (i) The eigenvalues of D_{fs} are bounded from zero uniformly in $s \leq T$. Also,

$$\begin{aligned}
\tilde{D}_{fs} - H_1' D_{fs} H_1 &= \sum_l \delta_l, \quad \text{where} \\
\delta_1 &= \frac{1}{N} (\tilde{\Lambda} - \Lambda H_1)' \text{diag}(X_s) M_{\tilde{\alpha}} \text{diag}(X_s) \tilde{\Lambda}, \\
\delta_2 &= \frac{1}{N} H_1' \Lambda' \text{diag}(X_s) M_{\tilde{\alpha}} \text{diag}(X_s) (\tilde{\Lambda} - \Lambda H_1) \\
\delta_3 &= \frac{1}{N} H_1' \Lambda' \text{diag}(X_s) (M_{\tilde{\alpha}} - M_{\alpha}) \text{diag}(X_s) \Lambda H_1. \quad (\text{G.4})
\end{aligned}$$

We now bound each term uniformly in $s \leq T$. The first term is

$$\frac{1}{N} (\tilde{\Lambda} - \Lambda H_1)' \text{diag}(X_s) M_{\tilde{\alpha}} \text{diag}(X_s) \tilde{\Lambda} \leq O_P(1) \frac{1}{\sqrt{N}} \|\tilde{\Lambda} - \Lambda H_1\|_F \max_{is} x_{is}^2 = o_P(1)$$

provided that $\max_{is} x_{is}^2 = o_P(C_{NT})$. The second term is bounded similarly. The third term is bounded by

$$O_P(1) \max_{is} x_{is}^2 \|M_{\tilde{\alpha}} - M_{\alpha}\| = O_P(1) \frac{1}{\sqrt{N}} \|\tilde{A} - A H_2\|_F \max_{is} x_{is}^2 = o_P(1).$$

This implies $\max_s \|\tilde{D}_{fs} - H'_1 D_{fs} H_1\| = o_P(1)$. In addition, because of the convergence of $\|\frac{1}{\sqrt{N}}(\tilde{\Lambda} - \Lambda H_1)\|_F$, we have $\min_j \psi_j(H'_1 H_1) \geq C \min_j \psi_j(\frac{1}{N} H'_1 \Lambda' \Lambda H_1)$, bounded away from zero. Thus $\min_s \min_j \psi_j(H'_1 D_{fs} H_1) \geq \min_s \min_j \psi_j(D_{fs})C$, bounded away from zero. This together with $\max_s \|\tilde{D}_{fs} - H'_1 D_{fs} H_1\| = o_P(1)$ imply $\min_s \|\tilde{D}_{fs}^{-1}\| = O_P(1)$.

(ii) By (G.4),

$$\begin{aligned}
& \frac{1}{T} \sum_{s \notin I} \|\tilde{D}_{fs}^{-1} - (H'_1 D_{fs} H_1)^{-1}\|^2 \\
& \leq \frac{1}{T} \sum_s \|\tilde{D}_{fs} - (H'_1 D_{fs} H_1)\|^2 \|(H'_1 D_{fs} H_1)^{-2}\| \max_s \|\tilde{D}_{fs}^{-2}\| \\
& \leq O_P(1) \frac{1}{T} \sum_s \|\tilde{D}_{fs} - (H'_1 D_{fs} H_1)\|^2 \\
& = O_P(1) \sum_{l=1}^3 \frac{1}{T} \sum_s \|\delta_l\|^2.
\end{aligned}$$

We bound each term below. Up to a $O_P(1)$ product,

$$\begin{aligned}
\frac{1}{T} \sum_{s \notin I} \|\delta_1\|^2 & \leq \frac{1}{N^2} \sum_{ij} \|\tilde{\lambda}_i - H'_1 \lambda_i\|^2 \|\tilde{\lambda}_j\|^2 \frac{1}{T} \sum_{s \notin I} x_{is}^2 x_{js}^2 \\
& \leq \frac{1}{N^2} \sum_{ij} \|\tilde{\lambda}_i - H'_1 \lambda_i\|^2 \|\tilde{\lambda}_j\|^2 \frac{1}{T} \sum_{s \notin I} (1 + e_{is}^2 e_{js}^2) \\
& \quad + \frac{1}{N^2} \sum_{ij} \|\tilde{\lambda}_i - H'_1 \lambda_i\|^2 \|\tilde{\lambda}_j\|^2 \left(\frac{1}{T} \sum_{s \notin I} e_{is}^4 \right)^{1/2}
\end{aligned}$$

$\{e_{is}\}$ is serially independent, so e_{is}, e_{js} are independent of $\|\tilde{\lambda}_i - H'_1 \lambda_i\|^2 \|\tilde{\lambda}_j - H'_1 \lambda_j\|^2$ for $s \notin I$. Take the conditional expectation $\mathbf{E}_I$.

$$\begin{aligned}
\frac{1}{T} \sum_{s \notin I} \|\delta_1\|^2 & \leq \frac{1}{N^2} \sum_{ij} \|\tilde{\lambda}_i - H'_1 \lambda_i\|^2 \|\tilde{\lambda}_j\|^2 \frac{1}{T} \sum_{s \notin I} \mathbf{E}_I(1 + e_{is}^2 e_{js}^2) \\
& \quad + \frac{1}{N^2} \sum_{ij} \|\tilde{\lambda}_i - H'_1 \lambda_i\|^2 \|\tilde{\lambda}_j\|^2 \mathbf{E}_I \left(\frac{1}{T} \sum_{s \notin I} e_{is}^4 \right)^{1/2} = O_P(C_{NT}^{-2}).
\end{aligned}$$

The term $\frac{1}{T} \sum_{s \notin I} \|\delta_2\|^2$ is bounded similarly.

$$\frac{1}{T} \sum_{s \notin I} \|\delta_3\|^2 \leq O_P(1) \|M_{\tilde{\alpha}} - M_{\alpha}\| = O_P(C_{NT}^{-2}).$$

Next,

$$\begin{aligned}
& \frac{1}{T} \sum_s \|D_{fs} - \bar{D}_{fs}\|^2 \leq \frac{1}{T} \sum_s \left\| \frac{1}{N} \sum_{ij} \lambda_i \lambda_j' M_{\alpha,ij} (x_{is} x_{js} - \mathbb{E} e_{is} e_{js} - l_i' w_s l_j' w_s) \right\|_F^2 \\
& \leq \frac{1}{T} \sum_s \left\| \frac{1}{N} \sum_{ij} \lambda_i \lambda_j' M_{\alpha,ij} (e_{is} e_{js} - \mathbb{E} e_{is} e_{js}) \right\|_F^2 \\
& \quad + \frac{2}{T} \sum_s \left\| \frac{1}{N} \sum_{ij} \lambda_i \lambda_j' M_{\alpha,ij} l_i' w_s l_j' w_s \right\|_F^2
\end{aligned}$$

We assume $\dim(\lambda_i) = \dim(p_i) = 1$. The first term is less than

$$\begin{aligned}
& \frac{1}{T} \sum_s \text{Var} \left(\frac{1}{N} \sum_{ij} \lambda_i \lambda_j' M_{\alpha,ij} e_{is} e_{js} \right) \\
& \leq \frac{1}{T} \sum_s \frac{1}{N^4} \sum_{ijkl} |\text{Cov}(e_{is} e_{js}, e_{ks} e_{ls})| = O(N^{-1})
\end{aligned}$$

provided that $\frac{1}{N^3} \sum_{ijkl} \text{Cov}(e_{is} e_{js}, e_{ks} e_{ls}) < C$. The second term is less than

$$O_P(1) \frac{1}{T} \sum_s \mathbb{E} w_s^2 \frac{1}{N^2} \sum_{ij} |\text{Cov}(e_{js}, e_{ls} | w_s)| = O(N^{-1})$$

provided that $\frac{1}{N} \sum_{ij} |\text{Cov}(e_{is}, e_{js} | w_s)| < \infty$ and $\|\mathbb{E} e_s e_s'\| < \infty$. So $\frac{1}{T} \sum_s \|D_{fs} - \bar{D}_{fs}\|^2 = O_P(N^{-1})$. □

Lemma G.4. Suppose $\max_{it} e_{it}^4 = O_P(\min\{N, T\})$. (i) $\frac{1}{T} \|\tilde{F} - F H_1^{-1'}\|_F^2 = O_P(C_{NT}^{-2}) = \frac{1}{T} \|\tilde{G} - G H_2^{-1'}\|_F^2$, and $\frac{1}{T} \sum_{t \notin I} \|\tilde{f}_t - H_1^{-1} f_t\|^2 e_{it}^2 u_{it}^2 = O_P(C_{NT}^{-2})$. (ii) $\max_i \|\tilde{D}_{\lambda_i}^{-1}\| = O_P(1)$. (iii) $\frac{1}{N} \sum_i \|\tilde{D}_{\lambda_i}^{-1} - (H_1^{-1} \bar{D}_{\lambda_i} H_1^{-1'})^{-1}\|^2 = O_P(C_{NT}^{-2})$.

Proof. (i) The proof is straightforward given the expansion of (G.7) and $\max_s \|\tilde{D}_{fs}^{-1}\| = O_P(1)$. We thus omit the details for brevity. (ii) Note that

$$\begin{aligned}
\tilde{D}_{\lambda_i} - H_1^{-1} D_{\lambda_i} H_1^{-1'} &= \sum_l \delta_l, \quad \text{where} \\
\delta_1 &= \frac{1}{T} (\tilde{F} - F H_1^{-1'})' \text{diag}(X_i) M_{\tilde{g}} \text{diag}(X_i) \tilde{F}, \\
\delta_2 &= \frac{1}{T} H_1^{-1} F' \text{diag}(X_i) M_{\tilde{g}} \text{diag}(X_i) (\tilde{F} - F H_1^{-1'}) \\
\delta_3 &= \frac{1}{T} H_1^{-1} F' \text{diag}(X_i) (M_{\tilde{g}} - M_g) \text{diag}(X_i) F H_1^{-1'}. \quad (\text{G.5})
\end{aligned}$$

The proof is very similar to that of Lemma G.3.

(iii)

$$\begin{aligned}
& \frac{1}{N} \sum_i \|\tilde{D}_{\lambda i}^{-1} - (H_1^{-1} \bar{D}_{\lambda i} H_1^{-1'})^{-1}\|^2 \\
& \leq \frac{1}{N} \sum_i \|\tilde{D}_{\lambda i} - (H_1^{-1} \bar{D}_{\lambda i} H_1^{-1'})\|^2 \max_i \|(H_1^{-1} \bar{D}_{\lambda i} H_1^{-1'})^{-2}\| \|\tilde{D}_{\lambda i}^{-2}\| \\
& \leq O_P(1) \frac{1}{N} \sum_i \|\tilde{D}_{\lambda i} - (H_1^{-1} \bar{D}_{\lambda i} H_1^{-1'})\|^2 \\
& = O_P(1) \sum_{l=1}^3 \frac{1}{N} \sum_i \|\delta_l\|^2 + O_P(1) \frac{1}{N} \sum_i \|D_{\lambda i} - \bar{D}_{\lambda i}\|^2.
\end{aligned}$$

We now bound each term. With the assumption that $\max_{it} x_{it}^4 = O_P(\min\{N, T\})$,

$$\begin{aligned}
\frac{1}{N} \sum_i \|\delta_1\|^2 & \leq \frac{1}{T} \|\tilde{F} - FH_1^{-1'}\|_F^2 \|M_{\tilde{g}} - M_g\|^2 \frac{1}{T} \|\tilde{F}\|_F^2 \frac{1}{N} \sum_i \|\text{diag}(X_i)\|^4 \\
& \quad + \frac{1}{T^2} \|\tilde{F} - FH_1^{-1'}\|_F^4 \frac{1}{N} \sum_i \|\text{diag}(X_i)\|^4 \\
& \quad + \frac{1}{N} \sum_i \left\| \frac{1}{T} (\tilde{F} - FH_1^{-1'})' \text{diag}(X_i) M_g \text{diag}(X_i) F H_1^{-1'} \right\|^2 \\
& \leq O_P(C_{NT}^{-4}) \max_{it} x_{it}^4 + \frac{1}{NT} \sum_i \sum_t x_{it}^2 \left(\sum_s M_{g,ts} x_{is} f_s \right)^2 O_P(C_{NT}^{-2}) \\
& \leq O_P(C_{NT}^{-2}).
\end{aligned}$$

$\frac{1}{N} \sum_i \|\delta_2\|^2$ is bounded similarly. $\frac{1}{N} \sum_i \|\delta_3\|^2 \leq O_P(1) \|M_{\tilde{g}} - M_g\|^2 = O_P(C_{NT}^{-2})$. Finally,

$$\begin{aligned}
\frac{1}{N} \sum_i \|D_{\lambda i} - \bar{D}_{\lambda i}\|^2 & \leq \frac{1}{N} \sum_i \left\| \frac{1}{T} \sum_{st} f_s f_t' M_{g,st} (x_{is} x_{it} - \mathbf{E} e_{is} e_{it} - l_i' w_s l_i' w_t) \right\|_F^2 \\
& \leq O_P(1) \frac{1}{N} \sum_i \left\| \frac{1}{T} \sum_{st} f_s f_t' M_{g,st} (e_{is} e_{it} - \mathbf{E} e_{is} e_{it}) \right\|_F^2 \\
& \quad + O_P(1) \frac{1}{N} \sum_i \left\| \frac{1}{T} \sum_{st} f_s f_t' M_{g,st} w_s e_{it} \right\|_F^2 \\
& \leq O_P(1) \frac{1}{N} \sum_i \left\| \frac{1}{T} \sum_t f_t^2 w_t e_{it} \right\|_F^2 + O_P(1) \frac{1}{N} \sum_i \left\| \frac{1}{T} \sum_t f_t g_t e_{it} \right\|_F^2 \\
& = O_P(C_{NT}^{-2}).
\end{aligned}$$

Lemma G.5. Suppose $\text{Var}(u_s | e_t, e_s, w_s) < C$ and $C_{NT}^{-1} \max_{is} |x_{is}|^2 = O_P(1)$.

(i) For each fixed $t \notin I$, $\frac{1}{N} \sum_i (H_1' \lambda_i - \dot{\lambda}_i) e_{it} = O_P(C_{NT}^{-2})$.

(ii) $\frac{1}{T} \sum_{t \notin I} \left\| \frac{1}{N} \sum_i c_i e_{it} (\dot{\lambda}_i - H_1' \lambda_i) \right\|^2 = O_P(C_{NT}^{-4})$ for any deterministic and bounded sequence c_i .

Proof. Given that the $|I^c| \times 1$ vector $y_i = G\alpha_i + \text{diag}(X_i)F\lambda_i + u_i$ where u_i is a $|I^c| \times 1$ vector and G is an $|I^c| \times K_1$ matrix, we have

$$\begin{aligned}\dot{\lambda}_i &= \tilde{D}_{\lambda i}^{-1} \frac{1}{T} \tilde{F}' \text{diag}(X_i) M_{\tilde{g}} y_i \\ &= H_1' \lambda_i + \tilde{D}_{\lambda i}^{-1} \frac{1}{T} \tilde{F}' \text{diag}(X_i) M_{\tilde{g}} (GH_2^{-1'} - \tilde{G}) H_2' \alpha_i \\ &\quad + \tilde{D}_{\lambda i}^{-1} \frac{1}{T} \tilde{F}' \text{diag}(X_i) M_{\tilde{g}} \text{diag}(X_i) (FH_1^{-1'} - \tilde{F}) H_1' \lambda_i + \tilde{D}_{\lambda i}^{-1} \frac{1}{T} \tilde{F}' \text{diag}(X_i) M_{\tilde{g}} u_i.\end{aligned}$$

(i)

Step 1: given $\mathbb{E} e_{it}^4 f_t^2 + e_{it}^2 f_t^2 w_t^2 < C$,

$$\begin{aligned}& \frac{1}{\sqrt{N}} \sum_i e_{it} (\tilde{D}_{\lambda i}^{-1} - (H_1^{-1} \bar{D}_{\lambda i} H_1^{-1'})^{-1}) \frac{1}{T} \tilde{F}' \text{diag}(X_i) M_{\tilde{g}} (GH_2^{-1'} - \tilde{G}) H_2' \alpha_i \\ & \leq O_P(\sqrt{N} C_{NT}^{-2}) + O_P(\sqrt{N} C_{NT}^{-2}) \left(\frac{1}{T} \sum_s (\tilde{f}_s - H_1^{-1} f_s)^2 \right)^{1/2} \max_{is} |x_{is}| \\ & \leq O_P(\sqrt{N} C_{NT}^{-2}) + O_P(\sqrt{N} C_{NT}^{-3}) \max_{is} |x_{is}| = O_P(\sqrt{N} C_{NT}^{-2}).\end{aligned}$$

Step 2: $\bar{D}_{\lambda i}$ is nonrandom given W, G, F ,

$$\begin{aligned}& \frac{1}{\sqrt{N}} \sum_i e_{it} (H_1^{-1} \bar{D}_{\lambda i} H_1^{-1'})^{-1} \frac{1}{T} \tilde{F}' \text{diag}(X_i) M_{\tilde{g}} (GH_2^{-1'} - \tilde{G}) H_2' \alpha_i \\ & \leq O_P(\sqrt{N} C_{NT}^{-1}) (a^{1/2} + b^{1/2}) \quad \text{where} \\ a &= \frac{1}{T} \sum_s (\tilde{f}_s - H_1^{-1} f_s)^2 \left(\frac{1}{N} \sum_i \alpha_i e_{it} \bar{D}_{\lambda i}^{-1} x_{is} \right)^2 \\ b &= \frac{1}{T} \sum_s f_s^2 \left(\frac{1}{N} \sum_i \alpha_i e_{it} \bar{D}_{\lambda i}^{-1} x_{is} \right)^2.\end{aligned}$$

We now bound each term. As for b , note that for each fixed t ,

$$\begin{aligned}& \mathbb{E} \left(\left(\frac{1}{N} \sum_i \alpha_i e_{it} \bar{D}_{\lambda i}^{-1} x_{is} \right)^2 \middle| F, G, W, u_s \right) \\ & \leq \mathbb{E} \left(\frac{1}{N} \sum_i \alpha_i e_{it} \bar{D}_{\lambda i}^{-1} e_{is} \middle| F, G, W, u_s \right)^2 + \mathbb{E} \left(\frac{1}{N} \sum_i \alpha_i e_{it} \bar{D}_{\lambda i}^{-1} l_i' w_s \middle| F, G, W, u_s \right)^2 \\ & \leq \frac{C}{N} + \frac{C \|w_s\|^2}{N} + \left(\mathbb{E} \frac{1}{N} \sum_i \alpha_i e_{it} e_{is} \bar{D}_{\lambda i}^{-1} \middle| F, G, W, u_s \right)^2\end{aligned} \tag{G.6}$$

with the assumption $\frac{1}{N} \sum_{ij} |\text{Cov}(e_{jt} e_{js}, e_{it} e_{is} | F, G, W, u_s)| < C$. Thus,

$$\mathbb{E} b \leq \frac{1}{T} \sum_s \mathbb{E} f_s^2 \left(\frac{1}{N} \sum_i \alpha_i e_{it} \bar{D}_{\lambda i}^{-1} x_{is} \right)^2 \leq O(C_{NT}^{-2}).$$

$$\begin{aligned}
a &= \frac{1}{T} \sum_s f_s^2 \left(\frac{1}{N} \tilde{\Lambda}' \text{diag}(X_s) M_{\tilde{\alpha}} \text{diag}(X_s) (\Lambda H_1 - \tilde{\Lambda}) \right)^2 \left(\frac{1}{N} \sum_i \alpha_i e_{it} \bar{D}_{\lambda i}^{-1} x_{is} \right)^2 \\
&\quad + \frac{1}{T} \sum_s (\tilde{D}_{fs}^{-1} \frac{1}{N} \tilde{\Lambda}' \text{diag}(X_s) M_{\tilde{\alpha}} \text{diag}(X_s) (\Lambda H_1 - \tilde{\Lambda}) H_1^{-1} f_s)^2 \left(\frac{1}{N} \sum_i \alpha_i e_{it} \bar{D}_{\lambda i}^{-1} x_{is} \right)^2 \\
&\quad + \frac{1}{T} \sum_s (\tilde{D}_{fs}^{-1} \frac{1}{N} \tilde{\Lambda}' \text{diag}(X_s) M_{\tilde{\alpha}} u_s)^2 \left(\frac{1}{N} \sum_i \alpha_i e_{it} \bar{D}_{\lambda i}^{-1} x_{is} \right)^2 \\
&\leq O_P(C_{NT}^{-2}) \max_{is} x_{is}^4 \frac{1}{T} \sum_s \mathbb{E} f_s^2 \left(\frac{1}{N} \sum_i \alpha_i e_{it} \bar{D}_{\lambda i}^{-1} x_{is} \right)^2 \\
&\quad + O_P(1) \frac{1}{T} \sum_s \left(\frac{1}{N} \tilde{\Lambda}' \text{diag}(X_s) M_{\tilde{\alpha}} u_s \right)^2 \left(\frac{1}{N} \sum_i \alpha_i e_{it} \bar{D}_{\lambda i}^{-1} x_{is} \right)^2 \\
&\leq O_P(C_{NT}^{-2}) + O_P(1) \frac{1}{T} \sum_s \mathbb{E} \left(\frac{1}{N} \Lambda' \text{diag}(X_s) M_{\alpha} u_s \right)^2 \left(\frac{1}{N} \sum_i \alpha_i e_{it} \bar{D}_{\lambda i}^{-1} x_{is} \right)^2 \\
&\quad + O_P(C_{NT}^{-2}) \max_{is} x_{is}^2 \frac{1}{TN} \sum_s \mathbb{E} \|u_s\|^2 \mathbb{E} \left(\left(\frac{1}{N} \sum_i \alpha_i e_{it} \bar{D}_{\lambda i}^{-1} x_{is} \right)^2 | u_s \right) \\
&:= O_P(C_{NT}^{-2}) + O_P(1)(a.1 + a.2).
\end{aligned}$$

We now respectively bound $a.1$ and $a.2$. As for $a.1$, note that $\text{Var}(u_s | e_t, e_s) < C$ almost surely, thus

$$\begin{aligned}
&\mathbb{E} \left(\frac{1}{N} \Lambda' \text{diag}(X_s) M_{\alpha} u_s | e_t, e_s \right)^2 = \frac{1}{N} \frac{1}{N} \Lambda' \text{diag}(X_s) M_{\alpha} \text{Var}(u_s | e_t, e_s) M_{\alpha} \text{diag}(X_s) \Lambda \\
&\leq C \frac{1}{N} \frac{1}{N} \Lambda' \text{diag}(X_s)^2 \Lambda.
\end{aligned}$$

As for $a.2$, we use (G.6). Thus,

$$\begin{aligned}
a.1 &\leq \frac{1}{T} \sum_s \mathbb{E} \left(\frac{1}{N} \sum_i \alpha_i e_{it} \bar{D}_{\lambda i}^{-1} x_{is} \right)^2 \mathbb{E} \left(\frac{1}{N} \Lambda' \text{diag}(X_s) M_{\alpha} u_s | e_t, e_s \right)^2 \\
&\leq \frac{C}{N} \frac{1}{T} \sum_s \mathbb{E} \left(\frac{1}{N} \sum_i \alpha_i e_{it} \bar{D}_{\lambda i}^{-1} x_{is} \right)^2 \frac{1}{N} \Lambda' \text{diag}(X_s)^2 \Lambda = O(N^{-1}). \\
a.2 &\leq C_{NT}^{-2} \max_{is} x_{is}^2 \frac{1}{TN} \sum_s \mathbb{E} \|u_s\|^2 \mathbb{E} \left(\left(\frac{1}{N} \sum_i \alpha_i e_{it} \bar{D}_{\lambda i}^{-1} x_{is} \right)^2 | u_s \right) \\
&\leq C_{NT}^{-2} \max_{is} x_{is}^2 \frac{1}{TN} \sum_s \mathbb{E} \|u_s\|^2 \frac{C}{N} \\
&\quad + C_{NT}^{-2} \max_{is} x_{is}^2 \frac{1}{TN} \sum_s \mathbb{E} \|u_s\|^2 \left(\mathbb{E} \frac{1}{N} \sum_i \alpha_i e_{it} e_{is} \bar{D}_{\lambda i}^{-1} | F, u_s \right)^2 \\
&\leq O(C_{NT}^{-2}).
\end{aligned}$$

Put together, $a^{1/2} + b^{1/2} = O(C_{NT}^{-1})$. The first term in the expansion of $\frac{1}{\sqrt{N}} \sum_i (H'_1 \lambda_i - \dot{\lambda}_i) e_{it}$ is then

$$\frac{1}{\sqrt{N}} \sum_i e_{it} \tilde{D}_{\lambda i}^{-1} \frac{1}{T} \tilde{F}' \text{diag}(X_i) M_{\tilde{g}} (G H_2^{-1'} - \tilde{G}) H_2' \alpha_i = O_P(\sqrt{N} C_{NT}^{-2}).$$

Step 3:

$$\begin{aligned} & \frac{1}{\sqrt{N}} \sum_i e_{it} (\tilde{D}_{\lambda i}^{-1} - (H_1^{-1} \bar{D}_{\lambda i} H_1^{-1'})^{-1}) \frac{1}{T} \tilde{F}' \text{diag}(X_i) M_{\tilde{g}} \text{diag}(X_i) (F H_1^{-1'} - \tilde{F}) H_1' \lambda_i \\ \leq & O_P(\sqrt{N} C_{NT}^{-2}) \left(\frac{1}{NT} \sum_i e_{it}^2 \|F' \text{diag}(X_i) M_g \text{diag}(X_i)\|^2 \right)^{1/2} \\ & + O_P(\sqrt{N} C_{NT}^{-2}) \left(\frac{1}{N} \sum_i e_{it}^2 \|\text{diag}(X_i)\|^4 \right)^{1/2} \left[\|M_g - M_{\tilde{g}}\| \frac{1}{\sqrt{T}} \|\tilde{F}\| + \frac{1}{\sqrt{T}} \|\tilde{F} - F H_1^{-1'}\| \right] \\ \leq & O_P(\sqrt{N} C_{NT}^{-2}) \left(\frac{1}{NT} \sum_i \sum_s e_{it}^2 f_s^2 x_{is}^4 \right)^{1/2} \\ & + O_P(\sqrt{N} C_{NT}^{-2}) \left(\frac{1}{NT} \sum_i \sum_s x_{is}^2 e_{it}^2 g_s^2 \left(\frac{1}{T} \sum_k f_k g_k x_{ik} \right)^2 \right)^{1/2} \\ & + O_P(\sqrt{N} C_{NT}^{-3}) \max_{it} x_{it}^2 = O_P(\sqrt{N} C_{NT}^{-2}). \end{aligned}$$

Step 4: Note $\frac{1}{T} \sum_s \left(\frac{1}{N} \sum_i \lambda_i e_{it} \bar{D}_{\lambda i}^{-1} e_{is}^2 \right)^2 = O_P(C_{NT}^{-2})$,

$$\begin{aligned} & \frac{1}{\sqrt{N}} \sum_i e_{it} (H_1^{-1} \bar{D}_{\lambda i} H_1^{-1'})^{-1} \frac{1}{T} (\tilde{F} - F H_1^{-1'})' \text{diag}(X_i) M_{\tilde{g}} \text{diag}(X_i) (F H_1^{-1'} - \tilde{F}) H_1' \lambda_i \\ \leq & O_P(\sqrt{N} C_{NT}^{-3}) \max_{it} x_{it}^2 \\ & + \frac{1}{\sqrt{N}} \sum_i e_{it} (H_1^{-1} \bar{D}_{\lambda i} H_1^{-1'})^{-1} \frac{1}{T} (\tilde{F} - F H_1^{-1'})' \text{diag}(X_i) M_g \text{diag}(X_i) (F H_1^{-1'} - \tilde{F}) H_1' \lambda_i \\ \leq & O_P(\sqrt{N} C_{NT}^{-2}) + \sqrt{N} \frac{1}{T} \sum_s (\tilde{f}_s - H_1^{-1} f_s)^2 \frac{1}{N} \sum_i \lambda_i e_{it} \bar{D}_{\lambda i}^{-1} x_{is}^2 \\ \leq & O_P(\sqrt{N} C_{NT}^{-2}) + \max_{is} x_{is}^2 O_P(\sqrt{N} C_{NT}^{-2}) \frac{1}{NT} \sum_s g_s^2 w_s^2 \sum_i \lambda_i e_{it} \bar{D}_{\lambda i}^{-1} l_i^2 \\ & + \max_{is} x_{is}^2 O_P(\sqrt{N} C_{NT}^{-2}) \frac{1}{T} \sum_s w_s^2 \frac{1}{N} \sum_j \lambda_j^2 x_{js}^2 f_s^2 \frac{1}{N} \sum_i \lambda_i e_{it} \bar{D}_{\lambda i}^{-1} l_i^2 \\ & + \sqrt{N} \frac{1}{T} \sum_s w_s^2 \left(\frac{1}{N} \Lambda' \text{diag}(X_s) M_{\alpha} u_s \right)^2 \frac{1}{N} \sum_i \lambda_i e_{it} \bar{D}_{\lambda i}^{-1} l_i^2 \\ \leq & O_P(\sqrt{N} C_{NT}^{-2}). \end{aligned}$$

Step 5: First we bound $\frac{1}{NT} \sum_i \sum_s (\tilde{f}_s - H_1^{-1} f_s)^2 e_{is}^2$. We have

$$\begin{aligned} \tilde{f}_s &= \tilde{D}_{fs}^{-1} \frac{1}{N} \tilde{\Lambda}' \text{diag}(X_s) M_{\tilde{\alpha}} y_s \\ &= H_1^{-1} f_s + \tilde{D}_{fs}^{-1} \frac{1}{N} \tilde{\Lambda}' \text{diag}(X_s) M_{\tilde{\alpha}} (A H_2 - \tilde{A}) H_2^{-1} g_s \\ &\quad + \tilde{D}_{fs}^{-1} \frac{1}{N} \tilde{\Lambda}' \text{diag}(X_s) M_{\tilde{\alpha}} \text{diag}(X_s) (\Lambda H_1 - \tilde{\Lambda}) H_1^{-1} f_s \\ &\quad + \tilde{D}_{fs}^{-1} \frac{1}{N} \tilde{\Lambda}' \text{diag}(X_s) M_{\tilde{\alpha}} u_s. \end{aligned}$$

Thus,

$$\begin{aligned} &\frac{1}{NT} \sum_i \sum_s (\tilde{f}_s - H_1^{-1} f_s)^2 e_{is}^2 \\ &= \frac{1}{NT} \sum_i \sum_{s \notin I} e_{is}^2 \frac{1}{N} \|\Lambda' \text{diag}(X_s)\|^2 [g_s^2 + \frac{1}{N} \|u_s\|^2] O_P(C_{NT}^{-2}) \\ &\quad + O_P(1) \frac{1}{N} \sum_j (H_1 \lambda_j - \tilde{\lambda}_j)^2 \max_j \frac{1}{NT} \sum_i \sum_{s \notin I} \mathbb{E}_I e_{is}^2 f_s^2 \frac{1}{N} \|\Lambda' \text{diag}(X_s)\|^2 x_{js}^2 \\ &\quad + \frac{1}{NT} \sum_i \sum_s e_{is}^2 \left(\frac{1}{N} \Lambda' \text{diag}(X_s) M_{\alpha} u_s \right)^2 \\ &\quad + \frac{1}{NT} \sum_i \sum_s e_{is}^2 \frac{1}{N} \|\text{diag}(X_s) M_{\alpha} u_s\|^2 O_P(C_{NT}^{-2}) \\ &= O_P(C_{NT}^{-2}). \end{aligned}$$

Therefore

$$\begin{aligned} &\frac{1}{\sqrt{N}} \sum_i e_{it} (H_1^{-1} \bar{D}_{\lambda i} H_1^{-1'})^{-1} \frac{1}{T} F' \text{diag}(X_i) (M_{\tilde{g}} - M_g) \text{diag}(X_i) (F H_1^{-1'} - \tilde{F}) H_1' \lambda_i \\ &\leq O_P(\sqrt{N} C_{NT}^{-1}) \left(\frac{1}{N} \sum_i \lambda_i^2 e_{it}^2 \frac{1}{T} \sum_k f_k^2 x_{ik}^2 \right)^{1/2} \left(\frac{1}{NT} \sum_i \sum_s (\tilde{f}_s - H_1^{-1} f_s)^2 e_{is}^2 \right)^{1/2} \\ &\leq O_P(\sqrt{N} C_{NT}^{-2}), \end{aligned}$$

where the last equality is due to $\frac{1}{NT} \sum_i \sum_{s \notin I} (\tilde{f}_s - H_1^{-1} f_s)^2 e_{is}^2 = O_P(C_{NT}^{-2})$.

Step 6:

$$\begin{aligned} &\frac{1}{\sqrt{N}} \sum_i e_{it} (H_1^{-1} \bar{D}_{\lambda i} H_1^{-1'})^{-1} \frac{1}{T} F' \text{diag}(X_i) M_g \text{diag}(X_i) (F H_1^{-1'} - \tilde{F}) H_1' \lambda_i \\ &\leq O_P(\sqrt{N} C_{NT}^{-1}) \left(\frac{1}{T} \sum_s f_s^2 \left(\frac{1}{N} \sum_i \lambda_i x_{is}^2 e_{it} \bar{D}_{\lambda i}^{-1} \right)^2 \right)^{1/2} \\ &\quad + O_P(\sqrt{N} C_{NT}^{-1}) \left(\frac{1}{T} \sum_s g_s^2 \left(\frac{1}{N} \sum_i x_{is} \lambda_i e_{it} \bar{D}_{\lambda i}^{-1} \frac{1}{T} \sum_k f_k x_{ik} g_k \right)^2 \right)^{1/2} \end{aligned}$$

$$= O_P(\sqrt{N}C_{NT}^{-1})(a^{1/2} + b^{1/2}).$$

We aim to show $a = O_P(C_{NT}^{-2}) = b$.

$$\begin{aligned} \mathbb{E}a &:= \frac{1}{T} \sum_s \mathbb{E}f_s^2 \left(\frac{1}{N} \sum_i \lambda_i x_{is}^2 e_{it} \bar{D}_{\lambda i}^{-1} \right)^2 \\ &\leq \frac{1}{T} \sum_s \frac{1}{N^2} \sum_{ij} \mathbb{E}f_s^2 \lambda_i \bar{D}_{\lambda i}^{-1} \lambda_j \bar{D}_{\lambda j}^{-1} \mathbb{E}(e_{is}^2 e_{js}^2 | F) \text{Cov}(e_{it}, e_{jt} | F, G, W) \\ &\quad + \frac{1}{T} \sum_s \mathbb{E}f_s^2 \frac{1}{N^2} \sum_{ij} \lambda_i l_i^2 w_s^2 \bar{D}_{\lambda i}^{-1} \lambda_j \mu_j^2 \bar{D}_{\lambda j}^{-1} \text{Cov}(e_{it}, e_{jt} | F, G, W) \\ &\quad + \frac{2}{T} \sum_s \mathbb{E}f_s^2 \frac{1}{N^2} \sum_{ij} \lambda_i l_i w_s \bar{D}_{\lambda i}^{-1} \bar{D}_{\lambda j}^{-1} \lambda_i l_j w_s \mathbb{E}(e_{js} e_{is} | F) \text{Cov}(e_{jt}, e_{it} | F, G, W) \\ &= O_P(N^{-1}). \\ \mathbb{E}b &:= \frac{1}{T} \sum_s \mathbb{E}g_s^2 \left(\frac{1}{N} \sum_i x_{is} \lambda_i e_{it} \bar{D}_{\lambda i}^{-1} \frac{1}{T} \sum_k f_k x_{ik} g_k \right)^2 \\ &= O_P(C_{NT}^{-2}). \end{aligned}$$

Therefore the second term is

$$\frac{1}{\sqrt{N}} \sum_i e_{it} \tilde{D}_{\lambda i}^{-1} \frac{1}{T} \tilde{F}' \text{diag}(X_i) M_{\tilde{g}} \text{diag}(X_i) (F H_1^{-1'} - \tilde{F}) H_1' \lambda_i = O_P(\sqrt{N}C_{NT}^{-2}).$$

Step 7: $\frac{1}{\sqrt{N}} \sum_i e_{it} \tilde{D}_{\lambda i}^{-1} \frac{1}{T} \tilde{F}' \text{diag}(X_i) M_{\tilde{g}} u_i$.

$$\begin{aligned} &\frac{1}{\sqrt{N}} \sum_i e_{it} (\tilde{D}_{\lambda i}^{-1} - (H_1^{-1} \bar{D}_{\lambda i} H_1^{-1'})^{-1}) \frac{1}{T} \tilde{F}' \text{diag}(X_i) M_{\tilde{g}} u_i \\ &\leq O_P(\sqrt{N}C_{NT}^{-2}) + O_P(1) \frac{1}{\sqrt{N}} \sum_i e_{it} (\tilde{D}_{\lambda i}^{-1} - (H_1^{-1} \bar{D}_{\lambda i} H_1^{-1'})^{-1}) \frac{1}{T} \tilde{F}' \text{diag}(X_i) M_{\tilde{g}} u_i \\ &\leq O_P(\sqrt{N}C_{NT}^{-2}) + O_P(\sqrt{N}C_{NT}^{-1}) \frac{1}{N} \sum_i |e_{it}| \left| \frac{1}{T} \sum_s f_s x_{is} u_{is} \right| \\ &\quad + O_P(\sqrt{N}C_{NT}^{-1}) \frac{1}{N} \sum_i |e_{it}| \left| \frac{1}{T} \sum_s f_s x_{is} g_s \right| \left| \frac{1}{T} \sum_k g_k u_{ik} \right| \\ &= O_P(\sqrt{N}C_{NT}^{-2}). \end{aligned}$$

Also,

$$\begin{aligned} &\frac{1}{\sqrt{N}} \sum_i e_{it} \bar{D}_{\lambda i}^{-1} \frac{1}{T} (\tilde{F} - F H_1^{-1'})' \text{diag}(X_i) M_{\tilde{g}} u_i \\ &\leq \frac{1}{\sqrt{N}} \sum_i e_{it} \bar{D}_{\lambda i}^{-1} \frac{1}{T} (\tilde{F} - F H_1^{-1'})' \text{diag}(X_i) M_{\tilde{g}} u_i \end{aligned}$$

$$\begin{aligned}
& +O_P(\sqrt{N}C_{NT}^{-1})(\frac{1}{NT} \sum_i e_{it}^2 \|u_i\|^2)^{1/2} (\frac{1}{NT} \sum_i \|(\tilde{F} - FH_1^{-1})' \text{diag}(X_i)\|^2)^{1/2} \\
& \leq O_P(\sqrt{N}C_{NT}^{-2}) + O_P(\sqrt{N}C_{NT}^{-1}) \|\frac{1}{N} \sum_i \frac{1}{\sqrt{T}} \text{diag}(X_i) M_g u_i e_{it} \bar{D}_{\lambda i}^{-1}\| \\
& \leq O_P(\sqrt{N}C_{NT}^{-2}) + O_P(\sqrt{N}C_{NT}^{-1}) (\frac{1}{T} \sum_s (\frac{1}{N} \sum_i x_{is} M'_{g,s} u_i e_{it} \bar{D}_{\lambda i}^{-1})^2)^{1/2} \\
& = O_P(\sqrt{N}C_{NT}^{-2}),
\end{aligned}$$

where the last equality is due to $\frac{1}{T} \sum_s (\frac{1}{N} \sum_i x_{is} M'_{g,s} u_i e_{it} \bar{D}_{\lambda i}^{-1})^2 = O_P(C_{NT}^{-2})$, proven as follows:

$$\begin{aligned}
& \frac{1}{T} \sum_s \mathbb{E}(\frac{1}{N} \sum_i x_{is} M'_{g,s} u_i e_{it} \bar{D}_{\lambda i}^{-1})^2 \\
& \leq \frac{1}{T} \sum_s \mathbb{E}(\frac{1}{N} \sum_i x_{is} u_{is} e_{it} \bar{D}_{\lambda i}^{-1})^2 + \frac{1}{T} \sum_s \mathbb{E}(\frac{1}{NT} \sum_i \sum_k x_{is} g_k g_s u_{ik} e_{it} \bar{D}_{\lambda i}^{-1})^2 \\
& \leq O(T^{-1}) + \frac{1}{T} \sum_{s \neq t} \frac{1}{N^2} \sum_{ij} \mathbb{E}(\mathbb{E}(e_{jt} e_{it} | F) \mathbb{E}(u_{is} u_{js} x_{js} x_{is} | F)) \\
& \quad + \frac{1}{T} \sum_s \mathbb{E} \frac{1}{NT} \sum_i \sum_k \bar{D}_{\lambda i}^{-1} \frac{1}{NT} \sum_j \sum_l \bar{D}_{\lambda j}^{-1} |x_{js} g_l g_s u_{jl} x_{is} g_k g_s u_{ik}| |\text{Cov}(e_{it}, e_{jt} | F)| \\
& \leq O(T^{-1}) + O(N^{-1})
\end{aligned}$$

where the last equality is due to $\max_j \sum_i |\text{Cov}(e_{it}, e_{jt} | F)| < C$.

Next,

$$\begin{aligned}
& \frac{1}{\sqrt{N}} \sum_i e_{it} \bar{D}_{\lambda i}^{-1} \frac{1}{T} F' \text{diag}(X_i) (M_{\tilde{g}} - M_g) u_i \\
& = \text{tr} \frac{1}{\sqrt{N}} \sum_i u_i e_{it} \bar{D}_{\lambda i}^{-1} \frac{1}{T} F' \text{diag}(X_i) (M_{\tilde{g}} - M_g) \\
& \leq O_P(\sqrt{N}C_{NT}^{-1}) \frac{1}{T} \|\frac{1}{N} \sum_i u_i e_{it} \bar{D}_{\lambda i}^{-1} F' \text{diag}(X_i)\|_F \\
& \leq O_P(\sqrt{N}C_{NT}^{-1}) (\frac{1}{T^2} \sum_{sk} \mathbb{E}(\frac{1}{N} \sum_i u_{is} e_{it} \bar{D}_{\lambda i}^{-1} f_k x_{ik})^2)^{1/2} = O_P(\sqrt{N}C_{NT}^{-2})
\end{aligned}$$

where the last equality is due to $\frac{1}{T^2} \sum_{sk} \mathbb{E}(\frac{1}{N} \sum_i u_{is} e_{it} \bar{D}_{\lambda i}^{-1} f_k x_{ik})^2 = O_P(C_{NT}^{-2})$.

$$\begin{aligned}
& \frac{1}{T^2} \sum_{sk} \mathbb{E}(\frac{1}{N} \sum_i u_{is} e_{it} \bar{D}_{\lambda i}^{-1} f_k x_{ik})^2 \\
& = \frac{1}{T^2} \sum_{sk} \mathbb{E} \frac{1}{N} \sum_i \bar{D}_{\lambda i}^{-1} \frac{1}{N} \sum_j \bar{D}_{\lambda j}^{-1} f_k^2 x_{jk} u_{is} x_{ik} u_{js} e_{jt} e_{it}
\end{aligned}$$

$$\begin{aligned}
&\leq O(T^{-1}) + \frac{1}{T^2} \sum_{sk} \frac{1}{N^2} \sum_{ij} \mathbb{E} \bar{D}_{\lambda_i}^{-1} \bar{D}_{\lambda_j}^{-1} f_k^2 \mathbb{E}(x_{jk} u_{is} x_{ik} u_{js} | F) \text{Cov}(e_{jt}, e_{it} | F) \\
&\leq O_P(C_{NT}^{-2}).
\end{aligned}$$

Step 8: Since u_{it} is conditionally serially independent given E, F ,

$$\begin{aligned}
&\mathbb{E} \left(\frac{1}{\sqrt{N}} \sum_i e_{it} \bar{D}_{\lambda_i}^{-1} \frac{1}{T} F' \text{diag}(X_i) M_g u_i \right)^2 \\
&= \mathbb{E} \frac{1}{N} \sum_{ij} \sum_s \bar{D}_{\lambda_j}^{-1} \bar{D}_{\lambda_i}^{-1} \frac{1}{T} (F' \text{diag}(X_i) M_{g,s})^2 e_{jt} e_{it} \frac{1}{T} u_{js} u_{is} \\
&\leq \frac{C}{T^2 N} \sum_{ij} \sum_s \mathbb{E} u_{js} u_{is} \mathbb{E} ((F' \text{diag}(X_i) M_{g,s})^2 e_{jt} e_{it} | U) \\
&\leq \frac{C}{T^2 N} \sum_{ij} \sum_s |\text{Cov}(u_{js}, u_{is})| = O(T^{-1}) = O_P(C_{NT}^{-2}).
\end{aligned}$$

Together, $\frac{1}{\sqrt{N}} \sum_i (H'_1 \lambda_i - \dot{\lambda}_i) e_{it} = O_P(\sqrt{T} C_{NT}^{-2})$.

(ii) The proof is the same as that of part (i), by substituting in the expansion of $H'_1 \lambda_i - \dot{\lambda}_i$, hence we ignore it for brevity. □

Lemma G.6. *For any bounded determinisitic sequence c_i ,*

$$\frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \left\| \frac{1}{T} \sum_{s \notin I} f_s w'_s e_{is} c'_i (\tilde{f}_s - H_1^{-1} f_s) \right\|^2 = O_P(C_{NT}^{-4}).$$

Proof. For $y_s = A g_s + \text{diag}(X_s) \Lambda f_s + u_s$,

$$\begin{aligned}
\tilde{f}_s &= \tilde{D}_{f_s}^{-1} \frac{1}{N} \tilde{\Lambda}' \text{diag}(X_s) M_{\tilde{\alpha}} y_s \\
&= H_1^{-1} f_s + \tilde{D}_{f_s}^{-1} \frac{1}{N} \tilde{\Lambda}' \text{diag}(X_s) M_{\tilde{\alpha}} (A H_2 - \tilde{A}) H_2^{-1} g_s \\
&\quad + \tilde{D}_{f_s}^{-1} \frac{1}{N} \tilde{\Lambda}' \text{diag}(X_s) M_{\tilde{\alpha}} \text{diag}(X_s) (\Lambda H_1 - \tilde{\Lambda}) H_1^{-1} f_s \\
&\quad + \tilde{D}_{f_s}^{-1} \frac{1}{N} \tilde{\Lambda}' \text{diag}(X_s) M_{\tilde{\alpha}} u_s.
\end{aligned} \tag{G.7}$$

Without loss of generality, we assume $\dim(g_s) = \dim(f_s) = 1$, as we can always work with their elements given that the number of factors is fixed. It suffices to prove, for $h_s := f_s w_s$,

$$\frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \left\| \frac{1}{T} \sum_{s \notin I} h_s e_{is} (\tilde{f}_s - H_1^{-1} f_s) \right\|^2 = O_P(C_{NT}^{-4}).$$

We plug in each term in the expansion of $\tilde{f}_s$:

$$\begin{aligned} & \frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \left\| \frac{1}{T} \sum_{s \notin I} h_s e_{is} \tilde{D}_{fs}^{-1} \frac{1}{N} \tilde{\Lambda}' \text{diag}(X_s) M_{\tilde{\alpha}} (A H_2 - \tilde{A}) H_2^{-1} g_s \right\|^2 \\ & + \frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \left\| \frac{1}{T} \sum_{s \notin I} h_s e_{is} \tilde{D}_{fs}^{-1} \frac{1}{N} \tilde{\Lambda}' \text{diag}(X_s) M_{\tilde{\alpha}} \text{diag}(X_s) (\Lambda H_1 - \tilde{\Lambda}) H_1^{-1} f_s \right\|^2 \\ & + \frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \left\| \frac{1}{T} \sum_{s \notin I} h_s e_{is} \tilde{D}_{fs}^{-1} \frac{1}{N} \tilde{\Lambda}' \text{diag}(X_s) M_{\tilde{\alpha}} u_s \right\|^2. \end{aligned}$$

First term: $\frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \left\| \frac{1}{T} \sum_{s \notin I} h_s e_{is} \tilde{D}_{fs}^{-1} \frac{1}{N} \tilde{\Lambda}' \text{diag}(X_s) M_{\tilde{\alpha}} (A H_2 - \tilde{A}) H_2^{-1} g_s \right\|^2$. By Lemma G.3,

$$\begin{aligned} & \frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \left\| \frac{1}{T} \sum_{s \notin I} h_s e_{is} (\tilde{D}_{fs}^{-1} - (H_1' \bar{D}_{fs} H_1)^{-1}) \frac{1}{N} \tilde{\Lambda}' \text{diag}(X_s) M_{\tilde{\alpha}} (A H_2 - \tilde{A}) H_2^{-1} g_s \right\|^2 \\ & \leq O_P(C_{NT}^{-4}) \frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \frac{1}{T} \sum_s \left\| \frac{1}{\sqrt{N}} \tilde{\Lambda}' \text{diag}(X_s) \right\|^2 \|g_s\|^2 \|h_s e_{is}\|^2 \\ & \leq O_P(C_{NT}^{-4}) \frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \frac{1}{T} \sum_s \left\| \frac{1}{\sqrt{N}} \tilde{\Lambda}' \text{diag}(X_s) \right\|^2 \|g_s\|^2 \|h_s e_{is}\|^2 = O_P(C_{NT}^{-4}). \\ & \frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \left\| \frac{1}{T} \sum_{s \notin I} h_s e_{is} (H_1' \bar{D}_{fs} H_1)^{-1} \frac{1}{N} \tilde{\Lambda}' \text{diag}(X_s) M_{\tilde{\alpha}} (A H_2 - \tilde{A}) H_2^{-1} g_s \right\|^2 \\ & \leq O_P(C_{NT}^{-2}) \frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \left\| \frac{1}{T \sqrt{N}} \sum_{s \notin I} h_s g_s e_{is} (H_1' \bar{D}_{fs} H_1)^{-1} \tilde{\Lambda}' \text{diag}(X_s) \right\|^2 \\ & \leq O_P(C_{NT}^{-2}) \frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \frac{1}{N} \sum_j \tilde{\lambda}_j^2 \left[\left(\frac{1}{T} \sum_{s \notin I} \bar{D}_{fs}^{-1} h_s g_s e_{is} e_{js} \right)^2 + \left(\frac{1}{T} \sum_{s \notin I} \bar{D}_{fs}^{-1} h_s g_s e_{is} w_s \right)^2 \right]. \end{aligned}$$

Because e_{is} is serially independent conditionally on (W, F, G) , the above is

$$\begin{aligned} & \leq O_P(C_{NT}^{-2}) \frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \left[\mathbb{E}_I \left(\frac{1}{T} \sum_{s \notin I} \bar{D}_{fs}^{-1} h_s g_s e_{is} e_{js} \right)^2 + \mathbb{E}_I \left(\frac{1}{T} \sum_{s \notin I} \bar{D}_{fs}^{-1} h_s g_s e_{is} w_s \right)^2 \right] \\ & = O_P(C_{NT}^{-4}). \end{aligned}$$

Put together, the first term is $O_P(C_{NT}^{-4})$.

Second term: $\frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \left\| \frac{1}{T} \sum_{s \notin I} h_s e_{is} \tilde{D}_{fs}^{-1} \frac{1}{N} \tilde{\Lambda}' \text{diag}(X_s) M_{\tilde{\alpha}} \text{diag}(X_s) (\Lambda H_1 - \tilde{\Lambda}) H_1^{-1} f_s \right\|^2$.

$$\begin{aligned} & \frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \left\| \frac{1}{T} \sum_{s \notin I} h_s e_{is} (\tilde{D}_{fs}^{-1} - (H_1' \bar{D}_{fs} H_1)^{-1}) \frac{1}{N} \tilde{\Lambda}' \text{diag}(X_s) M_{\tilde{\alpha}} \text{diag}(X_s) (\Lambda H_1 - \tilde{\Lambda}) H_1^{-1} f_s \right\|^2 \\ & \leq O_P(C_{NT}^{-2}) \frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \frac{1}{T} \sum_s \left\| \frac{1}{\sqrt{N}} \tilde{\Lambda}' \text{diag}(X_s) M_{\tilde{\alpha}} \text{diag}(X_s) \right\|^2 \|h_s\|^2 \|f_s e_{is}\|^2 = O_P(C_{NT}^{-2}). \end{aligned}$$

The same proof leads to

$$\begin{aligned} & \frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \left\| \frac{1}{T} \sum_{s \notin I} h_s e_{is} (H_1' \bar{D}_{fs} H_1)^{-1} \frac{1}{N} \tilde{\Lambda}' \text{diag}(X_s) (M_{\tilde{\alpha}} - M_{\alpha}) \text{diag}(X_s) (\Lambda H_1 - \tilde{\Lambda}) H_1^{-1} f_s \right\|^2 \\ & \leq O_P(C_{NT}^{-2}). \end{aligned}$$

Now let $M_{\tilde{\alpha},ij}$ be the (i, j) th component of $M_{\tilde{\alpha}}$. Let $z_{js} = \sum_k \tilde{\lambda}_k M_{\alpha,kj} x_{ks}$,

$$\begin{aligned} & \frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \left\| \frac{1}{T} \sum_{s \notin I} h_s e_{is} (H_1' \bar{D}_{fs} H_1)^{-1} \frac{1}{N} \tilde{\Lambda}' \text{diag}(X_s) M_{\alpha} \text{diag}(X_s) (\Lambda H_1 - \tilde{\Lambda}) H_1^{-1} f_s \right\|^2 \\ & \leq O_P(C_{NT}^{-2}) \frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \left\| \frac{1}{T} \sum_s f_s h_s e_{is} \bar{D}_{fs}^{-1} \frac{1}{\sqrt{N}} \tilde{\Lambda}' \text{diag}(X_s) M_{\alpha} \text{diag}(X_s) \right\|^2 \\ & \leq O_P(C_{NT}^{-2}) \frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \frac{1}{N} \sum_j \left(\frac{1}{T} \sum_s f_s h_s \bar{D}_{fs}^{-1} e_{is} z_{js} x_{js} \right)^2 \\ & \leq O_P(C_{NT}^{-2}) \frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \frac{1}{N} \sum_j \left(\frac{1}{T} \sum_s \mathbb{E}_I(f_s h_s \bar{D}_{fs}^{-1} e_{is} x_{js}^2) \right)^2 (\tilde{\lambda}_j M_{\alpha,jj})^2 \\ & \quad + O_P(C_{NT}^{-2}) \frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \frac{1}{N} \sum_j \frac{1}{N} \sum_{k \neq j} \left(\frac{1}{T} \sum_s \mathbb{E}_I f_s h_s \bar{D}_{fs}^{-1} x_{ks} e_{is} x_{js} \right)^2 \\ & \quad + O_P(C_{NT}^{-2}) \frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \frac{1}{N} \sum_j \text{Var}_I \left(\frac{1}{T} \sum_s f_s h_s \bar{D}_{fs}^{-1} e_{is} z_{js} x_{js} \right) \\ & \leq O_P(C_{NT}^{-2}) \frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \frac{1}{N} \sum_j \left(\frac{1}{T} \sum_s \mathbb{E}_I(f_s h_s \bar{D}_{fs}^{-1} e_{is} x_{js}^2) \right)^2 (\tilde{\lambda}_j - H_1' \lambda_j)^2 \\ & \quad + O_P(C_{NT}^{-2}) \frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \frac{1}{N} \sum_j \left(\frac{1}{T} \sum_s \mathbb{E}_I f_s h_s \bar{D}_{fs}^{-1} e_{is} e_{js}^2 \right)^2 \\ & \quad + O_P(C_{NT}^{-2}) \frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \frac{1}{N} \sum_j \left(\frac{1}{T} \sum_s \mathbb{E}_I f_s h_s \bar{D}_{fs}^{-1} e_{is} l_j w_s e_{js} \right)^2 \\ & \quad + O_P(C_{NT}^{-2}) \frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \frac{1}{N} \sum_j \frac{1}{N} \sum_{k \neq j} \left(\frac{1}{T} \sum_s \mathbb{E}_I f_s h_s \bar{D}_{fs}^{-1} e_{ks} e_{is} e_{js} \right)^2 \\ & \quad + O_P(C_{NT}^{-2}) \frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \frac{1}{N} \sum_j \frac{1}{N} \sum_{k \neq j} \left(\frac{1}{T} \sum_s \mathbb{E}_I f_s h_s \bar{D}_{fs}^{-1} e_{ks} e_{is} l_j w_s \right)^2 \\ & \quad + O_P(C_{NT}^{-2}) \frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \frac{1}{N} \sum_j \frac{1}{N} \sum_{k \neq j} \left(\frac{1}{T} \sum_s \mathbb{E}_I f_s h_s \bar{D}_{fs}^{-1} l_k w_s e_{is} e_{js} \right)^2 + O_P(C_{NT}^{-4}) = O_P(C_{NT}^{-4}) \end{aligned}$$

given that $\sum_j |\mathbb{E}_I(e_{is} e_{js} | f_s, w_s)| + \frac{1}{N} \sum_{k \neq j} |\mathbb{E}_I(e_{ks} e_{is} e_{js} | f_s, w_s)| < \infty$. Put together, the second term is $O_P(C_{NT}^{-4})$.

Third term: $\frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \left\| \frac{1}{T} \sum_{s \notin I} h_s e_{is} \tilde{D}_{fs}^{-1} \frac{1}{N} \tilde{\Lambda}' \text{diag}(X_s) M_{\tilde{\alpha}} u_s \right\|^2$.

$$\begin{aligned}
& \frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \left\| \frac{1}{T} \sum_{s \notin I} h_s e_{is} (\tilde{D}_{fs}^{-1} - (H_1' \bar{D}_{fs} H_1)^{-1}) \frac{1}{N} \tilde{\Lambda}' \text{diag}(X_s) M_{\tilde{\alpha}} u_s \right\|^2 \\
& \leq O_P(C_{NT}^{-2}) \frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \frac{1}{T} \sum_{s \notin I} |h_s e_{is} \frac{1}{N} \tilde{\Lambda}' \text{diag}(X_s) M_{\tilde{\alpha}} u_s|^2 \\
& \leq O_P(C_{NT}^{-2}) \frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \frac{1}{T} \sum_{s \notin I} \mathbb{E}_I |h_s e_{is} \frac{1}{N} \tilde{\Lambda}' \text{diag}(X_s) M_{\tilde{\alpha}} u_s|^2 \\
& = O_P(C_{NT}^{-2}) \frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \frac{1}{T} \sum_{s \notin I} \mathbb{E}_I h_s^2 e_{is}^2 \frac{1}{N^2} \tilde{\Lambda}' \text{diag}(X_s) M_{\tilde{\alpha}} \mathbb{E}_I(u_s u_s' | X_s, f_s) M_{\tilde{\alpha}} \text{diag}(X_s) \tilde{\Lambda} \\
& \leq O_P(C_{NT}^{-2} N^{-1}) \frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \frac{1}{T} \sum_{s \notin I} \mathbb{E}_I h_s^2 e_{is}^2 \frac{1}{N} \|\tilde{\Lambda}' \text{diag}(X_s)\|^2 \\
& = O_P(C_{NT}^{-2} N^{-1}) = O_P(C_{NT}^{-4}).
\end{aligned}$$

Next, due to $\mathbb{E}(u_s | f_s, D_I, e_s) = 0$, and e_s being conditionally serially independent given (f_s, u_s) ,

$$\begin{aligned}
& \frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \left\| \frac{1}{T} \sum_{s \notin I} h_s e_{is} (H_1' \bar{D}_{fs} H_1)^{-1} \frac{1}{N} \tilde{\Lambda}' \text{diag}(X_s) (M_{\tilde{\alpha}} - M_{\alpha}) u_s \right\|^2 \\
& = \frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} [\text{tr} \frac{1}{T} \sum_{s \notin I} u_s h_s e_{is} (H_1' \bar{D}_{fs} H_1)^{-1} \frac{1}{N} \tilde{\Lambda}' \text{diag}(X_s) (M_{\tilde{\alpha}} - M_{\alpha})]^2 \\
& \leq O_P(C_{NT}^{-2}) \frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \left\| \frac{1}{T} \sum_{s \notin I} u_s h_s e_{is} (H_1' \bar{D}_{fs}^{-1} H_1)^{-1} \frac{1}{N} \tilde{\Lambda}' \text{diag}(X_s) \right\|_F^2 \\
& \leq O_P(C_{NT}^{-2}) \frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \frac{1}{N} \sum_j \mathbb{E}_I \left\| \frac{1}{T} \sum_{s \notin I} \bar{D}_{fs}^{-1} u_s h_s e_{is} \frac{1}{\sqrt{N}} x_{js} \right\|_F^2 \tilde{\lambda}_j^2 \\
& \leq O_P(C_{NT}^{-2}) \max_{ij} \mathbb{E}_I \mathbb{E}_I \left(\left\| \frac{1}{T} \sum_{s \notin I} \bar{D}_{fs}^{-1} u_s h_s e_{is} \frac{1}{\sqrt{N}} x_{js} \right\|_F^2 | X, F, W \right) \\
& \leq O_P(C_{NT}^{-2}) \max_{ijk} \frac{1}{T} \text{Var}_I(\bar{D}_{fs}^{-1} u_{ks} h_s e_{is} x_{js}) \leq O_P(C_{NT}^{-2} T^{-1}) = O_P(C_{NT}^{-4}).
\end{aligned}$$

Next,

$$\begin{aligned}
& \frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \left\| \frac{1}{T} \sum_{s \notin I} h_s e_{is} (H_1' \bar{D}_{fs} H_1)^{-1} \frac{1}{N} (\tilde{\Lambda} - \Lambda H_1)' \text{diag}(X_s) M_{\alpha} u_s \right\|^2 \\
& \leq O_P(1) \frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \left\| \frac{1}{N} \sum_j (\tilde{\lambda}_j - H_1' \lambda_j) \frac{1}{T} \sum_{s \notin I} \bar{D}_{fs}^{-1} h_s e_{is} M'_{\alpha, j} u_s x_{js} \right\|^2 \\
& \leq O_P(C_{NT}^{-2}) \frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \frac{1}{N} \sum_j \left(\frac{1}{T} \sum_{s \notin I} \bar{D}_{fs}^{-1} h_s e_{is} M'_{\alpha, j} u_s x_{js} \right)^2
\end{aligned}$$

$$\begin{aligned}
&\leq O_P(C_{NT}^{-2}T^{-1}) \frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \frac{1}{N} \sum_j \frac{1}{T} \sum_{s \notin I} \mathbf{E}_I \bar{D}_{fs}^{-2} h_s^2 x_{js}^2 e_{is}^2 M'_{\alpha,j} \mathbf{Var}_I(u_s | e_s, w_s, f_s) M_{\alpha,j} \\
&\leq O_P(C_{NT}^{-2}T^{-1}) \frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \frac{1}{N} \sum_j \frac{1}{T} \sum_{s \notin I} \mathbf{E}_I \bar{D}_{fs}^{-2} h_s^2 x_{js}^2 e_{is}^2 \|M_{\alpha,j}\|^2 \\
&= O_P(C_{NT}^{-2}T^{-1}) = O_P(C_{NT}^{-4}).
\end{aligned}$$

Finally, due to the conditional serial independence,

$$\begin{aligned}
&\frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \left\| \frac{1}{T} \sum_{s \notin I} h_s e_{is} (H'_1 \bar{D}_{fs} H_1)^{-1} \frac{1}{N} H'_1 \Lambda' \text{diag}(X_s) M_\alpha u_s \right\|^2 \\
&\leq O_P(N^{-2}T^{-2}) \sum_{s \notin I} \mathbf{E}_I \bar{D}_{fs}^{-2} h_s^2 e_{is}^2 \Lambda' \text{diag}(X_s) M_\alpha \mathbf{E}_I(u_s u'_s | E, G, F) M_\alpha \text{diag}(X_s) \Lambda \\
&\leq O_P(N^{-2}T^{-1}) \mathbf{E}_I \bar{D}_{fs}^{-2} h_s^2 e_{is}^2 \|\Lambda' \text{diag}(X_s)\|^2 = O_P(N^{-1}T^{-1}) = O_P(C_{NT}^{-4}).
\end{aligned}$$

Put together, the third term is $O_P(C_{NT}^{-4})$. This finishes the proof. $\square$

G.3. Technical lemmas for $\hat{f}_t$.

Lemma G.7. Assume $\frac{1}{N} \sum_{ij} |\text{Cov}(e_{it}^2, e_{jt}^2)| < C$. For each fixed t ,

(i) $\hat{B}_t - B = O_P(C_{NT}^{-1})$.

(ii) The upper two blocks of $\hat{B}_t^{-1} \hat{S}_t - B^{-1} S$ are both $O_P(C_{NT}^{-2})$.

Proof. Throughout the proof, we assume $\dim(\alpha_i) = \dim(\lambda_i) = 1$ without loss of generality.

(i) $\hat{B}_t - B = b_1 + b_2$, where

$$\begin{aligned}
b_1 &= \frac{1}{N} \sum_i \begin{pmatrix} \tilde{\lambda}_i \tilde{\lambda}'_i \hat{e}_{it}^2 - H'_1 \lambda_i \lambda'_i H_1 e_{it}^2 & \tilde{\lambda}_i \tilde{\alpha}'_i \hat{e}_{it} - H'_1 \lambda_i \alpha_i H'_2 e_{it} \\ \tilde{\alpha}_i \tilde{\lambda}'_i \hat{e}_{it} - H'_2 \alpha_i \lambda_i H'_1 e_{it} & \tilde{\alpha}_i \tilde{\alpha}'_i - H'_2 \alpha_i \alpha'_i H_2 \end{pmatrix} \\
b_2 &= \frac{1}{N} \sum_i \begin{pmatrix} H'_1 \lambda_i \lambda'_i H_1 (e_{it}^2 - \mathbf{E} e_{it}^2) & H'_1 \lambda_i \alpha_i H'_2 e_{it} \\ H'_2 \alpha_i \lambda_i H'_1 e_{it} & 0 \end{pmatrix}.
\end{aligned}$$

To prove the convergence of b_1 , first note that

$$\begin{aligned}
&\frac{1}{N} \sum_i \tilde{\lambda}_i \tilde{\lambda}'_i (\hat{e}_{it}^2 - e_{it}^2) = \frac{1}{N} \sum_i \tilde{\lambda}_i \tilde{\lambda}'_i (\hat{e}_{it} - e_{it})^2 + \frac{2}{N} \sum_i \tilde{\lambda}_i \tilde{\lambda}'_i (\hat{e}_{it} - e_{it}) e_{it} \\
&\leq \frac{1}{N} \sum_i (\tilde{\lambda}_i - H'_1 \lambda_i) (\tilde{\lambda}_i - H'_1 \lambda_i)' (\hat{e}_{it} - e_{it})^2 + \frac{1}{N} \sum_i (\tilde{\lambda}_i - H'_1 \lambda_i) \lambda'_i H'_1 (\hat{e}_{it} - e_{it})^2 \\
&\quad + O_P(1) \frac{1}{N} \sum_i (\hat{e}_{it} - e_{it})^2 + \frac{2}{N} \sum_i (\tilde{\lambda}_i - H'_1 \lambda_i) (\tilde{\lambda}_i - H'_1 \lambda_i)' (\hat{e}_{it} - e_{it}) e_{it} \\
&\quad + \frac{2}{N} \sum_i (\tilde{\lambda}_i - H'_1 \lambda_i) \lambda'_i H'_1 (\hat{e}_{it} - e_{it}) e_{it} + O_P(1) \frac{1}{N} \sum_i \lambda_i \lambda'_i (\hat{e}_{it} - e_{it}) e_{it}
\end{aligned}$$

$$\begin{aligned}
&\leq O_P(C_{NT}^{-2}) \max_{it} |\widehat{e}_{it} - e_{it}| \max_{it} |e_{it}| + O_P(C_{NT}^{-1}) \left(\frac{1}{N} \sum_i (\widehat{e}_{it} - e_{it})^4 \right)^{1/2} \\
&\quad + O_P(1) \frac{1}{N} \sum_i (\widehat{e}_{it} - e_{it})^2 + O_P(C_{NT}^{-1}) \left(\frac{1}{N} \sum_i (\widehat{e}_{it} - e_{it})^2 e_{it}^2 \right)^{1/2} \\
&\quad + O_P(1) \frac{1}{N} \sum_i \lambda_i \lambda_i' (\widehat{e}_{it} - e_{it}) e_{it} = O_P(C_{NT}^{-2}), \tag{G.8}
\end{aligned}$$

by Lemma G.1. In addition, still by Lemma G.1,

$$\begin{aligned}
&\frac{1}{N} \sum_i \widetilde{\lambda}_i \widetilde{\alpha}_i' (\widehat{e}_{it} - e_{it}) = \frac{1}{N} \sum_i (\widetilde{\lambda}_i - H_1' \lambda_i) (\widetilde{\alpha}_i - H_2' \alpha_i)' (\widehat{e}_{it} - e_{it}) \\
&\quad + \frac{1}{N} \sum_i (\widetilde{\lambda}_i - H_1' \lambda_i) \alpha_i' H_2 (\widehat{e}_{it} - e_{it}) + O_P(1) \frac{1}{N} \sum_i \lambda_i \alpha_i' (\widehat{e}_{it} - e_{it}) \\
&\leq O_P(C_{NT}^{-2}) \max_{it} |\widehat{e}_{it} - e_{it}| + O_P(C_{NT}^{-1}) \left(\frac{1}{N} \sum_i (\widehat{e}_{it} - e_{it})^2 \right)^{1/2} \\
&\quad + O_P(1) \frac{1}{N} \sum_i \lambda_i \alpha_i' (\widehat{e}_{it} - e_{it}) = O_P(C_{NT}^{-2}). \tag{G.9}
\end{aligned}$$

Then the first term of b_1 is, due to the serial independence of e_{it}^2 ,

$$\begin{aligned}
&\frac{1}{N} \sum_i \widetilde{\lambda}_i \widetilde{\lambda}_i' \widehat{e}_{it}^2 - H_1' \lambda_i \lambda_i' H_1 e_{it}^2 \\
&\leq \frac{1}{N} \sum_i \widetilde{\lambda}_i \widetilde{\lambda}_i' (\widehat{e}_{it}^2 - e_{it}^2) + \frac{1}{N} \sum_i (\widetilde{\lambda}_i \widetilde{\lambda}_i' - H_1' \lambda_i \lambda_i' H_1) e_{it}^2 \\
&\leq O_P(C_{NT}^{-2}) + O_P(1) \frac{1}{N} \sum_i \|\widetilde{\lambda}_i \widetilde{\lambda}_i' - H_1' \lambda_i \lambda_i' H_1\|_F \mathbf{E}_I e_{it}^2 \\
&\leq O_P(C_{NT}^{-2}) + O_P(1) \frac{1}{N} \sum_i \|\widetilde{\lambda}_i \widetilde{\lambda}_i' - H_1' \lambda_i \lambda_i' H_1\|_F = O_P(C_{NT}^{-1}).
\end{aligned}$$

The second term is

$$\begin{aligned}
&\frac{1}{N} \sum_i \widetilde{\lambda}_i \widetilde{\alpha}_i' \widehat{e}_{it} - H_1' \lambda_i \alpha_i H_2' e_{it} \\
&\leq O_P(C_{NT}^{-1}) + O_P(1) \frac{1}{N} \sum_i \|\widetilde{\lambda}_i \widetilde{\alpha}_i' - H_1' \lambda_i \alpha_i H_2'\|_F \mathbf{E}_I |e_{it}| \leq O_P(C_{NT}^{-1}).
\end{aligned}$$

The third term of b_1 is bounded similarly. The last term of b_1 is easy to show to be $O_P(C_{NT}^{-1})$. As for b_2 , $\frac{1}{N} \sum_{ij} |\text{Cov}(e_{it}^2, e_{jt}^2)| < C$ by assumption, thus $b_2 = O_P(N^{-1/2})$. Hence $\widehat{B}_t - B = O_P(C_{NT}^{-1})$.

(ii) We have

$$B_t^{-1} \widehat{S}_t - B^{-1} S = (B_t^{-1} - B^{-1})(\widehat{S}_t - S) + (B_t^{-1} - B^{-1})S + B^{-1}(\widehat{S}_t - S).$$

We first bound the four blocks of $\widehat{S}_t - S$. We have $\widehat{S}_t - S = c_t + d_t$,

$$\begin{aligned} c_t &= \frac{1}{N} \sum_i \begin{pmatrix} \widetilde{\lambda}_i \lambda'_i H_1 e_{it} (\widehat{e}_{it} - e_{it}) + \widetilde{\lambda}_i \widetilde{\lambda}'_i (\widehat{e}_{it}^2 - e_{it}^2) & \widetilde{\lambda}_i (\widehat{e}_{it} - e_{it}) (\alpha'_i H_2 - \widetilde{\alpha}'_i) \\ \widetilde{\alpha}_i \lambda'_i H_1 (e_{it} - \widehat{e}_{it}) + \widetilde{\alpha}_i (\lambda'_i H_1 - \widetilde{\lambda}'_i) (\widehat{e}_{it} - e_{it}) & 0 \end{pmatrix} \\ d_t &= \frac{1}{N} \sum_i \begin{pmatrix} (\widetilde{\lambda}_i \lambda'_i H_1 - \widetilde{\lambda}_i \widetilde{\lambda}'_i) e_{it}^2 - H_1' \lambda_i (\lambda'_i H_1 - \widetilde{\lambda}'_i) \mathbb{E} e_{it}^2 & \widetilde{\lambda}_i e_{it} (\alpha'_i H_2 - \widetilde{\alpha}'_i) \\ \widetilde{\alpha}_i (\lambda'_i H_1 - \widetilde{\lambda}'_i) e_{it} & 0 \end{pmatrix}. \end{aligned}$$

Call each block of c_t to be $c_{t,1}, \dots, c_{t,4}$ in clockwise order. Note that $c_{t,4} = 0$.

As for $c_{t,1}$, it follows from Lemma G.1 that

$$\begin{aligned} \frac{1}{N} \sum_i \widetilde{\lambda}_i \lambda'_i H_1 e_{it} (\widehat{e}_{it} - e_{it}) &\leq \left(\frac{1}{N} \sum_i e_{it}^2 (\widehat{e}_{it} - e_{it})^2 \right)^{1/2} O_P(C_{NT}^{-1}) \\ &+ O_P(1) \frac{1}{N} \sum_i \lambda_i \lambda_i e_{it} (\widehat{e}_{it} - e_{it}) \leq O_P(C_{NT}^{-2}). \end{aligned}$$

We have also shown $\frac{1}{N} \sum_i \widetilde{\lambda}_i \widetilde{\lambda}'_i (\widehat{e}_{it}^2 - e_{it}^2) = O_P(C_{NT}^{-2})$. Thus $c_{t,1} = O_P(C_{NT}^{-2})$.

For $c_{t,2}$, from Lemma G.1,

$$\begin{aligned} &\frac{1}{N} \sum_i \widetilde{\lambda}_i (\widehat{e}_{it} - e_{it}) (\alpha'_i H_2 - \widetilde{\alpha}'_i) \\ &\leq O_P(C_{NT}^{-2}) \max_i |e_{it} - \widehat{e}_{it}| + O_P(C_{NT}^{-1}) \left(\frac{1}{N} \sum_i (\widehat{e}_{it} - e_{it})^2 \right)^{1/2} = O_P(C_{NT}^{-2}). \end{aligned}$$

For the third term of c_t , similarly, $\frac{1}{N} \sum_i \widetilde{\alpha}_i (\lambda'_i H_1 - \widetilde{\lambda}'_i) (\widehat{e}_{it} - e_{it}) = O_P(C_{NT}^{-2})$. Also, by Lemma G.1,

$$\begin{aligned} &\frac{1}{N} \sum_i \widetilde{\alpha}_i \lambda'_i H_1 (e_{it} - \widehat{e}_{it}) \\ &\leq O_P(C_{NT}^{-1}) \left(\frac{1}{N} \sum_i (\widehat{e}_{it} - e_{it})^2 \right)^{1/2} + O_P(1) \frac{1}{N} \sum_i \lambda_i \alpha'_i (\widehat{e}_{it} - e_{it}) \\ &\leq O_P(C_{NT}^{-1}). \end{aligned}$$

So $c_{t,3} = O_P(C_{NT}^{-1})$.

As for d_t , we first prove that $\frac{1}{N} \sum_i \lambda_i (\lambda'_i H_1 - \widetilde{\lambda}'_i) (e_{it}^2 - \mathbb{E} e_{it}^2) = O_P(C_{NT}^{-1} N^{-1/2})$. Note that $\mathbb{E}_I \frac{1}{N} \sum_i \lambda_i (\lambda'_i H_1 - \widetilde{\lambda}'_i) (e_{it}^2 - \mathbb{E} e_{it}^2) = 0$. Let Υ_t be an $N \times 1$ vector of e_{it}^2 , and $\text{diag}(\Lambda)$ be diagonal matrix consisting of elements of Λ . Then

$$\|\text{Var}(\Upsilon_t)\| \leq \max_i \sum_j |\text{Cov}(e_{it}^2, e_{jt}^2)| < C,$$

and $\frac{1}{N} \sum_i \lambda_i (\lambda'_i H_1 - \tilde{\lambda}'_i) e_{it}^2 = \frac{1}{N} (H_1 \Lambda - \tilde{\Lambda})' \text{diag}(\Lambda) \Upsilon_t$. So

$$\begin{aligned} \text{Var}_I \left(\frac{1}{N} \sum_i \lambda_i (\lambda'_i H_1 - \tilde{\lambda}'_i) e_{it}^2 \right) &= \frac{1}{N^2} (H_1 \Lambda - \tilde{\Lambda})' \text{diag}(\Lambda) \text{Var}(\Upsilon_t) \text{diag}(\Lambda) (H_1 \Lambda - \tilde{\Lambda}) \\ &\leq C \frac{1}{N^2} \|H_1 \Lambda - \tilde{\Lambda}\|_F^2 = O_P(C_{NT}^{-2} N^{-1}). \end{aligned}$$

This implies $\frac{1}{N} \sum_i \lambda_i (\lambda'_i H_1 - \tilde{\lambda}'_i) (e_{it}^2 - \mathbb{E} e_{it}^2) = O_P(C_{NT}^{-1} N^{-1/2})$.

Thus the first term of d_t is

$$\begin{aligned} &\frac{1}{N} \sum_i (\tilde{\lambda}_i \lambda'_i H_1 - \tilde{\lambda}_i \tilde{\lambda}'_i) e_{it}^2 - H'_1 \lambda_i (\lambda'_i H_1 - \tilde{\lambda}'_i) \mathbb{E} e_{it}^2 \\ &\leq \frac{1}{N} \sum_i (\tilde{\lambda}_i - H'_1 \lambda_i) (\lambda'_i H_1 - \tilde{\lambda}'_i) e_{it}^2 + \frac{1}{N} \sum_i H'_1 \lambda_i (\lambda'_i H_1 - \tilde{\lambda}'_i) (e_{it}^2 - \mathbb{E} e_{it}^2) \\ &= O_P(1) \frac{1}{N} \sum_i (\tilde{\lambda}_i - H'_1 \lambda_i) (\lambda'_i H_1 - \tilde{\lambda}'_i) \mathbb{E} e_{it}^2 + O_P(C_{NT}^{-1} N^{-1/2}) = O_P(C_{NT}^{-2}). \end{aligned}$$

As for the second term of d_t , $\frac{1}{N} \sum_i \tilde{\lambda}_i e_{it} (\alpha'_i H_2 - \tilde{\alpha}'_i)$, note that

$$\begin{aligned} &\frac{1}{N} \sum_i (\tilde{\lambda}_i - H'_1 \lambda_i) e_{it} (\alpha'_i H_2 - \tilde{\alpha}'_i) = O_P(C_{NT}^{-1}) \left(\frac{1}{N} \sum_i e_{it}^2 (\alpha'_i H_2 - \tilde{\alpha}'_i)^2 \right)^{1/2} \\ &\leq O_P(C_{NT}^{-1}) \left(\frac{1}{N} \sum_i (\alpha'_i H_2 - \tilde{\alpha}'_i)^2 \mathbb{E} e_{it}^2 \right)^{1/2} = O_P(C_{NT}^{-2}). \end{aligned}$$

And, $\mathbb{E}_I \frac{1}{N} \sum_i \tilde{\lambda}_i e_{it} (\alpha'_i H_2 - \tilde{\alpha}'_i) = 0$.

$$\begin{aligned} &\text{Var}_I \left(\frac{1}{N} \sum_i \lambda_i e_{it} (\alpha'_i H_2 - \tilde{\alpha}'_i) \right) = \frac{1}{N^2} \text{Var}_I ((AH_2 - \tilde{A})' \text{diag}(\Lambda) e_t) \\ &\leq \frac{1}{N^2} (AH_2 - \tilde{A})' \text{diag}(\Lambda) \text{Var}(e_t) \text{diag}(\Lambda) (AH_2 - \tilde{A}) = O_P(C_{NT}^{-2} N^{-1}), \end{aligned}$$

implying $\frac{1}{N} \sum_i \tilde{\lambda}_i e_{it} (\alpha'_i H_2 - \tilde{\alpha}'_i) = O_P(C_{NT}^{-2})$.

Finally the third term of d_t , $\frac{1}{N} \sum_i \tilde{\lambda}_i e_{it} (\alpha'_i H_2 - \tilde{\alpha}'_i)$, is bounded similarly. So $d_t = O_P(C_{NT}^{-2})$. Put together, we have: $\hat{S}_t - S = c_t + d_t = O_P(C_{NT}^{-1})$. The upper two blocks of $\hat{S}_t - S$ are $O_P(C_{NT}^{-2})$, determined by $c_{t,1}, c_{t,2}$ and the upper blocks of d_t . In addition, note that both B, S are block diagonal matrices, and the diagonal blocks of S are $O_P(C_{NT}^{-1})$. Due to

$$\hat{B}_t^{-1} \hat{S}_t - B^{-1} S = (\hat{B}_t^{-1} - B^{-1}) (\hat{S}_t - S) + (\hat{B}_t^{-1} - B^{-1}) S + B^{-1} (\hat{S}_t - S),$$

we have the upper blocks of $\hat{B}_t^{-1} \hat{S}_t - B^{-1} S$ are both $O_P(C_{NT}^{-2})$.

□

Lemma G.8. Suppose $\max_{t \leq T} |\frac{1}{N} \sum_i \alpha_i \lambda_i e_{it}| = O_P(1)$,
 $C_{NT}^{-1} \max_{it} |e_{it}|^2 + \max_t \|\frac{1}{N} \sum_i \lambda_i \lambda_i' \bar{e}_{it} e_{it}\|_F = o_P(1)$. Then
(i) $\max_{t \leq T} \|\hat{B}_t^{-1}\| = O_P(1)$, $\max_{t \leq T} |\hat{B}_t - B_t| = o_P(1)$.
(ii) $\frac{1}{T} \sum_{s \neq I} \|\hat{B}_s^{-1} - B^{-1}\|^2 = O_P(C_{NT}^{-2})$.
(iii) Write

$$\hat{S}_t - S = \begin{pmatrix} \Delta_{t1} \\ \Delta_{t2} \end{pmatrix},$$

whose partition matches with that of $(f_t', g_t')'$. Then $\frac{1}{T} \sum_{t \notin I} \|\Delta_{t1}\|^2 = O_P(C_{NT}^{-4})$
and $\frac{1}{T} \sum_{t \notin I} \|\Delta_{t2}\|^2 = O_P(C_{NT}^{-2})$.

(iv) $\max_{t \leq T} \|\hat{S}_t\| = O_P(1)$ and $\max_{t \leq T} \|\hat{S}_t - S\| = o_P(1)$.

Proof. Define

$$B_t = \frac{1}{N} \sum_i \begin{pmatrix} H_1' \lambda_i \lambda_i' H_1 e_{it}^2 & 0 \\ 0 & H_2' \alpha_i \alpha_i' H_2 \end{pmatrix}.$$

Then $\hat{B}_t - B_t = b_{1t} + b_{2t}$,

$$\begin{aligned} b_{1t} &= \frac{1}{N} \sum_i \begin{pmatrix} \tilde{\lambda}_i \tilde{\lambda}_i' \hat{e}_{it}^2 - H_1' \lambda_i \lambda_i' H_1 e_{it}^2 & \tilde{\lambda}_i \tilde{\alpha}_i' \hat{e}_{it} - H_1' \lambda_i \alpha_i H_2' e_{it} \\ \tilde{\alpha}_i \tilde{\lambda}_i' \hat{e}_{it} - H_2' \alpha_i \lambda_i H_1' e_{it} & \tilde{\alpha}_i \tilde{\alpha}_i' - H_2' \alpha_i \alpha_i' H_2 \end{pmatrix} \\ b_{2t} &= \frac{1}{N} \sum_i \begin{pmatrix} 0 & H_1' \lambda_i \alpha_i H_2' e_{it} \\ H_2' \alpha_i \lambda_i H_1' e_{it} & 0 \end{pmatrix}. \end{aligned}$$

(i) We now show $\max_{t \leq T} |b_{1t}| = o_P(1)$. In addition, by assumption $\max_{t \leq T} |b_{2t}| = o_P(1)$. Thus $\max_{t \leq T} |\hat{B}_t - B_t| = o_P(1)$, and thus $\max_{t \leq T} \|\hat{B}_t^{-1}\| = O_P(1) + \max_{t \leq T} \|B_t^{-1}\| = O_P(1)$, by the assumption that $\|B_t^{-1}\| = O_P(1)$ uniformly in t . To show $\max_{t \leq T} |b_{1t}| = o_P(1)$, note that:

First term:

$$\begin{aligned} & \max_t \left\| \frac{1}{N} \sum_i \tilde{\lambda}_i \tilde{\lambda}_i' (\hat{e}_{it}^2 - e_{it}^2) \right\| \\ & \leq O_P(1) \left(\frac{1}{N} \sum_i \|\tilde{\lambda}_i - H_1' \lambda_i\|^2 \right)^{1/2} \max_t \left[\frac{1}{N} \sum_i (\hat{e}_{it} - e_{it})^4 + (\hat{e}_{it} - e_{it})^2 e_{it}^2 \right]^{1/2} \\ & \quad + \frac{1}{N} \sum_i \|\tilde{\lambda}_i - H_1' \lambda_i\|^2 [2 \max_t |(\hat{e}_{it} - e_{it}) e_{it}| + \max_{t \leq T} (\hat{e}_{it} - e_{it})^2] \\ & \quad + O_P(1) \max_t \left\| \frac{1}{N} \sum_i \lambda_i \lambda_i' (\hat{e}_{it} - e_{it}) e_{it} \right\|_F \\ & \quad + O_P(1) \max_t \left\| \frac{1}{N} \sum_i \lambda_i \lambda_i' (\hat{e}_{it} - e_{it})^2 \right\|_F \end{aligned}$$

$$\begin{aligned}
&\leq O_P(1 + \max_{t \leq T} \|w_t\|^2 + b_{NT,2}^2) C_{NT}^{-2} + O_P(b_{NT,1}^2 + b_{NT,3}^2) \\
&\quad + O_P(1 + \max_{t \leq T} \|w_t\| + b_{NT,2}) \max_{it} |e_{it}| C_{NT}^{-1} + O_P(b_{NT,1} + b_{NT,3}) (\max_{t \leq T} \frac{1}{N} \sum_i e_{it}^2)^{1/2} C_{NT}^{-1} \\
&\quad + O_P(\phi_{NT}^2 C_{NT}^{-2}) + \phi_{NT} \max_{it} |e_{it}| C_{NT}^{-2} + \max_{t \leq T} \left\| \frac{1}{N} \sum_i c_i e_{it} l_i \right\|_F O_P(b_{NT,1} + b_{NT,3}) \\
&= o_P(1),
\end{aligned}$$

given assumptions $C_{NT}^{-1}(b_{NT,2} + \max_{t \leq T} \|w_t\|) \max_{it} |e_{it}| = o_P(1)$, $\phi_{NT} \max_{it} |e_{it}| = O_P(1)$, $\max_{t \leq T} \left\| \frac{1}{N} \sum_i e_{it} \alpha_i \lambda_i' \right\|_F = o_P(1)$, and $(b_{NT,1} + b_{NT,3})[(\max_{t \leq T} \frac{1}{N} \sum_i e_{it}^2)^{1/2} C_{NT}^{-1} + 1] = o_P(1)$.

In addition, $\max_{t \leq T} \left\| \frac{1}{N} \sum_i (\tilde{\lambda}_i \tilde{\lambda}_i' - H_1' \lambda_i \lambda_i' H_1) e_{it}^2 \right\| \leq O_P(C_{NT}^{-1}) \max_{it} e_{it}^2 = o_P(1)$. So the first term of $\max_{t \leq T} |b_{1t}|$ is $o_P(1)$.

Second term,

$$\begin{aligned}
&\max_t \left\| \frac{1}{N} \sum_i \tilde{\lambda}_i \tilde{\alpha}_i' (\hat{e}_{it} - e_{it}) \right\|_F \\
&\leq \max_{it} |\hat{e}_{it} - e_{it}| \frac{1}{N} \sum_i \|(\tilde{\lambda}_i - H_1' \lambda_i)(\tilde{\alpha}_i - H_2' \alpha_i)'\|_F \\
&\quad + O_P(1) \left(\frac{1}{N} \sum_i \|\tilde{\lambda}_i - H_1' \lambda_i\|^2 + \left(\frac{1}{N} \sum_i \|\tilde{\alpha}_i - H_2' \alpha_i\|^2 \right)^{1/2} \max_t \left(\frac{1}{N} \sum_i (\hat{e}_{it} - e_{it})^2 \right)^{1/2} \right. \\
&\quad \left. + O_P(1) \max_t \left\| \frac{1}{N} \sum_i \lambda_i \alpha_i' (\hat{e}_{it} - e_{it}) \right\|_F \right) \\
&\leq O_P(\phi_{NT} + 1 + \max_{t \leq T} \|w_t\|) C_{NT}^{-2} + O_P(b_{NT,1} + b_{NT,3} + C_{NT}^{-1} b_{NT,2}) = o_P(1).
\end{aligned}$$

Next,

$$\max_{t \leq T} \left\| \frac{1}{N} \sum_i (\tilde{\lambda}_i \tilde{\alpha}_i' - H_1' \lambda_i \alpha_i' H_2) e_{it} \right\|_F = O_P(C_{NT}^{-1}) \max_{it} |e_{it}| = o_P(1).$$

Thus, the second term of $\max_{t \leq T} |b_{1t}|$ is $o_P(1)$. Similarly, the third term is also $o_P(1)$. Finally, the last term $\left\| \frac{1}{N} \sum_i \tilde{\alpha}_i \tilde{\alpha}_i' - H_2' \alpha_i \alpha_i' H_2 \right\|_F = o_P(1)$.

(ii) Because we have proven $\max_{t \leq T} \|\hat{B}_t^{-1}\| = O_P(1)$, it suffices to prove $\frac{1}{T} \sum_{s \notin I} \|\hat{B}_s - B\|_F^2 = O_P(C_{NT}^{-2})$, or $\frac{1}{T} \sum_{s \notin I} \|b_{1t}\|_F^2 = O_P(C_{NT}^{-2}) = \frac{1}{T} \sum_{s \notin I} \|b_{2t}\|_F^2$.

First term of b_{1t} : By (G.8), the first term is bounded by, using Lemma G.2,

$$\begin{aligned}
&\frac{1}{T} \sum_{t \notin I} \left\| \frac{1}{N} \sum_i \tilde{\lambda}_i \tilde{\lambda}_i' \hat{e}_{it}^2 - H_1' \lambda_i \lambda_i' H_1 e_{it}^2 \right\|_F^2 \\
&\leq \frac{1}{T} \sum_{t \notin I} \left\| \frac{1}{N} \sum_i \tilde{\lambda}_i \tilde{\lambda}_i' (\hat{e}_{it}^2 - e_{it}^2) \right\|_F^2 + \frac{1}{T} \sum_{t \notin I} \left\| \frac{1}{N} \sum_i (\tilde{\lambda}_i \tilde{\lambda}_i' - H_1' \lambda_i \lambda_i' H_1) e_{it}^2 \right\|_F^2
\end{aligned}$$

$$\begin{aligned}
&\leq O_P(C_{NT}^{-4}) \max_{it} |\widehat{e}_{it} - e_{it}|^2 \max_{it} |e_{it}|^2 + O_P(C_{NT}^{-2}) \frac{1}{T} \sum_{t \notin I} \frac{1}{N} \sum_i (\widehat{e}_{it} - e_{it})^4 \\
&\quad + O_P(1) \frac{1}{T} \sum_{t \notin I} \left(\frac{1}{N} \sum_i (\widehat{e}_{it} - e_{it})^2 \right)^2 + O_P(C_{NT}^{-2}) \frac{1}{T} \sum_{t \notin I} \frac{1}{N} \sum_i (\widehat{e}_{it} - e_{it})^2 e_{it}^2 \\
&\quad + O_P(1) \frac{1}{T} \sum_{t \notin I} \left(\frac{1}{N} \sum_i \lambda_i \lambda'_i (\widehat{e}_{it} - e_{it}) e_{it} \right)^2 \\
&\quad + O_P(1) \text{tr} \left(\frac{1}{N^2} \sum_{ij} (\widetilde{\lambda}_i \widetilde{\lambda}'_i - H'_1 \lambda_i \lambda'_i H_1) (\widetilde{\lambda}_j \widetilde{\lambda}'_j - H'_1 \lambda_j \lambda'_j H_1) \right) \frac{1}{T} \sum_{t \notin I} \mathbb{E}_I e_{jt}^2 e_{it}^2 \\
&\leq O_P(C_{NT}^{-2}) + O_P(1) \left(\frac{1}{N} \sum_i \|\widetilde{\lambda}_i \widetilde{\lambda}'_i - H'_1 \lambda_i \lambda'_i H_1\|_F \right)^2 \\
&\leq O_P(C_{NT}^{-4} \phi_{NT}^2) \max_{it} |e_{it}|^2 + O_P(C_{NT}^{-2}) = O_P(C_{NT}^{-2}). \tag{G.10}
\end{aligned}$$

Second term of b_{1t} :

$$\begin{aligned}
&\frac{1}{T} \sum_{t \notin I} \left\| \frac{1}{N} \sum_i \widetilde{\lambda}_i \widetilde{\alpha}'_i \widehat{e}_{it} - H'_1 \lambda_i \alpha_i H'_2 e_{it} \right\|_F^2 \\
&\leq \frac{1}{T} \sum_{t \notin I} \left\| \frac{1}{N} \sum_i \widetilde{\lambda}_i \widetilde{\alpha}'_i (\widehat{e}_{it} - e_{it}) \right\|_F^2 + \frac{1}{T} \sum_{t \notin I} \left\| \frac{1}{N} \sum_i (\widetilde{\lambda}_i \widetilde{\alpha}'_i - H'_1 \lambda_i \alpha_i H'_2) e_{it} \right\|_F^2 \\
&\leq O_P(C_{NT}^{-4}) \max_i \frac{1}{T} \sum_t (\widehat{e}_{it} - e_{it})^2 + O_P(C_{NT}^{-2}) \frac{1}{NT} \sum_{it} (\widehat{e}_{it} - e_{it})^2 \\
&\quad + \frac{1}{T} \sum_{t \notin I} \left\| \frac{1}{N} \sum_i \lambda_i \alpha'_i (\widehat{e}_{it} - e_{it}) \right\|_F^2 \\
&\quad + O_P(1) \text{tr} \frac{1}{N^2} \sum_{ij} (\widetilde{\lambda}_i \widetilde{\alpha}'_i - H'_1 \lambda_i \alpha_i H'_2) (\widetilde{\lambda}_j \widetilde{\alpha}'_j - H'_1 \lambda_j \alpha_j H'_2) \frac{1}{T} \sum_{t \notin I} \mathbb{E}_I e_{it} e_{jt} \\
&\leq O_P(b_{NT,4}^2 + b_{NT,5}^2 + C_{NT}^{-2}) C_{NT}^{-4} + O_P(C_{NT}^{-2}) = O_P(C_{NT}^{-2}). \tag{G.11}
\end{aligned}$$

The third term of b_{1t} is bounded similarly. Finally, it is straightforward to see that fourth term of b_{1t} is $\left\| \frac{1}{N} \sum_i \widetilde{\alpha}_i \widetilde{\alpha}'_i - H'_2 \alpha_i \alpha'_i H_2 \right\|_F^2 = O_P(C_{NT}^{-2})$. Thus $\frac{1}{T} \sum_{s \notin I} \|b_{1t}\|_F^2 = O_P(C_{NT}^{-2})$.

As for b_{2t} , it suffices to prove

$$\frac{1}{T} \sum_{t \notin I} \left\| \frac{1}{N} \sum_i \alpha_i \lambda'_i e_{it} \right\|_F^2 = O_P(1) \frac{1}{T} \sum_{t \notin I} \mathbb{E} \left\| \frac{1}{N} \sum_i \alpha_i \lambda'_i e_{it} \right\|_F^2 = O_P(N^{-1}).$$

Thus $\frac{1}{T} \sum_{s \notin I} \|b_{2t}\|_F^2 = O_P(C_{NT}^{-2})$.

(iii) Note that $\widehat{S}_t - S = c_t + d_t$, where

$$c_t = \frac{1}{N} \sum_i \begin{pmatrix} \widetilde{\lambda}_i \lambda'_i H_1 e_{it} (\widehat{e}_{it} - e_{it}) + \widetilde{\lambda}_i \widetilde{\lambda}'_i (\widehat{e}_{it}^2 - e_{it}^2) & \widetilde{\lambda}_i (\widehat{e}_{it} - e_{it}) (\alpha'_i H_2 - \widetilde{\alpha}'_i) \\ \widetilde{\alpha}_i \lambda'_i H_1 (e_{it} - \widehat{e}_{it}) + \widetilde{\alpha}_i (\lambda'_i H_1 - \widetilde{\lambda}'_i) (\widehat{e}_{it} - e_{it}) & 0 \end{pmatrix}$$

$$d_t = \frac{1}{N} \sum_i \begin{pmatrix} (\tilde{\lambda}_i \lambda'_i H_1 - \tilde{\lambda}_i \tilde{\lambda}'_i) e_{it}^2 - H'_1 \lambda_i (\lambda'_i H_1 - \tilde{\lambda}'_i) \mathbb{E} e_{it}^2 & \tilde{\lambda}_i e_{it} (\alpha'_i H_2 - \tilde{\alpha}'_i) \\ \tilde{\alpha}_i (\lambda'_i H_1 - \tilde{\lambda}'_i) e_{it} & 0 \end{pmatrix}.$$

As for the upper blocks of c_t , first note that

$$\begin{aligned} & \frac{1}{T} \sum_{t \notin I} \left(\frac{1}{N} \sum_i \|\tilde{\lambda}_i - H'_1 \lambda_i\|^2 |e_{it}| \right)^2 \\ &= O_P(1) \frac{1}{N^2} \sum_{ij} \|\tilde{\lambda}_i - H'_1 \lambda_i\|^2 \|\tilde{\lambda}_j - H'_1 \lambda_j\|^2 \frac{1}{T} \sum_{t \notin I} \mathbb{E}_I |e_{jt} e_{it}| \\ &= O_P(1) \left(\frac{1}{N} \sum_i \|\tilde{\lambda}_i - H'_1 \lambda_i\|^2 \right)^2 = O_P(C_{NT}^{-4}). \end{aligned}$$

Then by Lemma G.2,

$$\begin{aligned} & \frac{1}{T} \sum_{t \notin I} \left\| \frac{1}{N} \sum_i \tilde{\lambda}_i \lambda'_i H_1 e_{it} (\hat{e}_{it} - e_{it}) \right\|_F^2 \\ &\leq O_P(1) \frac{1}{N} \sum_i \|\tilde{\lambda}_i - H'_1 \lambda_i\|^2 \frac{1}{NT} \sum_i \sum_{t \notin I} e_{it}^2 (\hat{e}_{it} - e_{it})^2 \\ &\quad + O_P(1) \frac{1}{T} \sum_{t \notin I} \left\| \frac{1}{N} \sum_i \lambda_i \lambda'_i e_{it} (\hat{e}_{it} - e_{it}) \right\|_F^2 \\ &= O_P(C_{NT}^{-4}) \\ &\quad + \frac{1}{T} \sum_{t \notin I} \left\| \frac{1}{N} \sum_i \tilde{\lambda}_i \tilde{\lambda}'_i (\hat{e}_{it}^2 - e_{it}^2) \right\|_F^2 \\ &\leq \left(\frac{1}{N} \sum_i \|\tilde{\lambda}_i - H'_1 \lambda_i\|^2 \right)^2 \max_{it} [(\hat{e}_{it} - e_{it})^4] \\ &\quad + \frac{1}{T} \sum_{t \notin I} \left(\frac{1}{N} \sum_i \|\tilde{\lambda}_i - H'_1 \lambda_i\|^2 |e_{it}| \right)^2 \max_{it} |\hat{e}_{it} - e_{it}|^2 \\ &\quad + \frac{1}{N} \sum_i \|\tilde{\lambda}_i - H'_1 \lambda_i\|^2 \frac{1}{NT} \sum_{t \notin I} \sum_i [(\hat{e}_{it} - e_{it})^4 + (\hat{e}_{it} - e_{it})^2 e_{it}^2] \\ &\quad + \frac{1}{T} \sum_{t \notin I} \left\| \frac{1}{N} \sum_i \lambda_i \lambda'_i (\hat{e}_{it} - e_{it}) e_{it} \right\|_F^2 + \frac{1}{T} \sum_{t \notin I} \left\| \frac{1}{N} \sum_i \lambda_i \lambda'_i (\hat{e}_{it} - e_{it})^2 \right\|_F^2 \\ &= O_P(C_{NT}^{-4}). \end{aligned}$$

Also,

$$\begin{aligned} & \frac{1}{T} \sum_{t \notin I} \left\| \frac{1}{N} \sum_i \tilde{\lambda}_i (\hat{e}_{it} - e_{it}) (\alpha'_i H_2 - \tilde{\alpha}'_i) \right\|_F^2 \\ &\leq \frac{1}{N} \sum_i \|\alpha'_i H_2 - \tilde{\alpha}'_i\|^2 \frac{1}{N} \sum_i \|\tilde{\lambda}_i - H'_1 \lambda_i\|^2 \max_{it} (\hat{e}_{it} - e_{it})^2 \end{aligned}$$

$$+O_P(1) \frac{1}{N} \sum_i \|\alpha'_i H_2 - \tilde{\alpha}'_i\|^2 \frac{1}{N} \sum_i \frac{1}{T} \sum_{t \notin I} (\hat{e}_{it} - e_{it})^2 = O_P(C_{NT}^{-4}).$$

Similarly, $\frac{1}{T} \sum_{t \notin I} \|\frac{1}{N} \sum_i \tilde{\alpha}_i (\lambda'_i H_1 - \tilde{\lambda}'_i) (\hat{e}_{it} - e_{it})\|_F^2 = O_P(C_{NT}^{-4})$. Finally,

$$\begin{aligned} \frac{1}{T} \sum_{t \notin I} \|\frac{1}{N} \sum_i \tilde{\alpha}_i \lambda'_i H_1 (e_{it} - \hat{e}_{it})\|_F^2 &= O_P(C_{NT}^{-4}) + O_P(1) \frac{1}{T} \sum_{t \notin I} \|\frac{1}{N} \sum_i \alpha_i \lambda'_i (e_{it} - \hat{e}_{it})\|_F^2 \\ &= O_P(C_{NT}^{-4}). \end{aligned}$$

If we let the two upper blocks of c_t be $c_{t,1}, c_{t,2}$, and the third block of c_t be $c_{t,3}$, then $\frac{1}{T} \sum_{t \notin I} \|c_{t,1}\|_F^2 + \frac{1}{T} \sum_{t \notin I} \|c_{t,2}\|_F^2 = O_P(C_{NT}^{-4})$ while $\frac{1}{T} \sum_{t \notin I} \|c_{t,3}\|_F^2 = O_P(C_{NT}^{-2})$.

As for d_t , let $d_{t,1}, \dots, d_{t,3}$ denote the nonzero blocks. Then $d_{t,1}$ depends on $\frac{1}{T} \sum_{t \notin I} \|\frac{1}{N} \sum_i \lambda_i (\lambda'_i H_1 - \tilde{\lambda}'_i) (e_{it}^2 - \mathbb{E} e_{it}^2)\|_F^2$. Let Υ_t be an $N \times 1$ vector of e_{it}^2 , and $\text{diag}(\Lambda)$ be diagonal matrix consisting of elements of Λ . Suppose $\dim(\lambda_i) = 1$ (focus on each element), then $\frac{1}{T} \sum_{t \notin I} \|d_{t,1}\|_F^2$ is bounded by

$$\begin{aligned} &\frac{1}{T} \sum_{t \notin I} \|\frac{1}{N} \sum_i (\tilde{\lambda}_i \lambda'_i H_1 - \tilde{\lambda}_i \tilde{\lambda}'_i) e_{it}^2 - H'_1 \lambda_i (\lambda'_i H_1 - \tilde{\lambda}'_i) \mathbb{E} e_{it}^2\|_F^2 \\ &\leq \frac{1}{T} \sum_{t \notin I} \|\frac{1}{N} \sum_i (\tilde{\lambda}_i - H'_1 \lambda_i) (\lambda'_i H_1 - \tilde{\lambda}'_i) e_{it}^2\|_F^2 \\ &\quad + \frac{1}{T} \sum_{t \notin I} \|\frac{1}{N} \sum_i H'_1 \lambda_i (\lambda'_i H_1 - \tilde{\lambda}'_i) (e_{it}^2 - \mathbb{E} e_{it}^2)\|_F^2 \\ &\leq O_P(1) \frac{1}{N^2} \sum_{ij} \|\tilde{\lambda}_i - H'_1 \lambda_i\|^2 \|\tilde{\lambda}_j - H'_1 \lambda_j\|^2 \frac{1}{T} \sum_{t \notin I} \mathbb{E} e_{jt}^2 e_{it}^2 \\ &\quad + O_P(1) \frac{1}{T} \sum_{t \notin I} \frac{1}{N^2} (\tilde{\Lambda} - \Lambda H_1)' \text{diag}(\Lambda) \text{Var}_I(\Upsilon_t) \text{diag}(\Lambda) (\tilde{\Lambda} - \Lambda H_1) \\ &= O_P(C_{NT}^{-4}). \end{aligned}$$

In addition, using the same technique, it is easy to show

$$\frac{1}{T} \sum_{t \notin I} \|\frac{1}{N} \sum_i \tilde{\lambda}_i e_{it} (\alpha'_i H_2 - \tilde{\alpha}'_i)\|_F^2 = O_P(C_{NT}^{-4}) = \frac{1}{T} \sum_{t \notin I} \|\frac{1}{N} \sum_i \tilde{\alpha}_i (\lambda'_i H_1 - \tilde{\lambda}'_i) e_{it}\|_F^2.$$

Hence $\frac{1}{T} \sum_{t \notin I} \|d_{t,k}\|_F^2 = O_P(C_{NT}^{-4})$ for $k = 1, 2, 3$. Together, we have

$$\frac{1}{T} \sum_{t \notin I} \|\Delta_{t1}\|^2 \leq \frac{1}{T} \sum_{t \notin I} \|c_{t,1} + d_{t,1}\|^2 + \frac{1}{T} \sum_{t \notin I} \|c_{t,2} + d_{t,2}\|^2 = O_P(C_{NT}^{-4})$$

and $\frac{1}{T} \sum_{t \notin I} \|\Delta_{t2}\|^2 = \frac{1}{T} \sum_{t \notin I} \|c_{t,3} + d_{t,3}\|^2 = O_P(C_{NT}^{-2})$.

(iv) Recall that $\widehat{S}_t - S = c_t + d_t$. It suffices to show $\max_{t \leq T} \|c_t\| = o_P(1) = \max_{t \leq T} \|d_t\|$.

$$\begin{aligned} \max_{t \leq T} \|c_t\| &\leq O_P(1) \max_{it} |\widehat{e}_{it} - e_{it}| (|e_{it}| + 1) = o_P(1). \\ \max_{t \leq T} \|d_t\| &\leq (\max_{it} |e_{it}| + \max_{it} |e_{it}|^2 + 1) O_P(C_{NT}^{-1}) = o_P(1). \end{aligned}$$

□

Lemma G.9. For terms defined in (F.2), and for each fixed $t \notin I$,

- (i) $\sum_{d=2}^5 A_{dt} = O_P(C_{NT}^{-2})$
- (ii) For the “upper block” of A_{6t} , $\frac{1}{N} \sum_i \lambda_i e_{it} (\ell'_i w_t \lambda'_i f_t - \widehat{\ell'_i w_t \lambda'_i f_t}) = O_P(C_{NT}^{-2})$.
- (iii) The upper block of A_{1t} is $O_P(C_{NT}^{-2})$.

Proof. (i) Term A_{2t} . Given $B_t^{-1} - B^{-1} = O_P(C_{NT}^{-1})$ and the cross-sectional weak correlations in u_{it} , it is easy to see $A_{2t} = O_P(C_{NT}^{-2})$.

Term A_{3t} . It suffices to prove:

$$\begin{aligned} \frac{1}{N} \sum_i (\widetilde{\lambda}_i \widehat{e}_{it} - H'_1 \lambda_i e_{it}) u_{it} &= O_P(C_{NT}^{-2}) \\ \frac{1}{N} \sum_i (\widetilde{\alpha}_i - H'_2 \alpha_i) u_{it} &= O_P(C_{NT}^{-2}). \end{aligned} \tag{G.12}$$

First, let Υ_t be an $N \times 1$ vector of $e_{it} u_{it}$. Due to the serial independence of (u_{it}, e_{it}) , we have

$$\begin{aligned} \mathbb{E}_I \frac{1}{N} \sum_i (\widetilde{\lambda}_i - H'_1 \lambda_i) e_{it} u_{it} &= 0, \\ \text{Var}_I \left(\frac{1}{N} \sum_i (\widetilde{\lambda}_i - H'_1 \lambda_i) e_{it} u_{it} \right) &= \frac{1}{N^2} (\widetilde{\Lambda} - \Lambda H'_1)' \text{Var}_I(\Upsilon_t) (\widetilde{\Lambda} - \Lambda H'_1) \\ &\leq O_P(C_{NT}^{-2} N^{-1}) \max_i \sum_j |\text{Cov}(e_{it} u_{it}, e_{jt} u_{jt})| = O_P(C_{NT}^{-2} N^{-1}). \end{aligned}$$

Similarly, $\mathbb{E}_I \frac{1}{N} \sum_i (\widetilde{\alpha}_i - H'_2 \alpha_i) u_{it} = 0$ and $\text{Var}_I \left(\frac{1}{N} \sum_i (\widetilde{\alpha}_i - H'_2 \alpha_i) u_{it} \right) = O_P(C_{NT}^{-2} N^{-1})$.

$$\begin{aligned} \frac{1}{N} \sum_i (\widetilde{\lambda}_i - H'_1 \lambda_i) e_{it} u_{it} &= O_P(C_{NT}^{-1} N^{-1/2}) \\ \frac{1}{N} \sum_i (\widetilde{\alpha}_i - H'_2 \alpha_i) u_{it} &= O_P(C_{NT}^{-1} N^{-1/2}) \\ \frac{1}{N} \sum_i \|\widetilde{\lambda}_i - H'_1 \lambda_i\|^2 u_{it}^2 &= O_P(1) \frac{1}{N} \sum_i \|\widetilde{\lambda}_i - H'_1 \lambda_i\|^2 \mathbb{E}_I u_{it}^2 = O_P(C_{NT}^{-2}). \end{aligned} \tag{G.13}$$

Thus, the first term of (G.12) is

$$\begin{aligned}
& \frac{1}{N} \sum_i (\tilde{\lambda}_i \hat{e}_{it} - H_1' \lambda_i e_{it}) u_{it} \leq \left(\frac{1}{N} \sum_i \|\tilde{\lambda}_i - H_1' \lambda_i\|^2 u_{it}^2 \right)^{1/2} \left(\frac{1}{N} \sum_i (\hat{e}_{it} - e_{it})^2 \right)^{1/2} \\
& + H_1' \frac{1}{N} \sum_i \lambda_i (\hat{e}_{it} - e_{it}) u_{it} + \frac{1}{N} \sum_i (\tilde{\lambda}_i - H_1' \lambda_i) e_{it} u_{it} \\
& \leq O_P(C_{NT}^{-1}) \left(\frac{1}{N} \sum_i (\hat{e}_{it} - e_{it})^2 \right)^{1/2} + O_P(C_{NT}^{-2}) + O_P(1) \frac{1}{N} \sum_i \lambda_i (\hat{e}_{it} - e_{it}) u_{it} \\
& = O_P(C_{NT}^{-2}),
\end{aligned}$$

where the last equality follows from Lemma G.1.

Term A_{4t} . Recall that $\mu_{it} = l_i' w_t$. It suffices to prove each of the following terms is $O_P(C_{NT}^{-2})$:

$$\begin{aligned}
C_{1t} &= \frac{1}{N} \sum_i (\tilde{\lambda}_i - H_1' \lambda_i) e_{it} (\hat{e}_{it} - e_{it}) (\dot{\lambda}_i - H_1' \lambda_i)' \tilde{f}_t \\
C_{2t} &= \frac{1}{N} \sum_i (\tilde{\lambda}_i - H_1' \lambda_i) (\hat{e}_{it} - e_{it})^2 (\dot{\lambda}_i - H_1' \lambda_i)' \tilde{f}_t \\
C_{3t} &= \frac{1}{N} \sum_i H_1' \lambda_i (\hat{e}_{it} - e_{it})^2 (\dot{\lambda}_i - H_1' \lambda_i)' \tilde{f}_t \\
C_{4t} &= \frac{1}{N} \sum_i (\tilde{\lambda}_i - H_1' \lambda_i) e_{it} (\hat{e}_{it} - e_{it}) \lambda_i' H_1 \tilde{f}_t \\
C_{5t} &= \frac{1}{N} \sum_i (\tilde{\lambda}_i - H_1' \lambda_i) (\hat{e}_{it} - e_{it})^2 \lambda_i' H_1 \tilde{f}_t \\
C_{6t} &= \frac{1}{N} \sum_i H_1' \lambda_i (\hat{e}_{it} - e_{it})^2 \lambda_i' H_1 \tilde{f}_t \\
C_{7t} &= \frac{1}{N} \sum_i (\tilde{\lambda}_i - H_1' \lambda_i) e_{it} \mu_{it} (\dot{\lambda}_i - H_1' \lambda_i)' \tilde{f}_t \\
C_{8t} &= \frac{1}{N} \sum_i (\tilde{\lambda}_i - H_1' \lambda_i) (\hat{e}_{it} - e_{it}) \mu_{it} (\dot{\lambda}_i - H_1' \lambda_i)' \tilde{f}_t \\
C_{9t} &= \frac{1}{N} \sum_i H_1' \lambda_i (\hat{e}_{it} - e_{it}) \mu_{it} (\dot{\lambda}_i - H_1' \lambda_i)' \tilde{f}_t \\
C_{10t} &= \frac{1}{N} \sum_i (\tilde{\lambda}_i - H_1' \lambda_i) e_{it} \mu_{it} \lambda_i' H_1 (\tilde{f}_t - H_1^{-1} f_t) \\
C_{11t} &= \frac{1}{N} \sum_i (\tilde{\lambda}_i - H_1' \lambda_i) (\hat{e}_{it} - e_{it}) \mu_{it} \lambda_i' H_1 (\tilde{f}_t - H_1^{-1} f_t) \\
C_{12t} &= \frac{1}{N} \sum_i H_1' \lambda_i (\hat{e}_{it} - e_{it}) \mu_{it} \lambda_i' H_1 (\tilde{f}_t - H_1^{-1} f_t)
\end{aligned}$$

$$\begin{aligned}
C_{13t} &= \frac{1}{N} \sum_i (\tilde{\alpha}_i - H'_2 \alpha_i) (\hat{e}_{it} - e_{it}) (\dot{\lambda}_i - H'_1 \lambda_i)' \tilde{f}_t \\
C_{14t} &= \frac{1}{N} \sum_i (\tilde{\alpha}_i - H'_2 \alpha_i) (\hat{e}_{it} - e_{it}) \lambda_i' H_1 \tilde{f}_t \\
C_{15t} &= \frac{1}{N} \sum_i (\tilde{\alpha}_i - H'_2 \alpha_i) \mu_{it} (\dot{\lambda}_i - H'_1 \lambda_i)' \tilde{f}_t \\
C_{16t} &= \frac{1}{N} \sum_i (\tilde{\alpha}_i - H'_2 \alpha_i) \mu_{it} \lambda_i' H_1 (\tilde{f}_t - H_1^{-1} f_t). \tag{G.14}
\end{aligned}$$

The proof follows from repeatedly applying the Cauchy-Schwarz inequality and is straightforward. In addition, it is also straightforward to apply Cauchy-Schwarz to prove that

$$\frac{1}{T} \sum_{t \notin I} \|C_{dt}\|^2 = O_P(C_{NT}^{-4}), \quad d = 1, \dots, 16. \tag{G.15}$$

Term A_{5t} . Note that

$$A_{5t} = (\hat{B}_t^{-1} - B^{-1}) \begin{pmatrix} H'_1 \sum_{d=1}^4 B_{dt} \\ H'_2 \sum_{d=5}^8 B_{dt} \end{pmatrix}$$

where B_{dt} are defined below. Given $B_t^{-1} - B^{-1} = O_P(C_{NT}^{-1})$, it suffices to prove the following terms are $O_P(C_{NT}^{-2})$.

$$\begin{aligned}
B_{1t} &= \frac{1}{N} \sum_i \lambda_i e_{it} (\hat{e}_{it} - e_{it}) (\dot{\lambda}_i - H'_1 \lambda_i)' \tilde{f}_t \\
B_{2t} &= \frac{1}{N} \sum_i \lambda_i e_{it} (\hat{e}_{it} - e_{it}) \lambda_i' H_1 \tilde{f}_t \\
B_{3t} &= \frac{1}{N} \sum_i \lambda_i e_{it} l_i' w_t (\dot{\lambda}_i - H'_1 \lambda_i)' \tilde{f}_t \\
B_{4t} &= \frac{1}{N} \sum_i \lambda_i e_{it} l_i' w_t \lambda_i' H_1 (\tilde{f}_t - H_1^{-1} f_t) \\
B_{5t} &= \frac{1}{N} \sum_i \alpha_i (\hat{e}_{it} - e_{it}) (\dot{\lambda}_i - H'_1 \lambda_i)' \tilde{f}_t \tag{G.16}
\end{aligned}$$

and that the following terms are $O_P(C_{NT}^{-1})$:

$$\begin{aligned}
B_{6t} &= \frac{1}{N} \sum_i \alpha_i (\hat{e}_{it} - e_{it}) \lambda_i' H_1 \tilde{f}_t \\
B_{7t} &= \frac{1}{N} \sum_i \alpha_i l_i' w_t (\dot{\lambda}_i - H'_1 \lambda_i)' \tilde{f}_t
\end{aligned}$$

$$B_{8t} = \frac{1}{N} \sum_i \alpha_i l'_i w_t \lambda'_i H_1 (\tilde{f}_t - H_1^{-1} f_t). \quad (\text{G.17})$$

In fact, $B_{1t}, B_{5,t} \sim B_{8t}$ follow immediately from the Cauchy-Schwarz inequality. $B_{2t} = O_P(C_{NT}^{-2})$ due to Lemma G.1. By Lemma G.5, $\frac{1}{\sqrt{N}} \sum_i (H'_1 \lambda_i - \dot{\lambda}_i) e_{it} = O_P(\sqrt{N} C_{NT}^{-2})$. Since $t \notin I$, $B_{3t} = O_P(1) \frac{1}{N} \sum_i \lambda_i e_{it} \mu_i (\dot{\lambda}_i - H'_1 \lambda_i)' = O_P(C_{NT}^{-2})$. In addition, for each fixed t , $\tilde{f}_t - H_1^{-1} f_t = O_P(C_{NT}^{-1})$. Thus

$$B_{4t} = O_P(C_{NT}^{-1}) \frac{1}{N} \sum_i \lambda_i e_{it} \mu_i \lambda'_i = O_P(C_{NT}^{-2}).$$

Thus, $A_{5t} = O_P(C_{NT}^{-2})$.

(ii) “upper block” of A_{6t} . Note that $l'_i w_t - \widehat{l'_i w_t} = \widehat{e}_{it} - e_{it}$.

$$A_{6t} = B^{-1} \begin{pmatrix} H'_1 \sum_{d=1}^4 B_{dt} \\ H'_2 \sum_{d=5}^8 B_{dt} \end{pmatrix}$$

So we only need to look at $\sum_{d=1}^4 B_{dt}$. From the proof of (i), we have $B_{dt} = O_P(C_{NT}^{-2})$, for $d = 1, \dots, 4$. It follows immediately that $\frac{1}{N} \sum_i \lambda_i e_{it} (l'_i w_t \lambda'_i f_t - \widehat{l'_i w_t} \dot{\lambda}_i \tilde{f}_t) = O_P(C_{NT}^{-2})$.

(iii) Lastly, note that the upper block of A_{1t} is determined by the upper blocks of $\widehat{B}_t^{-1} \widehat{S}_t - B^{-1} S$, and are both $O_P(C_{NT}^{-2})$ by Lemma G.7.

□

G.4. Technical lemmas for $\widehat{\lambda}_i$.

Lemma G.10. Suppose $\max_i \|\frac{1}{T} \sum_s f_s f'_s (e_{is}^2 - \mathbb{E} e_{is}^2)\| = o_P(1) = \max_i \|\frac{1}{T} \sum_s f_s g'_s e_{is}\|$.

- (i) $\frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \|\widehat{D}_i - D_i\|^2 = O_P(C_{NT}^{-2})$.
- (ii) $\max_{i \leq N} \|\widehat{D}_i^{-1} - D_i^{-1}\| = o_P(1)$.
- (iii) $\frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \|\widehat{D}_i^{-1} - D_i^{-1}\|^2 = O_P(C_{NT}^{-2})$.

Proof. $\widehat{D}_i - D_i$ is a two-by-two block matrix. The first block is

$$\begin{aligned} d_{1i} &:= \frac{1}{T_0} \sum_{s \notin I} \widehat{f}_s \widehat{f}'_s \widehat{e}_{is}^2 - H_f f_s f'_s H'_f \mathbb{E} e_{is}^2 \\ &= \frac{1}{T_0} \sum_{s \notin I} (\widehat{f}_s - H_f f_s) (\widehat{f}_s - H_f f_s)' [(\widehat{e}_{is} - e_{is})^2 + (\widehat{e}_{is} - e_{is}) e_{is} + e_{is}^2] \\ &\quad + \frac{1}{T_0} \sum_{s \notin I} (\widehat{f}_s - H_f f_s) f'_s H'_f [(\widehat{e}_{is} - e_{is})^2 + (\widehat{e}_{is} - e_{is}) e_{is} + e_{is}^2] \end{aligned}$$

$$\begin{aligned}
& + \frac{1}{T_0} \sum_{s \notin I} H_f f_s (\hat{f}_s - H_f f_s)' [(\hat{e}_{is} - e_{is})^2 + (\hat{e}_{is} - e_{is})e_{is} + e_{is}^2] \\
& + \frac{1}{T_0} \sum_{s \notin I} H_f f_s f_s' H_f' [(\hat{e}_{is} - e_{is})^2 + (\hat{e}_{is} - e_{is})e_{is}] + \frac{1}{T_0} \sum_{s \notin I} H_f f_s f_s' H_f' (e_{is}^2 - \mathbb{E} e_{is}^2).
\end{aligned}$$

It follows from Lemma G.2 that $\frac{1}{NT} \sum_{it} (\hat{e}_{it} - e_{it})^2 e_{it}^2 = O_P(C_{NT}^{-2})$, $\frac{1}{NT} \sum_{it} (\hat{e}_{it} - e_{it})^4 = O_P(C_{NT}^{-4})$, and from Lemma G.1 that $\max_{it} |\hat{e}_{it} - e_{it}| e_{it} = O_P(1)$.

$$\begin{aligned}
\frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \|d_{1i}\|^2 & \leq O_P(1) \frac{1}{T_0} \sum_{s \notin I} \|\hat{f}_s - H_f f_s\|^2 (1 + \|f_t\|^2 + \|f_t\|^2) \frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} e_{it}^4 \\
& + O_P(C_{NT}^{-2}) = O_P(C_{NT}^{-2}).
\end{aligned}$$

The second block is

$$\begin{aligned}
d_{2i} & := \frac{1}{T_0} \sum_{s \notin I} \hat{f}_s \hat{g}_s' \hat{e}_{is} = \frac{1}{T_0} \sum_{s \notin I} (\hat{f}_s - H_f f_s) (\hat{g}_s - H_g g_s)' \hat{e}_{is} \\
& + \frac{1}{T_0} \sum_{s \notin I} (\hat{f}_s - H_f f_s) g_s' H_g' \hat{e}_{is} + \frac{1}{T_0} \sum_{s \notin I} H_f f_s (\hat{g}_s - H_g g_s)' \hat{e}_{is} \\
& + \frac{1}{T_0} \sum_{s \notin I} H_f f_s g_s' H_g' (\hat{e}_{is} - e_{is}) + \frac{1}{T_0} \sum_{s \notin I} H_f f_s g_s' H_g' e_{is}.
\end{aligned}$$

Thus, $\frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \|d_{2i}\|^2 = O_P(C_{NT}^{-2})$. The third block is similar. The fourth block is $d_{4i} = \frac{1}{T_0} \sum_{s \notin I} \hat{g}_s \hat{g}_s' - H_g g_s g_s' H_g'$. So $\frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \|d_{4i}\|^2 = O_P(C_{NT}^{-2})$.

Hence, $\frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \|\hat{D}_i - D_i\|^2 = O_P(C_{NT}^{-2})$.

(ii) Note that

$$\begin{aligned}
\max_i \|d_{1i}\| & = o_P(1) + O_P(1) \max_i \left\| \frac{1}{T} \sum_s f_s f_s' (e_{is}^2 - \mathbb{E} e_{is}^2) \right\| = o_P(1) \\
\max_i \|d_{2i}\| & = o_P(1) + O_P(1) \max_i \left\| \frac{1}{T} \sum_s f_s g_s' e_{is} \right\| = o_P(1).
\end{aligned}$$

So $\max_i \|\hat{D}_i - D_i\| = o_P(1)$. Now because $\min_i \lambda_{\min}(D_i) \geq c$, with probability approaching one $\min_i \lambda_{\min}(\hat{D}_i) \geq \min_i \lambda_{\min}(D_i) - \max_i \|\hat{D}_i - D_i\| > c/2$. So $\max_i \|D_i^{-1}\| + \max_i \|\hat{D}_i^{-1}\| = O_P(1)$. So

$$\max_i \|\hat{D}_i^{-1} - D_i^{-1}\| \leq \max_i \|\hat{D}_i^{-1}\| \max_i \|D_i^{-1}\| \max_i \|\hat{D}_i - D_i\| = o_P(1).$$

(iii) Because $\min_i \lambda_{\min}(D_i) \geq c$,

$$\begin{aligned}
\frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \|\widehat{D}_i^{-1} - D_i^{-1}\|^2 &\leq \frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \|\widehat{D}_i^{-1}\|^2 \|D_i^{-1}\|^2 \|\widehat{D}_i - D_i\|^2 \\
&\leq \max_i [\|\widehat{D}_i^{-1} - D_i^{-1}\| + \|D_i^{-1}\|]^2 \frac{C}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \|\widehat{D}_i - D_i\|^2 \\
&= O_P(1) \frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \|\widehat{D}_i - D_i\|^2 = O_P(C_{NT}^{-2}).
\end{aligned}$$

□

Lemma G.11. $\frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \|\frac{1}{T} \sum_{s \notin I} (\widehat{f}_s - H_f f_s) e_{is} f'_s\|^2 = O_P(C_{NT}^{-4})$.

This lemma is needed to prove the performance for $\widehat{\lambda}_i$, which controls the effect of $\widehat{f}_s - H_f f_s$ on the estimation of λ_i .

Proof. Using (F.2), $\frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \|\frac{1}{T} \sum_{s \notin I} (\widehat{f}_s - H_f f_s) e_{is} f'_s\|^2$ is bounded by, up to a multiplier of order $O_P(1)$,

$$\frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \left\| \frac{1}{T} \sum_{t \notin I} \frac{1}{N} \sum_{j=1}^N \lambda_j e_{jt} u_{jt} e_{it} f'_t \right\|^2 + \sum_{d=1}^6 \frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \left\| \frac{1}{T} \sum_{t \notin I} a_{dt} e_{it} f'_t \right\|^2.$$

where a_{dt} is the upper block of A_{dt} . We shall assume $\dim(f_t) = 1$ without loss of generality. The proof for $\frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \|\frac{1}{T} \sum_{t \notin I} a_{1t} e_{it} f'_t\|^2$ is the hardest, and we shall present it last.

Step 1. First term:

Let $\nu_t = (\lambda_1 e_{1t}, \dots, \lambda_N e_{Nt})'$. Then

$$\begin{aligned}
&\mathbb{E} \frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \left\| \frac{1}{T} \sum_{t \notin I} \frac{1}{N} \sum_{j=1}^N \lambda_j e_{jt} u_{jt} e_{it} f'_t \right\|^2 = \frac{1}{N^2 T^2} \frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \sum_{t \notin I} \mathbb{E} (\nu'_t u_t e_{it} f_t)^2 \\
&\leq \frac{1}{N^2 T^2} \frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \sum_{t \notin I} \mathbb{E} f_t^2 \nu'_t \mathbb{E} (u_t u'_t | E, F) \nu_t e_{it}^2 \leq \frac{1}{N^2 T^2} \frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \sum_{t \notin I} \sum_{j=1}^N \mathbb{E} f_t^2 \lambda_j^2 e_{jt}^2 e_{it}^2 \\
&= O(N^{-1} T^{-1}) = O(C_{NT}^{-4}).
\end{aligned}$$

Step 2. Show $\sum_{d=2}^6 \frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \|\frac{1}{T} \sum_{t \notin I} a_{dt} e_{it} f'_t\|^2$.

Up to a multiplier of order $O_P(1)$, by Lemma G.8,

$$\frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \left\| \frac{1}{T} \sum_{t \notin I} a_{2t} e_{it} f'_t \right\|^2$$

$$\begin{aligned}
&\leq \frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \left| \frac{1}{T} \sum_{t \notin I} \|e_{it} f'_t\| \|\widehat{B}_t^{-1} - B^{-1}\| \left[\left\| \frac{1}{N} \sum_j \lambda_j e_{jt} u_{jt} \right\| + \left\| \frac{1}{N} \sum_j \alpha_j u_{jt} \right\| \right] \right|^2 \\
&\leq \frac{1}{T} \sum_{t \notin I} \|\widehat{B}_t - B\|^2 \frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \frac{1}{T} \sum_{t \notin I} \|e_{it} f'_t\|^2 \left[\left\| \frac{1}{N} \sum_j \lambda_j e_{jt} u_{jt} \right\| + \left\| \frac{1}{N} \sum_j \alpha_j u_{jt} \right\| \right]^2 \\
&= O_P(C_{NT}^{-4}) \\
&\quad \frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \left\| \frac{1}{T} \sum_{t \notin I} a_{3t} e_{it} f'_t \right\|^2 \\
&\leq \max_{t \leq T} \|\widehat{B}_t^{-1}\|^2 \frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \left(\frac{1}{T} \sum_{t \notin I} \|e_{it} f'_t\| \left\| \frac{1}{N} \sum_j (\tilde{\lambda}_j \widehat{e}_{jt} - H'_1 \lambda_j e_{jt}) u_{jt} \right\| \right)^2 \\
&\quad + \max_{t \leq T} \|\widehat{B}_t^{-1}\|^2 \frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \left(\frac{1}{T} \sum_{t \notin I} \|e_{it} f'_t\| \left\| \frac{1}{N} \sum_j (\tilde{\alpha}_j - H'_2 \alpha_j) u_{jt} \right\| \right)^2 \\
&\leq O_P(1) \frac{1}{T} \sum_{t \notin I} \left\| \frac{1}{N} \sum_j (\tilde{\lambda}_j - H'_1 \lambda_j) e_{jt} u_{jt} \right\|^2 + O_P(1) \frac{1}{T} \sum_{t \notin I} \left\| \frac{1}{N} \sum_j (\tilde{\alpha}_j - H'_2 \alpha_j) u_{jt} \right\|^2 \\
&\quad + O_P(1) \frac{1}{T} \sum_{t \notin I} \left\| \frac{1}{N} \sum_j (\tilde{\lambda}_j - H'_1 \lambda_j) (\widehat{e}_{jt} - e_{jt}) u_{jt} \right\|^2 \\
&\quad + O_P(1) \frac{1}{T} \sum_{t \notin I} \left\| \frac{1}{N} \sum_j \lambda_j (\widehat{e}_{jt} - e_{jt}) u_{jt} \right\|^2 = O_P(C_{NT}^{-4}).
\end{aligned}$$

The first term is bounded by, for $\omega_{it} = e_{it} u_{it}$,

$$O_P(1) \frac{1}{T} \sum_{t \notin I} \frac{1}{N^2} (\tilde{\Lambda} - \Lambda H_1)' \text{Var}_I(\omega_t) (\tilde{\Lambda} - \Lambda H_1) \leq O_P(1) \frac{1}{N^2} \|\tilde{\Lambda} - \Lambda H_1\|^2 = O_P(C_{NT}^{-4}).$$

The second term $\frac{1}{T} \sum_{t \notin I} \left\| \frac{1}{N} \sum_j (\tilde{\alpha}_j - H'_2 \alpha_j) u_{jt} \right\|^2$ is bounded similarly. The third term follows from Cauchy-Schwarz. The last term follows from Lemma G.2. Next,

$$\frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \left\| \frac{1}{T} \sum_{t \notin I} a_{4t} e_{it} f'_t \right\|^2 \leq \max_{t \leq T} \|\widehat{B}_t^{-1}\|^2 \sum_{d=1}^{16} \frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \left(\frac{1}{T} \sum_{t \notin I} \|e_{it} f'_t\| \|C_{dt}\| \right)^2 = O_P(C_{NT}^{-4}),$$

where C_{dt} 's are defined in the proof of Lemma G.9. Applying the simple Cauchy-Schwarz proves $\frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \left(\frac{1}{T} \sum_{t \notin I} \|e_{it} f'_t\| \|C_{dt}\| \right)^2 = O_P(C_{NT}^{-4})$ for all $d \leq 16$.

Next, for B_{dt} defined in the proof of Lemma G.9, repeatedly use Cauchy-Schwarz,

$$\frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \left\| \frac{1}{T} \sum_{t \notin I} a_{5t} e_{it} f'_t \right\|^2 \leq O_P(C_{NT}^{-2}) \sum_{d=1}^8 \frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \frac{1}{T} \sum_{t \notin I} \|B_{dt} e_{it} f'_t\|^2 = O_P(C_{NT}^{-4}).$$

Also,

$$\frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \left\| \frac{1}{T} \sum_{t \notin I} a_{6t} e_{it} f'_t \right\|^2 \leq \sum_{d=1}^4 \frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \frac{1}{T} \sum_{t \notin I} \|B_{dt} e_{it} f'_t\|^2 = O_P(C_{NT}^{-4}), \quad \text{where}$$

$$\begin{aligned}
\frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \frac{1}{T} \sum_{t \notin I} \|B_{1t} e_{it} f'_t\|^2 &= O_P(C_{NT}^{-4}) \text{ Cauchy-Schwarz} \\
\frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \frac{1}{T} \sum_{t \notin I} \|B_{2t} e_{it} f'_t\|^2 &\leq O_P(1) \frac{1}{T} \sum_{t \notin I} \left| \frac{1}{N} \sum_j \lambda_j^2 e_{jt} (\hat{e}_{jt} - e_{jt}) \right|^2 = O_P(C_{NT}^{-4}) \\
&\quad (\text{Lemma G.2}) \\
\frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \frac{1}{T} \sum_{t \notin I} \|B_{3t} e_{it} f'_t\|^2 &\leq O_P(1) \frac{1}{T} \sum_{t \notin I} \left\| \frac{1}{N} \sum_j l_j \lambda_j e_{jt} (\dot{\lambda}_j - H_1' \lambda_j) \right\|^2 = O_P(C_{NT}^{-4}) \\
&\quad (\text{Lemma G.5}) \\
\frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \frac{1}{T} \sum_{t \notin I} \|B_{4t} e_{it} f'_t\|^2 &\leq O_P(C_{NT}^{-2}) \frac{1}{T} \sum_{t \notin I} \left\| \frac{1}{N} \sum_j \lambda_j^2 l_j e_{jt} \right\|^2 = O_P(C_{NT}^{-4}).
\end{aligned}$$

Step 3. Finally, we bound $\frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \left\| \frac{1}{T} \sum_{t \notin I} a_{1t} e_{it} f'_t \right\|^2$.

Note that

$$A_{1t} = (\hat{B}_t^{-1} \hat{S}_t - B^{-1} S) \begin{pmatrix} H_1^{-1} f_t \\ H_2^{-1} g_t \end{pmatrix} := (A_{1t,a} + A_{1t,b} + A_{1t,c}) \begin{pmatrix} H_1^{-1} f_t \\ H_2^{-1} g_t \end{pmatrix},$$

where

$$\begin{aligned}
A_{1t,a} &= (\hat{B}_t^{-1} - B^{-1})(\hat{S}_t - S) \\
A_{1t,b} &= (\hat{B}_t^{-1} - B^{-1})S \\
A_{1t,c} &= B^{-1}(\hat{S}_t - S).
\end{aligned}$$

Let $(a_{1t,a}, a_{1t,b}, a_{1t,c})$ respectively be the upper blocks of $(A_{1t,a}, A_{1t,b}, A_{1t,c})$. Then

$$\frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \left\| \frac{1}{T} \sum_{t \notin I} a_{1t} e_{it} f'_t \right\|^2 \leq \frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \left(\frac{1}{T} \sum_{t \notin I} (\|a_{1t,a}\| + \|a_{1t,b}\| + \|a_{1t,c}\|) \|e_{it} f_t\| (\|f_t\| + \|g_t\|) \right)^2.$$

By the Cauchy-Schwarz and Lemma G.8, and B is a block diagonal matrix,

$$\begin{aligned}
&\frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \left[\frac{1}{T} \sum_{t \notin I} \|a_{1t,b}\| \|e_{it} f_t\| (\|f_t\| + \|g_t\|) \right]^2 \leq O_P(1) \frac{1}{T} \sum_{t \notin I} \|A_{1t,b}\|^2 \\
&\leq O_P(\|S\|^2) \frac{1}{T} \sum_{t \notin I} \|\hat{B}_t - B\|^2 = O_P(C_{NT}^{-4}). \\
&\frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \left[\frac{1}{T} \sum_{t \notin I} \|a_{1t,c}\| \|e_{it} f_t\| (\|f_t\| + \|g_t\|) \right]^2 \leq O_P(1) \frac{1}{T} \sum_{t \notin I} \|a_{1t,c}\|^2 \\
&\leq O_P(1) \frac{1}{T} \sum_{t \notin I} \|\Delta_{t1}\|^2 = O_P(C_{NT}^{-4}),
\end{aligned}$$

where Δ_{t1} is defined in Lemma G.8, the upper block of $\hat{S}_t - S$.

The treatment of $\frac{1}{T} \sum_{t \notin I} \|a_{1t,a}\| \|e_{it} f_t\| (\|f_t\| + \|g_t\|)$ is slightly different. Note that $\max_{t \leq T} \|\hat{B}_t^{-1}\| + \|B^{-1}\| = O_P(1)$, shown in Lemma G.8. Partition

$$\hat{S}_t - S = \begin{pmatrix} \Delta_{t1} \\ (\Delta_{t2,1} + \Delta_{t2,2}, 0) \end{pmatrix}$$

where the notation Δ_{t1} is defined in the proof of Lemma G.8. The proof of Lemma G.8 also gives

$$\begin{aligned} \Delta_{t2,1} &= \frac{1}{N} \sum_i \tilde{\alpha}_i (\lambda'_i H_1 - \tilde{\lambda}'_i) e_{it} + \frac{1}{N} \sum_i \tilde{\alpha}_i (\lambda'_i H_1 - \tilde{\lambda}'_i) (\hat{e}_{it} - e_{it}) \\ &\quad + \frac{1}{N} \sum_i (\tilde{\alpha}_i - H'_2 \alpha_i) \lambda'_i H_1 (e_{it} - \hat{e}_{it}) \\ \Delta_{t2,2} &= H'_2 \frac{1}{N} \sum_i \alpha_i \lambda'_i (e_{it} - \hat{e}_{it}) H_1. \end{aligned}$$

Therefore,

$$\begin{aligned} \|a_{1t,a}\| &\leq \|(\hat{B}_t^{-1} - B^{-1})(\hat{S}_t - S)\| \\ &\leq (\max_t \|\hat{B}_t^{-1}\| + \|B^{-1}\|)(\|\Delta_{t1}\| + \|\Delta_{t2,1}\|) + \|\hat{B}_t^{-1} - B^{-1}\| \|\Delta_{t2,2}\|. \end{aligned} \tag{G.18}$$

Note that the above bound treats Δ_{t1} and Δ_{t2} differently because by the proof of Lemma G.8, $\frac{1}{T} \sum_{t \notin I} \|\Delta_{t1}\|^2 = O_P(C_{NT}^{-4}) = \frac{1}{T} \sum_{t \notin I} \|\Delta_{t2,1}\|^2$ but the rate of convergence for $\frac{1}{T} \sum_{t \notin I} \|\Delta_{t2,2}\|^2$ is slower ($= O_P(C_{NT}^{-2})$).

Hence

$$\begin{aligned} &\frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \left[\frac{1}{T} \sum_{t \notin I} \|a_{1t,a}\| \|e_{it} f_t\| (\|f_t\| + \|g_t\|) \right]^2 \leq O_P(1) \frac{1}{T} \sum_{t \notin I} \|\Delta_{t1}\|^2 + \|\Delta_{t2,1}\|^2 \\ &\quad + \frac{1}{T} \sum_{t \notin I} \|\hat{B}_t^{-1} - B^{-1}\|^2 \frac{1}{T} \sum_{t \notin I} \|\Delta_{t2,2}\|^2 \frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \|e_{it} f_t\|^2 (\|f_t\| + \|g_t\|)^2 \\ &\leq^{(a)} O_P(C_{NT}^{-4}) + O_P(C_{NT}^{-2}) \frac{1}{T} \sum_{t \notin I} \|\Delta_{t2,2}\|^2 \frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \|e_{it} f_t\|^2 (\|f_t\| + \|g_t\|)^2 \\ &\leq^{(b)} O_P(C_{NT}^{-4}). \end{aligned}$$

where (a) follows from the proof of Lemma G.8, while (b) follows from Lemma G.1. Thus

$$\frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \left\| \frac{1}{T} \sum_{t \notin I} a_{1t} e_{it} f'_t \right\|^2 = O_P(C_{NT}^{-4}).$$

Also note that in the above proof, we have also proved (to be used later)

$$\frac{1}{T} \sum_{t \notin I} [\|a_{1t,b}\|^2 + \|a_{1t,c}\|^2 + \|\Delta_{t1}\|^2 + \|\Delta_{t2,1}\|^2] \leq O_P(C_{NT}^{-4}) \quad (\text{G.19})$$

□

Lemma G.12. for $\mu_{it} = l'_i w_t$, $\widehat{\mu}_{it} = \widehat{l'_i w_t}$,

$$\begin{aligned} \frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \left\| \frac{1}{T} \sum_{s \notin I} \widehat{f}_s \widehat{e}_{is} (\mu_{it} \lambda'_i f_s - \widehat{\mu}_{it} \dot{\lambda}'_i \widetilde{f}_s) \right\|^2 &= O_P(C_{NT}^{-4}) \\ \frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \left\| \frac{1}{T} \sum_{s \notin I} \widehat{g}_s (\mu_{it} \lambda'_i f_s - \widehat{\mu}_{it} \dot{\lambda}'_i \widetilde{f}_s) \right\|^2 &= O_P(C_{NT}^{-2}). \end{aligned}$$

Proof. It suffices to prove that the following statements:

$$\frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \|R_{3i,d}\|^2 = O_P(C_{NT}^{-4}), d = 1 \sim 6, \quad \frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \|R_{3i,7}\|^2 = O_P(C_{NT}^{-2}),$$

where

$$\begin{aligned} R_{3i,1} &:= \frac{1}{T} \sum_{s \notin I} (\widehat{f}_s - H_f f_s) (\widehat{e}_{is} - e_{is}) \mu_{it} \lambda'_i f_s + \frac{1}{T} \sum_{s \notin I} (\widehat{f}_s \widehat{e}_{is} - H_f f_s e_{is}) (\mu_{it} \lambda'_i f_s - \widehat{\mu}_{it} \dot{\lambda}'_i \widetilde{f}_s) \\ R_{3i,2} &:= \frac{1}{T} \sum_{s \notin I} f_s e_{is} (\widehat{e}_{is} - e_{is}) (\dot{\lambda}'_i \widetilde{f}_s - \lambda'_i f_s) + \frac{1}{T} \sum_{s \notin I} f_s e_{is} \mu_{is} (\dot{\lambda}_i - H'_1 \lambda_i)' (\widetilde{f}_s - H_1^{-1} f_s) \\ R_{3i,3} &:= \frac{1}{T} \sum_{s \notin I} f_s e_{is} \mu_{is} f'_s H_1^{-1'} (\dot{\lambda}_i - H'_1 \lambda_i) \\ R_{3i,4} &:= \frac{1}{T} \sum_{s \notin I} f_s e_{is} (\widehat{e}_{is} - e_{is}) \lambda'_i f_s \\ R_{3i,5} &:= \frac{1}{T} \sum_{s \notin I} f_s e_{is} \mu_{is} \lambda'_i (\widetilde{f}_s - H_1^{-1} f_s) \\ R_{3i,6} &:= \frac{1}{T} \sum_{s \notin I} (\widehat{f}_s - H_f f_s) e_{is} \mu_{is} \lambda'_i f_s \\ R_{3i,7} &:= \frac{1}{T} \sum_{s \notin I} \widehat{g}_s (\mu_{is} \lambda'_i f_s - \widehat{\mu}_{is} \dot{\lambda}'_i \widetilde{f}_s). \end{aligned} \quad (\text{G.20})$$

We have

$$\begin{aligned} \frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \|R_{3i,d}\|^2 &= O_P(C_{NT}^{-4}), \quad d = 1, 2, 3, \text{ by Cauchy-Schwarz} \\ \frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \|R_{3i,4}\|^2 &= O_P(C_{NT}^{-4}), \text{ Lemma G.2} \end{aligned}$$

$$\begin{aligned}
\frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \|R_{3i,5}\|^2 &= O_P(C_{NT}^{-4}), \text{ Lemma G.6} \\
\frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \|R_{3i,6}\|^2 &= O_P(C_{NT}^{-4}), \text{ Lemma G.11} \\
\frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \|R_{3i,7}\|^2 &= O_P(C_{NT}^{-2}), \text{ by Cauchy-Schwarz.}
\end{aligned}$$

□

Lemma G.13. *Let r_{5i} be the upper block of R_{5i} , where*

$$R_{5i} = \hat{D}_i^{-1} \frac{1}{T_0} \sum_{s \notin I} \begin{pmatrix} \hat{e}_{is} & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \hat{f}_s - H_f f_s \\ \hat{g}_s - H_g g_s \end{pmatrix} u_{is}.$$

Then $\frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \|r_{5i}\| = O_P(C_{NT}^{-2})$.

Proof. For any matrix A of the same size R_{5i} , we use $\mathcal{U}(A)$ to denote its upper block (the first $\dim(f_t)$ -submatrix of A). Then, in this notation, $r_{5i} = \mathcal{U}(R_{5i})$.

First, by Cauchy-Schwarz

$$\begin{aligned}
& \left[\frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \|\hat{D}_i^{-1} - D_i^{-1}\| \left(\left\| \frac{1}{T} \sum_{s \notin I} \hat{e}_{is} (\hat{f}_s - H_f f_s) u_{is} \right\| + \left\| \frac{1}{T} \sum_{s \notin I} (\hat{g}_s - H_g g_s) u_{is} \right\| \right) \right]^2 \\
& \leq O_P(C_{NT}^{-2}) \frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \left(\frac{1}{T} \sum_{s \notin I} (\hat{f}_s - H_f f_s)^2 \frac{1}{T} \sum_{s \notin I} u_{is}^2 \hat{e}_{is}^2 + \frac{1}{T} \sum_{s \notin I} (\hat{g}_s - H_g g_s)^2 \frac{1}{T} \sum_{s \notin I} u_{is}^2 \right) \\
& \leq O_P(C_{NT}^{-4}).
\end{aligned}$$

Now let $\bar{R}_{5i}$ be defined as R_{5i} but with $\hat{D}_i^{-1}$ replaced with D_i^{-1} . Substitute in the expansion (F.2), then

$$\bar{R}_{5i} = D_i^{-1} \frac{1}{T_0} \sum_{t \notin I} \begin{pmatrix} \hat{e}_{it} & 0 \\ 0 & 1 \end{pmatrix} B^{-1} \frac{1}{N} \sum_j \begin{pmatrix} H'_1 \lambda_j e_{jt} \\ H'_2 \alpha_j \end{pmatrix} u_{jt} u_{it} + \sum_{d=1}^6 D_i^{-1} \frac{1}{T_0} \sum_{t \notin I} \begin{pmatrix} \hat{e}_{it} & 0 \\ 0 & 1 \end{pmatrix} A_{dt} u_{it}.$$

Both B^{-1} and D_i^{-1} are block diagonal; let b, d_i respectively denote their first diagonal block. Then

$$\mathcal{U}(\bar{R}_{5i}) = \frac{1}{T_0} \sum_{t \neq I} d_i \hat{e}_{it} b H'_1 \frac{1}{N} \sum_j \lambda_j e_{jt} u_{jt} u_{it} + \sum_{d=1}^6 d_i \frac{1}{T_0} \sum_{t \neq I} \hat{e}_{it} u_{it} \mathcal{U}(A_{dt}).$$

The goal is now to prove $\frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \|\mathcal{U}(\bar{R}_{5i})\| = O_P(C_{NT}^{-2})$.

First, let $\nu_t = (\lambda_j u_{jt} : j \leq N)$, a column vector.

$$\begin{aligned}
& \left[\frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \left\| \frac{1}{T_0} \sum_{t \neq I} d_i (\hat{e}_{it} - e_{it}) b H_1' \frac{1}{N} \sum_j \lambda_j e_{jt} u_{jt} u_{it} \right\| \right]^2 \\
& \leq O_P(C_{NT}^{-2}) \frac{1}{T_0} \sum_{t \neq I} \frac{1}{N^2} \mathbb{E} \text{Var}(\nu_t' e_t u_{it} | U) \leq O_P(C_{NT}^{-2}) \frac{1}{T_0} \sum_{t \neq I} \frac{1}{N^2} \mathbb{E} u_{it}^2 \nu_t' \text{Var}(e_t | U) \nu_t \\
& \leq O_P(C_{NT}^{-2} N^{-1}) \frac{1}{T_0} \sum_{t \neq I} \mathbb{E} u_{it}^2 \frac{1}{N} \sum_{j=1}^N \lambda_j^2 u_{jt}^2 = O_P(C_{NT}^{-4}).
\end{aligned}$$

Next, for $\omega_{it} = e_{it} u_{it}$,

$$\begin{aligned}
& \left[\frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \left\| \frac{1}{T_0} \sum_{t \neq I} d_i e_{it} b H_1' \frac{1}{N} \sum_j \lambda_j e_{jt} u_{jt} u_{it} \right\| \right]^2 \\
& \leq O_P(1) \frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \mathbb{E} \left| \frac{1}{NT} \sum_t \sum_j e_{it} \lambda_j e_{jt} u_{jt} u_{it} \right|^2 \\
& \leq O_P(1) \frac{1}{N^2} (\max_i \sum_j |\mathbb{E} \omega_{jt} \omega_{it}|)^2 + O_P(1) \frac{1}{N^2 T^2} \frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \sum_t \text{Var}(\sum_j \lambda_j \omega_{jt} \omega_{it}) \\
& \leq O_P(N^{-2}) + O_P(1) \frac{1}{N^2 T^2} \frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \sum_t \sum_{jk} |\text{Cov}(\omega_{jt} \omega_{it}, \omega_{kt} \omega_{it})| \\
& = O_P(N^{-2} + N^{-1} T^{-1}) = O_P(C_{NT}^{-4}).
\end{aligned}$$

We now show $\sum_{d=1}^6 \frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \|d_i \frac{1}{T_0} \sum_{t \neq I} \hat{e}_{it} u_{it} \mathcal{U}(A_{dt})\| = O_P(C_{NT}^{-2})$. For each $d \leq 6$,

$$\begin{aligned}
& \left[\frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \left\| d_i \frac{1}{T_0} \sum_{t \neq I} (\hat{e}_{it} - e_{it}) u_{it} \mathcal{U}(A_{dt}) \right\| \right]^2 \\
& \leq \frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \frac{1}{T_0} \sum_{t \neq I} (\hat{e}_{it} - e_{it})^2 u_{it}^2 d_i^2 \frac{1}{T_0} \sum_{t \neq I} \|A_{dt}\|^2 = O_P(C_{NT}^{-4}),
\end{aligned}$$

where it follows from applying Cauchz-Schwarz that $\frac{1}{T_0} \sum_{t \neq I} \|A_{dt}\|^2 = O_P(C_{NT}^{-2})$.

It remains to prove $\sum_{d=1}^6 \frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \|d_i \frac{1}{T_0} \sum_{t \neq I} \omega_{it} \mathcal{U}(A_{dt})\| = O_P(C_{NT}^{-2})$.

Term $\mathcal{U}(A_{1t})$: Using the notation of the proof of Lemma G.11, we have

$$\mathcal{U}(A_{1t}) := (a_{1t,a} + a_{1t,b} + a_{1t,c}) \begin{pmatrix} H_1^{-1} f_t \\ H_2^{-1} g_t \end{pmatrix},$$

Also, (G.18) and (G.19) yield, for $m_{it} := \|d_i \omega_{it}\|(\|f_t\| + \|g_t\|)$,

$$\begin{aligned}
& \frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \|d_i \frac{1}{T_0} \sum_{t \neq I} \omega_{it} \mathcal{U}(A_{dt})\| \\
& \leq \left[\frac{1}{T_0} \sum_{t \neq I} \|a_{1t,b} + a_{1t,c}\|^2 \right]^{1/2} \left[\frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \frac{1}{T_0} \sum_{t \neq I} m_{it}^2 \right]^{1/2} + \frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \frac{1}{T_0} \sum_{t \neq I} \|m_{it} a_{1t,a}\| \\
& \leq O_P(C_{NT}^{-2}) + \left[\frac{1}{T_0} \sum_{t \neq I} \|\Delta_{t1} + \Delta_{t2,1}\|^2 \right]^{1/2} \left[\frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \frac{1}{T_0} \sum_{t \neq I} m_{it}^2 \right]^{1/2} \\
& \quad + \frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \frac{1}{T_0} \sum_{t \neq I} m_{it} \|\hat{B}_t^{-1} - B^{-1}\| \|\Delta_{t2,2}\| \\
& \leq O_P(C_{NT}^{-2}) + O_P(C_{NT}^{-1}) \left[\frac{1}{T} \sum_t \|\Delta_{t2,2}\|^4 \right]^{1/4} \\
& \leq O_P(C_{NT}^{-2}) + O_P(C_{NT}^{-1}) \left[\frac{1}{T} \sum_t \frac{1}{N} \sum_i (e_{it} - \hat{e}_{it})^4 \right]^{1/4} = O_P(C_{NT}^{-2}).
\end{aligned}$$

Terms $\mathcal{U}(A_{2t}), \mathcal{U}(A_{4t}), \mathcal{U}(A_{5t})$: $\frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \|d_i \frac{1}{T_0} \sum_{t \neq I} \omega_{it} \mathcal{U}(A_{dt})\| = O_P(C_{NT}^{-2})$,
 $d = 2, 4, 5$ follow from the simple Cauchy-Schwarz.

Term $\mathcal{U}(A_{3t})$: Because $\max_{t \leq T} \|\hat{B}_t^{-1}\| = O_P(1)$,

$$\begin{aligned}
& \frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \|d_i \frac{1}{T_0} \sum_{t \neq I} \omega_{it} \mathcal{U}(A_{3t})\| \leq \frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \|d_i \frac{1}{T_0} \sum_{t \neq I} \omega_{it} A_{3t}\| \\
& \leq \left[\frac{1}{T_0} \sum_{t \neq I} \left\| \frac{1}{N} \sum_j \lambda_i u_{jt} (\hat{e}_{jt} - e_{jt}) \right\|^2 \right]^{1/2} + \left[\frac{1}{T_0} \sum_{t \neq I} \left\| \frac{1}{N} \sum_j u_{jt} (\tilde{\lambda}_j - H'_1 \lambda_j) (\hat{e}_{jt} - e_{jt}) \right\|^2 \right]^{1/2} \\
& \quad + \left[\frac{1}{T_0} \sum_{t \neq I} \left\| \frac{1}{N} \sum_j \omega_{jt} (\tilde{\lambda}_j - H'_1 \lambda_j) \right\|^2 \right]^{1/2} + \left[\frac{1}{T_0} \sum_{t \neq I} \left\| \frac{1}{N} \sum_j u_{jt} (\tilde{\alpha}_j - H'_2 \alpha_j) \right\|^2 \right]^{1/2} \\
& = O_P(C_{NT}^{-2}).
\end{aligned}$$

The first term follows from Lemma G.2. The second term follows from Cauchy-Schwarz. The third and fourth are due to, for instance,

$$\begin{aligned}
& \frac{1}{T_0} \sum_{t \neq I} \mathbb{E}_I \left\| \frac{1}{N} \sum_j \omega_{jt} (\tilde{\lambda}_j - H'_1 \lambda_j) \right\|^2 \\
& \leq \frac{1}{T_0} \sum_{t \neq I} \frac{1}{N^2} \text{Var}_I(\omega'_t (\tilde{\Lambda} - \Lambda H_1)) \leq \frac{1}{T_0} \sum_{t \neq I} \frac{1}{N^2} \|\tilde{\Lambda} - \Lambda H_1\|^2 \|\text{Var}_I(\omega_t)\| \\
& \leq O(C_{NT}^{-4}).
\end{aligned}$$

Term $\mathcal{U}(A_{6t})$: Note that B^{-1} is block diagonal. Let b be the first block of B^{-1} . Recall that Lemma G.9 shows

$$\mathcal{U}(A_{6t}) = bH'_1 \sum_{d=1}^4 B_{dt}.$$

So $\frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \|d_i \frac{1}{T_0} \sum_{t \neq I} \omega_{it} \mathcal{U}(A_{6t})\| \leq O_P(1) \sum_{d=1}^4 \frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \|d_i \frac{1}{T_0} \sum_{t \neq I} \omega_{it} B_{dt}\|$.

For $d = 1, 4$, we apply Cauchy-Schwarz,

$$\frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \|d_i \frac{1}{T_0} \sum_{t \neq I} \omega_{it} B_{dt}\| \leq O_P(1) \left[\frac{1}{T} \sum_t \|B_{dt}\|^2 \right]^{1/2} = O_P(C_{NT}^{-2}).$$

For $d = 2$, we still apply Cauchy-Schwarz and Lemma G.2 that

$$\begin{aligned} \frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \|d_i \frac{1}{T_0} \sum_{t \neq I} \omega_{it} B_{2t}\| &\leq O_P(C_{NT}^{-2}) + O_P(1) \left[\frac{1}{T} \sum_t \left(\frac{1}{N} \sum_i \lambda_i^2 e_{it} (\hat{e}_{it} - e_{it}) \right)^2 \right]^{1/2} \\ &= O_P(C_{NT}^{-2}). \end{aligned}$$

Finally, we bound for $d = 3$.

$$\begin{aligned} \left[\frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \|d_i \frac{1}{T_0} \sum_{t \neq I} \omega_{it} B_{3t}\| \right]^2 &\leq O_P(1) \frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \left\| \frac{1}{T_0} \sum_{t \neq I} \omega_{it} B_{3t} \right\|^2 \\ &\leq O_P(1) \frac{1}{|\mathcal{G}|_0} \sum_{j \in \mathcal{G}} \left\| \frac{1}{T_0} \sum_{t \neq I} \omega_{jt} \frac{1}{N} \sum_i l_i \lambda_i e_{it} w_t (\dot{\lambda}_i - H'_1 \lambda_i)' \tilde{f}_t \right\|^2 \\ &\leq O_P(C_{NT}^{-2}) \frac{1}{|\mathcal{G}|_0} \sum_{j \in \mathcal{G}} \frac{1}{N} \sum_i \left\| \frac{1}{T_0} \sum_{t \neq I} \omega_{jt} e_{it} w_t (\tilde{f}_t - H_1^{-1} f_t) \right\|^2 \\ &\quad + O_P(C_{NT}^{-2}) \frac{1}{|\mathcal{G}|_0} \sum_{j \in \mathcal{G}} \frac{1}{N} \sum_i \left\| \frac{1}{T_0} \sum_{t \neq I} \omega_{jt} e_{it} w_t f_t \right\|^2 \\ &= O_P(C_{NT}^{-4}) + O_P(C_{NT}^{-2}) \frac{1}{|\mathcal{G}|_0} \sum_{j \in \mathcal{G}} \frac{1}{N} \sum_i \left\| \frac{1}{T_0} \sum_{t \neq I} e_{jt} u_{jt} e_{it} w_t f_t \right\|^2 \\ &= O_P(C_{NT}^{-4}). \end{aligned}$$

The last equality is due to the conditional serial independence of (e_t, u_t) and that $E(u_t | E, F, W) = 0$.

□

Lemma G.14. Let r_{6i} be the upper block of R_{6i} , where

$$R_{6i} = \hat{D}_i^{-1} \frac{1}{T_0} \sum_{s \notin I} \begin{pmatrix} \hat{f}_s \hat{e}_{is} \\ \hat{g}_s \end{pmatrix} (\lambda'_i H_f^{-1}, \alpha'_i H_g^{-1}) \begin{pmatrix} \hat{e}_{is} & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \hat{f}_s - H_f f_s \\ \hat{g}_s - H_g g_s \end{pmatrix}.$$

Then $\frac{1}{|\mathcal{G}|_0} \sum_{i \in \mathcal{G}} \|r_{6i}\| = O_P(C_{NT}^{-2})$.

Proof. Write $B^{-1} = \text{diag}(b_1, b_2)$,

$$\begin{aligned}\Gamma^j &:= \Gamma_0^j + \Gamma_1^j + \dots + \Gamma_6^j, \\ \Gamma_0^j &:= \frac{1}{T_0} \sum_{t \notin I} \begin{pmatrix} \widehat{f}_t \widehat{e}_{jt} \\ \widehat{g}_t \end{pmatrix} (\lambda'_j H_f^{-1} \widehat{e}_{jt} b_1, \alpha'_j H_g^{-1} b_2) \frac{1}{N} \sum_i \begin{pmatrix} H'_1 \lambda_i e_{it} \\ H'_2 \alpha_i \end{pmatrix} u_{it} \\ \Gamma_d^j &:= \frac{1}{T_0} \sum_{t \notin I} \begin{pmatrix} \widehat{f}_t \widehat{e}_{jt} \\ \widehat{g}_t \end{pmatrix} (\lambda'_j H_f^{-1} \widehat{e}_{jt}, \alpha'_j H_g^{-1}) A_{dt}, \quad d = 1, \dots, 6.\end{aligned}$$

Then $R_{6j} = \widehat{D}_j^{-1} \Gamma^j$. Let d_j be the first diagonal block of D_j^{-1} (which is a block diagonal matrix). We have

$$\begin{aligned}[\frac{1}{|\mathcal{G}|_0} \sum_{j \in \mathcal{G}} \|r_{6j}\|]^2 &\leq \frac{1}{|\mathcal{G}|_0} \sum_{j \in \mathcal{G}} \|\widehat{D}_j^{-1} - D_j^{-1}\|^2 \frac{1}{|\mathcal{G}|_0} \sum_{j \in \mathcal{G}} \|\Gamma^j\|^2 + [\frac{1}{|\mathcal{G}|_0} \sum_{j \in \mathcal{G}} \|d_j \mathcal{U}(\Gamma^j)\|]^2 \\ &\leq O_P(C_{NT}^{-2}) \frac{1}{|\mathcal{G}|_0} \sum_{j \in \mathcal{G}} \|\Gamma^j\|^2 + \frac{1}{|\mathcal{G}|_0} \sum_{j \in \mathcal{G}} \|\mathcal{U}(\Gamma^j)\|^2 \\ &\leq O_P(1) \sum_{d=2}^5 \frac{1}{|\mathcal{G}|_0} \sum_{j \in \mathcal{G}} \|\Gamma_d^j\|^2 + O_P(1) \frac{1}{|\mathcal{G}|_0} \sum_{j \in \mathcal{G}} \|\Gamma_0^j\|^2 \\ &\quad + O_P(C_{NT}^{-2}) \frac{1}{|\mathcal{G}|_0} \sum_{j \in \mathcal{G}} \|\Gamma_1^j\|^2 + \frac{1}{|\mathcal{G}|_0} \sum_{j \in \mathcal{G}} \|\mathcal{U}(\Gamma_1^j)\|^2 \\ &\quad + O_P(C_{NT}^{-2}) \frac{1}{|\mathcal{G}|_0} \sum_{j \in \mathcal{G}} \|\Gamma_6^j\|^2 + \frac{1}{|\mathcal{G}|_0} \sum_{j \in \mathcal{G}} \|\mathcal{U}(\Gamma_6^j)\|^2.\end{aligned}$$

In the above inequalities, we treat each Γ_d^j differently. This is because we aim to show the following:

$$\begin{aligned}\frac{1}{|\mathcal{G}|_0} \sum_{j \in \mathcal{G}} \|\Gamma_d^j\|^2 &= O_P(C_{NT}^{-4}), \quad d = 0, 2, 3, 4, 5, \\ \frac{1}{|\mathcal{G}|_0} \sum_{j \in \mathcal{G}} \|\Gamma_d^j\|^2 &= O_P(C_{NT}^{-2}), \quad d = 1, 6, \\ \frac{1}{|\mathcal{G}|_0} \sum_{j \in \mathcal{G}} \|\mathcal{U}(\Gamma_d^j)\|^2 &= O_P(C_{NT}^{-4}), \quad d = 1, 6.\end{aligned}$$

That is, while proving $\frac{1}{|\mathcal{G}|_0} \sum_{j \in \mathcal{G}} \|\Gamma_d^j\|^2$ are $O_P(C_{NT}^{-4})$ is unlikely for $d = 1, 6$, it can be proved that their upper blocks are. Here we use $\mathcal{U}(A)$ to denote the upper block (the first $\dim(f_t)$ -submatrix of A).

Step 1: $\frac{1}{|\mathcal{G}|_0} \sum_{j \in \mathcal{G}} \|\Gamma_d^j\|^2$ for $d = 0, 2 \sim 5$. Let

$$m_t = \max_{jt} |\widehat{e}_{jt}^4 - e_{jt}^4| + \frac{1}{|\mathcal{G}|_0} \sum_{j \in \mathcal{G}} e_{jt}^4 + f_t^2 + g_t^2 + \max_{jt} |\widehat{e}_{jt}^2 - e_{jt}^2| + \frac{1}{|\mathcal{G}|_0} \sum_{j \in \mathcal{G}} e_{jt}^2 + 1.$$

$$\begin{aligned} \frac{1}{|\mathcal{G}|_0} \sum_{j \in \mathcal{G}} \|\Gamma_0^j\|^2 &\leq \frac{1}{|\mathcal{G}|_0} \sum_{j \in \mathcal{G}} \left\| \frac{1}{T} \sum_{t \notin I} \widehat{f}_t \frac{1}{N} \sum_i \widehat{e}_{jt}^2 u_{it} \lambda_j \lambda_i e_{it} \right\|^2 + \frac{1}{|\mathcal{G}|_0} \sum_{j \in \mathcal{G}} \left\| \frac{1}{T} \sum_{t \notin I} \widehat{f}_t \frac{1}{N} \sum_i \widehat{e}_{jt} u_{it} \alpha_j \alpha_i \right\|^2 \\ &\quad + \frac{1}{|\mathcal{G}|_0} \sum_{j \in \mathcal{G}} \left\| \frac{1}{T} \sum_{t \notin I} \widehat{g}_t \frac{1}{N} \sum_i u_{it} \lambda_j \lambda_i \widehat{e}_{jt} e_{it} \right\|^2 + \frac{1}{|\mathcal{G}|_0} \sum_{j \in \mathcal{G}} \left\| \frac{1}{T} \sum_{t \notin I} \widehat{g}_t \frac{1}{N} \sum_i u_{it} \alpha_j \alpha_i \right\|^2 \\ &\leq O_P(C_{NT}^{-2}) \frac{1}{T} \sum_{t \notin I} m_t \left\| \frac{1}{N} \sum_i u_{it} \lambda_i e_{it} \right\|^2 + O_P(C_{NT}^{-2}) \frac{1}{T} \sum_{t \notin I} m_t \left\| \frac{1}{N} \sum_i u_{it} \alpha_i \right\|^2 \\ &\quad + \frac{1}{|\mathcal{G}|_0} \sum_{j \in \mathcal{G}} \left\| \frac{1}{T} \sum_{t \notin I} g_t \frac{1}{N} \sum_i u_{it} \lambda_j \lambda_i e_{jt} e_{it} \right\|^2 + \frac{1}{|\mathcal{G}|_0} \sum_{j \in \mathcal{G}} \left\| \frac{1}{T} \sum_{t \notin I} g_t \frac{1}{N} \sum_i u_{it} \alpha_j \alpha_i \right\|^2 \\ &\quad + \frac{1}{|\mathcal{G}|_0} \sum_{j \in \mathcal{G}} \left\| \frac{1}{T} \sum_{t \notin I} e_{jt}^2 f_t \frac{1}{N} \sum_i u_{it} \lambda_j \lambda_i e_{it} \right\|^2 + \frac{1}{|\mathcal{G}|_0} \sum_{j \in \mathcal{G}} \left\| \frac{1}{T} \sum_{t \notin I} e_{jt} f_t \frac{1}{N} \sum_i u_{it} \alpha_j \alpha_i \right\|^2 \\ &\leq O_P(C_{NT}^{-4}). \end{aligned}$$

Now let

$$\mathcal{D} = \frac{1}{T_0} \sum_{t \in I^c} \|\widehat{f}_t - H_f f_t\|^2 + \|\widehat{g}_t - H_g g_t\|^2 = O_P(C_{NT}^{-2}).$$

$$\begin{aligned} &\frac{1}{|\mathcal{G}|_0} \sum_{j \in \mathcal{G}} \|\Gamma_2^j\|^2 \\ &= \frac{1}{|\mathcal{G}|_0} \sum_{j \in \mathcal{G}} \left\| \frac{1}{T_0} \sum_{t \in I^c} \begin{pmatrix} \widehat{f}_t \widehat{e}_{jt} \\ \widehat{g}_t \end{pmatrix} (\lambda'_j H_f^{-1} \widehat{e}_{jt}, \alpha'_j H_g^{-1}) (\widehat{B}_t^{-1} - B^{-1}) \frac{1}{N} \sum_i \begin{pmatrix} H'_1 \lambda_i e_{it} \\ H'_2 \alpha_i \end{pmatrix} u_{it} \right\|^2 \\ &\leq \frac{1}{|\mathcal{G}|_0} \sum_{j \in \mathcal{G}} \mathcal{D} \frac{1}{T_0} \sum_{t \in I^c} (|\widehat{e}_{jt}|^2 + 1)^2 \left\| \frac{1}{N} \sum_i \begin{pmatrix} \lambda_i e_{it} \\ \alpha_i \end{pmatrix} u_{it} \right\|^2 \max_{t \leq T} \|\widehat{B}_t^{-1} - B^{-1}\| \\ &\quad + \frac{1}{T_0} \sum_{t \in I^c} \|\widehat{B}_t^{-1} - B^{-1}\|^2 \frac{1}{|\mathcal{G}|_0} \sum_{j \in \mathcal{G}} \frac{1}{T_0} \sum_{t \in I^c} \left\| \begin{pmatrix} \widehat{f}_t \widehat{e}_{jt} \\ g_t \end{pmatrix} \right\|^2 (|\widehat{e}_{jt}| + 1)^2 \left\| \frac{1}{N} \sum_i \begin{pmatrix} \lambda_i e_{it} \\ \alpha_i \end{pmatrix} u_{it} \right\|^2 \\ &\leq O_P(C_{NT}^{-2}) \frac{1}{T_0} \sum_{t \in I^c} n_t \left\| \frac{1}{N} \sum_i \begin{pmatrix} \lambda_i e_{it} \\ \alpha_i \end{pmatrix} u_{it} \right\|^2 = O_P(C_{NT}^{-4}), \end{aligned}$$

where

$$n_t = \max_{t \leq T} \|\widehat{B}_t^{-1} - B^{-1}\| + \frac{1}{|\mathcal{G}|_0} \sum_{j \in \mathcal{G}} \frac{1}{T_0} \sum_{t \in I^c} (|e_{jt}| + 1 + \max_{jt} |e_{jt} - \widehat{e}_{jt}|)^2 [\|f_t\|^2 + \|g_t\|^2 + \|f_t e_{jt}\|^2].$$

Next, let

$$\mathcal{A}_t = \frac{1}{N} \sum_i \begin{pmatrix} \tilde{\lambda}_i \hat{e}_{it} - H'_1 \lambda_i e_{it} \\ \tilde{\alpha}_i - H'_2 \alpha_i \end{pmatrix} u_{it}.$$

$$\begin{aligned} \frac{1}{|\mathcal{G}|_0} \sum_{j \in \mathcal{G}} \|\Gamma_3^j\|^2 &= \frac{1}{|\mathcal{G}|_0} \sum_{j \in \mathcal{G}} \left\| \frac{1}{T_0} \sum_{t \in I^c} \begin{pmatrix} \hat{f}_t \hat{e}_{jt} \\ \hat{g}_t \end{pmatrix} (\lambda'_j H_f^{-1} \hat{e}_{jt}, \alpha'_j H_g^{-1}) \hat{B}_t^{-1} \mathcal{A}_t \right\|^2 \\ &\leq \frac{1}{T_0} \sum_{t \in I^c} \mathcal{D} \frac{1}{|\mathcal{G}|_0} \sum_{j \in \mathcal{G}} (e_{jt}^2 + 1)^2 \|\mathcal{A}_t\|^2 + O_P(1) \frac{1}{T_0} \sum_{t \in I^c} \|\mathcal{A}_t\|^2 \\ &\leq O_P(C_{NT}^{-4}) + \frac{1}{T_0} \sum_{t \in I^c} \left\| \frac{1}{N} \sum_i (\tilde{\alpha}_i - H'_2 \alpha_i) u_{it} \right\|^2 \\ &\quad + \frac{1}{T_0} \sum_{t \in I^c} \left\| \frac{1}{N} \sum_i (\tilde{\lambda}_i - H'_1 \lambda_i) e_{it} u_{it} \right\|^2 + \frac{1}{T_0} \sum_{t \in I^c} \left\| \frac{1}{N} \sum_i \lambda_i (\hat{e}_{it} - e_{it}) u_{it} \right\|^2 \\ &\leq O_P(C_{NT}^{-4}), \end{aligned}$$

where the last equality is due to Lemma G.2 and :

$$\begin{aligned} &\frac{1}{T_0} \sum_{t \in I^c} \mathbb{E} \left\| \frac{1}{N} \sum_i (\tilde{\lambda}_i - H'_1 \lambda_i) e_{it} u_{it} \right\|^2 |E, I| \\ &= \frac{1}{T_0} \sum_{t \in I^c} \frac{1}{N^2} (\tilde{\Lambda} - \Lambda H_1)' \text{diag}(e_t) \text{Var}(u_t | E, I) \text{diag}(e_t) (\tilde{\Lambda} - \Lambda H_1) \\ &\leq O_P(1) \frac{1}{N^2} \sum_i (\tilde{\lambda}_i - H'_1 \lambda_i)^2 \frac{1}{T_0} \sum_{t \in I^c} \mathbb{E}_I e_{it}^2 = O_P(C_{NT}^{-4}). \end{aligned}$$

Next, for C_{dt} defined in the proof of Lemma G.9,

$$\begin{aligned} \frac{1}{|\mathcal{G}|_0} \sum_{j \in \mathcal{G}} \|\Gamma_4^j\|^2 &= \sum_{d=1}^{16} \frac{1}{|\mathcal{G}|_0} \sum_{j \in \mathcal{G}} \left\| \frac{1}{T_0} \sum_{t \in I^c} \begin{pmatrix} \hat{f}_t \hat{e}_{jt} \\ \hat{g}_t \end{pmatrix} (\lambda'_j H_f^{-1} \hat{e}_{jt}, \alpha'_j H_g^{-1}) \hat{B}_t^{-1} C_{dt} \right\|^2 \\ &\leq \sum_{d=1}^{16} \mathcal{D} \frac{1}{|\mathcal{G}|_0} \sum_{j \in \mathcal{G}} \frac{1}{T_0} \sum_{t \in I^c} |e_{jt}|^4 \|C_{dt}\|^2 + O_P(1) \sum_{d=1}^{16} \frac{1}{T_0} \sum_{t \in I^c} \|C_{dt}\|^2 \\ &= O_P(C_{NT}^{-4}). \end{aligned}$$

For B_{dt} defined in the proof of Lemma G.9, and

$$n_t = \|f_t\|^2 \frac{1}{|\mathcal{G}|_0} \sum_{j \in \mathcal{G}} (e_{jt}^2 + |e_{jt}|)^2 + \|g_t\|^2 \frac{1}{|\mathcal{G}|_0} \sum_{j \in \mathcal{G}} (1 + |e_{jt}|)^2 + \frac{1}{|\mathcal{G}|_0} \sum_{j \in \mathcal{G}} (|e_{jt}|^2 + 1)^2,$$

$$\frac{1}{|\mathcal{G}|_0} \sum_{j \in \mathcal{G}} \|\Gamma_5^j\|^2 = \frac{1}{|\mathcal{G}|_0} \sum_{j \in \mathcal{G}} \left\| \sum_{d=1}^8 \frac{1}{T_0} \sum_{t \in I^c} \begin{pmatrix} \hat{f}_t \hat{e}_{jt} \\ \hat{g}_t \end{pmatrix} (\lambda'_j H_f^{-1} \hat{e}_{jt}, \alpha'_j H_g^{-1}) (\hat{B}_t^{-1} - B^{-1}) B_{dt} \right\|^2$$

$$\begin{aligned}
&\leq [\max_{t \leq T} \|\hat{B}_t^{-1} - B^{-1}\| \mathcal{D} + O_P(C_{NT}^{-2})] \sum_{d=1}^8 \frac{1}{T_0} \sum_{t \in I^c} n_t \|B_{dt}\|^2 \\
&\leq O_P(C_{NT}^{-2}) \sum_{d=1}^8 \frac{1}{T_0} \sum_{t \in I^c} n_t \|B_{dt}\|^2 = O_P(C_{NT}^{-4}),
\end{aligned}$$

where the last equality follows from simply applying Cauchy-Schwarz to show $\frac{1}{T_0} \sum_{t \in I^c} n_t \|B_{dt}\|^2 = O_P(C_{NT}^{-2})$ for $d = 1 \sim 8$.

Step 2: $\frac{1}{|\mathcal{G}|_0} \sum_{j \in \mathcal{G}} \|\Gamma_d^j\|^2$ for $d = 1, 6$.

By Lemma G.8, $\max_{t \leq T} [\|\hat{B}_t^{-1} + \|\hat{S}_t\|] = O_P(1)$, $\frac{1}{T} \sum_t \|\hat{B}_t^{-1} \hat{S}_t - B^{-1} S\|^2 = O_P(C_{NT}^{-2})$.

$$\begin{aligned}
&\frac{1}{|\mathcal{G}|_0} \sum_{j \in \mathcal{G}} \|\Gamma_1^j\|^2 \\
&= \frac{1}{|\mathcal{G}|_0} \sum_{j \in \mathcal{G}} \left\| \frac{1}{T_0} \sum_{t \in I^c} \begin{pmatrix} \hat{f}_t \hat{e}_{jt} \\ \hat{g}_t \end{pmatrix} (\lambda'_j H_f^{-1} \hat{e}_{jt}, \alpha'_j H_g^{-1}) (\hat{B}_t^{-1} \hat{S}_t - B^{-1} S) \begin{pmatrix} H_1^{-1} f_t \\ H_2^{-1} g_t \end{pmatrix} \right\|^2 \\
&\leq O_P(C_{NT}^{-2}) \frac{1}{|\mathcal{G}|_0} \sum_{j \in \mathcal{G}} \frac{1}{T_0} \sum_{t \in I^c} |\hat{e}_{jt}|^2 (|\hat{e}_{jt}|^2 + 1) \left\| \begin{pmatrix} H_1^{-1} f_t \\ H_2^{-1} g_t \end{pmatrix} \right\|^2 \\
&\quad + \frac{1}{T} \sum_t \|(\hat{B}_t^{-1} \hat{S}_t - B^{-1} S)\|^2 \frac{1}{|\mathcal{G}|_0} \sum_{j \in \mathcal{G}} \frac{1}{T_0} \sum_{t \in I^c} \left\| \begin{pmatrix} \hat{f}_t \hat{e}_{jt} \\ \hat{g}_t \end{pmatrix} (\lambda'_j H_f^{-1} \hat{e}_{jt}, \alpha'_j H_g^{-1}) \right\|^2 \left\| \begin{pmatrix} H_1^{-1} f_t \\ H_2^{-1} g_t \end{pmatrix} \right\|^2 \\
&\leq O_P(C_{NT}^{-2}). \\
&\frac{1}{|\mathcal{G}|_0} \sum_{j \in \mathcal{G}} \|\Gamma_6^j\|^2 \\
&= \frac{1}{|\mathcal{G}|_0} \sum_{j \in \mathcal{G}} \left\| \sum_{d=1}^6 \frac{1}{T_0} \sum_{t \notin I} \begin{pmatrix} \hat{f}_t \hat{e}_{jt} \\ \hat{g}_t \end{pmatrix} (\lambda'_j H_f^{-1} \hat{e}_{jt}, \alpha'_j H_g^{-1}) B^{-1} \begin{pmatrix} H_1' \sum_{d=1}^4 B_{dt} \\ H_2' \sum_{d=5}^8 B_{dt} \end{pmatrix} \right\|^2 \\
&\leq O_P(C_{NT}^{-2}) \sum_{d=1}^8 \frac{1}{T_0} \sum_{t \in I^c} n_t \|B_{dt}\|^2 \\
&\quad + \sum_{d=1}^8 \frac{1}{T} \sum_t \|B_{dt}\|^2 \frac{1}{|\mathcal{G}|_0} \sum_{j \in \mathcal{G}} \frac{1}{T_0} \sum_{t \in I^c} \left\| \begin{pmatrix} \hat{f}_t \hat{e}_{jt} \\ \hat{g}_t \end{pmatrix} (\lambda'_j H_f^{-1} \hat{e}_{jt}, \alpha'_j H_g^{-1}) \right\|^2 = O_P(C_{NT}^{-2}).
\end{aligned}$$

Step 3: $\frac{1}{|\mathcal{G}|_0} \sum_{j \in \mathcal{G}} \|\mathcal{U}(\Gamma_d^j)\|^2$ for $d = 1, 6$. Let

$$q_t = \frac{1}{|\mathcal{G}|_0} \sum_{j \in \mathcal{G}} \sqrt{e_{jt}^4 + e_{jt}^2 (\|f_t\|^2 + \|g_t\|^2) (\|f_t\|^2 + 1)}$$

$$\begin{aligned}
\frac{1}{|\mathcal{G}|_0} \sum_{j \in \mathcal{G}} \|\mathcal{U}(\Gamma_1^j)\|^2 &= \frac{1}{|\mathcal{G}|_0} \sum_{j \in \mathcal{G}} \left\| \frac{1}{T_0} \sum_{t \in I^c} \widehat{f}_t \widehat{e}_{jt} (\lambda'_j H_f^{-1} \widehat{e}_{jt}, \alpha'_j H_g^{-1}) (A_{1t,a} + A_{1t,b} + A_{1t,c}) \begin{pmatrix} H_1^{-1} f_t \\ H_2^{-1} g_t \end{pmatrix} \right\|^2 \\
&\leq O_P(C_{NT}^{-4}) + O_P(C_{NT}^{-2}) \frac{1}{T_0} \sum_{t \in I^c} q_t \|\widehat{S}_t - S\|^2 \\
&\quad + O_P(1) \frac{1}{T_0} \sum_{t \in I^c} [\|\Delta_{t1,1}\|^2 + \|\Delta_{t1,2}\|^2] + O_P(1) \frac{1}{|\mathcal{G}|_0} \sum_{j \in \mathcal{G}} \left\| \frac{1}{T_0} \sum_{t \in I^c} f_t^2 e_{jt} \Delta_{t2,1} \right\|^2
\end{aligned}$$

where $A_{1t,a} = (\widehat{B}_t^{-1} - B^{-1})(\widehat{S}_t - S)$, $A_{1t,b} = (\widehat{B}_t^{-1} - B^{-1})S$, $A_{1t,c} = B^{-1}(\widehat{S}_t - S)$,

$$\widehat{S}_t - S = \begin{pmatrix} \Delta_{t1,1} & \Delta_{t1,2} \\ \Delta_{t2,1} & 0 \end{pmatrix}$$

$$\begin{aligned}
\Delta_{t1,1} &= \frac{1}{N} \sum_i \widetilde{\lambda}_i \lambda'_i H_1 e_{it} (\widehat{e}_{it} - e_{it}) + \widetilde{\lambda}_i \widetilde{\lambda}'_i (\widehat{e}_{it}^2 - e_{it}^2) + (\widetilde{\lambda}_i \lambda'_i H_1 - \widetilde{\lambda}_i \widetilde{\lambda}'_i) e_{it}^2 - H_1' \lambda_i (\lambda'_i H_1 - \widetilde{\lambda}'_i) \mathbb{E} e_{it}^2 \\
\Delta_{t1,2} &= \frac{1}{N} \sum_i \widetilde{\lambda}_i (\widehat{e}_{it} - e_{it}) (\alpha'_i H_2 - \widetilde{\alpha}'_i) + \widetilde{\lambda}_i e_{it} (\alpha'_i H_2 - \widetilde{\alpha}'_i) \\
\Delta_{t2,1} &= \frac{1}{N} \sum_i \widetilde{\alpha}_i \lambda'_i H_1 (e_{it} - \widehat{e}_{it}) + \widetilde{\alpha}_i (\lambda'_i H_1 - \widetilde{\lambda}'_i) (\widehat{e}_{it} - e_{it}) + \widetilde{\alpha}_i (\lambda'_i H_1 - \widetilde{\lambda}'_i) e_{it}
\end{aligned}$$

By Lemma G.8, $\frac{1}{T_0} \sum_{t \in I^c} [\|\Delta_{t1,1}\|^2 + \|\Delta_{t1,2}\|^2] = O_P(C_{NT}^{-4})$. It is straightforward to use Cauchy-Schwarz to show $\frac{1}{T_0} \sum_{t \in I^c} q_t \|\widehat{S}_t - S\|^2 = O_P(C_{NT}^{-2})$. In addition,

$$\begin{aligned}
\frac{1}{|\mathcal{G}|_0} \sum_{j \in \mathcal{G}} \left\| \frac{1}{T_0} \sum_{t \in I^c} f_t^2 e_{jt} \Delta_{t2,1} \right\|^2 &\leq \frac{1}{|\mathcal{G}|_0} \sum_{j \in \mathcal{G}} \left\| \frac{1}{T_0} \sum_{t \in I^c} \frac{1}{N} \sum_i \alpha_i \lambda_i f_t^2 e_{jt} (e_{it} - \widehat{e}_{it}) \right\|^2 \\
&+ \frac{1}{|\mathcal{G}|_0} \sum_{j \in \mathcal{G}} \left\| \frac{1}{N} \sum_i \frac{1}{T_0} \sum_{t \in I^c} f_t^2 e_{jt} \alpha_i (\lambda'_i H_1 - \widetilde{\lambda}'_i) (\widehat{e}_{it} - e_{it}) \right\|^2 \\
&+ \frac{1}{|\mathcal{G}|_0} \sum_{j \in \mathcal{G}} \left\| \frac{1}{N} \sum_i \frac{1}{T_0} \sum_{t \in I^c} f_t^2 \alpha_i (\lambda'_i H_1 - \widetilde{\lambda}'_i) (e_{it} e_{jt} - \mathbb{E} e_{it} e_{jt}) \right\|^2 \\
&+ \frac{1}{|\mathcal{G}|_0} \sum_{j \in \mathcal{G}} \left\| \frac{1}{N} \sum_i \frac{1}{T_0} \sum_{t \in I^c} f_t^2 \alpha_i (\lambda'_i H_1 - \widetilde{\lambda}'_i) \mathbb{E} e_{it} e_{jt} \right\|^2 + O_P(C_{NT}^{-4}).
\end{aligned}$$

By Lemma G.2, $\frac{1}{|\mathcal{G}|_0} \sum_{j \in \mathcal{G}} \left\| \frac{1}{T_0} \sum_{t \in I^c} \frac{1}{N} \sum_i \alpha_i \lambda_i f_t^2 e_{jt} (e_{it} - \widehat{e}_{it}) \right\|^2 = O_P(C_{NT}^{-4})$. The middle two terms follow from Cauchy-Schwarz. Also, by Cauchy-Schwarz

$$\begin{aligned}
&\frac{1}{|\mathcal{G}|_0} \sum_{j \in \mathcal{G}} \left\| \frac{1}{N} \sum_i \alpha_i (\lambda'_i H_1 - \widetilde{\lambda}'_i) \frac{1}{T_0} \sum_{t \in I^c} f_t^2 \mathbb{E} e_{it} e_{jt} \right\|^2 \\
&= O_P(C_{NT}^{-4}) \max_j \sum_{i=1}^N |\mathbb{E} e_{it} e_{jt}| = O_P(C_{NT}^{-4}).
\end{aligned}$$

Hence $\frac{1}{|\mathcal{G}|_0} \sum_{j \in \mathcal{G}} \left\| \frac{1}{T_0} \sum_{t \in I^c} f_t^2 e_{jt} \Delta_{t2,1} \right\|^2 = O_P(C_{NT}^{-4})$. Thus $\frac{1}{|\mathcal{G}|_0} \sum_{j \in \mathcal{G}} \|\mathcal{U}(\Gamma_1^j)\|^2 = O_P(C_{NT}^{-4})$.

Finally,

$$\frac{1}{|\mathcal{G}|_0} \sum_{j \in \mathcal{G}} \|\mathcal{U}(\Gamma_6^j)\|^2 = \frac{1}{|\mathcal{G}|_0} \sum_{j \in \mathcal{G}} \left\| \frac{1}{T_0} \sum_{t \in I^c} \hat{f}_t \hat{e}_{jt} (\hat{e}_{jt} \lambda'_j H_f^{-1}, \alpha'_j H_g^{-1}) B^{-1} \begin{pmatrix} H'_1 & 0 \\ 0 & H_2 \end{pmatrix} \begin{pmatrix} \sum_{d=1}^4 B_{dt} \\ \sum_{d=5}^8 B_{dt} \end{pmatrix} \right\|^2$$

It suffices to prove

$$\begin{aligned} \frac{1}{|\mathcal{G}|_0} \sum_{j \in \mathcal{G}} \left\| \frac{1}{T_0} \sum_{t \in I^c} \hat{f}_t \hat{e}_{jt}^2 B_{dt} \right\|^2 &= O_P(C_{NT}^{-4}), \quad d = 1 \sim 4 \\ \frac{1}{|\mathcal{G}|_0} \sum_{j \in \mathcal{G}} \left\| \frac{1}{T_0} \sum_{t \in I^c} \hat{f}_t \hat{e}_{jt} B_{dt} \right\|^2 &= O_P(C_{NT}^{-4}), \quad d = 5 \sim 8. \end{aligned}$$

This is proved by Lemma G.15. □

Lemma G.15. For B_{dt} defined in (G.16) and (G.17),

$$\begin{aligned} \frac{1}{|\mathcal{G}|_0} \sum_{j \in \mathcal{G}} \left\| \frac{1}{T_0} \sum_{t \in I^c} \hat{f}_t \hat{e}_{jt}^2 B_{dt} \right\|^2 &= O_P(C_{NT}^{-4}), \quad d = 1 \sim 4 \\ \frac{1}{|\mathcal{G}|_0} \sum_{j \in \mathcal{G}} \left\| \frac{1}{T_0} \sum_{t \in I^c} \hat{f}_t \hat{e}_{jt} B_{dt} \right\|^2 &= O_P(C_{NT}^{-4}), \quad d = 5 \sim 8. \end{aligned}$$

Proof. For $r = 1, 2$ and all $d = 1, \dots, 8$,

$$\begin{aligned} &\frac{1}{|\mathcal{G}|_0} \sum_{j \in \mathcal{G}} \left\| \frac{1}{T_0} \sum_{t \in I^c} \hat{f}_t \hat{e}_{jt}^r B_{dt} \right\|^2 \leq \max_{jt} |\hat{e}_{jt}^r - e_{jt}^r| \frac{1}{T_0} \sum_{t \in I^c} \|\hat{f}_t - H_f f_t\|^2 \frac{1}{T_0} \sum_{t \in I^c} \|B_{dt}\|^2 \\ &+ \frac{1}{|\mathcal{G}|_0} \sum_{j \in \mathcal{G}} \frac{1}{T_0} \sum_{t \in I^c} \|\hat{f}_t - H_f f_t\|^2 e_{jt}^{2r} \frac{1}{T_0} \sum_{t \in I^c} \|B_{dt}\|^2 + \frac{1}{|\mathcal{G}|_0} \sum_{j \in \mathcal{G}} \left\| \frac{1}{T_0} \sum_{t \in I^c} f_t \hat{e}_{jt}^r B_{dt} \right\|^2 \\ &\leq O_P(C_{NT}^{-4}) + \frac{1}{|\mathcal{G}|_0} \sum_{j \in \mathcal{G}} \left\| \frac{1}{T_0} \sum_{t \in I^c} f_t e_{jt}^r B_{dt} \right\|^2 + \frac{1}{|\mathcal{G}|_0} \sum_{j \in \mathcal{G}} \left\| \frac{1}{T_0} \sum_{t \in I^c} f_t (\hat{e}_{jt}^r - e_{jt}^r) B_{dt} \right\|^2. \end{aligned}$$

It remains to show the following three steps.

Step 1. Show $\frac{1}{|\mathcal{G}|_0} \sum_{j \in \mathcal{G}} \left\| \frac{1}{T_0} \sum_{t \in I^c} f_t e_{jt}^2 B_{dt} \right\|^2 = O_P(C_{NT}^{-4})$ for $d = 1 \sim 4$.

When $d = 1, 4$, it follows from Cauchy-Schwarz.

$$\frac{1}{|\mathcal{G}|_0} \sum_{j \in \mathcal{G}} \left\| \frac{1}{T_0} \sum_{t \in I^c} f_t e_{jt}^2 B_{2t} \right\|^2 = O_P(C_{NT}^{-4}) + \frac{1}{|\mathcal{G}|_0} \sum_{j \in \mathcal{G}} \left\| \frac{1}{T_0} \sum_{t \in I^c} f_t^2 e_{jt}^2 \frac{1}{N} \sum_i \lambda_i^2 e_{it} (\hat{e}_{it} - e_{it}) \right\|^2,$$

which is $O_P(C_{NT}^{-4})$ due to Lemma G.2.

Next, by Lemma G.5, for any fixed sequence c_i , $\frac{1}{T} \sum_{t \notin I} \|\frac{1}{N} \sum_i c_i e_{it} (\dot{\lambda}_i - H'_1 \lambda_i)\|^2 = O_P(C_{NT}^{-4})$. So

$$\begin{aligned} \frac{1}{|\mathcal{G}|_0} \sum_{j \in \mathcal{G}} \left\| \frac{1}{T_0} \sum_{t \in I^c} f_t e_{jt}^2 B_{3t} \right\|^2 &= O_P(C_{NT}^{-4}) + \frac{1}{|\mathcal{G}|_0} \sum_{j \in \mathcal{G}} \left\| \frac{1}{T_0} \sum_{t \in I^c} f_t^2 e_{jt}^2 \frac{1}{N} \sum_i \lambda_i e_{it} l'_i w_t (\dot{\lambda}_i - H'_1 \lambda_i)' \right\|^2 \\ &\leq O_P(C_{NT}^{-4}). \end{aligned}$$

Step 2. Show $\frac{1}{|\mathcal{G}|_0} \sum_{j \in \mathcal{G}} \left\| \frac{1}{T_0} \sum_{t \in I^c} f_t e_{jt} B_{dt} \right\|^2 = O_P(C_{NT}^{-4})$ for $d = 5 \sim 8$.

When $d = 5$, it follows from Cauchy-Schwarz.

Next by Lemma G.2,

$$\begin{aligned} \frac{1}{|\mathcal{G}|_0} \sum_{j \in \mathcal{G}} \left\| \frac{1}{T_0} \sum_{t \in I^c} f_t e_{jt} B_{6t} \right\|^2 &= \frac{1}{|\mathcal{G}|_0} \sum_{j \in \mathcal{G}} \left\| \frac{1}{T_0} \sum_{t \in I^c} f_t^2 e_{jt} \frac{1}{N} \sum_i \alpha_i (\hat{e}_{it} - e_{it}) \lambda_i \right\|^2 + O_P(C_{NT}^{-4}) \\ &\leq O_P(C_{NT}^{-4}). \\ \frac{1}{|\mathcal{G}|_0} \sum_{j \in \mathcal{G}} \left\| \frac{1}{T_0} \sum_{t \in I^c} f_t e_{jt} B_{7t} \right\|^2 &= \frac{1}{|\mathcal{G}|_0} \sum_{j \in \mathcal{G}} \left\| \frac{1}{T_0} \sum_{t \in I^c} f_t^2 e_{jt} w_t \right\|^2 \left\| \frac{1}{N} \sum_i \alpha_i l_i (\dot{\lambda}_i - H'_1 \lambda_i) \right\|^2 + O_P(C_{NT}^{-4}) \\ &\leq O_P(C_{NT}^{-4}). \end{aligned}$$

Next, by Lemma G.6,

$$\begin{aligned} \frac{1}{|\mathcal{G}|_0} \sum_{j \in \mathcal{G}} \left\| \frac{1}{T_0} \sum_{t \in I^c} f_t e_{jt} B_{8t} \right\|^2 &= \frac{1}{|\mathcal{G}|_0} \sum_{j \in \mathcal{G}} \left\| \frac{1}{T_0} \sum_{t \in I^c} f_t e_{jt} (\tilde{f}_t - H_1^{-1} f_t) w_t \right\|^2 \left\| \frac{1}{N} \sum_i \alpha_i l_i \lambda_i \right\|^2 \\ &\leq O_P(C_{NT}^{-4}). \end{aligned}$$

Step 3. Show $\frac{1}{|\mathcal{G}|_0} \sum_{j \in \mathcal{G}} \left\| \frac{1}{T_0} \sum_{t \in I^c} f_t (\hat{e}_{jt}^r - e_{jt}^r) B_{dt} \right\|^2 = O_P(C_{NT}^{-4})$ for $d = 1 \sim 8$.

It is bounded by Cauchy-Schwarz.

□

G.5. Technical lemmas for covariance estimation.

Lemma G.16. Suppose uniformly over $i \leq N$, the following terms are $o_P(1)$: $q \|\frac{1}{T} \sum_t f_t e_{it} u_{it}\|$, $q \|\frac{1}{T} \sum_t g_t u_{it}\|$, $q \|\frac{1}{T} \sum_s f_s w'_s e_{is}\|$, $q \|\frac{1}{T} \sum_{tj} c_j (m_{jt} m_{it} - \mathbb{E} m_{jt} m_{it})\|$ for $m_{it} \in \{e_{it} u_{it}, u_{it}\}$, $|\frac{1}{T} \sum_t e_{it}^2 - \mathbb{E} e_{it}^2|$. Suppose $q \max_{t \leq T} \|\frac{1}{N} \sum_i \lambda_i \omega_{it} + \alpha_i u_{it}\| = o_P(1)$ and $q C_{NT}^{-1} (\max_{it} |e_{it} u_{it}|) = o_P(1)$ (which are satisfied given our assumptions), where

$$q := \max_{it} (\|f_t x_{it}\| + 1 + |x_{it}|).$$

Then (i) $\max_i [\|\hat{\lambda}_i - H_f'^{-1} \lambda_i\| + \|\hat{\alpha}_i - H_g'^{-1} \alpha_i\|] q = o_P(1)$.

(ii) $\max_{t \leq T} [\|\hat{f}_t - H_f f_t\| + \|\hat{g}_t - H_g g_t\|] q = o_P(1)$.

(iii) $\max_{is} |\hat{u}_{is} - u_{is}| = o_P(1)$,

(iv) $\max_i |\hat{\sigma}_i^2 - \sigma_i^2| = o_P(1)$, where $\sigma_i^2 = \mathbb{E}e_{it}^2$ and $\hat{\sigma}_i^2 = \frac{1}{T} \sum_t \hat{e}_{it}^2$.

Proof. (i) In view of (F.6), it suffices to show $\max_i \|R_{di}\|_q = o_P(1)$ for all $d = 1 \sim 6$. First, by our assumption, $q \max_i \|R_{1i}\| = o_P(1)$. Also by Lemma G.10, $\max_i \|\hat{D}_i^{-1} - D_i^{-1}\| = o_P(1)$. Also, $q \max_i \|R_{2i}\| = o_P(1)$ follows from $q \max_{it} |\hat{e}_{it} - e_{it}| |e_{it} + u_{it}| = o_P(1)$.

Term $\max_i [\|R_{3i}\| + \|R_{4i}\|]$. It suffices to prove $q \max_i \|R_{3i,d}\| = o_P(1)$ for $R_{3i,d}$ defined in Lemma G.12.

$$\begin{aligned} \max_i \|R_{3i,1}\|_q &\leq o_P(1) + q \max_i \left\| \frac{1}{T} \sum_s (\hat{f}_s - H_f f_s) (\hat{e}_{is} - e_{is}) (\mu_{it} \lambda'_i f_s - \hat{\mu}_{it} \dot{\lambda}'_i \tilde{f}_s) \right\| \\ &\quad + q \max_i \left\| \frac{1}{T} \sum_s (\hat{f}_s - H_f f_s) e_{is} (\mu_{it} \lambda'_i f_s - \hat{\mu}_{it} \dot{\lambda}'_i \tilde{f}_s) \right\| \\ &\quad + q O_P(1) \max_i \left\| \frac{1}{T} \sum_s f_s (\hat{e}_{is} - e_{is}) (\mu_{it} \lambda'_i f_s - \hat{\mu}_{it} \dot{\lambda}'_i \tilde{f}_s) \right\| \\ &= o_P(1) \end{aligned}$$

following from the Cauchy-Schwarz, $q \max_{it} |\hat{e}_{it} - e_{it}| |e_{it}| = o_P(1)$, $q C_{NT}^{-1} \max_{it} e_{it}^2 = o_P(1)$ and Lemma G.17. Similarly, $\max_i \|R_{3i,d}\|_q = o_P(1)$ for all $d = 2 \sim 7$.

Term $\max_i \|R_{5i}\|$. By the assumption:

$$q \max_i \left\| \frac{1}{T} \sum_{tj} \lambda_j (\omega_{jt} \omega_{it} - \mathbb{E} \omega_{jt} \omega_{it}) \right\| + q \max_i \left\| \frac{1}{T} \sum_{tj} \alpha_j (u_{jt} u_{it} - \mathbb{E} u_{jt} u_{it}) \right\| = o_P(1).$$

It suffices to prove for A_{dt} defined in (F.2),

$$q \sum_{d=1}^6 \max_i \left\| \frac{1}{T} \sum_t \begin{pmatrix} \hat{e}_{it} & 0 \\ 0 & 1 \end{pmatrix} A_{dt} u_{it} \right\| = o_P(1).$$

First, for $A_{1t,a} = (\hat{B}_t^{-1} - B^{-1})(\hat{S}_t - S)$, $A_{1t,b} = (\hat{B}_t^{-1} - B^{-1})S$, $A_{1t,c} = B^{-1}(\hat{S}_t - S)$,

$$\begin{aligned} q \max_i \left\| \frac{1}{T} \sum_t \begin{pmatrix} \hat{e}_{it} & 0 \\ 0 & 1 \end{pmatrix} A_{1t} u_{it} \right\| &\leq q \max_i \left\| \frac{1}{T} \sum_t \begin{pmatrix} e_{it} & 0 \\ 0 & 1 \end{pmatrix} A_{1t,a} \begin{pmatrix} H_1^{-1} f_t \\ H_2^{-1} g_t \end{pmatrix} u_{it} \right\| \\ &\quad + q \max_i \left\| \frac{1}{T} \sum_t \begin{pmatrix} e_{it} & 0 \\ 0 & 1 \end{pmatrix} A_{1t,b} \begin{pmatrix} H_1^{-1} f_t \\ H_2^{-1} g_t \end{pmatrix} u_{it} \right\| \\ &\quad + q \max_i \left\| \frac{1}{T} \sum_t \begin{pmatrix} e_{it} & 0 \\ 0 & 1 \end{pmatrix} A_{1t,c} \begin{pmatrix} H_1^{-1} f_t \\ H_2^{-1} g_t \end{pmatrix} u_{it} \right\| + o_P(1) \\ &= q O_P(C_{NT}^{-1}) (\max_{it} |e_{it} u_{it}|) + o_P(1) = o_P(1). \end{aligned}$$

Similarly, $d = 2 \sim 6$, Cauchy-Schwarz and $qO_P(C_{NT}^{-1})(\max_{it} |e_{it}u_{it}|) = o_P(1)$ imply

$$q \max_i \left\| \frac{1}{T} \sum_t \begin{pmatrix} \widehat{e}_{it} & 0 \\ 0 & 1 \end{pmatrix} A_{dt} u_{it} \right\| \leq o_P(1).$$

Term $\max_i \|R_{6i}\|$. Lemma G.10 shows $\max_i \|\widehat{D}_i^{-1}\| = O_P(1)$. It suffices to prove $q \max_i \|\Gamma_d^i\| = o_P(1)$, where Γ_d^i is as defined in the proof of Lemma G.14.

$$\begin{aligned} q \max_j \|\Gamma_0^j\| &= q \max_j \left\| \frac{1}{T_0} \sum_{t \notin I} \begin{pmatrix} \widehat{f}_t \widehat{e}_{jt} \\ \widehat{g}_t \end{pmatrix} (\lambda'_j H_f^{-1} \widehat{e}_{jt} b_1, \alpha'_j H_g^{-1} b_2) \frac{1}{N} \sum_i \begin{pmatrix} H'_1 \lambda_i e_{it} \\ H'_2 \alpha_i \end{pmatrix} u_{it} \right\| \\ &= qO_P(C_{NT}^{-1}) \max_j |\widehat{e}_{jt}^2 + \widehat{e}_{jt}| = o_P(1) \\ \max_j \|\Gamma_d^j\| &= q \max_j \left\| \frac{1}{T_0} \sum_{t \notin I} \begin{pmatrix} \widehat{f}_t \widehat{e}_{jt} \\ \widehat{g}_t \end{pmatrix} (\lambda'_j H_f^{-1} \widehat{e}_{jt}, \alpha'_j H_g^{-1}) A_{dt} \right\| = o_P(1) \end{aligned}$$

due to $O_P(C_{NT}^{-1}) \max_j |\widehat{e}_{jt}^2 + \widehat{e}_{jt}| = o_P(1)$ and $\max_i \|\dot{\lambda}_i - H'_1 \lambda_i\| = o_P(1)$.

(ii) In view of (F.2), we have $\max_{t \leq T} \left\| \frac{1}{N} \sum_i \lambda_i \omega_{it} + \alpha_i u_{it} \right\| = o_P(1)$. So it suffices to prove $\max_{t \leq T} \|A_{dt}\|q = o_P(1)$ for $d = 1 \sim 8$.

The conclusion of Lemma G.8 can be strengthened to ensure that

$q \max_{t \leq T} (\|f_t\| + \|g_t\|) \max_{t \leq T} [\|\widehat{S}_t - S\| + \|\widehat{B}_t^{-1} - B^{-1}\|] = o_P(1)$. So this is true for $d = 1, 2$. Next, $C_{NT}^{-1}q = o_P(1)$ and Lemma G.17 shows $q \max_t \|\widetilde{f}_t - H_1^{-1} f_t\| = o_P(1)$. So it is easy to verify using the Cauchy-Schwarz that it also holds true for $d = 3 \sim 8$.

(iii) Uniformly in i, s , by part (ii),

$$\begin{aligned} |\widehat{u}_{is} - u_{is}| &\leq |x_{it}| \|\widehat{\lambda}_i - \lambda_i\| \|\widehat{f}_t\| + C |x_{it}| \|\widehat{f}_t - f_t\| \\ &\quad + \|\widehat{\alpha}_i - \alpha_i\| \|\widehat{g}_t\| + C \|\widehat{g}_t - g_t\| = o_P(1). \end{aligned}$$

(iv) Uniformly in $i \leq N$,

$$\frac{1}{T} \sum_t \widehat{e}_{it}^2 - \mathbb{E} e_{it}^2 \leq \left| \frac{1}{T} \sum_t \widehat{e}_{it}^2 - e_{it}^2 \right| + \frac{1}{T} \sum_t e_{it}^2 - \mathbb{E} e_{it}^2 = o_P(1).$$

□

Lemma G.17. Suppose the following are $o_P(1)$ (which are satisfied given our assumptions): $\max_{it} |\widehat{e}_{it} - e_{it}| |e_{it} + u_{it}|$, $C_{NT}^{-1} \max_{it} |x_{it}|^3 \|f_t\|$, $\max_i \left| \frac{1}{T} \sum_t u_{it}^2 - \mathbb{E} u_{it}^2 \right|$, $q \max_i \left\| \frac{1}{T} \sum_t f_t e_{it} u_{it} \right\|$, $q \max_i \left\| \frac{1}{T} \sum_t f_t w_t u_{it} \right\|$, $q \max_{it} |x_{it}| \max_i \left\| \frac{1}{T} \sum_t u_{it} g_t \right\|$, $\max_i \left\| \frac{1}{T} \sum_t \omega_{it}^2 - \mathbb{E} \omega_{it}^2 \right\| + \max_i \left\| \frac{1}{T} \sum_t u_{it}^2 w_t^2 - \mathbb{E} u_{it}^2 w_t^2 \right\|$,

$$\max_{t \leq T} |\frac{1}{N} \sum_i u_{it}^2 - \mathbb{E} u_{it}^2|, q \max_{t \leq T} [\|\frac{1}{N} \sum_i \lambda_i e_{it} u_{it}\| + \|\frac{1}{N} \sum_i \lambda_i w_t u_{it}\|], q \max_{it} \max_{t \leq T} \|\frac{1}{N} \sum_i u_{it} \alpha_i\|, \\ \max_{t \leq T} |\frac{1}{N} \sum_i \omega_{it}^2 - \mathbb{E} \omega_{it}^2|, \max_i \|\frac{1}{N} \sum_i u_{it}^2 w_t w'_t - \mathbb{E} u_{it}^2 w_t w'_t\|,$$

where

$$q := \max_{it} (\|f_t x_{it}\| + 1 + |x_{it}|).$$

Then (i) $q \max_{it} \|\dot{\lambda}_i - H'_1 \lambda_i\| = o_P(1)$. (ii) $q \max_t \|\tilde{f}_t - H_1^{-1} f_t\| = o_P(1)$.

Proof. Recall

$$\begin{aligned} \dot{\lambda}_i &= \tilde{D}_{\lambda i}^{-1} \frac{1}{T} \tilde{F}' \text{diag}(X_i) M_{\tilde{g}} y_i \\ &= H'_1 \lambda_i + \tilde{D}_{\lambda i}^{-1} \frac{1}{T} \tilde{F}' \text{diag}(X_i) M_{\tilde{g}} (G H_2^{-1'} - \tilde{G}) H'_2 \alpha_i \\ &\quad + \tilde{D}_{\lambda i}^{-1} \frac{1}{T} \tilde{F}' \text{diag}(X_i) M_{\tilde{g}} \text{diag}(X_i) (F H_1^{-1'} - \tilde{F}) H'_1 \lambda_i + \tilde{D}_{\lambda i}^{-1} \frac{1}{T} \tilde{F}' \text{diag}(X_i) M_{\tilde{g}} u_i. \end{aligned}$$

(i) By Lemma G.4, $\max_i \|\tilde{D}_{\lambda i}^{-1}\| = O_P(1)$.

First we show $q \max_i \|\frac{1}{T} \tilde{F}' \text{diag}(X_i) M_{\tilde{g}} (G H_2^{-1'} - \tilde{G})\| = o_P(1)$. It is in fact bounded by

$$O_P(C_{NT}^{-1}) \max_{it} |x_{it}| q = o_P(1).$$

Next, $q \max_i \|\frac{1}{T} \tilde{F}' \text{diag}(X_i) M_{\tilde{g}} \text{diag}(X_i) (F H_1^{-1'} - \tilde{F}) H'_1 \lambda_i\|$ is bounded by

$$O_P(C_{NT}^{-1}) q \max_{it} |x_{it}|^2 = o_P(1)$$

Finally, note that $\max_i \frac{1}{T} \|u_i\|^2 = \max_i |\frac{1}{T} \sum_t u_{it}^2 - \mathbb{E} u_{it}^2| + O_P(1) = O_P(1)$. So

$$\begin{aligned} & q \max_i \|\frac{1}{T} \tilde{F}' \text{diag}(X_i) M_{\tilde{g}} u_i\| \\ & \leq q O_P(C_{NT}^{-1}) \max_{it} |x_{it}| \max_i \frac{1}{\sqrt{T}} \|u_i\| + q \max_i \|\frac{1}{T} \sum_t \tilde{f}_t x_{it} u_{it}\| \\ & \quad + O_P(1) q \max_i \|\frac{1}{T} \tilde{F}' \text{diag}(X_i) G\| \frac{1}{T} \max_i \|G' u_i\| \\ & \leq o_P(1) + O_P(C_{NT}^{-1}) q \max_i (\frac{1}{T} \sum_t u_{it}^2 x_{it}^2)^{1/2} \\ & \quad + O_P(1) q \max_i [\|\frac{1}{T} \sum_t f_t e_{it} u_{it}\| + \|\frac{1}{T} \sum_t f_t w_t u_{it}\|] \\ & \quad + O_P(1) q \max_{it} |x_{it}| \max_i \|\frac{1}{T} \sum_t u_{it} g_t\| \\ & \leq o_P(1) + O_P(C_{NT}^{-1}) q \max_i (\frac{1}{T} \sum_t \omega_{it}^2 - \mathbb{E} \omega_{it}^2)^{1/2} \end{aligned}$$

$$+O_P(C_{NT}^{-1})q \max_i \left(\frac{1}{T} \sum_t u_{it}^2 w_t^2 - \mathbb{E} u_{it}^2 w_t^2 \right)^{1/2} = o_P(1).$$

(ii) Recall that

$$\begin{aligned} \tilde{f}_s &= \tilde{D}_{fs}^{-1} \frac{1}{N} \tilde{\Lambda}' \text{diag}(X_s) M_{\tilde{\alpha}} y_s \\ &= H_1^{-1} f_s + \tilde{D}_{fs}^{-1} \frac{1}{N} \tilde{\Lambda}' \text{diag}(X_s) M_{\tilde{\alpha}} (A H_2 - \tilde{A}) H_2^{-1} g_s \\ &\quad + \tilde{D}_{fs}^{-1} \frac{1}{N} \tilde{\Lambda}' \text{diag}(X_s) M_{\tilde{\alpha}} \text{diag}(X_s) (\Lambda H_1 - \tilde{\Lambda}) H_1^{-1} f_s \\ &\quad + \tilde{D}_{fs}^{-1} \frac{1}{N} \tilde{\Lambda}' \text{diag}(X_s) M_{\tilde{\alpha}} u_s. \end{aligned}$$

By Lemma G.3, $\max_s \|\tilde{D}_{fs}^{-1}\| = O_P(1)$. The rest of the proof is very similar to part (i), so we omit details to avoid repetitions. The only extra assumption, parallel to those of part (i), is that the following terms are $o_P(1)$: $\max_{t \leq T} |\frac{1}{N} \sum_i u_{it}^2 - \mathbb{E} u_{it}^2|$, $q \max_{t \leq T} [\|\frac{1}{N} \sum_i \lambda_i e_{it} u_{it}\| + \|\frac{1}{N} \sum_i \lambda_i w_{it} u_{it}\|]$, $q \max_{it} \max_{t \leq T} \|\frac{1}{N} \sum_i u_{it} \alpha_i\|$, $\max_{t \leq T} |\frac{1}{N} \sum_i \omega_{it}^2 - \mathbb{E} \omega_{it}^2|$, $\max_i \|\frac{1}{N} \sum_i u_{it}^2 w_t w'_t - \mathbb{E} u_{it}^2 w_t w'_t\|$.

□

APPENDIX H. WHEN x_{it} FOLLOWS MANY-MEAN MODEL

Consider the decomposition

$$x_{it} = \mu_{it} + e_{it}$$

where μ_{it} is the “mean process” of x_{it} . As we stated in the main paper, we require μ_{it} be well estimable to apply the partialing-out method. The main text considers the case when μ_{it} follows a factor structure. In this section, we consider another example known as the simple “many-mean” model:

$$\mu_{it} = \mu_i \quad \forall i, t$$

for some unknown non-random mean μ_i . This structure admits simple estimators: $\bar{x}_i = \frac{1}{T} \sum_t x_{it}$ and $\hat{e}_{it} = x_{it} - \bar{x}_i$. The main body of the proof goes through to show that $\hat{e}_{it}$ is a high quality estimator for e_{it} , and the inferential theory carries over in this case.

The inferential theory follows from combining Proposition H.1, whose proof is lengthy but very similar to the proof of the case when μ_{it} admits a factor model. So we omit the proof for brevity.

Proposition H.1. *We have*

$$\widehat{f}_t - H_f f_t = H_f \left(\frac{1}{N} \sum_i \lambda_i \lambda'_i \mathbf{E} e_{it}^2 \right)^{-1} \frac{1}{N} \sum_i \lambda_i e_{it} u_{it} + O_P(C_{NT}^{-2}).$$

$$\widehat{\lambda}_i = H_f'^{-1} \lambda_i + H_f'^{-1} \left(\frac{1}{T_0} \sum_{s \in I^c} f_s f'_s \mathbf{E} e_{is}^2 \right)^{-1} \frac{1}{T_0} \sum_{s \in I^c} f_s e_{is} u_{is} + O_P(C_{NT}^{-2}).$$

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